

Liquidity Illusion: The General Equilibrium Theory of Private Capital Valuation

Graduate Finance Course · 32 Sessions · Spring 2027

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1 The General Equilibrium Theory of Private Capital Valuation

Graduate Finance Course · Spring 2027 · 32 Sessions · 3 Credit Hours

Samir Asaf, PhD, CFA, CMA, CTP, CM&AA
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1.1 Why This Course Exists

The most dangerous phrase in finance is: “*assume the liquidity premium is constant.*” It is not a simplification — it is a mistake with large, measurable, and correctable consequences for managing trillions in institutional capital.

This course teaches the GE-LAV® framework — the first continuous-time general equilibrium model of liquidity pricing in private markets. Students will learn why conventional Discounted Cash Flow (DCF) analysis structurally misvalues private assets, what a correct theory must deliver, and how to implement liquidity-adjusted valuation across private equity, infrastructure, private credit, and real estate.

1.2 What You Will Build

By the end of the course, every student will have:

- A working command of the GE-LAV® valuation hierarchy (DCF → LAV → GE-LAV)
- The ability to identify and quantify the Jensen convexity bias in real private-market portfolios
- A complete liquidity-adjusted valuation of a real private market asset of their choice (the term project)
- Familiarity with the secondary market evidence that exposes the liquidity illusion
- Working knowledge of the GE-LAV® computational platform

Track 2 students will additionally:

- Derive the optimal exit boundary from first principles using stochastic control
- Prove the existence of a mean-field equilibrium under McKean-Vlasov dynamics
- Calibrate the Ornstein-Uhlenbeck liquidity process to empirical data
- Implement the Pigouvian exit tax under stress scenarios

1.3 Two-Track Design

This course recognizes that graduate finance students arrive with different mathematical backgrounds and different career trajectories. Rather than teaching to the lowest common denominator (which would frustrate quantitative students) or the highest (which would exclude practitioner-oriented students), the course is structured as **two parallel tracks merging into a common core**.

	Track 1 (Practitioner)	Track 2 (Researcher)
Sessions 1–24	Common core: 24 sessions identical for both tracks	
Sessions 25–31	Applied case workshops	Full mathematical derivations
Session 32	Project presentations (both tracks present)	
Math required	Asset pricing, basic calculus	Stochastic calculus, measure theory
Problem sets	4 applied (real data)	4 mathematical (proofs + numerics)
Best for	MBA / MSF / applied master's students	MFE / quant master's / PhD-track students

See [Tracks](#) for full declaration policy and decision guidance.

1.4 Course Structure at a Glance

- **Unit 1 (Sessions 1–5):** Why private market valuation is broken — DCF failures, the liquidity illusion, term structure
- **Unit 2 (Sessions 6–10):** Measurement and theory — IRR/PME, what a correct theory must deliver, midterm
- **Unit 3 (Sessions 11–18):** Exit timing, portfolio construction, regulation, implementation
- **Unit 4 (Sessions 19–24):** Math intuition bridges — OU, HJB, MFG, GE, Jensen, Pigouvian (concepts, not proofs)
- **Unit 5 (Sessions 25–31): Split track** — applied vs. mathematical deep dives
- **Capstone (Session 32):** Project presentations

See the [full 32-session schedule](#).

1.5 Assessments

Component	Weight
Midterm exam (Session 10)	20%
4 problem sets (track-differentiated)	25%
Term project (proposal + draft + presentation)	50%
Class participation	5%

Component	Weight
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There is no final exam. Session 32 is reserved for project presentations.

See the [project specification](#), [midterm blueprint](#), and [problem sets overview](#) for details.

1.6 Primary Text

Asaf, S. (2026). *Liquidity Illusion: The General Equilibrium Theory of Private Capital Valuation*. Oxford University Press / Regent Financial LLC, New York. 421 pp.

The book is structured in two parts that mirror the course's two tracks:

- **Part 1 (Chapters 1–9): The Practitioner's Guide** — required for all students
- **Part 2 (Chapters 10–20): The Researcher's Framework** — required for Track 2; recommended optional for Track 1
- **Chapter 21:** Future research directions

1.7 Instructor

Samir Asaf is Senior Partner at Regent Financial, a private markets advisory firm in New York. He holds a PhD in Finance, an MSc (Economics) from LSE, a PGDip in Mathematical Economics from Birkbeck, and a BSc (Economics) from Boston College. He is a CFA, CMA, CTP, CM&AA, and Chartered Director (FIoD, UK).

His research explores stochastic control, McKean-Vlasov mean-field games, and their application to private markets. The GE-LAV® framework introduced in the primary text is his main contribution to financial economics. He has advised leading institutional investors and fund managers on private equity, infrastructure, private credit, and cross-border M&A across North America, Europe, Asia-Pacific, and the Middle East.

His earlier work, *Executive Corporate Finance: The Business of Enhancing Shareholder Value* (Prentice Hall, 2004), has been adopted worldwide for MBA and executive education programs.

This course website is built with [Quarto](#) and is the canonical source for syllabus, schedule, lecture notes, problem sets, and the term project. All updates appear here first.

Part I

Course Foundations

2 Syllabus

2.1 Course Information

Course Title	Liquidity Illusion: The General Equilibrium Theory of Private Capital Valuation
Course Code	FIN [TBD] (graduate elective)
Term	Spring 2027
Credit Hours	3
Sessions	32 sessions $\times$ 75 minutes (16 weeks $\times$ 2 sessions/week)
Format	In-person with synchronous online option
Instructor	Samir Asaf, PhD, CFA, CMA, CTP, CM&AA
Office Hours	[Day, time] · Office: [TBD] · or by appointment
TA	[TBD]
Email	drsamirasaf@gmail.com
Course site	[course URL]

2.2 Course Description

This course teaches the GE-LAV® framework — a continuous-time general equilibrium model of liquidity-adjusted valuation in private capital markets. Students learn why conventional Discounted Cash Flow (DCF) methodology, IRR, and Public Market Equivalent (PME) systematically misvalue private assets when liquidity premia are stochastic and collectively determined.

The course covers the full valuation hierarchy DCF → LAV → GE-LAV, with applications to private equity, venture capital, infrastructure, private credit, and real estate. Students will use the GE-LAV® computational platform (liquidityillusion.com) to calibrate and apply the model to real private market data.

The course is offered in two parallel tracks: a Practitioner track (Track 1) and a Researcher track (Track 2). The two tracks share a 24-session common core and diverge for sessions 25–31 before reunifying at session 32 for project presentations.

2.3 Prerequisites

For both tracks:

- Graduate-level Corporate Finance or Asset Pricing
- Comfort with discounted cash flow analysis, NPV, IRR, and CAPM
- Working knowledge of Excel or Python for financial modeling
- Basic probability and statistics (random variables, expectation, variance, normal distribution)

Additional for Track 2 only:

- Stochastic processes or stochastic calculus (one course)
- Measure-theoretic probability OR willingness to do supplementary reading from the course appendix during Sessions 1–4

If you are uncertain whether your background qualifies you for Track 2, contact the instructor before the second week of class. Track selection is final after Session 5.

2.4 Learning Objectives

By the end of this course, students will be able to:

2.4.1 Common Learning Objectives (Both Tracks)

1. **Diagnose** the three structural failures of DCF in private market valuation and quantify their magnitude using secondary market evidence.
2. **Explain** the Jensen convexity bias and compute its magnitude for a given asset class, horizon, and liquidity state.
3. **Identify** when an investor should hold, sell, or delay exit from an illiquid position using the GE-LAV exit boundary.
4. **Construct** a liquidity-adjusted portfolio that accounts for hedge demand against aggregate liquidity shocks.
5. **Evaluate** the regulatory implications of liquidity-adjusted valuation for Solvency II, AIFMD, IFRS 13, and Basel III.
6. **Apply** the GE-LAV® platform to calibrate model parameters and produce valuations for real private market assets.
7. **Critique** standard performance metrics (IRR, PME) and propose liquidity-adjusted alternatives (LA-IRR, LA-PME).
8. **Communicate** GE-LAV results to non-technical audiences (investment committees, boards, regulators).

2.4.2 Track 2 Additional Learning Objectives

T2.1. **Derive** the Hamilton-Jacobi-Bellman equation for the LP's optimal stopping problem and identify the smooth pasting condition at the exit boundary.

T2.2. **Prove** the existence and uniqueness of a McKean-Vlasov mean-field equilibrium under the GE-LAV stability condition.

T2.3. **Solve** the Fokker-Planck equation for the stationary distribution of the Ornstein-Uhlenbeck liquidity process.

- T2.4. **Implement** numerical solution of the HJB master equation on Wasserstein space.
- T2.5. **Calibrate** the GE-LAV stochastic parameters to empirical secondary market data using maximum likelihood estimation.
- T2.6. **Derive** the Pigouvian exit tax that implements the constrained social optimum.

2.5 Primary Text

Asaf, S. (2026). *Liquidity Illusion: The General Equilibrium Theory of Private Capital Valuation*. Oxford University Press / Regent Financial LLC. ISBN 9798881716103. **Required.**

The book is structured for both audiences:

- **Part 1 (Chapters 1–9):** The Practitioner’s Guide — required for both tracks
- **Part 2 (Chapters 10–20):** The Researcher’s Framework — required for Track 2; recommended optional for Track 1
- **Chapter 21:** Future research directions

2.6 Supplementary Readings

A full reading list appears in [Appendix B](#). Selected supplementary readings include:

For practitioner context:

- Stulz, R. M. (2007). Hedge funds: Past, present, and future. *Journal of Economic Perspectives*.
- Phalippou, L. (2020). *Private Equity Laid Bare*. The Enterprising Investor.
- IPEV Valuation Guidelines (2022). International Private Equity and Venture Capital Valuation Guidelines.

For mathematical context (Track 2):

- Shreve, S. (2004). *Stochastic Calculus for Finance II*. Springer.
- Karatzas, I. and Shreve, S. (1991). *Brownian Motion and Stochastic Calculus*. Springer.
- Carmona, R. and Delarue, F. (2018). *Probabilistic Theory of Mean-Field Games*. Springer.

2.7 Assessments and Grading

Component	Weight	Track 1	Track 2
Midterm exam (Session 10)	20%	Same exam	Same exam
Problem sets (4)	25%	Applied	Mathematical
Term project — proposal (Session 14)	5%		
Term project — draft (Session 30)	15%		
Term project — presentation (Session 32)	30%		
Class participation	5%		
Total	100%		

Component	Weight	Track 1	Track 2
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2.7.1 Grading Scale

Letter	Percentage	GPA
A	93–100	4.0
A–	90–92	3.7
B+	87–89	3.3
B	83–86	3.0
B–	80–82	2.7
C+	77–79	2.3
C	73–76	2.0
C–	70–72	1.7
D	60–69	1.0
F	< 60	0.0

2.7.2 Midterm Exam

The midterm exam is held in **Session 10** and covers Sessions 1–9 (book chapters 1–4 and selected concepts from chapter 5). The exam is identical for both tracks. The midterm is **closed-book, 75 minutes, with one 8.5×11” double-sided handwritten formula sheet permitted**. See [midterm blueprint](#) for topic weighting and sample questions.

2.7.3 Problem Sets

Four problem sets total, **track-differentiated**:

- **Track 1 PSets:** Applied work using the GE-LAV® platform and real private market data (secondary market discount series, fund cash flows, PE database).
- **Track 2 PSets:** Mathematical derivations, proofs, and numerical implementations.

PSet	Drops	Due	Topics
PS1	Session 1	Session 8	Diagnosis, OU process, secondary market evidence
PS2	Session 9	Session 16	Exit timing, portfolio construction, regulation
PS3	Session 17	Session 24	Math bridges + Jensen bias + Pigouvian
PS4	Session 25	Session 31	Advanced applications (track-specific)

2.7.4 Term Project (50% of grade)

The term project is the integrative deliverable of the course. Students apply the full GE-LAV framework to a **real private-market asset of their choosing** (a specific PE fund, VC secondary, infrastructure asset, private credit deal, or real estate position).

Track 1 projects emphasize empirical implementation, regulatory analysis, and investment committee communication.

Track 2 projects emphasize theoretical extension, novel calibration, or mathematical analysis of a specific GE-LAV result.

See the [full project specification and rubric](#).

Project milestones:

- **Session 14 — Proposal due (5%):** 2-page proposal, asset selection, data plan, methodology outline
- **Session 30 — Draft due (15%):** 15-page draft paper with preliminary results
- **Session 32 — Presentation (30%):** 20-minute presentation + 5-minute Q&A

2.7.5 Class Participation (5%)

Participation is graded based on substantive contribution to in-class discussion, not mere attendance. Quality matters more than quantity. Students who join via synchronous online attendance are evaluated identically.

2.8 Course Policies

2.8.1 Attendance

Attendance is expected at every session. Students missing more than 4 sessions without documented cause forfeit participation credit. Sessions are recorded and posted to the course site within 24 hours.

2.8.2 Late Work

- Problem sets: 10% penalty per 24-hour period late, up to 72 hours. No credit after 72 hours.
- Project proposal: No late submissions; missed proposals receive 0.
- Project draft: 24-hour grace period without penalty; thereafter as PSets.
- Project presentation: Slot is fixed; missed presentations receive 0 unless documented emergency.

2.8.3 Academic Integrity

All submitted work must be your own. Collaboration on problem sets is permitted in groups of up to 3, but each student must write and submit their own solutions. Collaboration must be disclosed at the top of the submitted PSet. The term project is **individual work only** — no group projects.

Plagiarism, fabrication of results, and unauthorized collaboration are violations of the university's academic integrity policy and will result in a grade of F for the course and referral to the academic conduct office.

2.8.4 Use of AI Tools

This course adopts the following AI use policy:

- **Permitted with disclosure:** Using LLMs (ChatGPT, Claude, Gemini, etc.) for brainstorming, explanation of concepts, code debugging, and writing feedback. You must disclose AI use in submitted work with a brief note describing how AI was used.
- **Permitted with care:** Using the GE-LAV® platform's AI features (calibration suggestions, scenario generation). The platform is designed for this; use it freely.
- **Not permitted:** Submitting AI-generated work as your own. Specifically: AI-drafted problem set solutions, AI-written project sections, AI-generated proofs.
- **For the term project:** AI may be used as a research and writing aid. You must (a) understand every word, equation, and result in your submission; (b) be prepared to defend any part of your project orally during the presentation; (c) disclose AI use in a methodology note.

Bright-line test: If you cannot explain or reproduce a portion of your submission without AI assistance, that portion was not your work, and submitting it is an integrity violation. The Q&A at the project presentation is designed to test this.

2.8.5 Accessibility

Students with documented disabilities should contact the Disability Services office and notify the instructor by Session 3. Reasonable accommodations will be provided.

2.8.6 Religious Observance and Personal Emergencies

Notify the instructor as early as possible if you anticipate missing a session for religious observance, family emergency, or medical reasons. Reasonable accommodations will be made for the midterm, PSet deadlines, and project milestones with appropriate documentation.

2.8.7 Diversity and Inclusion

This course welcomes students of all backgrounds, identities, and perspectives. Disagreement on substantive academic content is encouraged; disrespectful or discriminatory behavior is not tolerated.

2.9 Office Hours and Communication

- **Office hours:** [Day, time], in-person and synchronous online via [platform]
- **Email response time:** 24 hours on weekdays, 48 hours on weekends
- **For urgent matters:** Course site message board (TA monitors during the week)
- **For project guidance:** Schedule a 1:1 meeting through the course calendar; do not save substantive project questions for office hours alone

2.10 Important Dates

Event	Session	Date
Course begins	1	[Date]
Track declaration deadline	end of 5	[Date]
PS1 due	8	[Date]
Midterm	10	[Date]
Project proposal due	14	[Date]
PS2 due	16	[Date]
PS3 due	24	[Date]
Split track begins	25	[Date]
Project draft due	30	[Date]
PS4 due	31	[Date]
Project presentations	32	[Date]
Final grades posted		[Date]

This syllabus may be updated during the semester. All material changes will be announced in class and posted on the course site. Continued enrollment after Session 2 constitutes acceptance of the syllabus terms.

3 32-Session Schedule

This page lists all 32 sessions with topic, book chapter, slide source, and assessment milestones. For full lecture notes on each session, click the session link.

3.1 Architecture Summary

- **Sessions 1–24:** Common core (all students attend identical sessions)
- **Sessions 25–31:** Split track (Track 1 = applied workshops, Track 2 = mathematical derivations)
- **Session 32:** Capstone — project presentations (both tracks present)

3.2 Unit Structure

Unit	Sessions	Topic	Book Chapters
1: Why DCF Fails	1–5	Diagnosis, illusion, term structure	Ch. 1–3
2: Measurement & Theory	6–10	IRR/PME, correct theory, midterm	Ch. 4–5
3: Decision & Application	11–18	Exit timing, portfolio, regulation, implementation	Ch. 6–9
4: Math Bridges	19–24	OU, HJB, MFG, GE, Jensen, Pigouvian (intuition)	Ch. 10–20 (selected)
5: Split Track	25–31	Applied workshops (T1) / Math derivations (T2)	Ch. 6–20
Capstone	32	Project presentations	—

3.3 Full 32-Session Schedule

3.3.1 Unit 1: Why Private Market Valuation Is Broken (Sessions 1–5)

#	Topic	Chapter	Slides	Notes
1	Course intro · Why DCF fails · Valuation hierarchy preview	Ch. 1	New (60%) + reuse from Class 9 (40%)	PS1 drops
2	Three structural failures · LA-CAPM	Ch. 1.5–1.7	New + reuse from Class 9	
3	The liquidity illusion · Secondary market evidence	Ch. 2	New + reuse from Class 21–22	
4	OU process for liquidity (intuition only)	Ch. 2.8–2.9	Adapt Class 5 (strip derivations)	
5	Term structure of private capital returns	Ch. 3	All new	Track declaration deadline

3.3.2 Unit 2: Measurement and Theory (Sessions 6–10)

#	Topic	Chapter	Slides	Notes
6	IRR: definition, limitations, biases	Ch. 4.2–4.4	All new	
7	PME: correct and incorrect implementations · LA-IRR · LA-PME	Ch. 4.5–4.9	All new	
8	What a correct theory must deliver · 5-result preview	Ch. 5	All new	PS1 due
9	Midterm review · Q&A · synthesis of Sessions 1–8	Ch. 1–5	All new (review slides)	
10	MIDTERM EXAM	Ch. 1–5	—	Midterm

3.3.3 Unit 3: Decision and Application (Sessions 11–18)

#	Topic	Chapter	Slides	Notes
11	Exit timing — the trapped investor problem	Ch. 6.1–6.4	Reuse Class 13 (PE decomp) + new	
12	The exit boundary $L^*(t)$ — numerical examples, intuition	Ch. 6.5–6.7	Reuse Class 13 + new	
13	Liquidity traps · Historical performance of GE-LAV exit rule	Ch. 6.7–6.10	Reuse Class 13 + new	
14	Portfolio construction · Liquidity hedge demand	Ch. 7.1–7.3	Reuse Class 24 (Portfolio Construction)	Project proposal due
15	Pacing · Tactical allocation · Stress scenarios	Ch. 7.4–7.6	Reuse Class 24 + new	
16	Regulatory implications: Solvency II, AIFMD, FSB · Crisis dynamics	Ch. 8	Reuse Class 19 (GFC) + new	PS2 due
17	GE-LAV platform — architecture, data, calibration workflow	Ch. 9.1–9.5	Reuse Class 10, 12, 25	
18	Platform demo: live walkthrough · validation against stress episodes	Ch. 9.6–9.10	Reuse Class 10, 12, 25 + new	

3.3.4 Unit 4: Math Intuition Bridges (Sessions 19–24)

These sessions teach the conceptual content of Part 2 of the book — what the math *means* and *does*, without the full derivations. Track 2 students will revisit this content with full proofs in Sessions 25–31.

#	Topic	Chapter	Slides	Notes
19	Brownian motion, Itô, the OU process (concepts only)	Ch. 10, 12	All new (intuition-stripped from Classes 2–5)	
20	Stochastic control, HJB (concepts) — what optimal stopping means	Ch. 11	All new (intuition from Class 7)	
21	Mean-field games — what collective interaction does to pricing	Ch. 13	All new (intuition from Class 8)	
22	Fokker-Planck, master equation — distributions over time	Ch. 14, 15	All new (intuition from Class 6)	
23	The LAV operator · GE equilibrium · market clearing	Ch. 16, 17	New + Class 17 (market clearing)	
24	Jensen bias · Pigouvian tax · welfare gap (results, not proofs)	Ch. 18, 19	New + Class 11/16/18/20 (intuition extracts)	PS3 due

3.3.5 Unit 5: Split Track (Sessions 25–31)

Track 1 (Practitioner) and Track 2 (Researcher) diverge. Both tracks meet at the same scheduled time but in different rooms (or different breakout rooms in synchronous online format).

#	Track 1: Applied Workshop	Track 2: Mathematical Derivation
25	PE buyout case workshop — apply GE-LAV to a Bain Capital fund. Reuse Class 21	Full HJB derivation — stochastic control, smooth pasting, optimal stopping proofs. Reuse Classes 7 & 14
26	VC + secondaries valuation lab — apply to a real VC secondary. Reuse Class 27 (Multi-Asset)	McKean-Vlasov MFG proofs — propagation of chaos, fixed-point existence, stability conditions. Reuse Class 8 & 15

#	Track 1: Applied Workshop	Track 2: Mathematical Derivation
27	Infrastructure case — the 57% illusion (Class 22). Reuse Class 22	Fokker-Planck & master equation — well-posedness, Wasserstein space. Reuse Class 6
28	Private credit case — spread decomposition (Class 23). Reuse Class 23	GE equilibrium existence — market clearing proof, Schauder fixed point. Reuse Class 17
29	Digital assets / climate risk hybrid case — GE-LAV applied to crypto and 50-year infra. Reuse Classes 30, 31	Jensen bias proof + Pigouvian welfare derivation — full theorems. Reuse Classes 11, 18, 20
30	Project draft workshop — peer review, feedback, refinement	Valuation hierarchy proof + project draft workshop — DCF LAV GE-LAV strict nesting proof
31	Future research · open practitioner questions — Class 26 + Class 28-29 hybrid	Future research · open theoretical questions — Class 26 + neural SDE (Class 29) for advanced students

3.3.6 Capstone (Session 32)

#	Topic	Notes
32	Project presentations — both tracks, 20 min presentation + 5 min Q&A each	Project final due

3.4 Slide Reuse Heat Map

Of the 485 existing slides in your 6 PPTX files:

Reuse pattern	Count	Where they go
Direct reuse (minimal adaptation)	~190 slides	Sessions 11–18, 25–31, 32
Conceptual adaptation (strip math)	~80 slides	Sessions 4, 19–24
Track 2 deep dive (full math)	~140 slides	Sessions 25–31 Track 2
Appendix / reference only	~75 slides	Appendix A (Math Foundations)

New slides needed: approximately 200–280 slides, primarily for Sessions 1–3, 5–10, and 19–24. Slide content briefs for these new slides are provided in the [slide-briefs](#) folder, one per session.

3.5 Visual Timeline

```
gantt
    title GE-LAV Course Timeline (16 weeks, 32 sessions)
    dateFormat YYYY-MM-DD
    axisFormat %b %d

    section Unit 1
    Sessions 1-5 (Why DCF Fails)           :u1, 2027-01-19, 2w

    section Unit 2
    Sessions 6-9 (Measurement)             :u2a, 2027-02-02, 11d
    Midterm (Session 10)                   :milestone, m1, 2027-02-12, 0d

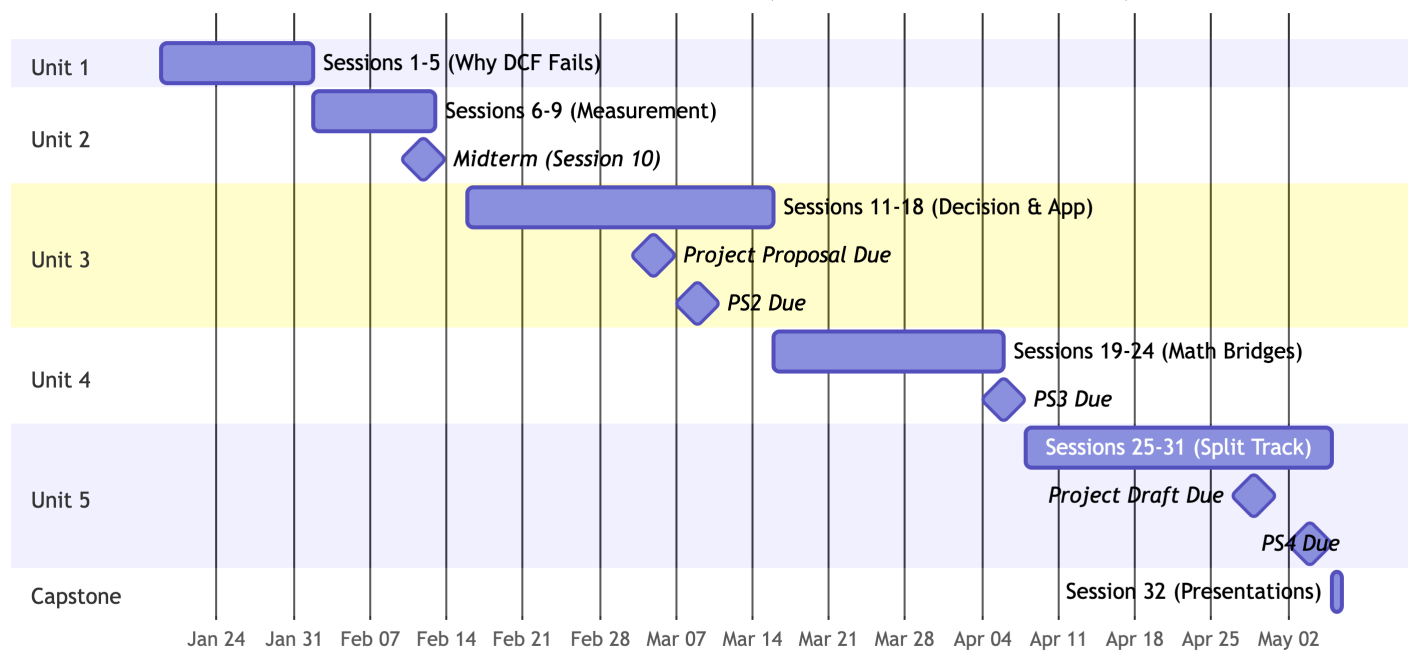
    section Unit 3
    Sessions 11-18 (Decision & App)        :u3, 2027-02-16, 4w
    Project Proposal Due                   :milestone, m2, 2027-03-05, 0d
    PS2 Due                                :milestone, m3, 2027-03-09, 0d

    section Unit 4
    Sessions 19-24 (Math Bridges)          :u4, 2027-03-16, 3w
    PS3 Due                                :milestone, m4, 2027-04-06, 0d

    section Unit 5
    Sessions 25-31 (Split Track)           :u5, 2027-04-08, 4w
    Project Draft Due                      :milestone, m5, 2027-04-29, 0d
    PS4 Due                                :milestone, m6, 2027-05-04, 0d

    section Capstone
    Session 32 (Presentations)             :u6, 2027-05-06, 1d
```

GE-LAV Course Timeline (16 weeks, 32 sessions)



Dates above are illustrative; actual dates will be set per the university's Spring 2027 academic calendar.

Next: [Tracks](#) →

4 Track Declaration

This course is offered in two tracks. **All students attend the same 24 common-core sessions (Sessions 1–24).** From Session 25 onward, students separate into Track 1 (Practitioner) or Track 2 (Researcher). Both tracks reunite at Session 32 for project presentations.

4.1 Track at a Glance

	Track 1: Practitioner	Track 2: Researcher
Best fit for	MBA, MSF (applied), MS Finance, professionals returning to school	MFE, MS Quantitative Finance, MS Financial Mathematics, PhD-track
Math prerequisite	Asset pricing + basic calculus	Stochastic calculus + measure-theoretic probability
Sessions 25–31	Applied case workshops with real data	Full mathematical derivations and proofs
Problem sets	Empirical, platform-based (real fund data)	Mathematical (proofs + numerical implementation)
Project orientation	Apply GE-LAV to a real asset, present to a hypothetical IC	Extend the theory, novel calibration, or mathematical critique
Career trajectory	LP allocator, GP investment professional, fund consultant, regulator	Quant researcher, academic, PhD applicant, model validation, central bank research

4.2 Why Two Tracks?

This course covers material at two very different levels of mathematical sophistication. The book itself is structured for two audiences: Part 1 (Chapters 1–9) is the Practitioner’s Guide, requiring no advanced math. Part 2 (Chapters 10–20) is the Researcher’s Framework, requiring stochastic calculus and measure theory.

Forcing all students into one track has two failure modes:

- **Lowest-common-denominator track:** Teaches only Part 1. MFE/PhD students are bored and don’t get the math they need. The course feels too easy for them.
- **Highest-common-denominator track:** Teaches both parts at full rigor. MBA/MSF students cannot follow the math and either drop the course or memorize without understanding.

The two-track design solves this by **sharing a common 24-session core** (which covers Part 1 plus the *intuition* of Part 2 results) and then **diverging for 7 sessions** during which Track 1 students do applied workshops and Track 2 students do the math.

4.3 How to Decide

4.3.1 Choose Track 1 (Practitioner) if you:

- Plan a career as an LP allocator, GP investment professional, fund consultant, advisor, or regulator
- Want to use GE-LAV in real investment work, not extend the theory
- Have strong asset pricing and applied finance background, but limited or no stochastic calculus
- Are more excited by “how do I apply this to a real fund?” than “how do I prove this theorem?”
- Want to spend more time on PE, VC, infrastructure, and private credit case work

4.3.2 Choose Track 2 (Researcher) if you:

- Plan a career as a quant researcher, academic, PhD candidate, or model validator
- Have completed (or are currently taking) graduate-level stochastic processes / stochastic calculus
- Are comfortable with measure-theoretic probability (σ -algebras, filtrations, Radon-Nikodym derivatives)
- Want to engage with the full math of GE-LAV — HJB equations, McKean-Vlasov SDEs, mean-field games
- May want to extend or critique the GE-LAV framework for a thesis, dissertation, or research paper

4.3.3 Still unsure?

The first 4 sessions are designed to help you decide. **Session 4 (OU Process Intuition)** is a deliberate diagnostic — if the OU process feels challenging at the intuition level, you should probably take Track 1. If you find yourself wanting *more* math depth than Session 4 provides, Track 2 is for you.

You may also schedule a 15-minute meeting with the instructor between Session 2 and Session 5 to discuss your background and goals.

4.4 Declaration Policy

Declaration deadline: end of Session 5.

Track choice is made via the course site form. After the deadline, track changes require instructor approval and are granted only in exceptional cases (e.g., the student took a stochastic processes course in parallel that wasn’t on their transcript when they enrolled).

Default track: If you do not declare by the deadline, you are assigned to Track 1 (Practitioner). This protects students who underestimate the math requirement.

4.5 What Stays the Same Across Tracks

- All 24 common-core sessions (Sessions 1–24)
- The midterm exam (Session 10) — identical for both tracks
- The book reading schedule for Part 1 (Ch 1–9)
- The lecture recordings — both tracks can watch each other’s sessions afterward
- The term project structure — both tracks complete the same 3 milestones (proposal, draft, presentation)
- Office hours and TA support
- Class participation expectations

4.6 What Differs Across Tracks

- **Sessions 25–31:** Different content in parallel (Track 1 in one room, Track 2 in another)
- **Problem sets:** Same drop dates, different question content
- **Book reading:** Part 2 (Ch 10–20) is required for Track 2 and recommended-optional for Track 1
- **Project rubric weighting:** Slightly different emphasis (applied implementation vs. theoretical depth) — see [project specification](#)

4.7 Cross-Track Audit Policy

Students enrolled in one track may **audit** sessions from the other track for free (synchronously online or asynchronously via recordings). Some Track 1 students will want to watch the Session 25 HJB derivation lecture even if they’re not assessed on it; some Track 2 students will benefit from watching the Session 25 PE buyout case workshop. **This is encouraged.**

You may also choose to do the other track’s problem sets for your own learning (not for credit). The instructor will provide solutions to both PSet streams.

4.8 Two-Track Honor System

The course depends on students self-selecting honestly. Choosing Track 1 when your background can handle Track 2 means you’ll get an easier course (and a likely higher grade), but you’ll miss the math that makes GE-LAV intellectually distinctive. Choosing Track 2 when you don’t have the math prep means you’ll struggle through proofs without genuine understanding, and your project quality will suffer. Choose honestly.

Next: Slide Reuse Map →

5 Slide Reuse Map

This page documents how your existing 485 slides (across 6 PPTX files, originally architected as a math-first course) are redeployed across the 32 sessions of the Path C two-track architecture.

5.1 Summary

Total existing slides	485
Direct reuse (minimal adaptation)	~190 (39%)
Conceptual adaptation (strip math/derivations)	~80 (16%)
Track 2 deep dive (full math, original form)	~140 (29%)
Appendix / reference only	~75 (15%)
New slides to create	~200–280
Estimated build time for new slides	80–120 hours

5.2 Mapping by Existing Class

The table below maps each existing class (Classes 1–32 as currently designed) to its new home in the Path C architecture.

Existing Class	Existing Title	Slides	New Home	Reuse Type
Class 1	Probability Spaces & Measure Theory	20	Appendix A · Track 2 prep for S25	Reference
Class 2	Stochastic Processes & Brownian Motion	20	Appendix A · Track 2 prep for S25	Reference
Class 3	Itô’s Lemma	19	Appendix A · Track 2 prep for S25	Reference
Class 4	SDEs / Girsanov	19	Appendix A · Track 2 prep for S25	Reference
Class 5	OU Process Complete Derivation	19	Session 4 (intuition extract) + Track 2 S25 (full)	Split

Existing Class	Existing Title	Slides	New Home	Reuse Type
Class 6	Fokker-Planck Equations	16	Session 22 (intuition) + Track 2 S27 (full)	Split
Class 7	Stochastic Control & HJB	15	Session 20 (intuition) + Track 2 S25 (full)	Split
Class 8	McKean-Vlasov MFG	15	Session 21 (intuition) + Track 2 S26 (full)	Split
Class 9	PC Markets: Scale, Structure, Frictions	14	Sessions 1–2 (heavy reuse)	Direct
Class 10	Calibrating GE-LAV v2.5	15	Sessions 17–18 (platform implementation)	Direct
Class 11	Jensen Bias Four-Channel Derivation	20	Session 24 (intuition) + Track 2 S29 (full proof)	Split
Class 12	Quadratic Liquidity Premium Calibration	16	Sessions 17–18	Direct
Class 13	PE Return Decomposition (5-component)	14	Sessions 11–13 (exit timing block)	Direct
Class 14	HJB Numerical $h(L)$	15	Track 2 S25 (full)	Direct
Class 15	Mean-Field Equilibrium Fixed Point	18	Track 2 S26 (full)	Direct
Class 16	Empirical Validation of Jensen Bias	15	Session 24 (common core)	Direct
Class 17	GE Market Clearing & Endogenous Pricing	14	Session 23 (intuition) + Track 2 S28 (full)	Split
Class 18	Welfare Analysis / PE Externality	13	Session 24 + Track 2 S29 (full derivation)	Split
Class 19	Crisis Dynamics & GFC Amplification	14	Session 16 (regulatory + crisis)	Direct

Existing Class	Existing Title	Slides	New Home	Reuse Type
Class 20	Pigouvian Tax Design	15	Session 24 (intuition) + Track 2 S29 (full)	Split
Class 21	PE Buyout Full GE-LAV Valuation	13	Track 1 S25 (PE workshop)	Direct
Class 22	Infrastructure: The 57% Illusion	13	Track 1 S27 (Infra case)	Direct
Class 23	Private Credit Spread Decomposition	12	Track 1 S28 (Private credit case)	Direct
Class 24	Portfolio Construction with GE-LAV	13	Sessions 14–15 (Ch 7 common core)	Direct
Class 25	Empirical Testing Pipeline	12	Sessions 17–18 (platform + validation)	Direct
Class 26	Research Frontiers & Course Synthesis	14	Session 31 (both tracks)	Direct
Class 27	Multi-Asset GE-LAV	13	Track 1 S26 (VC + secondaries adaptation)	Direct
Class 28	Network Topology & Heterogeneity	11	Track 2 S30 (advanced)	Direct
Class 29	Neural SDE Calibration	12	Track 2 S30 (advanced)	Direct
Class 30	GE-LAV & Digital Asset Markets	11	Track 1 S29 (hybrid case)	Direct
Class 31	Climate Risk & Long-Duration Assets	12	Track 1 S29 (hybrid case)	Direct
Class 32	Final Exam Review & Synthesis	17	Session 32 (project presentations) — content largely discarded since no final	Discard most

5.3 Mapping by New Session

Reverse view — for each new session, what existing slides feed it and what is new.

5.3.1 Sessions 1–10 (Mostly New)

New Session	Topic	Source Slides	New Slides Needed
1	Course intro · DCF fails	Class 9 (selected)	~15
2	Three structural failures · LA-CAPM	Class 9 (remainder)	~12
3	Liquidity illusion · Secondary markets	Class 21–22 (excerpts)	~12
4	OU process for liquidity (intuition)	Class 5 (intuition-stripped)	~10
5	Term structure of private capital	None	~15
6	IRR: definition, biases	None	~15
7	PME · LA-IRR · LA-PME	None	~15
8	What a correct theory must deliver	None	~12
9	Midterm review	None	~10
10	MIDTERM EXAM	—	—
Subtotal		~50 slides reused	~116 new

5.3.2 Sessions 11–18 (Mostly Reuse — Sweet Spot)

New Session	Topic	Source Slides	New Slides Needed
11	Exit timing — trapped investor	Class 13 (partial)	~5
12	Exit boundary $L^*(t)$ numerical	Class 13 (partial)	~5
13	Liquidity traps · Historical performance	Class 13 (remainder)	~5
14	Portfolio construction	Class 24 (partial)	~3
15	Pacing · Tactical allocation	Class 24 (remainder)	~5
16	Regulatory · Crisis dynamics	Class 19	~5
17	GE-LAV platform — architecture	Classes 10, 12, 25 (partial)	~5
18	Platform demo · validation	Classes 10, 12, 25 (remainder)	~5
Subtotal		~85 slides reused	~38 new

5.3.3 Sessions 19–24 (Math Intuition Bridges — Mostly New)

New Session	Topic	Source Slides	New Slides Needed
19	BM · Itô · OU (concepts)	Classes 2–5 (heavy stripping)	~15
20	Stochastic control · HJB (concepts)	Class 7 (intuition extract)	~15
21	Mean-field games (concepts)	Class 8 (intuition extract)	~15
22	Fokker-Planck · master equation	Class 6 (intuition extract)	~12
23	LAV operator · GE clearing	Class 17 (intuition extract)	~12
24	Jensen · Pigouvian · welfare	Classes 11, 16, 18, 20 (intuition)	~15
Subtotal		~30 slides reused	~84 new

5.3.4 Sessions 25–31 (Split Track — Mostly Reuse)

New Session	Track 1 Source	Track 2 Source	New Slides Needed
25 T1	Class 21 (PE buyout)	—	~3
25 T2	—	Classes 7, 14 (HJB full)	~5
26 T1	Class 27 (Multi-asset, adapt for VC)	—	~5
26 T2	—	Classes 8, 15 (MFG full)	~3
27 T1	Class 22 (Infrastructure)	—	~2
27 T2	—	Class 6 (Fokker-Planck full)	~3
28 T1	Class 23 (Private credit)	—	~3
28 T2	—	Class 17 (GE full)	~3
29 T1	Classes 30, 31 (Digital + Climate)	—	~5
29 T2	—	Classes 11, 18, 20 (Jensen + Pigouvian full)	~3
30 T1	None	None	Workshop format ~5
30 T2	—	None	Workshop format ~5
31	Class 26 (Research frontiers)	Class 26 + Classes 28, 29 (advanced)	~5
Subtotal			~50 new

5.3.5 Session 32 (Capstone)

New Session	Topic	Source	New
32	Project presentations	None — Class 32's “Final Review” content largely discarded	~5 (intro frame)

5.4 Total Count

Category	Slides
Existing slides directly reused (minimal adaptation)	~165
Existing slides reused with conceptual adaptation	~80
Existing slides reused in Track 2 (original form)	~140
Existing slides moved to Appendix A	~75 (Classes 1–4)
Existing slides discarded	~25 (mostly Class 32)
New slides needed	~243

5.5 What Gets Built When

For a Spring 2027 delivery with 14 months of runway, suggested build sequence:

Month	Build target	Hours estimate
1–2	Sessions 1–8 new content (all of Unit 1 + most of Unit 2)	40 hr
3	Sessions 19–24 math intuition bridges	30 hr
4	Adapt Sessions 11–18 (light editing of existing slides)	10 hr
5	Track-specific Session 25–31 prep (mostly reuse)	15 hr
6	Problem set writing (8 PSets)	20 hr
7	Midterm and project rubric refinement	8 hr
8–10	Pilot dry-run + iteration on weak sessions	20 hr
11–14	Polish, recording prep, platform integration	15 hr
Total		~158 hr

Spread across 14 months at ~3 hr/week, this is realistic alongside your other work.

Next: Project Specification →

Part II

Assessments

6 Term Project Specification

The term project is **50% of your course grade** and the integrative deliverable of the course. It replaces a traditional final exam. Every student completes the project; track determines orientation, not requirement.

6.1 Project Statement

Apply the full GE-LAV® framework to a real private market asset of your choosing, producing a complete liquidity-adjusted valuation that you would defend to an investment committee (Track 1) or to a thesis defense (Track 2).

The project must include:

1. **Asset selection and motivation** — Why this asset? What's the valuation question?
2. **Data acquisition and OU calibration** — Secondary market data, fund cash flows, or public proxies
3. **Standard valuation baseline** — DCF, IRR, PME using current industry practice
4. **GE-LAV valuation** — Full pipeline through the GE-LAV® platform
5. **Comparison and economic interpretation** — Where does the valuation gap come from? Jensen bias? Externality? Crisis amplification?
6. **Sensitivity / robustness analysis** — How do conclusions change under stress scenarios?
7. **Recommendation** — Buy/sell/hold (T1) or research extension (T2) with justification

6.2 Deliverables and Weighting

Milestone	Due	Weight	Format
Proposal	Session 14	5%	2-page PDF + 1-page data plan
Draft paper	Session 30	15%	15-page PDF + supporting code/spreadsheets
Final presentation	Session 32	25%	20-minute presentation + 5-minute Q&A
Final paper	Session 32	5%	20-page PDF + supporting materials
Total		50%	

6.3 Proposal (Session 14, 5%)

What to submit (2 pages + 1-page data plan):

- **Asset description (1 paragraph):** Name the specific fund / asset / deal. Provide vintage year, asset class, geography, AUM if known.
- **Valuation question (1 paragraph):** What specifically are you trying to value? A specific LP interest? A potential secondary purchase? A regulatory exposure? Be precise.
- **Why GE-LAV is the right tool (1 paragraph):** What makes this asset's valuation particularly sensitive to liquidity premium dynamics? Why would DCF be wrong here?
- **Methodology outline (1 page):** Which GE-LAV results will you use (Jensen bias, exit boundary, Pigouvian tax, etc.)? How will you calibrate? What data?
- **Data plan (1 page):** What data do you need? Where are you getting it? What are the gaps? How will you address them? Cite specific sources (Preqin, PitchBook, Capital IQ, Bloomberg, Burgiss, Cliffwater, Pevara, secondary market data providers).
- **Track:** State explicitly which track you are on and how the project reflects that orientation.

Grading (5%):

Criterion	Weight
Asset is specific and accessible (not “PE in general”)	1%
Valuation question is precise and answerable	1%
Methodology choice is appropriate	1%
Data plan is realistic	1%
Track orientation is clear	1%

6.4 Draft Paper (Session 30, 15%)

What to submit (15-page PDF + appendices + supporting materials):

A draft of the full project paper. This is a working draft, not a polished final — but it must contain real preliminary results.

Required sections:

1. **Introduction** (1–2 pages) — Asset, question, why GE-LAV
2. **Data and calibration** (2–3 pages) — Sources, processing, OU parameters
3. **DCF baseline** (1–2 pages) — Standard valuation with documented assumptions
4. **GE-LAV implementation** (3–4 pages) — Full pipeline, parameters, intermediate results
5. **Results and comparison** (2–3 pages) — Where DCF and GE-LAV differ, by how much, decomposition
6. **Preliminary robustness** (1 page) — At least 2 sensitivity analyses
7. **Open questions and next steps** (1 page) — What needs work before final
8. **References** (1 page)

Required supporting materials:

- Spreadsheets or notebooks (Excel, Python, or R) showing all calculations

- GE-LAV® platform outputs (CSV exports or screenshots)
- Data sources documented (file or URL)

Grading (15%):

Criterion	Weight
All required sections present with real content	2%
Data and calibration are sound	3%
GE-LAV implementation is correct	4%
Comparison is well-decomposed (which channel matters?)	3%
Writing is clear and quantitative	2%
Supporting materials are complete and reproducible	1%

Peer feedback: Drafts are exchanged with one other student in Session 30 (workshop session). Each student provides a 1-page peer review of their partner's draft. Peer review quality factors into the final paper grade.

6.5 Final Presentation (Session 32, 25%)

Format: 20 minutes presentation + 5 minutes Q&A. Slides required. Live or pre-recorded with live Q&A both acceptable.

Required slide content:

- **Slide 1:** Title, your name, track, asset
- **Slides 2–3:** Motivation — why this asset, why GE-LAV
- **Slides 4–5:** Data and calibration
- **Slide 6:** DCF baseline
- **Slides 7–10:** GE-LAV results
- **Slide 11:** Comparison and decomposition
- **Slide 12:** Sensitivity / robustness
- **Slide 13:** Recommendation
- **Slide 14:** Limitations and next steps
- **Slide 15:** Q&A

Q&A is graded. Questions will probe whether you understand your own work. If you cannot answer a basic question about your own calibration, your own results, or your own conclusions, that is a significant penalty.

Grading (25%):

Criterion	Weight
Clarity of motivation and question	3%
Quality of empirical work and calibration	5%
Correct application of GE-LAV framework	6%
Quality of comparison and economic interpretation	4%

Criterion	Weight
Quality of recommendation/conclusion	3%
Q&A defense	4%

6.6 Final Paper (Session 32, 5%)

What to submit: A polished 20-page paper (excluding references and appendices), submitted by the end of Session 32.

This is the draft paper, revised based on:

- Peer feedback from Session 30
- Instructor feedback on the draft
- Your own further work between Session 30 and Session 32

Grading (5%):

Criterion	Weight
Improvement over draft	2%
Writing quality and polish	2%
Final results are correct	1%

6.7 Track-Specific Project Orientation

6.7.1 Track 1 (Practitioner) Projects

Track 1 projects emphasize **empirical realism, investment decision relevance, and communication to non-technical audiences**. Strong Track 1 projects:

- Pick a specific, real asset (a named PE fund, a specific infrastructure asset, a specific private credit deal)
- Use real or realistic data from industry sources (Preqin, PitchBook, Burgiss, etc.)
- Frame the question as one an investment committee or LP allocator would ask
- Compare DCF vs. GE-LAV with clear dollar magnitudes
- End with a defensible recommendation an actual professional could act on
- Make the Q&A defense feel like an investment committee meeting

Examples of strong Track 1 project questions:

- “Should CalPERS commit \$200M to Blackstone Capital Partners VIII in 2027? Standard DCF says yes; what does GE-LAV say?”
- “Is the secondary market quote of −18% on an Apollo IX LP interest fair value, distress, or opportunity?”
- “Solvency II requires Insurer X to hold capital against its €500M private debt portfolio. Standard methodology gives a Y% charge; what does GE-LAV-adjusted methodology give?”

6.7.2 Track 2 (Researcher) Projects

Track 2 projects emphasize **theoretical depth, mathematical rigor, or novel methodological contribution**. Strong Track 2 projects:

- Extend, generalize, or critique a specific GE-LAV result
- Add genuine mathematical content (a proof, a derivation, a numerical method)
- Could plausibly become a working paper or thesis chapter
- May calibrate to less-standard data (cryptocurrency, climate-exposed infrastructure, NFTs, art)
- Make the Q&A defense feel like a thesis defense

Examples of strong Track 2 project questions:

- “Extend GE-LAV with jump-diffusion dynamics. Does the Jensen bias formula remain affine? What changes?”
- “Calibrate the McKean-Vlasov mean-field game equilibrium to the 2023 venture capital secondaries data. Does the GE-LAV stability condition hold? What does it imply?”
- “Derive the Pigouvian tax under heterogeneous beta. Does optimal tax design require investor-specific rates?”

6.8 Asset Selection Guidance

Acceptable asset categories:

- Private equity buyout funds (mega, mid-cap, growth)
- Venture capital funds and individual VC secondaries
- Infrastructure funds and specific infrastructure assets
- Private credit / direct lending funds
- Real estate funds (open-end and closed-end)
- Private market fund-of-funds
- LP interests in any of the above
- GP commitment / co-investment vehicles
- Insurance company private market portfolios (for regulatory framing)
- Pension fund private market allocations
- Family office private market allocations

Unacceptable assets (because GE-LAV machinery is not the right tool):

- Public equities or fixed income — these are not illiquid in the GE-LAV sense
- Bank deposits or money market funds
- Crypto spot tokens (acceptable in modified form — derivative positions or token venture funds, yes; spot BTC, no — see Track 2 examples)

If unsure whether your asset is acceptable, discuss with the instructor before submitting the proposal.

6.9 Use of the GE-LAV® Platform

The GE-LAV® platform at liquidityillusion.com is the canonical tool for this project. All students will receive access credentials during Session 17.

The platform provides:

- OU calibration interface
- Jensen bias calculator
- Exit boundary computation
- Pigouvian tax calculator
- Portfolio liquidity-adjusted weights
- Stress scenario library

Track 1 students: Use the platform heavily — it's designed for your use case.

Track 2 students: Use the platform for baseline, but may also implement extensions or alternative numerics in Python / R / Julia. Disclose any deviation from the platform in your methodology section.

6.10 Honor Code and AI Use

Re: AI use — see the [syllabus AI policy](#). For the project specifically:

- AI may be used as a research and writing aid
- AI may **not** generate your core analytical work — calibration, results, or interpretation
- Every word, equation, and result in your submission must be something you can defend orally
- The Q&A in Session 32 is designed to test this

Re: collaboration — the project is **individual work only**. You may discuss your project with classmates informally, but no shared analysis, no shared code, no shared writing. If you and a classmate happen to pick the same asset, your work must still be independent.

6.11 Office Hours and Project Support

- **Drop-in office hours:** Standard course office hours; bring project questions
- **Project-specific 1:1s:** Schedule a 30-minute slot via the course calendar in Sessions 14–30; designed for substantive project discussion
- **Peer workshop:** Session 30 is dedicated to peer review of drafts
- **GE-LAV® platform support:** Technical issues — email Regent Financial; methodological issues — bring to office hours

6.12 Examples of Past Projects (For Future Iterations)

This section will be populated after the first delivery of the course.

[Next: Midterm Blueprint →](#)

7 Midterm Exam Blueprint

The midterm exam is held in **Session 10**, is **identical for both tracks**, and is worth **20% of the course grade**.

7.1 Format

- **Duration:** 75 minutes
- **Location:** [TBD]
- **Open/closed:** Closed-book, closed-laptop
- **Permitted aids:** One 8.5×11” double-sided **handwritten** formula sheet (no typed sheets; no photocopies)
- **Calculators:** Non-programmable scientific calculators permitted; no phones, no graphing calculators with internet
- **Honor code:** University standard

7.2 Coverage

The midterm covers **Sessions 1–9 (book Chapters 1–5)**:

Unit	Sessions	Book Chapters	Weight on Exam
Why DCF Fails	1–5	Ch. 1–3	50%
Measurement & Theory	6–9	Ch. 4–5	50%

Specifically out of scope (not on midterm):

- Exit timing and the trapped investor (Ch. 6)
- Portfolio construction (Ch. 7)
- Regulatory implications (Ch. 8)
- GE-LAV platform implementation (Ch. 9)
- All Part 2 math content (Ch. 10–20)

7.3 Topic Weighting

Topic	Approximate % of exam
The three structural failures of DCF	12%
Valuation hierarchy (DCF LAV GE-LAV)	8%
Secondary market evidence and the liquidity illusion	10%
OU process — intuition, parameters, what it captures	10%
Term structure of private capital returns	10%
IRR — definition, two biases, when it fails	12%
PME — Kaplan-Schoar vs. Long-Nickels vs. Direct Alpha	8%
LA-IRR and LA-PME — what they fix	8%
Jensen convexity bias — intuition and magnitude	12%
Five requirements of a correct valuation theory (Ch. 5)	10%

7.4 Question Types

The exam contains four question types:

Type	Count	Points each	Total
Multiple choice / short answer	8	3	24
Quantitative problems	4	9	36
Conceptual short essays (200 words)	2	10	20
Integrative case analysis	1	20	20
Total	15		100

7.5 Sample Questions

7.5.1 Sample Multiple Choice (3 points)

Q. The Jensen convexity bias arises because:

- (a) IRR is mathematically equivalent to a constant discount rate
- (b) The expected value of a nonlinear function of liquidity differs from the function evaluated at the expected liquidity
- (c) Secondary market discounts are mean-reverting
- (d) The OU process has fat tails

Correct answer: (b)

7.5.2 Sample Short Answer (3 points)

Q. State the GE-LAV stationary distribution of the liquidity state L under the OU calibration $\alpha = 0.45$, $\beta = 0.32$, $L = 1.0$. Show units.

Expected answer: $L_\infty \sim N(L, \beta^2/(2\alpha)) = N(1.0, 0.1138)$. Variance is dimensionless; mean is in units of the liquidity index.

7.5.3 Sample Quantitative Problem (9 points)

Q. An infrastructure asset has a 10-year hold horizon. The OU process for its liquidity state is calibrated to $\alpha = 0.30$, $\beta = 0.28$, $L = 1.00$, with current state $L_0 = 0.85$ (slightly stressed). The DCF risk-free rate is 4.5%, the equity risk premium is 6%, the beta is 0.7, and the constant illiquidity premium assumed by the fund is 3%.

- (a) Compute the DCF discount rate. (2 pts)
- (b) Estimate the LAV-adjusted discount rate using the linear approximation $r_{LAV}(L) = r_{DCF} + \beta \cdot (1 - L)$. Assume $\alpha = 0.04$. (3 pts)
- (c) Approximately what is the Jensen bias contribution to the LAV vs. DCF valuation gap over a 10-year horizon, using the affine bias formula $B(T) = A \cdot T + C$, with $A = 1.4\%$ per year and $C = 0.5\%$? (2 pts)
- (d) Comment briefly on which channel (level vs. convexity) dominates the gap in this case. (2 pts)

Solution sketch: (a) $r_{DCF} = 4.5\% + 0.7 \cdot 6\% + 3\% = 11.7\%$ (b) $r_{LAV}(0.85) = 11.7\% + 0.04 \cdot (1 - 0.85) = 11.7\% + 0.6\% = 12.3\%$ (c) $B(10) = 1.4\% \cdot 10 + 0.5\% = 14.5\%$ (d) The Jensen bias (convexity channel, 14.5%) dominates the level adjustment (0.6%). This is typical for long-horizon assets where the OU process has time to generate path-dependent valuation.

7.5.4 Sample Conceptual Essay (10 points)

Q. Explain in 200 words why the standard PME (Public Market Equivalent) methodology systematically misvalues private market performance when liquidity premia vary over time. In your answer, distinguish between Kaplan-Schoar PME and Long-Nickels PME, identify which is more vulnerable to the liquidity illusion, and propose what LA-PME does differently.

Marking guide: - 2 pts: Correctly characterize PME as comparing private-market cash flows against discounted public benchmark cash flows - 2 pts: Identify that PME embeds a *constant* discount assumption that mirrors the constant-rate DCF problem - 2 pts: Distinguish Kaplan-Schoar (ratio of FVs) from Long-Nickels (ICM with reinvestment) - 2 pts: Argue that Long-Nickels is somewhat more vulnerable because it amplifies timing distortions - 2 pts: Explain that LA-PME uses a stochastic discount path L_t -conditional rather than a constant public-benchmark return

7.5.5 Sample Integrative Case (20 points)

Case: Trapped LP in a 2007-vintage Buyout Fund

An LP in a 2007-vintage US buyout fund holds an interest with NAV of \$50M. The fund is approaching year 10 of a 10+2 structure. The GP has indicated they will request an extension; the LP is unsure whether to consent or to exit via the secondary market.

Current observed conditions (Q4 2025):

- Secondary market quote: -15% to NAV (so \$42.5M cash today)
- Estimated time to natural liquidation if extension is granted: 2 more years
- Liquidity state L_t (from secondary market spreads): 0.82 (stressed but not crisis)
- OU parameters: $\alpha = 0.40$, $\beta = 0.30$, $L = 1.0$
- Expected residual NAV growth (gross, under GP plan): 6% per year over 2 years

Answer the following:

- (a) **(6 pts)** Using DCF reasoning with a 12% discount rate, what is the LP's expected present value if they hold to liquidation? Should they accept the -15% secondary offer or hold?
- (b) **(6 pts)** Using the LAV adjustment with $\alpha = 0.04$ (so $r_{LAV}(0.82) = 12\% + 0.04 \cdot 0.18 = 12.7\%$), how does your conclusion in (a) change? Show your work.
- (c) **(4 pts)** Identify which channel (level vs. convexity) is doing the most work in shifting your conclusion. Quantify roughly.
- (d) **(4 pts)** Discuss in 1–2 paragraphs what additional GE-LAV considerations (exit boundary, externality, regime dependence) the LP should think about that DCF alone does not capture. You do not need to compute these — just explain why they matter.

7.6 Studying for the Midterm

Best preparation strategy:

1. **Re-read book Chapters 1–5 carefully.** The exam tracks the book; if you understand the book, you can pass the exam.
2. **Work all PS1 problems** — even ones not yet graded. PS1 is designed to be a study aid for the midterm.
3. **Practice with the sample questions above** under exam conditions (75 minutes, closed book, formula sheet only).
4. **Make your formula sheet now**, not the night before. The act of compressing the course onto one page is itself a study technique. Recommended formulas to include: OU process equation and stationary distribution; standard DCF and LAV discount rate forms; the affine Jensen bias formula $B(T) = A \cdot T + C$; PME definitions (Kaplan-Schoar, Long-Nickels); the five requirements from Ch. 5.
5. **Attend Session 9 (Midterm Review).** Bring questions.

What NOT to spend time on:

- Memorizing proofs (no proofs on the midterm)
- Memorizing platform UI (platform comes after the midterm)
- Part 2 math content (out of scope)
- Specific calibration numbers from the book (you'll be given parameters)

7.7 Logistics

- Bring: pencil, pen, eraser, calculator, formula sheet, ID
 - Arrive 10 minutes early
 - No phones at the desk — power off and stowed
 - Bathroom breaks permitted; one at a time
 - Submit formula sheet with exam (it does not affect your grade — this is honor-code documentation)
-

[Next: Problem Sets Overview →](#)

8 Problem Sets Overview

This course assigns **four problem sets**, each worth **6.25% of your grade (25% total)**. Problem sets are **track-differentiated** — Track 1 students submit applied/empirical work; Track 2 students submit mathematical work. Same due dates, same topical coverage; different question content.

8.1 PSet Schedule

PSet	Drops	Due	Sessions Covered	Topics
PS1	Session 1	Session 8	1–7	Diagnosis · OU · Term structure · IRR/PME
PS2	Session 9	Session 16	11–15	Exit timing · Portfolio · Regulatory
PS3	Session 17	Session 24	17–23	Platform · Math intuition · Jensen · Pigouvian
PS4	Session 25	Session 31	25–30	Track-specific advanced topics

8.2 Submission

- Submitted via course site as a **single PDF**
- Supporting files (Excel, Python notebooks, R scripts) submitted as a zip
- Cover page must include: name, track, collaborators (if any, max 3), AI use disclosure
- Late penalty: 10% per 24-hour period, up to 72 hours; no credit thereafter

8.3 Collaboration Policy

- Permitted: groups of up to 3 students discussing approach and method
- **Each student must independently write and submit their own solutions** — no shared documents, no copy-paste between submissions
- Disclose collaborators on cover page
- Track 1 and Track 2 students may discuss general concepts but not specific problem solutions (since the questions differ)

8.4 PSet Structure Standards

Every PSet contains:

- **Conceptual question (15%)** — Test understanding of recent lecture material with no math required
- **Applied / mathematical core (60%)** — The substantive work for the PSet, track-differentiated
- **Stretch question (15%)** — Designed to be genuinely challenging; partial credit available
- **Reflection (10%)** — 200-word reflection: What did you learn? Where did you get stuck? How does this connect to your project?

8.5 Sample PSet (Brief Sketches)

8.5.1 PS1 — Track 1 (Applied)

1. **Conceptual (15 pts):** Explain in 200 words why the IRR’s two biases (reinvestment assumption + sign-pattern dependence) make it unsuitable for assessing vintage-year liquidity effects.
2. **Applied core (60 pts):**
 - (20) Using the provided dataset of Prequin secondary market discounts (2003–2024), compute the implied liquidity state L_t for each quarter using the linear discount-to- L mapping from Ch. 2.
 - (20) Fit the OU process $(\cdot, \cdot, L)$ to your computed L_t series using MLE. Report your point estimates with 95% confidence intervals. Compare to the book’s calibration.
 - (20) Compute the Jensen bias $B(T)$ for $T = 5$ and $T = 10$ using your calibration. Comment on the difference vs. the book’s calibration.
3. **Stretch (15 pts):** Identify and characterize the four “regimes” of the secondary market discount series (2003–2024). Are they consistent with the book’s regime classification? Where do they differ?
4. **Reflection (10 pts):** As specified above.

8.5.2 PS1 — Track 2 (Mathematical)

1. **Conceptual (15 pts):** Explain in 200 words why the Jensen convexity bias must be strictly positive whenever the discount factor is a strictly convex function of the liquidity state.
2. **Mathematical core (60 pts):**
 - (15) Derive the stationary distribution of the OU process $dL_t = (L - L_t)dt + \sigma dW_t$ from first principles using the Fokker-Planck equation. Show your work.
 - (15) Prove that $E[L_t] \rightarrow L$ as $t \rightarrow \infty$ and $\text{Var}[L_t] \rightarrow \sigma^2/(2\kappa)$ regardless of L_0 .
 - (15) Derive the conditional moments $E[L_t | L_0]$ and $\text{Var}[L_t | L_0]$. State the result and prove it.

- (15) Using your derivation, prove that the Jensen bias $B(T)$ under a quadratic discount rate $r(L) = \bar{r} + (1-L) + (1-L)^2$ is affine in T : $B(T) = A \cdot T + C$. Identify A and C explicitly.
3. **Stretch (15 pts):** Extend the OU process to a jump-diffusion with compensated Poisson jumps. Derive the new stationary distribution. Under what conditions does it remain Gaussian?
 4. **Reflection (10 pts):** As specified above.

8.6 PS2, PS3, PS4 — Topic Outlines

Full question sets will be released at PSet drop. Outlines below.

8.6.1 PS2 (Sessions 11–15)

Track 1 topics:

- Exit boundary computation for a real PE fund position
- Portfolio construction with hedge demand — applied to a \$1B endowment allocation
- Regulatory case: compute the Solvency II SCR for a real insurance company's private market portfolio under both standard and LA approaches

Track 2 topics:

- Numerical solution of the HJB equation for $h(L)$ using finite differences
- Mean-field equilibrium fixed-point iteration — implement and verify convergence
- Welfare gap derivation under heterogeneous beta

8.6.2 PS3 (Sessions 17–23)

Track 1 topics:

- Full GE-LAV platform pipeline: take a real fund, calibrate, value, stress-test
- Conceptual: explain the Pigouvian tax to a non-economist (writing exercise)
- Apply LA-IRR and LA-PME to a real fund and compare to standard metrics

Track 2 topics:

- Prove a property of the LAV operator (e.g., path-dependence)
- McKean-Vlasov mean-field game — finite-N approximation and rate of convergence
- Derive the Pigouvian tax formula explicitly from the welfare-gap analysis

8.6.3 PS4 (Sessions 25–30)

Track 1 topics:

- Multi-asset GE-LAV applied to a real diversified private market portfolio
- Stress scenario analysis: 2008 crisis, 2020 COVID, hypothetical 2030 climate event
- Communication exercise: 1-page IC memo summarizing GE-LAV vs. DCF for a specific deal

Track 2 topics:

- Network topology with heterogeneous — analytical and numerical
- Neural SDE calibration to GE-LAV — implement and benchmark
- Extension proposal: a 5-page research note on a novel GE-LAV extension

8.7 Grading Rubric (Common Across PSets)

Score range	Meaning
90–100%	Excellent work — correct, well-explained, with insight beyond the minimum
80–89%	Solid work — correct conclusions; some minor errors or missing detail
70–79%	Acceptable work — main results obtained; some confusion in derivation or interpretation
60–69%	Weak work — partial credit for effort and partial correctness
Below 60%	Insufficient — significant misunderstanding or incomplete submission

For mathematical work (Track 2): Show your steps. Final answer only earns ~40% credit even if correct.

For applied work (Track 1): Document your assumptions, your data, and your calculations. Final number only earns ~40% credit.

8.8 AI Use Policy for PSets

See the [syllabus AI policy](#). For PSets specifically:

- **Permitted:** Using AI to explain concepts, debug code, suggest derivation strategies, check intermediate steps
- **Required:** You must understand and be able to defend every step of your submission
- **Disclosure:** Brief note on cover page describing how AI was used (e.g., “Used Claude to debug Python code in Q2; verified all derivations independently”)

Forbidden: Submitting AI-generated solutions verbatim. The PSets are designed so that AI alone cannot produce a correct submission — most questions require your specific dataset, your specific calibration, or your specific interpretation.

8.9 Solutions Release

- Solutions are released **48 hours after the deadline** on the course site
- Both tracks' solutions are released to all students (encouraging cross-track learning)
- Solutions explain not just the answer but also *common errors* and *good vs. bad approaches*

Next: [Session 1](#) →

Part III

Lectures: Common Core (Sessions 1–24)

9 Session 1: Course Intro · Why Private Market Valuation Is Broken

Unit	1 — Why DCF Fails
Book Chapter	1 (sections 1.1–1.4, 1.10–1.16)
Slide source	New (60%) + existing Class 9 (40%)
Slides needed	~15 new + ~6 adapted from Class 9
Track	Common core (both tracks)
Assessment milestone	PS1 drops

9.1 Learning Objectives

By the end of this session, students will be able to:

1. **State the scale** of the private capital market (\$13 trillion AUM, ~10,000+ funds) and articulate why valuation methodology matters at this scale.
2. **Describe** how DCF is applied to private market assets in current industry practice, including the typical structure of a discount rate (risk-free + ERP + size + illiquidity).
3. **Preview** the GE-LAV® valuation hierarchy (DCF → LAV → GE-LAV) and the five main results of the GE-LAV framework.
4. **Identify** that the illiquidity premium is the critical and problematic component of the DCF discount rate.
5. **Commit** to a course working agreement: track preview, assessment expectations, AI use policy, project orientation.

9.2 Pre-Class Assignment

- **Read:** Book Preface + Chapter 1, sections 1.1–1.4 (~25 pages)
- **Read:** [Course syllabus](#) and [Tracks page](#) in full
- **Optional:** Skim Preqin Global Private Capital Secondary Market Report Q4 2024 (link on course site)

9.3 In-Class Outline (75 minutes)

Time	Segment	Format
0:00–0:10	Course welcome · instructor intro · agenda for the semester	Lecture
0:10–0:20	The scale of the problem: why \$13T at stake	Lecture + visuals
0:20–0:35	How DCF is applied to private assets in current practice	Lecture
0:35–0:45	Why the illiquidity premium is the problematic component	Lecture + discussion
0:45–0:55	Preview: the valuation hierarchy and 5 main results	Lecture
0:55–1:05	Course logistics: tracks, assessments, AI policy, project	Lecture + Q&A
1:05–1:15	Discussion: “Where in your career will you encounter this?”	Discussion

9.4 Slide Content Briefs (New Slides Needed)

9.4.1 Slide 1.1 — Title Slide

- **Title:** “Why Private Market Valuation Is Broken”
- **Subtitle:** “Session 1 · Unit 1 · Liquidity Illusion · GE-LAV® Framework”
- **Visual:** Photo or stylized image of a private market context (no copyrighted material — consider a generic chart showing PE NAV growth 1990–2025)
- **Footer:** Course name, instructor name, school

9.4.2 Slide 1.2 — Course Agenda

- **Title:** “What we’ll do today”
- **Bullets:**
 - The scale of the problem
 - How DCF is applied to private assets
 - Why the illiquidity premium is the weak link
 - Preview: the GE-LAV framework
 - Course logistics: tracks, assessment, project
- **Speaker notes:** Frame the day as motivational — students should leave wanting to take the rest of the course.

9.4.3 Slide 1.3 — The Scale of the Problem

- **Title:** “Private capital is now too big for valuation errors to be tolerable”
- **Stat callouts (large numbers):**

- **\$13T** Global private capital AUM (2025)
- **~10,000+** Active private market funds
- **\$2.5T** Public pension fund private market allocations
- **15-25%** Typical private market allocation for endowments and large pensions
- **Bottom-of-slide line:** “A 5% systematic valuation error implies \$650B mismeasured. This is not a rounding issue.”
- **Visual:** Single bar showing PE+VC+Infra+RE+PC AUM growth 2000–2025 (use stylized bars; do not reproduce copyrighted images)
- **Speaker notes:** This is the motivation. Students need to feel that this matters. Reference specific allocators (CalPERS, CalSTRS, Yale endowment) by way of example.

9.4.4 Slide 1.4 — Who Cares About This?

- **Title:** “Who has skin in the game”
- **Format:** Four-quadrant or icon-grid layout
- **Items:**
 - **LPs** (pensions, endowments, sovereign wealth funds, insurers): need correct NAVs for allocation, rebalancing, and reporting
 - **GPs** (PE/VC/RE/PC fund managers): need correct valuations for IC review, board reporting, and carry calculation
 - **Regulators** (insurance, banking, securities): need correct valuations for capital adequacy and systemic risk
 - **Auditors / valuation firms:** need defensible methodology for fair value reporting
- **Speaker notes:** Career relevance — wherever students end up, this matters. Ask: “Who here is going to work for an LP? GP? Regulator? Auditor?” — get a hand count.

9.4.5 Slide 1.5 — How DCF Is Applied (Standard Practice)

- **Title:** “DCF in current private market practice”
- **Layout:** Equation with annotated components
- **Equation:** $V_0 = \sum_{t=1}^T \frac{CF_t}{(1+r)^t}$
- **Discount rate decomposition (boxed):** $r = r_f + \beta \cdot ERP + \text{Size premium} + \text{Illiquidity premium}$
- **Typical PE buyout calibration:**
 - $r_f = 4.0\%$ (SOFR + term)
 - $\beta \cdot ERP = 3\text{--}4\%$
 - Size premium = 1–2%
 - **Illiquidity premium = 3–4%**
 - Total = **11–14%**, held constant for fund life
- **Speaker notes:** The whole rest of the lecture is about that bottom row. Why is it 3–4%? Why is it constant? Both questions are wrong.

9.4.6 Slide 1.6 — Where the Illiquidity Premium Comes From

- **Title:** “The 3% illiquidity premium: history, not theory”
- **Bullets:**
 - Amihud & Mendelson (1986): illiquidity premium = bid-ask spread (for public markets)
 - Private market application: averaged historical excess return of PE over public equity (~3.5%)
 - IPEV valuation guidelines (2022): endorse this practice
 - Solvency II (insurance): embeds similar fixed premium structure
 - **Result:** the 3.5% is a *historical average*, not a *theoretical equilibrium*
- **Speaker notes:** Stress that this is “what the data showed on average” — but averages mask everything that matters about liquidity.

9.4.7 Slide 1.7 — Secondary Market Evidence (Tease)

- **Title:** “What the secondary market shows”
- **Visual:** Time series chart of secondary market discount to NAV, 2003–2024 (you’ll need to build or recreate this; cite Preqin data)
- **Annotations on chart:**
 - 2007 peak: 3–8% discount
 - 2009 GFC trough: 35–50% discount
 - 2020 COVID: 15–25% discount briefly
 - 2022 rate shock: 15–20% discount sustained
- **Bottom-line:** “If the premium is fixed at 3.5%, where do these moves come from?”
- **Speaker notes:** This is the *first appearance* of the chart that will dominate Session 3. Tease it here; don’t unpack fully.

9.4.8 Slide 1.8 — The Three Structural Failures (Preview)

- **Title:** “Three reasons DCF fails for private assets”
- **Layout:** Three columns or three icons
- **Failure 1:** “The premium is stochastic, not constant”
- **Failure 2:** “Premium variability creates Jensen bias”
- **Failure 3:** “Liquidity is collectively determined (McKean-Vlasov externality)”
- **Speaker notes:** This is a preview. Session 2 unpacks each failure in detail. Don’t drill down here.

9.4.9 Slide 1.9 — The Valuation Hierarchy

- **Title:** “Three nested theories”
- **Visual:** Three nested rectangles or set-theoretic diagram showing **DCF** ⊂ **LAV** ⊂ **GE-LAV**
- **Annotations:**
 - DCF: constant premium, partial equilibrium, deterministic discount factor

- LAV: stochastic premium (OU process), partial equilibrium, Jensen-corrected discount factor
- GE-LAV: stochastic premium, **general equilibrium** (market clearing), McKean-Vlasov collective dynamics
- **Speaker notes:** This diagram appears throughout the course. Establish it now. Strict containment matters — DCF is a special case of LAV which is a special case of GE-LAV.

9.4.10 Slide 1.10 — The Five Main Results (Preview)

- **Title:** “What GE-LAV delivers”
- **Numbered list:**
 1. **Jensen convexity bias:** closed-form formula, calibrated 0.8–3.6% by asset class
 2. **Optimal exit boundary:** when to hold, sell, or delay
 3. **Welfare gap:** quantified cost of the McKean-Vlasov externality (~2.3%/year)
 4. **Pigouvian exit tax:** the corrective instrument
 5. **Valuation hierarchy:** strict nesting DCF LAV GE-LAV
- **Speaker notes:** Foreshadowing. Each of these is a session or set of sessions later. Tell students they will be able to compute or apply each by mid-semester.

9.4.11 Slide 1.11 — A Worked Number to Anchor the Course

- **Title:** “The \$73 million gap”
- **Setup:** “A pension fund holds a 20-year infrastructure asset”
- **DCF mark:** \$103.6M (constant 7.5% discount, 3.5% illiquidity premium)
- **GE-LAV equilibrium at GFC depth ($L = -1.5$):** \$30.9M
- **Gap:** \$73M, or 70.1%
- **Bottom-line:** “This is not a mark-to-market fluctuation. It is the correct equilibrium price at which the market would clear.”
- **Visual:** Two bars side-by-side — DCF vs. GE-LAV, with the gap shaded
- **Speaker notes:** This is the single most powerful number in the course. Anchor everything around it. Refer back when students ask “but how big is the error really?”

9.4.12 Slide 1.12 — Course Roadmap

- **Title:** “Where we go from here”
- **Visual:** Sankey or flow diagram showing Sessions 1–32 organized by unit
- **Highlight:** Sessions 1–10 (diagnosis + measurement), 11–18 (decision + application), 19–24 (math intuition), 25–31 (split track), 32 (capstone)
- **Speaker notes:** This is in the syllabus, but show it visually now. Students need to know what they’re committing to.

9.4.13 Slide 1.13 — Tracks Preview

- **Title:** “Track 1 (Practitioner) vs. Track 2 (Researcher)”
- **Visual:** Side-by-side comparison table
- **Key differentiators:** math depth in sessions 25–31, PSet content, project orientation
- **Decision point:** “Choose by end of Session 5”
- **Speaker notes:** Reassure students that the default is Track 1 if they don’t choose. Tell them Session 4 will be a deliberate self-test.

9.4.14 Slide 1.14 — Assessment

- **Title:** “How you’ll be graded”
- **Pie chart or table:** Midterm 20% / PSets 25% / Project 50% / Participation 5%
- **Key dates:** PS1 due Session 8, Midterm Session 10, Project proposal Session 14, Project presentations Session 32
- **Speaker notes:** Highlight that there is no final exam. The project replaces it. Project is 50% — invest accordingly.

9.4.15 Slide 1.15 — Working Agreement: Discussion Questions

- **Title:** “Pair discussion: 5 minutes”
- **Prompts:**
 - Where in your career do you expect to encounter private market valuation?
 - What’s one thing you already believe about DCF that this course might change?
 - Which track are you leaning toward and why?
- **Speaker notes:** Use this for the last 10 minutes. Walk around, listen, get a sense of the room. This is also informally diagnostic of student backgrounds.

9.5 Existing Slides to Reuse from Class 9 (“Private Capital Markets: Scale, Structure & Core Frictions”)

From your existing Deck 2 (Classes 6–10), slides covering Class 9 — pull approximately 6 slides:

- Class 9 slide on scale of private capital (\$13T figure)
- Class 9 slide on PE/VC/Infra/PC AUM breakdown
- Class 9 slide on LP/GP frictions (information asymmetry, illiquidity, agency)
- Class 9 slide on secondary market structure (Lazard, Evercore, etc.)
- Class 9 slide on capital call / distribution timing (J-curve intuition)
- Class 9 slide on regulatory landscape (IPEV, IFRS 13, Solvency II at high level)

Adaptation needed: Mostly editorial — strip any math-heavy content, keep the institutional and structural content. These slides become slides 1.3, 1.4, and parts of 1.6.

9.6 Discussion Questions

For the last 10 minutes of class. Choose 2–3 depending on time:

1. The Black-Scholes-Merton option pricing formula was published in 1973 but not adopted by major investment banks until the late 1980s. What lessons does this hold for GE-LAV's path to adoption?
2. If you were a CIO of a \$50B pension fund and learned that GE-LAV implies your private market portfolio is overvalued by 15% in current conditions, what would you do? What would the constraints be?
3. IPEV guidelines and IFRS 13 embed DCF-style methodology. Is changing the methodology a technical problem or a political/coordination problem? Both?

9.7 Worked Example for the Lecture

The Infrastructure Mispricing Example. A 20-year infrastructure asset with \$50M annual contracted cash flows.

Method	Discount rate	PV
DCF (constant 7.5%, illiq premium 3.5%)	$r = 7.5\%$	\$103.6M
LAV (Jensen-corrected, normal markets)	$r = 7.7\%$	~\$108M
GE-LAV (normal markets)	$r = 8.4\%$	~\$95M
GE-LAV (GFC depth, $L = -1.5$)	$r = 32.3\%$	\$30.9M

Teaching note: Walk through *why* GFC GE-LAV is so far from DCF: it's the McKean-Vlasov collective externality — when everyone tries to exit simultaneously, the equilibrium price clears at a much higher rate than the GP-assumed constant. This is the preview of what comes in later sessions.

9.8 Common Student Confusions (Instructor Notes)

- **“If GE-LAV says NAV is lower, isn't that just a markdown?”** No — the GE-LAV value is the equilibrium clearing price *given collective behavior*. It's the correct value, not a discount on the correct value. The DCF “value” is the wrong value.
- **“If everyone uses DCF, doesn't that make it the right value?”** No — convention isn't correctness. The market clearing price (secondary market) reveals the true equilibrium; DCF is a model that fails to compute it correctly.
- **“Why hasn't the industry adopted this already?”** Practitioner consensus trap (covered in section 1.4 of book) — being the first GP to mark down to GE-LAV creates LP pushback. Coordination problem.
- **“Is this an academic exercise or does it matter for real portfolios?”** \$73M on one \$103M asset matters. Magnify across a \$13T market and the implications are systemic.

9.9 What to Expect Next Session

Session 2 unpacks each of the three structural failures in detail. We'll cover:

- The secondary market data that proves the premium is stochastic
- The mathematics of Jensen's inequality and why convex discount factors create systematic bias
- The McKean-Vlasov externality — what it is and why it makes liquidity a *collective* phenomenon, not an individual asset property
- The Liquidity-Adjusted CAPM as a connection to the portfolio theory students already know

Reading: Book Chapter 1, sections 1.5–1.7. Skim sections 1.8–1.10 (the worked example).

[← Back to Schedule](#) | [Next: Session 2 →](#)

10 Session 2: The Three Structural Failures · LA-CAPM

Unit	1 — Why DCF Fails
Book Chapter	1 (sections 1.5–1.7)
Slide source	New (~12 slides) + Class 9 leftovers
Slides needed	~12 new
Track	Common core (both tracks)

10.1 Learning Objectives

By the end of this session, students will be able to:

1. **State and explain** the three structural failures of DCF for private market valuation.
2. **Articulate Jensen’s inequality** at an intuition level — why $E[f(X)] \neq f(E[X])$ when f is nonlinear — and apply it to the discount factor e^{-rT} .
3. **Explain** the McKean-Vlasov externality as a collective-action problem in liquidity provision.
4. **Position** GE-LAV within the existing illiquidity premium literature (Amihud-Mendelson, Pastor-Stambaugh, Acharya-Pedersen).
5. **State** the Liquidity-Adjusted CAPM (LA-CAPM) and how it generalizes standard CAPM.

10.2 Pre-Class Assignment

- **Read:** Book Chapter 1, sections 1.5–1.7 (~15 pages)
- **Optional:** Acharya & Pedersen (2005), “Asset Pricing with Liquidity Risk,” *Journal of Financial Economics* — abstract and conclusion only

10.3 In-Class Outline (75 minutes)

Time	Segment	Format
0:00–0:05	Recap Session 1: scale, DCF, illiquidity premium	Lecture
0:05–0:25	Failure 1: The premium is stochastic — secondary market evidence	Lecture

Time	Segment	Format
0:25–0:45	Failure 2: Jensen convexity bias from stochastic premia	Lecture + worked numerical example
0:45–1:05	Failure 3: Liquidity is collectively determined — McKean-Vlasov	Lecture + thought experiment
1:05–1:15	LA-CAPM — connection to portfolio theory you already know	Lecture + Q&A

10.4 Slide Content Briefs (New Slides Needed)

10.4.1 Slide 2.1 — Title and Recap

- **Title:** “Three Structural Failures of DCF”
- **Visual:** Carry forward the valuation hierarchy diagram from Session 1
- **One-line recap:** “Last session: the illiquidity premium is the weak link. Today: three specific ways it breaks DCF.”

10.4.2 Slide 2.2 — Failure 1: The Premium Is Stochastic

- **Title:** “Failure 1: A constant premium contradicts the data”
- **Layout:** Headline + key evidence
- **Headline:** “Secondary market discounts move 10× the magnitude of the assumed constant premium”
- **Evidence (table):**
 - 2007 peak: secondary discount 3–8% → implied premium near floor
 - 2009 GFC: discount 35–50% → implied premium ~28%
 - 2020 COVID: discount 15–25% → implied premium ~18%
 - 2022 rate shock: discount 15–20% sustained → premium ~12%
 - “Long-run mean: ~3.5%. Volatility: 0.34”
- **Bottom-line:** “If the assumption is 3.5% constant, the data is screaming ‘no.’”
- **Speaker notes:** Establish the credibility of the data. Mention Preqin, Setter Capital, and broker reports.

10.4.3 Slide 2.3 — The OU Calibration (Numbers)

- **Title:** “The Ornstein-Uhlenbeck calibration”
- **Equation (large, centered):** $dL_t = \kappa(\bar{L} - L_t) dt + \sigma_L dW_t$
- **Calibrated parameters (boxed):**
 - $\kappa = 0.45/\text{yr}$ (mean reversion speed; half-life 18 months)
 - $\bar{L} = 1.0$ (normalized long-run mean liquidity state)

- $\sigma_L = 0.32$ (diffusion)
- Stationary std: $\sqrt{\sigma_L^2/(2\kappa)} \approx 0.34$
- **Visual:** Simulated paths of OU process — 5 sample paths overlaid, all converging to $\bar{L}$ but with stochastic fluctuation
- **Bottom-line:** “This is the engine that drives everything in the course. Memorize the form.”
- **Speaker notes:** Don’t derive yet — that’s Session 4. Just establish the process and the numbers. Connect to a half-life concept (similar to interest rate models — Vasicek).

10.4.4 Slide 2.4 — Failure 2: Jensen Convexity Bias (Setup)

- **Title:** “Failure 2: Even if you got the mean right, the math fails”
- **Setup (text):** “Suppose by miracle you knew $E[r] = 7.5\%$. Is your DCF correct?”
- **Statement of Jensen’s inequality:**
 - For any **strictly convex** function f and any **non-degenerate** random variable X :

$$E[f(X)] > f(E[X])$$

- **Visual:** Standard Jensen’s inequality diagram — convex function with two points and a chord
- **Speaker notes:** Many students will have seen Jensen in stats class but never in finance. Make the connection.

10.4.5 Slide 2.5 — Why Discount Factors Are Convex

- **Title:** “The discount factor is strictly convex in the rate”
- **Math display:**
 - $f(r) = e^{-rT}$
 - $f'(r) = -Te^{-rT}$
 - $f''(r) = T^2e^{-rT} > 0 \quad \forall r, T > 0$
- **Implication:** “ $E[e^{-rT}] > e^{-E[r]T}$ whenever r is genuinely stochastic”
- **Visual:** Plot of e^{-rT} as function of r for $T = 10$ years, with the chord vs. curve gap shaded
- **Speaker notes:** Convexity gets larger with T — this is why long-horizon assets have larger Jensen bias.

10.4.6 Slide 2.6 — The Magnitude of the Jensen Bias

- **Title:** “Jensen bias by asset class and horizon”
- **Layout:** Table
- **Data:**

Asset Class	Horizon T	Jensen Bias B(T)
5-year buyout	5 yr	0.8%
10-year growth equity	10 yr	1.7%
15-year secondaries	15 yr	2.5%

Asset Class	Horizon T	Jensen Bias B(T)
20-year infrastructure	20 yr	3.6%

- **Bottom-line:** “Bias scales approximately linearly with T at calibrated parameters.”
- **Speaker notes:** These are *closed-form* results derived in Chapter 18 of the book. Track 2 students will see the proof in Session 29.

10.4.7 Slide 2.7 — Worked Numerical Example: 20-Year Infrastructure

- **Title:** “Worked: a \$1B infrastructure fund”
- **Layout:** Comparison table
- **Data:**
 - DCF mark (constant 7.5% discount): \$1,000M
 - LAV mark (Jensen-corrected, normal markets): \$1,036M
 - **Gap: \$36M = 3.6% on Jensen bias alone**
 - GE-LAV mark (normal markets): can be lower or higher depending on regime
 - GE-LAV mark (GFC depth): ~\$300M
- **Speaker notes:** This is the building block of the larger story. The Jensen bias is *one* of the three corrections. The McKean-Vlasov correction (Failure 3) is the bigger one, especially in stressed regimes.

10.4.8 Slide 2.8 — Failure 3: McKean-Vlasov Externality (Setup)

- **Title:** “Failure 3: Liquidity is collective”
- **Thought experiment:**
 - A single LP wants to exit a \$50M PE position
 - Lazard quotes 5% discount on volume; the LP transacts; no aggregate effect
 - Now 100 large pension funds each want to exit \$500M positions simultaneously
 - The total \$50B is suddenly trying to find buyers
 - Equilibrium clearing price drops; discount widens to 25–40%
- **Punchline:** “The clearing price depends on *everyone’s* behavior, not just yours.”
- **Speaker notes:** This is the central conceptual move of GE-LAV. McKean-Vlasov is the mathematical formalization of this dependence — but the *idea* is intuitive.

10.4.9 Slide 2.9 — McKean-Vlasov: The Math Preview (Intuition Only)

- **Title:** “How math formalizes ‘liquidity depends on everyone’ ”
- **Layout:** Side-by-side
- **Left (standard SDE):** $dL_t = \kappa(\bar{L} - L_t) dt + \sigma_L dW_t$ — *just one investor’s view*
- **Right (McKean-Vlasov):** $dL_t = \kappa(\bar{L} - L_t) dt + \gamma \int \phi(L_t, l') \mu_t(dl') dt + \sigma_L dW_t$ — *includes coupling to distribution μ_t of all investors*
- **Intuition:** “ μ_t is the *aggregate* distribution of liquidity states across all investors. Your dynamics depend on it. Theirs depends on yours.”

- **Bottom-line:** “This is a fixed point: behavior depends on premium; premium depends on behavior.”
- **Speaker notes:** Do NOT derive. Just point at the structure. Track 2 students see this for real in Session 26.

10.4.10 Slide 2.10 — Why DCF Misses This Entirely

- **Title:** “DCF cannot see the externality”
- **Bullets:**
 - DCF takes the illiquidity premium as a fixed parameter
 - It cannot ask: “What premium will clear the market given other investors’ positions?”
 - It is, mathematically, a *partial equilibrium* calculation
 - The 2008 crisis revealed this: every DCF mark was wrong because they ignored coordinated exit behavior
- **Speaker notes:** Connect to 2008. Many students will know the story of “the markdown that wasn’t.” This is the formalization.

10.4.11 Slide 2.11 — The Three Failures Together

- **Title:** “Failures compound”
- **Layout:** Three-row table
- **Rows:**
 - **Failure 1:** stochastic premium → adds 2–4% premium volatility to base 3.5%
 - **Failure 2:** Jensen bias → adds 1–4% systematic positive bias in valuation
 - **Failure 3:** McKean-Vlasov externality → adds 5–20% in normal markets, 50–70% in crisis
- **Bottom-line:** “Each failure compounds the others. GE-LAV is the framework that handles all three simultaneously.”
- **Speaker notes:** This slide motivates the rest of the course. Each failure gets its own session(s).

10.4.12 Slide 2.12 — Liquidity-Adjusted CAPM

- **Title:** “LA-CAPM: connecting to portfolio theory you know”
- **Layout:** Standard CAPM → LA-CAPM transformation
- **Standard CAPM:** $E[r_i] - r_f = \beta_i(E[r_m] - r_f)$
- **LA-CAPM (Acharya-Pedersen 2005):** $E[r_i] - r_f = E[c_i] + \beta_i^1 \cdot (\text{premium}_m) + \beta_i^2 \cdot (\text{premium}_L) - \beta_i^3 \cdot (\text{cov with } r_m, L)$
- **Annotation:** Four channels — (1) expected illiquidity cost; (2) standard market beta; (3) liquidity beta (sensitivity to aggregate liquidity); (4) cross-effects
- **GE-LAV connection:** “GE-LAV refines the *aggregate liquidity factor* into a stochastic, equilibrium-determined process”
- **Speaker notes:** This is the bridge from standard finance theory to GE-LAV. Students with prior asset pricing background will recognize the structure.

10.5 Discussion Questions

For the last 10 minutes if time permits:

1. Which of the three failures do you think most practitioners are aware of? Which are they unaware of? Why?
2. The Pastor-Stambaugh (2003) liquidity factor is widely used in factor models. Is it a satisfactory response to the three failures we covered today? What does it miss?
3. Suppose you were the regulator overseeing private market fund NAVs. Which of the three failures would you prioritize fixing first? Why?

10.6 Worked Example: Quantifying Each Failure

Setup: A 10-year private equity fund, current NAV \$100M.

Failure 1 (stochastic premium): If the OU stationary std is 0.34 and current state is at the long-run mean, the *expected* premium next year has standard deviation $\sim 3.4\%$. A constant premium of 3.5% has 0 std. Failure: ignoring 3.4 percentage points of uncertainty.

Failure 2 (Jensen bias): For $T = 10$ and the calibrated parameters, $B(10) \approx 1.7\%$. So LAV value $101.7M$ vs. DCF $100M$. **Failure:** $1.7M$ systematic undervaluation in DCF.

Failure 3 (McKean-Vlasov): In normal markets, GE-LAV equilibrium rate exceeds DCF assumed rate by $\sim 1\%$, so GE-LAV value $\sim 98M$. In GFC depth, equilibrium rate is 32.3% vs. DCF 7.5% $\rightarrow$ GE-LAV value $\sim 30M$.

Walk through: “Are the corrections all in the same direction? Sometimes (Failure 2 always positive). Sometimes they offset (Failure 2 positive, Failure 3 can be negative in crisis). GE-LAV handles them simultaneously and correctly.”

10.7 Common Student Confusions (Instructor Notes)

- “Jensen bias seems small. Why do we care?” Two reasons: (1) it’s *always positive and systematic* — every long-horizon asset is underpriced by DCF, every quarter, forever; (2) cumulative across the \$13T market, the bias is $\sim \$300B$ – $\$500B$ mispricing.
- “McKean-Vlasov sounds like a panic-selling story. Isn’t that already in CAPM via correlations?” No — CAPM uses *covariance* (a linear concept). The MV externality is *fixed-point structure*: clearing price depends on coordinated behavior, not just correlation. Different math entirely.
- “If LA-CAPM exists, why do we need GE-LAV?” LA-CAPM (Acharya-Pedersen) is a static factor model. GE-LAV is a *continuous-time general equilibrium* with endogenous, stochastic, collectively-determined premium dynamics. LA-CAPM is the “linearized snapshot”; GE-LAV is the dynamic system.

10.8 What to Expect Next Session

Session 3 dives deep into Failure 1: the secondary market evidence. We'll examine the data carefully — what the secondary market quotes mean, who participates, the four regimes (2003–2007 boom, 2008–2010 GFC, 2011–2019 normalization, 2020–present multi-shock era), and what the discount-to-NAV ratio actually measures. This is the **empirical** session of Unit 1.

Reading: Book Chapter 2, all sections (~30 pages). Pay particular attention to sections 2.3–2.6.

[← Session 1](#) | [Schedule](#) | [Next: Session 3](#) →

11 Session 3: The Liquidity Illusion · Secondary Market Evidence

Unit	1 — Why DCF Fails
Book Chapter	2 (sections 2.1–2.7)
Slide source	New (~12 slides) + Classes 21–22 excerpts
Slides needed	~12 new
Track	Common core (both tracks)

11.1 Learning Objectives

By the end of this session, students will be able to:

1. **Describe the structure** of the private market secondary market — who trades, why, through which brokers, at what volume.
2. **Interpret** the discount-to-NAV ratio as a market-implied measurement of the aggregate liquidity premium.
3. **Identify the four regimes** of secondary market discounts 2003–2024 and describe what economic conditions characterize each.
4. **Define the liquidity illusion** rigorously: the gap between GP-reported NAV and economic clearing value.
5. **Distinguish** level illusion (level of NAV is wrong) from convexity illusion (NAV’s sensitivity to liquidity state is wrong).

11.2 Pre-Class Assignment

- **Read:** Book Chapter 2, all sections (~30 pages)
- **View:** Pre-class video (5 min) — “What is the PE secondary market?” (link on course site)
- **Skim:** Most recent Preqin or Setter Capital secondary market report (link on course site)

11.3 In-Class Outline (75 minutes)

Time	Segment	Format
0:00–0:05	Recap Session 2: three failures, focus today on Failure 1 evidence	Lecture
0:05–0:25	Secondary market structure — who, what, how, why	Lecture + diagram
0:25–0:50	The four regimes 2003–2024: empirical walkthrough	Lecture + chart deep-dive
0:50–1:05	Level illusion vs. convexity illusion	Lecture
1:05–1:15	Implications for portfolio mark-to-market	Discussion + Q&A

11.4 Slide Content Briefs (New Slides Needed)

11.4.1 Slide 3.1 — Title and Recap

- **Title:** “The Liquidity Illusion: What Secondary Markets Reveal”
- **Tag line:** “Failure 1, examined empirically”
- **Speaker notes:** Today is the *evidence* day. Session 2 told them the premium is stochastic. Today we look at the data.

11.4.2 Slide 3.2 — Secondary Market: Structure

- **Title:** “The PE secondary market — who participates”
- **Layout:** Diagram or three-column grid
- **Sellers:**
 - LPs needing liquidity (denominator effect, funding obligations, portfolio rebalancing)
 - LPs exiting underperforming relationships
 - Institutional restructurings (insurance run-offs, sovereign wealth reallocation)
- **Buyers:**
 - Dedicated secondaries funds (Ardian, Lexington, Coller, HarbourVest, Pantheon)
 - Strategic buyers (other GPs acquiring stakes in companies they know)
 - LP-led tender offers
- **Brokers/intermediaries:** Lazard, Evercore, Goldman Sachs Asset Management, Hamilton Lane, Setter Capital, PJT Park Hill
- **Speaker notes:** Students may not know how this market is organized. Establish basic plumbing.

11.4.3 Slide 3.3 — Market Scale

- **Title:** “Secondary market in numbers”
- **Stat callouts:**
 - **\$130B+** Annual transaction volume (2024)
 - **~1,200** Transactions per year tracked by Preqin
 - **5–8%** of total PE NAV trades through secondaries each year
 - **Growing 12–15%/yr** since 2015
- **Bottom-line:** “Large enough to be informationally relevant; structured enough to extract a price signal”
- **Speaker notes:** This is *not* a distressed market. It’s a real, deep, growing market.

11.4.4 Slide 3.4 — The Discount-to-NAV Ratio

- **Title:** “Reading the price signal”
- **Equation:** $\text{Discount} = 1 - \frac{\text{Secondary transaction price}}{\text{GP-reported NAV}}$
- **Worked example:**
 - GP reports NAV of \$100M for an LP position
 - Secondary buyer pays \$85M cash
 - Discount = 15%
- **Translation to liquidity premium:** “For a 5-year hold position with normal mean reversion, a 15% discount corresponds to roughly a +5% addition to the implied annual discount rate over the hold period”
- **Speaker notes:** This is the data we observe. Everything that follows is interpreting this number across time and assets.

11.4.5 Slide 3.5 — The Four Regimes (Overview)

- **Title:** “Four regimes of secondary discounts (2003–2024)”
- **Visual:** Full time series chart 2003–2024, shaded by regime
- **Regimes:**
 - **2003–2007:** Boom (discounts 0–10%)
 - **2008–2010:** Crisis (discounts 25–50%)
 - **2011–2019:** Normalization (discounts 5–15%)
 - **2020–2024:** Multi-shock era (discounts 10–25% with volatility)
- **Speaker notes:** Each regime gets one slide of detail.

11.4.6 Slide 3.6 — Regime 1: The Boom (2003–2007)

- **Title:** “The boom: cheap money, no shocks”
- **Key features:**

- Tight discounts (often 0–5%)
- High deal volume but driven by GP-side restructurings, not LP distress
- Liquidity state L_t persistently above $\bar{L}$
- Implied premium: 1–3% (vs. long-run 3.5%)
- **Speaker notes:** This is the regime DCF was calibrated against. The “constant premium” assumption is *least wrong* here.

11.4.7 Slide 3.7 — Regime 2: The Crisis (2008–2010)

- **Title:** “The crisis: the constant-premium assumption breaks”
- **Key features:**
 - Discounts widen to 35–50%
 - Volume *initially collapses* (no buyers willing to bid), then recovers in mid-2009 as opportunistic buyers enter
 - Liquidity state L_t at -1.5 or worse
 - Implied premium peak: ~28%
- **Anecdote:** “In Q1 2009, some endowments sold PE positions at 60% of NAV to meet liquidity needs. CalPERS publicly disclosed selling secondaries at large discounts to rebalance.”
- **Speaker notes:** This is the regime that should have killed the constant-premium assumption. It didn’t, because there was no theoretical alternative.

11.4.8 Slide 3.8 — Regime 3: Normalization (2011–2019)

- **Title:** “The normalization: false comfort”
- **Key features:**
 - Discounts settle at 5–15%
 - Deal volume grows steadily
 - GP-led secondaries (continuation funds) become common
 - Implied premium: 4–7%
- **Speaker notes:** Practitioners drew the wrong lesson from this period. “The crisis was a one-off; the normal state is stable.” Wrong.

11.4.9 Slide 3.9 — Regime 4: Multi-Shock Era (2020–2024)

- **Title:** “The multi-shock era: volatility is the new normal”
- **Key shocks:**
 - **March 2020:** COVID liquidity shock — discounts spike to 15–25%, then recover
 - **2022:** Rate shock + denominator effect — discounts widen to 15–20% sustained
 - **2023–24:** Persistent elevated discounts as rates stayed high
- **Implied premium:** 8–15% sustained
- **Speaker notes:** This regime is the strongest argument for GE-LAV. Repeated shocks, persistent dislocation, no return to pre-2020 calm.

11.4.10 Slide 3.10 — The Liquidity Illusion Defined

- **Title:** “What is the liquidity illusion?”
- **Definition (boxed):** “The systematic gap between GP-reported NAV (computed via DCF with constant premium) and the economic clearing value (computed via GE-LAV equilibrium).”
- **Decomposition:**
 - **Level illusion:** Level of GP-reported NAV is wrong (Failure 1: stochastic premium)
 - **Convexity illusion:** Sensitivity of NAV to changes in liquidity state is wrong (Failure 2: Jensen bias)
- **Bottom-line:** “Both illusions persist because the GP’s mark uses a model that cannot see liquidity state dynamics”
- **Speaker notes:** The term “illusion” is deliberate. NAVs *appear* stable but mask large, regime-dependent valuation gaps.

11.4.11 Slide 3.11 — Level Illusion: Numerical Example

- **Title:** “Level illusion: a 25-year infrastructure asset”
- **Setup:**
 - GP-reported NAV (DCF, constant 7.5% rate): \$100M
 - True economic value:
 - * In Regime 3 (normalization): ~\$95M (close to GP mark)
 - * In Regime 2 (crisis): ~\$60M (40% gap)
 - * In Regime 4 (multi-shock): ~\$80M (20% gap)
- **Bottom-line:** “GP mark of \$100M is approximately correct ONLY in normalization regimes. Across the full cycle, it’s wrong by an average of ~12%.”
- **Speaker notes:** Anchor this with the dollar magnitudes. Make the gap viscerally real.

11.4.12 Slide 3.12 — Implications for Portfolio Mark-to-Market

- **Title:** “What this means for your portfolio”
- **Bullets:**
 - GP-reported NAVs **smooth and lag** true economic value
 - During crisis: GP marks *overstate* value (no markdown despite secondary market signal)
 - During boom: GP marks *understate* value (no markup despite tight discounts)
 - The smoothing effect is what makes private markets *appear* uncorrelated with public markets
 - **The diversification benefit of private markets is partly an artifact of stale marks**
- **Bottom-line:** “If you mark to market correctly using secondary market evidence, the diversification benefit is smaller than commonly believed.”
- **Speaker notes:** This is the controversial slide. It directly challenges a core marketing pitch of private markets — “uncorrelated returns.” Be prepared for pushback. Lock-step with academic literature (Stafford, Phalippou) that has made this point.

11.5 Existing Slides to Reuse

From your existing Deck 5 (Classes 21–22 PE Buyout and Infrastructure):

- Class 21 slides showing real secondary market discount data with annotations
- Class 22 infrastructure slides showing long-horizon valuation gaps
- Adapt these to fit the regime-based narrative

Approximately 4–6 slides from these can be brought forward with minimal modification.

11.6 Discussion Questions

1. The 2008 crisis revealed the constant-premium assumption was wrong. Why did it take 17+ years (and counting) for the industry to seriously revisit DCF methodology?
2. If GP-reported NAVs smooth and lag, is private market “diversification” partly an illusion? How big is the illusion?
3. The denominator effect (LPs selling secondaries because public market declines push private allocations over limits) is mechanical, not a true liquidity signal. How would you adjust the discount-to-NAV interpretation to account for this?

11.7 Worked Numerical Example

Computing the implied liquidity state from a secondary discount

Setup:

- Secondary discount: 18%
- Remaining fund life: 6 years
- Risk-free rate: 4%, $\text{ERP} \times \text{size} = 4\%$, size = 1%
- GP-assumed illiquidity premium: 3.5%
- DCF total rate: 12.5%

Step 1: Implied total discount rate from secondary:

For a perpetuity approximation: $\text{Discount} \approx 1 - e^{-r \cdot T_{\text{eff}}}$ where T_{eff} is the duration-weighted hold.

With 6 years, the implied rate r_{implied} satisfies $0.18 = 1 - e^{-r_{\text{implied}} \cdot 6}$, giving $r_{\text{implied}} \approx 3.3\%$ premium above the DCF rate.

Step 2: Implied excess premium:

Excess above GP-assumed: ~3.3 percentage points. So implied total illiquidity premium = 3.5% + 3.3% = **6.8%**, roughly double the GP assumption.

Step 3: Liquidity state inversion:

Using $\pi(L) = \pi_0 - \pi_1 L + \pi_2 L^2$ with calibrated $\pi_0 = 0.045$, $\pi_1 = 0.025$, $\pi_2 = 0.021$, and current premium 6.8%, solve for L :

$$0.068 = 0.045 - 0.025L + 0.021L^2$$

Quadratic in L : $0.021L^2 - 0.025L - 0.023 = 0$

Solutions: $L = (0.025 \pm \sqrt{0.000625 + 0.001932})/0.042 = (0.025 \pm 0.0506)/0.042$

Taking the negative root (since L should drop below $\bar{L}$ when premium rises): $L \approx -0.61$, or **0.61 below the long-run mean**.

Interpretation: Current secondary market is pricing in a moderate-stress liquidity state. Not crisis (which would be -1.5 or worse), but not normalization either.

11.8 Common Student Confusions (Instructor Notes)

- **“Aren’t secondary discounts noisy / not representative?”** Volume of \$130B+/yr with \$1,200+ transactions is more than adequate for statistical inference. The Preqin and Setter aggregations smooth idiosyncratic noise.
- **“Doesn’t the denominator effect contaminate the signal?”** Partially yes, but the regime-level signal is robust. Even excluding pure denominator-driven sales, the 2022 widening was clear.
- **“How is this different from the bond yield curve telling us about expected default rates?”** Conceptually similar. The secondary market is doing for private equity what corporate bond spreads do for credit — extracting market-implied premia. But private markets had no such signal before secondaries scaled up.
- **“If GP NAVs smooth, doesn’t reported portfolio variance underestimate true variance?”** Yes — Stafford (2022) and others document this. Once you correct for smoothing, the Sharpe ratio of PE is much closer to public equity than gross numbers suggest.

11.9 What to Expect Next Session

Session 4 introduces the **Ornstein-Uhlenbeck process** that governs the liquidity state L_t — but at the **intuition level only**. We’ll cover:

- What an OU process is (the simplest mean-reverting random process)
- Why it’s the right model for liquidity dynamics
- The parameters μ , $\bar{L}$, σ and what they mean economically
- How to simulate paths
- Connection to Vasicek interest rate models (for students who know them)

Session 4 is a deliberate self-diagnostic for track selection. If the OU process feels intuitive and you want more depth, you may be a Track 2 candidate. If you find it manageable but don’t want the proofs, Track 1 is right for you.

Reading: Book Chapter 2, sections 2.8–2.9 (OU calibration and properties).

[← Session 2](#) | [Schedule](#) | [Next: Session 4](#) →

12 Session 4: The OU Process for Liquidity (Intuition Only)

Unit	1 — Why DCF Fails
Book Chapter	2 (sections 2.8–2.9)
Slide source	Adapted from existing Class 5 (stripped of full derivation) + new intuition slides
Slides needed	~10 new + adapt 6–8 slides from Class 5
Track	Common core (both tracks) — TRACK DIAGNOSTIC SESSION

! Track Diagnostic

This session is deliberately positioned as a self-test for track selection. The OU process is taught at the **intuition level only** — no derivations, no proofs, no measure theory. Students who find this content easy and want more depth are Track 2 candidates. Students who find this content appropriate and don't want proofs are Track 1 candidates. The full mathematical derivation of the OU process appears in **Session 25 (Track 2 only)**, drawing on the original Class 5 slides.

12.1 Learning Objectives

By the end of this session, students will be able to:

1. **Describe** an Ornstein-Uhlenbeck process in plain English and write its SDE form.
2. **Identify** the three OU parameters κ , $\bar{L}$, σ and explain what each controls economically.
3. **Compute** the OU stationary distribution given the parameters.
4. **Simulate** an OU path in their head (or on paper) using the discretized recursion.
5. **Connect** OU dynamics to interest rate models (Vasicek) and to mean-reverting volatility.
6. **Assess their own readiness** for Track 2 based on comfort with this material.

12.2 Pre-Class Assignment

- **Read:** Book Chapter 2, sections 2.8–2.9 (~12 pages)
- **Watch:** 8-minute video introducing mean reversion (link on course site)
- **Optional (Track 2 candidates):** Skim Karatzas-Shreve Section 5.6 (Ornstein-Uhlenbeck process)

12.3 In-Class Outline (75 minutes)

Time	Segment	Format
0:00–0:05	Recap Session 3 · Why we need a process model	Lecture
0:05–0:20	What is a stochastic process? (intuition: BM, random walk)	Lecture
0:20–0:40	The OU process: SDE form, what each parameter does	Lecture + live simulation demo
0:40–0:55	Stationary distribution and what it tells us	Lecture
0:55–1:05	Connection to Vasicek and other mean-reverting models	Lecture
1:05–1:15	Self-diagnostic for track selection	Activity

12.4 Slide Content Briefs (New Slides Needed)

12.4.1 Slide 4.1 — Title

- **Title:** “The Ornstein-Uhlenbeck Process for Liquidity”
- **Subtitle:** “Intuition only — the math itself is in Session 25 (Track 2)”
- **Note:** “If today feels easy and you want more depth, consider Track 2”

12.4.2 Slide 4.2 — Why We Need a Process Model

- **Title:** “What we have, what we need”
- **What we have (from Session 3):**
 - Time series of secondary discounts 2003–2024
 - Implied liquidity state L_t at quarterly frequency
 - Four regimes with distinct statistical character
- **What we need:**
 - A *model* of how L_t evolves
 - Closed-form properties (mean, variance, distribution)
 - Ability to forecast and price
- **Speaker notes:** Without a process model, we can describe the past but cannot value future cash flows.

12.4.3 Slide 4.3 — Three Properties Our Model Must Have

- **Title:** “What properties does L_t have?”
- **Empirical properties (from data):**
 1. **Mean reversion** — discounts widen in crises but return toward a long-run average
 2. **Stochastic shocks** — period-to-period changes are partly random
 3. **Bounded variance** — L_t doesn’t drift indefinitely; the unconditional distribution has finite variance
- **Speaker notes:** These three properties together rule out Brownian motion (which has unbounded variance) and rule in mean-reverting processes.

12.4.4 Slide 4.4 — Introducing the OU Process

- **Title:** “The Ornstein-Uhlenbeck process”
- **SDE form (large, centered):** $dL_t = \kappa(\bar{L} - L_t) dt + \sigma dW_t$
- **Verbal translation:** “ L_t tends to revert toward $\bar{L}$ at speed κ , with random shocks of magnitude σ ”
- **History:** “Originally derived by Ornstein and Uhlenbeck in 1930 to model the velocity of a particle undergoing Brownian motion in a viscous fluid. Same math applies to mean-reverting financial variables.”
- **Speaker notes:** Don’t over-explain notation. Students will get used to it through repetition. The verbal translation is more important than the symbols.

12.4.5 Slide 4.5 — What Each Parameter Does (κ)

- **Title:** “ κ controls mean reversion *speed*”
- **Visual:** Three OU paths overlaid:
 - Low κ ($= 0.1$): slow return to mean, looks almost like a random walk
 - Calibrated κ ($= 0.45$): moderate return, typical of liquidity data
 - High κ ($= 2.0$): fast return, looks like noise around mean
- **Half-life formula:** “Half-life $= \ln(2)/\kappa$ ”
 - $\kappa = 0.1 \rightarrow$ half-life 6.9 years
 - $\kappa = 0.45 \rightarrow$ half-life 1.5 years (matches data)
 - $\kappa = 2.0 \rightarrow$ half-life 0.35 years
- **Speaker notes:** Connect to a familiar concept: radioactive decay. Same math. Half-life is the time to revert halfway to mean.

12.4.6 Slide 4.6 — What Each Parameter Does ($\bar{L}$, σ)

- **Title:** “ $\bar{L}$ and σ ”
- **$\bar{L}$ (long-run mean):**
 - The value L_t reverts *toward*

- In the GE-LAV calibration, $\bar{L} = 1.0$ by normalization
- Economic meaning: the “normal” liquidity state
- σ (volatility / diffusion coefficient):
 - Controls the *size of random shocks*
 - Calibrated $\sigma = 0.32$
 - Higher $\sigma \rightarrow$ more dispersion around mean
- **Visual:** Two paths overlaid with different σ values
- **Speaker notes:** Students sometimes confuse σ (diffusion) with stationary std (which is different). Set that up for the next slide.

12.4.7 Slide 4.7 — Stationary Distribution

- **Title:** “Where does L_t end up in the long run?”
- **Result (boxed):** $L_\infty \sim \mathcal{N}\left(\bar{L}, \frac{\sigma^2}{2\kappa}\right)$
- **Calibrated values:**
 - Mean: $\bar{L} = 1.0$
 - Variance: $\sigma^2/(2\kappa) = 0.32^2/0.90 = 0.1138$
 - Standard deviation: $\sqrt{0.1138} = 0.337$
- **Visual:** Histogram of OU path values over a long simulation, overlaid with Gaussian density
- **Speaker notes:** This is the *only* result this lecture proves (informally). Track 2 students will see the full derivation via Fokker-Planck in Session 27.

12.4.8 Slide 4.8 — What “Stationary” Means

- **Title:** “Stationary = unconditional distribution”
- **Bullets:**
 - If you pick a random year far in the future, the distribution of L_t in that year is $\mathcal{N}(1.0, 0.1138)$
 - The *transition density* (where will I be next year given where I am now) is different — it’s a function of current state
 - As $t \rightarrow \infty$, transition density $\rightarrow$ stationary density
- **Connection to data:** “The empirical histogram of L_t across 2003–2024 is approximately consistent with this stationary density — modulo regime changes.”
- **Speaker notes:** Avoid the word “ergodic” unless asked. Just say “in the long run, OU forgets where it started.”

12.4.9 Slide 4.9 — Path Simulation (Discrete Recursion)

- **Title:** “How to simulate an OU path”
- **Discretization:** Using small time step Δt (e.g., 1 month = 1/12 year):
 - $L_{t+\Delta t} = L_t + \kappa(\bar{L} - L_t)\Delta t + \sigma\sqrt{\Delta t} \cdot Z$

- where $Z \sim \mathcal{N}(0, 1)$ is a random draw
- **Live demo:** In Excel or Python, simulate 100 paths and show their distribution
- **Speaker notes:** This is genuinely useful for students even at intuition level. They should be able to do this in Excel. Track 1 students will need it for PS1.

12.4.10 Slide 4.10 — Connection to Vasicek (For Students Who Know Rate Models)

- **Title:** “If you’ve seen Vasicek, you’ve seen OU”
- **Vasicek interest rate model:** $dr_t = a(b - r_t) dt + \sigma dW_t$
- **GE-LAV liquidity model:** $dL_t = \kappa(\bar{L} - L_t) dt + \sigma dW_t$
- **Same equation.** Only the variable interpretation differs:
 - Vasicek: r_t is the short interest rate
 - GE-LAV: L_t is the liquidity state
- **Other examples:** Hull-White (interest rates), Heston (volatility), commodity convenience yield models
- **Speaker notes:** This slide helps students with prior fixed income or derivatives knowledge plug into the framework quickly.

12.4.11 Slide 4.11 — What OU Cannot Do

- **Title:** “Limitations of pure OU”
- **Bullets:**
 - **No jumps:** OU paths are continuous; real liquidity has discrete shocks (March 2020, GFC)
 - **No fat tails:** Stationary distribution is Gaussian; real data has more extreme observations
 - **No regime switching:** Single κ , $\bar{L}$, σ across all conditions
- **Resolution in GE-LAV:** “Chapter 13 of the book extends OU to jump-diffusion. We adopt that for advanced work in Session 28+.”
- **Speaker notes:** Honesty about model limitations. Students will appreciate that you’re not selling them perfection.

12.4.12 Slide 4.12 — Track Self-Diagnostic

- **Title:** “How did today feel?”
- **Three columns:**
 - **“Felt easy, I wanted more”:** Likely Track 2 candidate. Read Karatzas-Shreve Ch. 5 by Session 5
 - **“Felt right, comfortable level”:** Likely Track 1 — focus on application
 - **“Felt hard but I got through”:** Stay on Track 1. Session 25 math depth would be a stretch
- **Speaker notes:** Use the last 10 minutes for this. Have students fill out a brief course form indicating their preliminary track choice and any questions they have about the math.

12.5 Existing Slides to Reuse from Class 5

From Deck 1, Class 5 “The Ornstein-Uhlenbeck Process: Complete Derivation” — pull approximately 6–8 slides, **with substantial editing**:

- Slide showing OU SDE with parameter labels (keep)
- Slide showing simulated paths (keep)
- Slide showing stationary distribution result (keep — but **strip the derivation**)
- Slide showing calibration numbers ($\kappa = 0.45$, etc.) (keep)
- Slide showing connection to Vasicek (keep)
- **Strip out:** Stochastic differential proofs, Itô formula applications, Riccati-style derivations — all of that goes to Session 25 (Track 2)

Adaptation summary: Take the Class 5 slides, remove every proof step, replace with intuition annotations and visuals. Reduce ~19 slides to ~10 usable in Session 4.

12.6 Discussion Questions

1. The OU process is Gaussian (stationary distribution is normal). The empirical secondary market discount data has fat tails (more extreme observations than Gaussian predicts). How much does this limitation matter for valuation?
2. The half-life of liquidity shocks is ~18 months at $\kappa = 0.45$. What’s the implication for an LP who plans to exit in 12 months vs. 36 months?
3. Vasicek (1977) introduced the OU process to model interest rates. Why did it take ~50 years for the same framework to be applied to liquidity in private markets?

12.7 Worked Example: OU Path Simulation in Plain English

Setup: Start at $L_0 = 0.85$ (slightly stressed, like Q3 2024). Use $\kappa = 0.45$, $\bar{L} = 1.0$, $\sigma = 0.32$. Simulate quarterly ($\Delta t = 0.25$).

Step 1 (next quarter): Mean reversion pulls toward 1.0: - Drift contribution: $0.45 \times (1.0 - 0.85) \times 0.25 = 0.017$ - Random shock: $\sigma\sqrt{\Delta t} \cdot Z = 0.32 \times 0.5 \times Z = 0.16Z$ - If $Z = +1$ (one std above): $L_{0.25} = 0.85 + 0.017 + 0.16 = 1.03$ - If $Z = -1$: $L_{0.25} = 0.85 + 0.017 - 0.16 = 0.71$ - If $Z = 0$: $L_{0.25} = 0.867$

Step 2 (second quarter): Repeat. Now starting from a new value.

Step 3 (long-run): After many quarters, the distribution of L_t converges to $\mathcal{N}(1.0, 0.1138)$ regardless of starting point.

Teaching note: Have students do one step by hand. This grounds the abstract math in arithmetic they can do in 90 seconds.

12.8 Common Student Confusions (Instructor Notes)

- **“Why is the stationary variance $\sigma^2/(2\kappa)$ and not just σ^2 ?”** Because mean reversion *fights* the random shocks. Faster reversion (larger κ) means smaller stationary variance. This is the most-asked question of the session.
- **“How do you actually calibrate κ , $\bar{L}$, σ from data?”** Tease the answer: it’s maximum likelihood estimation on the time series. Track 1 sees the recipe in Session 17; Track 2 sees the math in Session 25.
- **“Is the Gaussian assumption realistic?”** No — empirical data has fatter tails. Jump-diffusion extension (Ch 13) handles this. For valuation purposes, the bias from Gaussianity is small except in extreme regimes.
- **“Does L_t stay positive?”** No — under OU, L_t can go negative. In the model, $L_t = 0$ represents the long-run mean liquidity (by normalization), and negative values represent stressed regimes. This is just notational; it’s not a constraint violation.

12.9 What to Expect Next Session

Session 5 covers the **term structure of private capital returns** — how the OU dynamics interact with investment horizon to produce horizon-dependent valuation effects. Key topics:

- The empirical term structure of PE returns by vintage and asset class
- How Jensen bias grows with T
- Why long-horizon assets (infrastructure, core real estate) are *most* affected by the liquidity illusion
- DLOM (Discount for Lack of Marketability) vs. GE-LAV — comparing methodologies

Reading: Book Chapter 3, all sections (~25 pages).

Track declaration: End of Session 5 is the track selection deadline. If you’re undecided, schedule a 15-min meeting with me this week.

[← Session 3](#) | [Schedule](#) | [Next: Session 5](#) →

13 Session 5: Term Structure of Private Capital Returns

Unit	1 — Why DCF Fails
Book Chapter	3 (all sections)
Slide source	All new (no existing slides on this topic)
Slides needed	~15 new
Track	Common core (both tracks)
Assessment milestone	Track declaration deadline (end of session)

13.1 Learning Objectives

By the end of this session, students will be able to:

1. **Describe** the empirical term structure of private capital returns by asset class (PE buyout, growth equity, infrastructure, real estate, private credit).
2. **Explain** why long-horizon assets are disproportionately exposed to the liquidity illusion.
3. **Compute** the Jensen bias $B(T)$ for a given asset class and horizon using the affine formula.
4. **Compare** the GE-LAV liquidity-adjusted term structure to the standard DLOM (Discount for Lack of Marketability) methodology.
5. **State and justify** their Track selection by the end of class.

13.2 Pre-Class Assignment

- **Read:** Book Chapter 3, all sections (~25 pages)
- **Optional:** Section 1 of Phalippou & Gottschalg (2009), “The Performance of Private Equity Funds,” *Review of Financial Studies*

13.3 In-Class Outline (75 minutes)

Time	Segment	Format
0:00–0:05	Recap: OU dynamics → today: how dynamics interact with horizon	Lecture

Time	Segment	Format
0:05–0:20	Empirical term structure of private capital returns	Lecture + chart deep-dive
0:20–0:40	The liquidity risk term structure	Lecture + worked formula
0:40–0:55	Asset class implications: which assets are most exposed	Lecture
0:55–1:05	DLOM vs. GE-LAV — comparative framework	Lecture
1:05–1:15	Track declaration · Q&A · final track guidance	Activity

13.4 Slide Content Briefs (New Slides Needed)

13.4.1 Slide 5.1 — Title

- **Title:** “The Term Structure of Private Capital Returns”
- **Tag line:** “Why long-horizon assets are most affected by the liquidity illusion”
- **Reminder banner:** “Track declaration deadline today”

13.4.2 Slide 5.2 — What Is a Term Structure?

- **Title:** “Term structures everyone knows”
- **Examples:**
 - **Bond yield curve:** yield as a function of maturity (1Y, 2Y, 5Y, 10Y, 30Y)
 - **Volatility term structure:** implied vol as a function of option maturity
 - **Credit spread term structure:** spread by tenor
- **New idea:** “What does the *risk* of private capital look like as a function of holding period?”
- **Speaker notes:** Connect to fixed income students already know. Then pivot to the novel object.

13.4.3 Slide 5.3 — Empirical Term Structure: PE Returns by Vintage

- **Title:** “PE returns by vintage and time-to-liquidation”
- **Visual:** Heatmap or 3D surface of pooled IRR by vintage year $\times$ years to liquidation
- **Key observations:**
 - 2006–2008 vintages show large vintage-year effects (returns depend on entry conditions)
 - Mid-vintage years (2010–2015) cluster near long-run averages
 - 2018–2020 vintages still uncertain (recent)
- **Speaker notes:** This is the *empirical* term structure. It’s noisy. The theoretical question is: what does GE-LAV predict?

13.4.4 Slide 5.4 — The Liquidity Risk Term Structure

- **Title:** “Risk grows with horizon — but not the way DCF says”
- **DCF prediction:** Risk scales as $\sigma\sqrt{T}$ (standard random walk scaling)
- **GE-LAV prediction (OU + Jensen):** Risk has two components:
 - **Diffusion component:** scales as $\sqrt{T}$ for short T , plateaus for long T (mean reversion)
 - **Jensen bias component:** grows linearly with T
- **Visual:** Two curves overlaid:
 - DCF risk: straight $\sqrt{T}$ line
 - GE-LAV risk: rises, plateaus, then rises again (Jensen kicking in)
- **Speaker notes:** This is the key conceptual contribution of Chapter 3. The shape is qualitatively different from DCF.

13.4.5 Slide 5.5 — The Jensen Bias Formula

- **Title:** “The Jensen bias as a function of horizon”
- **Formula (boxed):** $B(T) = A \cdot T + C$ (affine in T , to leading order)
- **Where:**
 - A = annual Jensen bias accrual rate (calibrated 0.1–0.4% per year by asset class)
 - C = small constant offset
- **Calibrated values by asset class:**

Asset Class	A (per year)	C
PE Buyout	0.16%	0.0%
Growth Equity	0.17%	0.1%
Real Estate	0.18%	0.0%
Infrastructure	0.18%	0.0%
Private Credit	0.10%	0.0%

- **Speaker notes:** This is a *closed-form* result. Track 2 students will derive it in Session 29.

13.4.6 Slide 5.6 — Worked Example: Jensen Bias by Horizon

- **Title:** “Computing $B(T)$ for typical horizons”
- **Table:**

Asset	Hold T	A	$B(T)$
5-yr Buyout	5	0.16%	0.8%
10-yr Growth	10	0.17%	1.7%
15-yr Secondary	15	0.17%	2.5%
20-yr Infrastructure	20	0.18%	3.6%

Asset	Hold T	A	$B(T)$
25-yr Core RE	25	0.18%	4.5%
30-yr PPP Asset	30	0.18%	5.4%

- **Bottom-line:** “30-year PPP assets carry Jensen bias $7\times$ larger than 5-year buyouts.”
- **Speaker notes:** This is *just* the Jensen bias channel. The GE-LAV equilibrium correction can be much larger.

13.4.7 Slide 5.7 — Why Long-Horizon Assets Are Most Affected

- **Title:** “Why infrastructure is more affected than buyout”
- **Three reasons:**
 1. **Jensen bias scales with T** — longer hold means more cumulative convexity effect
 2. **More time for liquidity state to drift** — variance grows with T (up to stationary limit)
 3. **Discount factor is more sensitive to rate changes at long T** — modified duration effect
- **Visual:** Three small charts showing each effect
- **Speaker notes:** This is *the* implication for infrastructure and long-hold strategies. Pension funds and insurers — the biggest holders of long-duration private assets — face the largest valuation errors.

13.4.8 Slide 5.8 — The Lévy Extension

- **Title:** “GE-LAV term structure with jumps”
- **Setup:** “Pure OU is Gaussian. Real data has jumps (March 2020, GFC). The Lévy extension adds compound Poisson jumps to OU.”
- **Effect on term structure:**
 - Short T : jumps add fat tails to short-horizon variance
 - Long T : jump effects average out, term structure resembles pure-OU result
- **Calibrated jump intensity:** ~ 0.3 jumps per year, mean jump size -0.4
- **Speaker notes:** Brief mention. Full Lévy treatment is in Track 2 advanced topics.

13.4.9 Slide 5.9 — DLOM: The Industry Alternative

- **Title:** “DLOM: Discount for Lack of Marketability”
- **What it is:**
 - Industry-standard approach to adjusting valuations for illiquidity
 - Typical magnitudes: 15–40% reduction in computed DCF value
 - Methods: Restricted stock studies, pre-IPO studies, option-based models (Finnerty, Longstaff)
- **How it works:** Apply a *flat percentage discount* to the DCF mark

- **Common assumption:** “DLOM is constant for an asset class” (e.g., 25% for PE, 30% for VC)
- **Speaker notes:** Most students will have encountered DLOM in valuation classes or CFA. Establish it as the current best industry alternative, then critique.

13.4.10 Slide 5.10 — DLOM by Asset Class (Industry Convention)

- **Title:** “Typical DLOM ranges”
- **Table:**

Asset Class	Typical DLOM
Late-stage private equity	15–25%
Early-stage venture	30–45%
Real estate private equity	20–30%
Infrastructure	10–20%
Private credit	5–15%

- **Speaker notes:** Source: AICPA Valuation Guidelines, various restricted stock studies. Vary across appraisers and contexts.

13.4.11 Slide 5.11 — DLOM vs. GE-LAV: Comparison

- **Title:** “Same problem, two answers”
- **Side-by-side table:**

Feature	DLOM	GE-LAV
Treatment of premium	Flat percentage discount	Stochastic OU dynamics
Regime dependence	None	Endogenous to liquidity state
Horizon dependence	Flat	Affine in T
Theoretical foundation	Empirical (restricted stock studies)	First-principles asset pricing
Crisis behavior	Static	Amplifies via McKean-Vlasov
Calibration	Subjective / appraiser judgment	OLS / MLE on secondary data

- **Bottom-line:** “DLOM was the best available method when secondary market data didn’t exist. GE-LAV is the first method that *can* use the secondary market data correctly.”
- **Speaker notes:** Be respectful of DLOM. It was a reasonable approach. Now we have better tools.

13.4.12 Slide 5.12 — DLOM vs. GE-LAV: Numerical Comparison

- **Title:** “A 20-year infrastructure asset, three methods”
- **Three columns:**

- **DCF:** \$100M (constant 7.5%, premium 3.5%)
- **DCF + DLOM (15%):** \$85M (just shave 15% off)
- **GE-LAV (normal markets):** \$95M (Jensen-corrected + small GE adjustment)
- **GE-LAV (Regime 4 multi-shock):** \$80M
- **GE-LAV (Regime 2 GFC depth):** \$30M
- **Punchline:** “DLOM is a *constant adjustment*. GE-LAV is a *regime-dependent calculation*. In normal markets, they’re close. In crises, they’re completely different.”

13.4.13 Slide 5.13 — Performance Measurement Implications (Tease)

- **Title:** “What does this mean for performance measurement?”
- **Setup:** “If valuations are wrong systematically, returns are wrong systematically”
- **Implications:**
 - IRR is computed against wrong valuations
 - PME comparisons are biased
 - Vintage-year analysis is contaminated
- **Speaker notes:** This is the bridge to Session 6 (IRR) and Session 7 (PME).

13.4.14 Slide 5.14 — Track Declaration

- **Title:** “Today is decision day”
- **Reminders:**
 - Default is Track 1 if you don’t choose
 - Track 2 requires stochastic calculus background or self-taught equivalent
 - Cross-track audit is allowed and encouraged
 - Track 2 PSets are mathematical; Track 1 PSets are applied
- **Self-test:** “Did Session 4’s OU material feel intuitive? If yes, Track 2 is feasible.”
- **Logistics:** Submit track choice via course form by end of class

13.4.15 Slide 5.15 — Where We Go From Here

- **Title:** “Unit 1 wraps. Unit 2 begins.”
- **Session 6:** IRR — what it is, what it isn’t, where it fails
- **Session 7:** PME and the measurement problem
- **Session 8:** What a correct theory must deliver (Ch 5 of book)
- **Session 9:** Midterm review
- **Session 10:** Midterm exam
- **Speaker notes:** Frame Unit 1 as the “diagnostic” — what’s wrong with current practice. Unit 2 finishes the diagnosis with measurement issues. Then Unit 3 builds the alternative.

13.5 Discussion Questions

1. The Jensen bias for a 30-year PPP asset is ~5.4%. The DLOM convention for infrastructure is 10–20%. Are these incompatible? How should they be reconciled?
2. Pension funds and insurers hold most of the long-duration private assets where Jensen bias is largest. If GE-LAV becomes adopted methodology, what happens to reported PE returns at large pension funds?
3. Some private credit assets have short effective duration (1–3 years for direct lending). Jensen bias is small there. Does that mean private credit is “GE-LAV-immune”? What other GE-LAV channels might still matter?

13.6 Worked Numerical Example: Choosing Between DLOM and GE-LAV

Setup: Insurance company holds a 15-year core infrastructure asset (regulated water utility). DCF mark = \$500M.

Method 1: DLOM at 15% - Adjusted value: $\$500M \times 0.85 = \$425M$ - Applied uniformly across all market conditions - “Always 15% off DCF”

Method 2: GE-LAV (current state, normalization regime) - Jensen bias for $T = 15$: $A \cdot 15 + C = 0.18\% \cdot 15 = 2.7\%$ - LAV value (just Jensen): $\$500M \times (1 + 0.027) = 513.5M$ - GE-LAV (normal regime, small equilibrium correction): ~\$495M - “About flat with DCF in normal markets”

Method 3: GE-LAV (current state, stressed regime, $L = -0.8$) - Equilibrium rate: $r_{\text{GE-LAV}}(L = -0.8) \approx 12\%$ (vs. DCF 7.5%) - GE-LAV value: $\$500M \times e^{\{-(12\% - 7.5\%) \cdot 15\}} = \$250M$ - “50% off DCF in stress”

Method 4: GE-LAV (GFC depth, $L = -1.5$) - Equilibrium rate: ~32% - GE-LAV value: ~\$150M - “70% off DCF at GFC depth”

Comparison: DLOM gives one answer (\$425M). GE-LAV gives a *menu* of answers conditional on regime. The insurance regulator should care about the GFC-depth number (\$150M) for solvency purposes.

13.7 Common Student Confusions (Instructor Notes)

- “If Jensen bias is small (1–4%), why is it a big deal?” It compounds across the \$13T market = \$130-520B in systematic underreporting. Also it’s the *only* always-positive correction — the GE-LAV equilibrium correction can be positive or negative depending on regime.
- “DLOM at 15-25% is much bigger than Jensen bias at 2-4%. Doesn’t DLOM dominate?” DLOM and Jensen are not the same thing. DLOM is trying to capture the entire illiquidity adjustment in one number. Jensen is just one channel. GE-LAV’s full correction (Jensen + GE equilibrium) can match or exceed DLOM in stressed regimes.

- **“How do we know the calibration is right?”** Maximum likelihood on 22 years of secondary market data. Reasonably robust. Confidence intervals are tight enough that the *qualitative* conclusions don’t change under reasonable parameter variation.
- **“What about private credit? Spread compensation already exists.”** Private credit has its own GE-LAV variant (Ch 23). Spread is partly compensation for credit risk, partly for illiquidity. Decomposing these requires GE-LAV.

13.8 What to Expect Next Session

Session 6 begins Unit 2 (Measurement and Theory) with a deep look at **IRR**. Why IRR is the dominant performance metric in private markets, what its two structural biases are, and why a constant-rate metric cannot correctly measure performance when rates are stochastic.

Reading: Book Chapter 4, sections 4.2–4.4 (~10 pages).

For Track 2 students who declared today: Begin reading Karatzas-Shreve Ch. 5 (Brownian motion and stochastic calculus). You’ll need this background by Session 25.

[← Session 4](#) | [Schedule](#) | [Next: Session 6](#) →

14 Session 6: IRR · The Internal Rate of Return Problem

Unit	2 — Measurement and Theory
Book Chapter	4 (sections 4.2–4.4)
Slide source	All new
Slides needed	~15 new
Track	Common core (both tracks)

14.1 Learning Objectives

By the end of this session, students will be able to:

1. **Define IRR** precisely and state its mathematical relationship to NPV.
2. **Identify** the two structural biases of IRR (reinvestment assumption + sign-pattern dependence).
3. **Distinguish** money-weighted from time-weighted returns and explain when each is appropriate.
4. **Compute IRR** for a simple cash flow pattern and identify when multiple IRRs exist.
5. **Explain** why a constant-rate metric cannot correctly summarize performance when discount rates are stochastic.
6. **Prepare** for Session 7's PME discussion by understanding why IRR's failures motivated PME's development.

14.2 Pre-Class Assignment

- **Read:** Book Chapter 4, sections 4.2–4.4 (~10 pages)
- **Optional:** Phalippou (2008), “The Hazards of Using IRR to Measure Performance: The Case of Private Equity,” *Journal of Performance Measurement*

14.3 In-Class Outline (75 minutes)

Time	Segment	Format
0:00–0:05	Why measurement matters: returns based on bad valuations are bad returns	Lecture

Time	Segment	Format
0:05–0:20	What IRR is, mathematically and conceptually	Lecture + math
0:20–0:35	Money-weighted vs. time-weighted returns	Lecture + worked example
0:35–0:50	Two IRR biases: reinvestment + sign-pattern	Lecture + worked examples
0:50–1:05	Multiple IRRs and the descending-discount problem	Lecture + numerical examples
1:05–1:15	The constant-rate fallacy: IRR under stochastic discount rates	Lecture

14.4 Slide Content Briefs (New Slides Needed)

14.4.1 Slide 6.1 — Title

- **Title:** “The IRR Problem”
- **Tag line:** “Why the most-used metric in private markets is structurally broken”
- **Speaker notes:** Frame Unit 2 as the *measurement* counterpart to Unit 1’s *valuation* critique. Bad valuations cause bad returns calculations.

14.4.2 Slide 6.2 — IRR: Definition

- **Title:** “IRR formally”
- **Definition:** “IRR is the discount rate r that makes $NPV = 0$ ”
- **Equation:** $0 = \sum_{t=0}^T \frac{CF_t}{(1+r)^t}$
- **Solving:** Implicit equation; typically solved numerically (Newton-Raphson or Excel’s IRR function)
- **Standard interpretation:** “Annualized return on capital invested”
- **Speaker notes:** Set up the math cleanly. Students will know IRR intuitively; the goal is to make them careful about it.

14.4.3 Slide 6.3 — IRR: Numerical Example

- **Title:** “A clean IRR computation”
- **Cash flows:**
 - $t = 0$: $-\$100M$ (initial commitment, drawn over 3 years)
 - $t = 4$: $+\$25M$
 - $t = 6$: $+\$50M$
 - $t = 8$: $+\$75M$
 - $t = 10$: $+\$30M$
- **IRR:** Solve $-100 + 25(1+r)^{-4} + 50(1+r)^{-6} + 75(1+r)^{-8} + 30(1+r)^{-10} = 0$

- **Solution:** $r \approx 14.2\%$
- **Speaker notes:** Walk through one Newton iteration on the board if time. Students should see *how* IRR is solved, not just trust Excel.

14.4.4 Slide 6.4 — Money-Weighted vs. Time-Weighted

- **Title:** “Two different questions, two different metrics”
- **Money-weighted return (= IRR):**
 - “What return did *the capital* earn?”
 - Heavily influenced by timing of large cash flows
 - Standard in private markets
- **Time-weighted return:**
 - “What return did *the manager* generate?”
 - Independent of investor cash flow timing
 - Standard in public markets (mutual funds, hedge funds)
- **Why PE uses MWR (= IRR):**
 - GP controls capital call timing
 - LP cares about return on dollars *actually deployed*
- **Speaker notes:** This distinction matters for comparing PE returns to public benchmarks. PME (Session 7) is designed to bridge MWR-based PE returns with TWR-based public benchmarks.

14.4.5 Slide 6.5 — Worked: MWR vs. TWR Difference

- **Title:** “When MWR and TWR diverge”
- **Setup:** Two funds with identical underlying portfolio performance but different cash flow timing
- **Fund A:** Calls capital early in fund life, distributes late
- **Fund B:** Calls capital late, distributes early
- **Result:** Same TWR (both managed the portfolio the same way), but Fund B has higher IRR (capital was deployed for less time on the larger amounts)
- **Speaker notes:** This is a classic teaching example. Make sure students understand: the same investment skill can produce different IRRs depending on cash flow timing alone.

14.4.6 Slide 6.6 — IRR Bias #1: The Reinvestment Assumption

- **Title:** “Bias 1: IRR assumes intermediate cash flows are reinvested at IRR itself”
- **Setup:** “When $IRR = 18\%$, the math implicitly assumes every distribution is reinvested at 18% for the remainder of the fund life”
- **Reality:** “LPs typically receive distributions and either spend them or invest at much lower public market rates ($\sim 7\text{--}10\%$ for diversified equities)”
- **Implication:** “IRR systematically overstates the realized return when distributions are large and early”

- **Magnitude:** “Can overstate by 2-5 percentage points for typical PE fund cash flow patterns”
- **Speaker notes:** This is well-known but under-appreciated. The Modified IRR (MIRR) is a partial fix that uses a separate reinvestment rate, but isn’t standard in PE reporting.

14.4.7 Slide 6.7 — IRR Bias #2: Sign-Pattern Dependence

- **Title:** “Bias 2: IRR is sensitive to the sign pattern of cash flows”
- **Standard case** (-, 0, ..., 0, +): Single IRR, well-defined
- **Mixed signs** (-, +, -, +): Multiple IRRs can exist (Descartes’ rule of signs)
- **PE example:** Continuation fund recapitalizations create intermediate negative cash flows (LP rolls over into new vehicle, takes some cash out) → multiple IRRs possible
- **Speaker notes:** This is *very* common in modern PE with continuation funds and GP-led secondaries. Students will encounter this in practice.

14.4.8 Slide 6.8 — Multiple IRRs: A Concrete Example

- **Title:** “When IRR isn’t unique”
- **Cash flows:**
 - $t = 0$: -\$100
 - $t = 1$: +\$230
 - $t = 2$: -\$132
- **NPV equation:** $-100 + 230/(1+r) - 132/(1+r)^2 = 0$
- **Two solutions:** $r = 10\%$ and $r = 20\%$
- **Which is “the IRR”?** Both, mathematically. Neither, economically.
- **Recommended approach:** “Use NPV at a fixed discount rate; report IRR only with caveats about sign pattern”
- **Speaker notes:** This blows students’ minds the first time. It’s textbook material but rarely emphasized.

14.4.9 Slide 6.9 — IRR Under Stochastic Discount Rates

- **Title:** “Where IRR fundamentally breaks: when rates are stochastic”
- **Setup:** “IRR by construction is a *single number* — one constant rate”
- **GE-LAV reality:** Discount rate $r(L_t)$ varies with liquidity state
- **The mismatch:**
 - IRR averages out the rate variation
 - But the *correct* present value uses the full path of L_t
 - These give different answers (Jensen’s inequality again)
- **Speaker notes:** Connect to Session 2’s Jensen discussion. IRR’s constant-rate assumption embeds the same convexity error as DCF’s constant-premium assumption.

14.4.10 Slide 6.10 — The Constant-Rate Fallacy

- **Title:** “The Constant-Rate Fallacy”
- **Boxed statement:** “Any metric that compresses time-varying discount rates into a single constant rate inherits the systematic bias of DCF.”
- **Implications:**
 - IRR is biased
 - Money-weighted return is biased (same as IRR)
 - Hurdle rate comparisons are biased
 - Carried interest computations are biased
- **The cure:** Liquidity-adjusted metrics (LA-IRR, LA-PME) — covered Session 7
- **Speaker notes:** This is the bridge to Session 7. IRR’s problem is the same as DCF’s problem, restated.

14.4.11 Slide 6.11 — Empirical Magnitudes: How Wrong Is IRR?

- **Title:** “How wrong is IRR in practice?”
- **Data:** Pooled IRR of US PE buyout funds 1990–2024 (Preqin, Kaplan-Schoar)
- **Reported pooled IRR:** ~12.5% per year
- **Phalippou (2009) adjustment for reinvestment assumption:** ~9.5%
- **Stafford (2022) adjustment for valuation smoothing:** ~7.5%
- **GE-LAV-adjusted estimate (LA-IRR):** ~8.5% (intermediate)
- **Speaker notes:** This is *the* number for students to remember. Reported PE returns are 3-5 percentage points overstated by methodology choices.

14.4.12 Slide 6.12 — IRR and Vintage-Year Effects

- **Title:** “Why vintage matters more than skill”
- **Empirical finding:** “Vintage year (when the fund started investing) explains 40-60% of variation in fund-level IRR across managers”
- **Implication:** “Top-quartile vintage 2009 outperforms top-quartile vintage 2007 — not because managers were better, but because market entry conditions were”
- **Connection to GE-LAV:** “Vintage effect is partly a *liquidity state* effect — funds starting in low-L states (high implied premium) benefit when L mean-reverts back to $\bar{L}$ ”
- **Speaker notes:** Students often think of PE returns as manager-driven. Vintage effects dominate. GE-LAV explains why mathematically.

14.4.13 Slide 6.13 — IRR and Carry: Why Practitioners Defend IRR

- **Title:** “Why IRR survives despite known flaws”
- **The institutional reasons:**
 - **Carried interest is computed on IRR** — moving away from IRR moves the carry calculation
 - **GP compensation is anchored to IRR hurdles** — typically 8% preferred return

- **LP allocators benchmark on IRR** — switching benchmarks is painful
- **No regulatory mandate to change** — IRR is in IPEV, ILPA, and Solvency II guidance
- **Speaker notes:** This is the *political economy* of IRR persistence. Students entering the industry should understand this.

14.4.14 Slide 6.14 — Proposed Solutions Overview (Tease for Session 7)

- **Title:** “Three responses to the IRR problem”
- **Three columns:**
 1. **Reformed IRR:** MIRR with explicit reinvestment rate; sign-pattern caveats; still constant-rate
 2. **PME family:** Compare to public benchmark — Kaplan-Schoar, Long-Nickels, Direct Alpha
 3. **LA-IRR / LA-PME:** Liquidity-adjusted version that handles stochastic rates correctly
- **Speaker notes:** Session 7 covers all three families in detail.

14.4.15 Slide 6.15 — Summary

- **Title:** “IRR: where it works, where it fails”
- **Works:**
 - Simple sign pattern (single negative followed by positives)
 - Stable market conditions
 - Comparing across short hold periods
- **Fails:**
 - Mixed sign patterns (continuation funds, recapitalizations)
 - Stochastic discount rates (always, in private markets)
 - Cross-vintage comparisons
 - Long-horizon assets where Jensen bias accumulates
- **Speaker notes:** Honest summary. IRR isn’t useless; it’s incomplete.

14.5 Discussion Questions

1. The GP-LP relationship has IRR baked into the carried interest formula. If GE-LAV becomes adopted methodology, does the carry formula need to change? How?
2. Public mutual funds use TWR. Why has PE never moved to TWR despite known IRR flaws?
3. Two PE funds have identical IRRs of 18% but very different cash flow timing patterns. How would you distinguish them from each other in selecting which to commit capital to?

14.6 Worked Numerical Example: When IRR Misleads

Setup: Two PE funds, both reporting $IRR = 18\%$.

Fund A (front-loaded distributions): - $t = 0$: $-\$100M$ - $t = 2$: $+\$80M$ - $t = 4$: $+\$40M$ - $t = 6$: $+\$20M$ - IRR: 18% (capital out quickly)

Fund B (back-loaded distributions): - $t = 0$: $-\$100M$ - $t = 2$: $+\$0M$ - $t = 4$: $+\$0M$ - $t = 8$: $+\$200M$ - IRR: 18% (capital out only at end)

The reinvestment problem:

If LP can reinvest at 7% public market rate:

- **Fund A realized return:** $\$80M$ from year 2 grows at 7% to $(80 \cdot 1.07^6) + (40 \cdot 1.07^4) + 20 = \$120M + \$52M + \$20M = \$192M$ at year 8 — total return 92% over 8 years = **8.4% annualized**
- **Fund B realized return:** $\$200M$ at year 8 — total return $100\% = \mathbf{9.05\% \text{ annualized}}$

Interpretation: Same reported IRR (18%), but Fund B's realized return to the LP is higher because the IRR overstatement is larger for Fund A.

Lesson: Reported IRR can rank funds in the wrong order when reinvestment assumption matters.

14.7 Common Student Confusions (Instructor Notes)

- “Isn't MIRR the standard fix?” MIRR exists but isn't standard in PE reporting. ILPA and IPEV use plain IRR. Industry inertia.
- “Can't I just use the Modified Dietz return?” That's a TWR computation. It tells you a different thing (manager skill) than IRR (return on dollars deployed). Both have a role.
- “If IRR is so flawed, why do all PE funds report it?” Because LPs demand it, regulators sanction it, and carry computation depends on it. Network effects.
- “What is Direct Alpha vs. Long-Nickels PME?” Tease — covered Session 7.

14.8 What to Expect Next Session

Session 7 dives into **PME (Public Market Equivalent)** and its variants. We'll cover: - The Kaplan-Schoar PME and what it computes - Long-Nickels PME (with implicit reinvestment in the benchmark) - Direct Alpha (Gredil-Griffiths-Stucke) - **LA-IRR and LA-PME** — the GE-LAV-corrected versions - Which method to use in which context

Reading: Book Chapter 4, sections 4.5–4.10 (~12 pages).

[← Session 5](#) | [Schedule](#) | [Next: Session 7](#) →

15 Session 7: PME · LA-IRR · LA-PME

Unit	2 — Measurement and Theory
Book Chapter	4 (sections 4.5–4.10)
Slide source	All new
Slides needed	~15 new
Track	Common core (both tracks)

15.1 Learning Objectives

By the end of this session, students will be able to:

1. **Compute** the Kaplan-Schoar PME for a given fund's cash flows against a chosen public benchmark.
2. **Distinguish** Kaplan-Schoar PME, Long-Nickels PME, and Direct Alpha — and state when each is appropriate.
3. **Articulate** why all standard PME variants share the same constant-rate weakness as DCF and IRR.
4. **State** the LA-IRR and LA-PME formulas and explain what corrections they apply.
5. **Apply** LA-PME to a worked example and interpret the result against a standard PME.

15.2 Pre-Class Assignment

- **Read:** Book Chapter 4, sections 4.5–4.10 (~12 pages)
- **Optional:** Kaplan & Schoar (2005), “Private Equity Performance: Returns, Persistence, and Capital Flows,” *Journal of Finance*; Gredil, Griffiths, & Stucke (2014), “Benchmarking Private Equity: The Direct Alpha Method”

15.3 In-Class Outline (75 minutes)

Time	Segment	Format
0:00–0:05	Recap: IRR's problems → need a different metric	Lecture
0:05–0:20	Kaplan-Schoar PME: ratio of FVs	Lecture + numerical example

Time	Segment	Format
0:20–0:35	Long-Nickels PME: implicit reinvestment	Lecture + numerical example
0:35–0:45	Direct Alpha: extracting excess return rate	Lecture
0:45–1:00	LA-IRR and LA-PME: GE-LAV corrections	Lecture + math
1:00–1:15	Worked comparison + when to use which	Worked example + discussion

15.4 Slide Content Briefs (New Slides Needed)

15.4.1 Slide 7.1 — Title

- **Title:** “PME, LA-IRR, LA-PME”
- **Tag line:** “Comparing private to public, correctly”

15.4.2 Slide 7.2 — The PME Question

- **Title:** “What does PME try to answer?”
- **The question:** “If I had invested the same cash flows in the S&P 500 instead of in this PE fund, how would I have done?”
- **Why this matters:**
 - Allocators need benchmarks
 - Carry hurdles are *absolute* (8% pref); PME is *relative* (vs. market)
 - Academic and regulatory work needs cross-asset-class comparison
- **Speaker notes:** Frame PME as the response to IRR’s weaknesses. PME doesn’t fix IRR; it gives a complementary lens.

15.4.3 Slide 7.3 — Kaplan-Schoar PME (Definition)

- **Title:** “Kaplan-Schoar (KS) PME”
- **Formula:** $PME_{KS} = \frac{\sum_t D_t/B_t}{\sum_t C_t/B_t}$
- **Where:**
 - D_t = distributions in period t
 - C_t = capital calls in period t
 - B_t = public benchmark cumulative return from inception to period t
- **Interpretation:**
 - $PME_{KS} > 1$: PE fund outperformed the benchmark
 - $PME_{KS} = 1$: PE fund matched the benchmark

- $PME_{KS} < 1$: PE fund underperformed
- **Speaker notes:** Most-used PME in practice. Simple, intuitive, easy to compute.

15.4.4 Slide 7.4 — Kaplan-Schoar: Worked Example

- **Title:** “KS PME: a worked example”
- **Setup:** 10-year PE fund vs. S&P 500
- **Cash flows and benchmark levels:**

Year	Call (M) <i>Distribution</i> (M)	S&P Index Level
0	30	100
1	40	108
2	30	118
5	0	145
8	0	180
10	0	200

- **Numerator (sum of distributions discounted by benchmark):** $80/(145/100) + 90/(180/100) + 60/(200/100) = 55.2 + 50.0 + 30.0 = 135.2$
- **Denominator (sum of calls discounted by benchmark):** $30/1.00 + 40/(108/100) + 30/(118/100) = 30 + 37.0 + 25.4 = 92.4$
- **KS PME:** $135.2/92.4 = 1.46 \rightarrow$ PE fund outperformed S&P 500 by 46%

15.4.5 Slide 7.5 — Long-Nickels PME (Definition)

- **Title:** “Long-Nickels (LN) PME”
- **Idea:** “Build a hypothetical portfolio that mirrors the PE fund’s cash flow timing in the public benchmark, then compute its IRR”
- **Mechanism:**
 - When PE calls X from LP, hypothetical portfolio buys X of benchmark
 - When PE distributes Y to LP, hypothetical portfolio sells Y of benchmark
 - Track the implied “synthetic public fund” NAV over time
 - Compute its IRR; compare to PE fund’s IRR
- **Output:** Two IRRs (PE fund vs. synthetic public); difference is the alpha
- **Speaker notes:** More sophisticated than KS PME. Handles the “what would the public benchmark have done with my exact cash flow pattern” question directly.

15.4.6 Slide 7.6 — KS vs. LN: When They Diverge

- **Title:** “KS and LN: same idea, different formulations”
- **Both compare PE returns to public benchmark with same cash flow timing**
- **KS gives a multiple (1.46 $\times$); LN gives an IRR-style annualized rate**
- **They can disagree when:**

- Cash flow patterns are unusual (continuation funds, recap distributions)
- Long fund lives where reinvestment matters
- Multiple IRRs in the synthetic fund (rare but possible)
- **Speaker notes:** For most simple cases, KS and LN agree. For corner cases, LN is more nuanced.

15.4.7 Slide 7.7 — Direct Alpha (Gredil-Griffiths-Stucke 2014)

- **Title:** “Direct Alpha: a cleaner formulation”
- **Idea:** “Compute the constant excess discount rate α such that adjusted PE cash flows have zero NPV against the benchmark”
- **Formula:** Find α solving $\sum_t (D_t - C_t) \cdot e^{-\alpha t} / B_t = 0$
- **Output:** A single number α interpreted as the “alpha vs. the benchmark”
- **Advantages over KS / LN:**
 - Returns an interpretable annualized rate
 - Handles complex cash flow patterns cleanly
 - More robust to vintage-year effects
- **Speaker notes:** Direct Alpha is the academic favorite. Practitioners still mostly use KS.

15.4.8 Slide 7.8 — The Constant-Rate Weakness (PME’s Achilles Heel)

- **Title:** “Why all standard PMEs share IRR’s weakness”
- **The common assumption:**
 - KS, LN, and Direct Alpha all treat the public benchmark as the *risk-adjusted reference*
 - They embed the implicit assumption that *any extra return beyond the benchmark* is fair compensation for liquidity, vintage, and selection
 - This is a *constant-rate* assumption about the illiquidity premium
- **The mismatch:**
 - GE-LAV says the illiquidity premium is *stochastic and regime-dependent*
 - PMEs cannot see this — they compress everything into a single number
- **Speaker notes:** This is the conceptual handoff to LA-PME.

15.4.9 Slide 7.9 — LA-IRR: Liquidity-Adjusted IRR (Definition)

- **Title:** “LA-IRR: GE-LAV’s correction to IRR”
- **The idea:** “Compute IRR using a *stochastic* discount rate path $r(L_t)$ instead of a constant rate”
- **Mechanics:**
 - Calibrate L_t path from secondary market data (or simulate from OU)
 - Compute period-by-period discount factors $D_t = \exp(-\int_0^t r(L_s) ds)$
 - Find the *effective single-rate equivalent* that matches the cash flow PV

- **Output:** A liquidity-adjusted single-rate summary, comparable to standard IRR but corrected
- **Magnitude:** “For typical PE fund cash flow patterns, LA-IRR is typically 2–4 percentage points *below* reported IRR”
- **Speaker notes:** This is the *measurement counterpart* to LAV’s valuation correction. Same underlying logic.

15.4.10 Slide 7.10 — LA-PME: Liquidity-Adjusted PME

- **Title:** “LA-PME: GE-LAV’s correction to PME”
- **The idea:** “Use the public benchmark *adjusted for the contemporaneous liquidity state* as the comparison”
- **Mechanics:**
 - At each period, compute the benchmark return *adjusted* for the implied illiquidity premium gap
 - $B_t^{LA} = B_t \cdot \exp(\int_0^t \pi(L_s) ds)$
 - Apply KS / LN / Direct Alpha methodology to B_t^{LA} instead of raw B_t
- **Output:** PME that compares PE to a “public benchmark plus correctly-priced liquidity premium”
- **Magnitude:** “Typical PE fund LA-PME is 5-15% below reported KS PME”
- **Speaker notes:** This is a meaningful magnitude. Allocators caring about absolute outperformance should use LA-PME.

15.4.11 Slide 7.11 — Comparing LA-PME to Standard PME: Worked Example

- **Title:** “Worked: 2010-vintage buyout fund”
- **Setup:**
 - Reported KS PME (vs. S&P 500): 1.32 — “32% outperformance”
 - Adjusted for L_t path 2010-2024 (mostly Regimes 3, 4): LA-PME = 1.18 — “18% outperformance”
- **Decomposition:**
 - Of the 32 percentage points of reported KS outperformance, **~14 points are compensation for time-varying illiquidity premium** (not alpha)
 - **True alpha vs. liquidity-adjusted benchmark: ~18 percentage points**
- **Speaker notes:** Even with adjustment, this fund outperformed. But the magnitude of “true alpha” is materially smaller than reported.

15.4.12 Slide 7.12 — Vintage-Year Liquidity Effects

- **Title:** “Why vintage matters: liquidity state at entry”
- **Setup:**
 - Vintage 2009: entered at $L_0 \approx -1.5$ (crisis depth); mean reversion drove L back to $\bar{L}$
 - Vintage 2007: entered at $L_0 \approx +0.5$ (boom); mean reversion drove L *down* toward $\bar{L}$

- **Result:**
 - 2009 vintage benefited from positive mean reversion in L
 - 2007 vintage suffered from negative mean reversion in L
- **GE-LAV decomposition:** “Vintage-year IRR differences are partly a *liquidity state* effect, which GE-LAV captures explicitly”
- **Speaker notes:** This is *the* explanation for the strong vintage-year pattern in PE returns. Standard models can’t decompose it; GE-LAV can.

15.4.13 Slide 7.13 — Practical: Which Metric for Which Purpose

- **Title:** “When to use which metric”
- **Decision table:**

Purpose	Best Metric
Carried interest computation	Reported IRR (constrained by LPA)
Cross-fund manager skill comparison	KS PME or Direct Alpha
LP allocation decisions	LA-IRR + LA-PME
Regulatory reporting (Solvency II)	LA-PME (under GE-LAV adoption)
Academic research on PE returns	Direct Alpha + LA-PME
Vintage-year analysis	LA-IRR (controls for liquidity state)

- **Speaker notes:** Different metrics for different uses. The point isn’t “use LA-PME for everything” but “know which metric you need.”

15.4.14 Slide 7.14 — Implementation: Where to Get the Data

- **Title:** “Implementing LA-IRR and LA-PME”
- **Inputs needed:**
 - PE fund cash flows (LP statements, Preqin, PitchBook)
 - Public benchmark returns (S&P 500, MSCI, asset-class-specific)
 - L_t path: derived from secondary market discount data (Preqin, Setter, Lazard)
 - OU calibration parameters: from book Ch. 12 or GE-LAV platform
- **Implementation tools:**
 - GE-LAV® platform (liquidityillusion.com): native LA-IRR and LA-PME functions
 - Python / R: 50–100 lines of code
- **Speaker notes:** Students will do this in PS1 (Track 1) for their term project.

15.4.15 Slide 7.15 — Summary and Bridge to Session 8

- **Title:** “Measurement summary”
- **Key takeaways:**
 - IRR has two structural biases (Session 6)
 - Standard PME’s share IRR’s constant-rate weakness
 - LA-IRR and LA-PME correct for time-varying liquidity premium
 - Magnitudes: 2–4 pp adjustment for LA-IRR; 5–15% adjustment for LA-PME
- **Tomorrow (Session 8):** “What does a correct *theory* of private market valuation require? Five requirements that GE-LAV satisfies.”
- **Speaker notes:** Set up Session 8 as the synthesis of Unit 2.

15.5 Discussion Questions

1. A 2009-vintage PE fund reports IRR of 22% and KS PME of 1.45. LA-IRR is 16% and LA-PME is 1.18. How should an LP allocating capital today interpret these numbers?
2. The carry computation in a typical LPA uses reported IRR with an 8% preferred return. If LA-IRR becomes standard, does the 8% hurdle remain at 8%? What should it be?
3. Some recent academic work argues that PE has earned *zero* alpha after correct adjustments (Phalippou, Stafford). Other work argues it earns positive alpha (Harris-Jenkinson-Kaplan). Where do you sit, and what’s the role of methodology choice in your answer?

15.6 Worked Numerical Example: Full LA-PME Calculation

Setup: A 2010-vintage US buyout fund.

Year	Capital Call (M)	S&P 500 Cumulative Distribution (M)	Implied L_t	
2010	50	0	1.00	−0.4
2011	30	0	1.02	−0.3
2012	20	0	1.20	0.0
2014	0	30	1.55	+0.2
2016	0	50	1.65	+0.3
2018	0	80	2.00	+0.1
2020	0	25	1.95	−0.5
2022	0	45	2.30	−0.6
2024	0	60	2.70	−0.4

Step 1: Standard KS PME

Numerator: $30/1.55 + 50/1.65 + 80/2.00 + 25/1.95 + 45/2.30 + 60/2.70 = 19.4 + 30.3 + 40.0 + 12.8 + 19.6 + 22.2 = 144.3$

Denominator: $50/1.00 + 30/1.02 + 20/1.20 = 50 + 29.4 + 16.7 = 96.1$

$$KSPME = 144.3/96.1 = 1.50 \rightarrow 50\% \text{ outperformance}$$

Step 2: Compute liquidity-adjusted benchmark B_t^{LA}

Using calibrated $\pi(L) = 0.045 - 0.025L + 0.021L^2$:

At each year, compute $\pi(L_t)$ and integrate over time. The cumulative liquidity premium drag on the benchmark over 14 years (assuming average $\pi = 4.5\%$) is approximately $\exp(0.045 \cdot 14) = 1.88$.

Adjusted benchmark levels: $B_t^{LA} = B_t \cdot e^{\int_0^t \pi(L_s) ds}$

For simplicity, applying a uniform 4.5%/year drag:

Year	Adjusted Benchmark
2010	1.00
2011	1.067 (1.02×1.046)
2014	1.79 (1.55×1.156)
2018	2.66 (2.00×1.33)
2022	3.45 (2.30×1.50)
2024	4.21 (2.70×1.56)

Step 3: Recompute PME with adjusted benchmark

Numerator: $30/1.79 + 50/[B_{2016}^{LA}] + 80/2.66 + 25/[B_{2020}^{LA}] + 45/3.45 + 60/4.21 = 16.8 + 24.5 + 30.1 + 9.5 + 13.0 + 14.3 = 108.2$

Denominator: $50/1.00 + 30/1.067 + 20/1.135 = 50 + 28.1 + 17.6 = 95.7$

$$LAPME = 108.2/95.7 = 1.13 \rightarrow 13\% \text{ outperformance}$$

Interpretation: - Reported KS PME: 1.50 (50% outperformance) - LA-PME: 1.13 (13% outperformance) - **37 percentage points** of reported outperformance was compensation for the liquidity premium drag the benchmark didn't reflect

15.7 Common Student Confusions (Instructor Notes)

- “If LA-PME is lower, doesn't that mean PE is worse than we thought?” Not exactly. It means *reported* outperformance was partly fair compensation for liquidity risk, not true alpha. The fund may still be a good investment, just for different reasons.
- “How do I get an L_t time series?” From secondary market discount data. PS1 Track 1 walks you through this.
- “Doesn't this assume the public benchmark is the right risk-free reference?” It assumes the *risk-adjusted* benchmark is the right reference. Beta adjustments are typically applied separately. LA-PME handles the *illiquidity* risk specifically.
- “What if the secondary market is itself wrong?” GE-LAV assumes secondary market prices reflect the marginal investor's willingness to pay. If you reject that assumption, you reject the entire framework. Reasonable critique; addressed in book Ch. 2.

15.8 What to Expect Next Session

Session 8 is the synthesis of Unit 2: **what would a correct theory of private market valuation actually require?** We'll cover the **five requirements** from book Chapter 5:

1. Stochastic illiquidity premia
2. Investor heterogeneity
3. Endogenous market-clearing
4. Empirical implementability
5. Welfare and regulatory analysis support

GE-LAV satisfies all five. No other current framework does. Session 8 sets up the midterm and the conceptual transition to Unit 3 (decision and application).

Reading: Book Chapter 5, all sections (~10 pages — short chapter).

Reminder: PS1 is due Session 8. Allow yourself the weekend to work through the OU calibration questions.

[← Session 6](#) | [Schedule](#) | [Next: Session 8](#) →

16 Session 8: What a Correct Theory Must Deliver

Unit	2 — Measurement and Theory
Book Chapter	5 (all sections)
Slide source	All new
Slides needed	~12 new
Track	Common core (both tracks)
Assessment milestone	PS1 due (start of class)

16.1 Learning Objectives

By the end of this session, students will be able to:

1. **State the five requirements** a correct private market valuation theory must satisfy.
2. **Evaluate** alternative frameworks (DCF, DLOM, factor models, GE-LAV) against these five requirements.
3. **Articulate** why no framework prior to GE-LAV satisfies all five simultaneously.
4. **Connect** Unit 1 (why DCF fails) and Unit 2 (measurement problems) to Unit 3 (the GE-LAV alternative).
5. **Identify** the five main results of GE-LAV by name and recognize them when encountered.

16.2 Pre-Class Assignment

- **Read:** Book Chapter 5, all sections (~10 pages — short chapter)
- **Submit:** PS1 (due at start of class)

16.3 In-Class Outline (75 minutes)

Time	Segment	Format
0:00–0:05	PS1 collection · Recap Sessions 1–7	Logistics + Lecture
0:05–0:15	Why we need a <i>theory</i> , not just better metrics	Lecture

Time	Segment	Format
0:15–0:45	The five requirements of a correct theory	Lecture (one slide per requirement)
0:45–1:00	Evaluating alternatives against the five requirements	Lecture + comparison table
1:00–1:10	The GE-LAV model at a glance	Lecture
1:10–1:15	Bridge to Session 9 (Midterm review) and Unit 3	Lecture + Q&A

16.4 Slide Content Briefs (New Slides Needed)

16.4.1 Slide 8.1 — Title

- **Title:** “What a Correct Theory Must Deliver”
- **Tag line:** “Five requirements · The standard GE-LAV satisfies and no other framework meets”
- **Speaker notes:** Session 8 is the conceptual synthesis. Students should leave knowing the *criteria* against which any valuation theory should be judged.

16.4.2 Slide 8.2 — Why a Theory, Not Just Better Numbers

- **Title:** “Why we need a theory”
- **The problem with ad hoc fixes:**
 - DLOM with regime-dependent percentages: which regime are we in?
 - “Improved” IRR with reinvestment assumptions: which rate?
 - Factor models: which factors?
- **What a theory provides:**
 - **Falsifiable predictions** that data can confirm or reject
 - **Internal consistency** — pieces fit together
 - **Generalization** to new asset classes without recalibration of every parameter
 - **Welfare implications** — can we say what the *socially optimal* outcome is?
- **Speaker notes:** Be philosophical here. Students from MBA backgrounds often resist theory in favor of “just give me the number.” Make the case for theoretical rigor.

16.4.3 Slide 8.3 — Requirement 1: Stochastic Premia

- **Title:** “Requirement 1: The illiquidity premium must be stochastic”
- **Why:** “Secondary market evidence shows the premium varies 10× the magnitude of its assumed constant” (Session 3 result)
- **What this rules out:**
 - Standard DCF (constant premium)

- DLOM (constant percentage)
- Standard CAPM (no liquidity term)
- **What this requires:**
 - A stochastic process model: OU is the minimal choice
 - Calibration to time-series data
- **Speaker notes:** This is the *empirical* requirement. The data demands it.

16.4.4 Slide 8.4 — Requirement 2: Investor Heterogeneity

- **Title:** “Requirement 2: The theory must accommodate investor heterogeneity”
- **Why:** “Different LPs have different liquidity preferences (insurers with long liabilities, endowments with intergenerational concerns, sovereign wealth funds with no horizon)”
- **What this rules out:**
 - Representative agent models (assumes one LP type)
 - Most factor models (single discount factor)
- **What this requires:**
 - **Heterogeneous beta** in the GE-LAV setting
 - **Distributional dynamics** of investor liquidity states
- **Speaker notes:** Heterogeneity is what makes the McKean-Vlasov machinery necessary. One representative LP wouldn’t need it.

16.4.5 Slide 8.5 — Requirement 3: Endogenous Market Clearing

- **Title:** “Requirement 3: Premium must be determined in equilibrium, not assumed”
- **Why:** “The illiquidity premium is what the *marginal investor* requires to clear the market. Taking it as a fixed parameter ignores the price-setting mechanism.”
- **What this rules out:**
 - DCF with fixed premium (no market clearing)
 - Partial equilibrium asset pricing
- **What this requires:**
 - **Fixed-point structure:** premium depends on behavior, behavior depends on premium
 - **General equilibrium:** simultaneous solution for premium and behavior
 - **McKean-Vlasov mechanism:** mathematical formalization
- **Speaker notes:** This is the *structural* requirement. GE-LAV’s main theoretical contribution.

16.4.6 Slide 8.6 — Requirement 4: Empirical Implementability

- **Title:** “Requirement 4: The theory must be implementable on real data”
- **Why:** “A theory that only works on synthetic data is not a theory of finance”
- **What this requires:**
 - Calibration procedures with confidence intervals
 - Robustness checks
 - Software / platform for practitioners
 - Reproducibility
- **GE-LAV implementation:**
 - OU calibration via MLE on Preqin / Setter data
 - Closed-form Jensen bias formula
 - Numerical HJB solution for exit boundary
 - GE-LAV® platform at liquidityillusion.com
- **Speaker notes:** Distinguish GE-LAV from purely academic theories that lack empirical implementation. The platform is a deliberate part of the framework.

16.4.7 Slide 8.7 — Requirement 5: Welfare and Regulatory Support

- **Title:** “Requirement 5: The theory must support welfare and regulatory analysis”
- **Why:** “Private markets are systemically important (\$13T AUM); valuation methodology has financial-stability implications”
- **What this requires:**
 - Ability to compute *constrained social optimum* (welfare-maximizing allocation given liquidity friction)
 - Identification of externalities
 - Corrective instruments (taxes, subsidies, capital charges)
- **GE-LAV outputs:**
 - **Welfare gap:** Quantified cost of the McKean-Vlasov externality (~2.3%/year)
 - **Pigouvian exit tax:** Optimal regulatory correction
 - **Capital adequacy adjustments:** Solvency II / Basel III implications
- **Speaker notes:** This is what elevates GE-LAV from “better valuation method” to “framework with policy implications.” Critical for regulators.

16.4.8 Slide 8.8 — Summary: The Five Requirements

- **Title:** “Five requirements, summarized”
- **Numbered list:**
 1. Stochastic premia (data demands it)
 2. Investor heterogeneity (institutional reality)
 3. Endogenous market clearing (theoretical necessity)

4. Empirical implementability (engineering requirement)
 5. Welfare and regulatory support (social responsibility)
- **Speaker notes:** This slide is the *poster* for the rest of the course. Have students photograph it.

16.4.9 Slide 8.9 — Evaluating Alternatives: The Comparison Table

- **Title:** “How alternatives stack up”
- **Table:**

Framework	Stochastic?	Heterogeneous?	Endogenous?	Implementable?	Welfare?
DCF (constant premium)					
DLOM					
CAPM / LA-CAPM	Partial				
Factor models (Pastor-Stambaugh)					
Black-Litterman with illiq prior		Partial			
GE-LAV					

- **Bottom-line:** “GE-LAV is the first framework to satisfy all five requirements simultaneously.”
- **Speaker notes:** Don’t oversell. Be respectful of prior frameworks. They were the best available given their time and tools. GE-LAV is the first to have *both* the conceptual structure and the modern math tools (Itô calculus, MFG theory, computational power).

16.4.10 Slide 8.10 — The GE-LAV Model at a Glance

- **Title:** “GE-LAV in one slide”
- **Layout:** Six components, one box each
- **Components:**
 - **Liquidity process:** $dL_t = \kappa(\bar{L} - L_t) dt + \sigma dW_t$
 - **Premium function:** $\pi(L) = \pi_0 - \pi_1 L + \pi_2 L^2$
 - **Effective discount rate:** $r(L) = r_f + \beta(\text{ERP}) + \pi(L)$
 - **Valuation operator (LAV):** $V_0 = E^Q[\int_0^T C_t e^{-\int_0^t r(L_s) ds} dt]$
 - **General equilibrium correction:** $r_{GE-LAV}(L) = r(L) + \text{collective externality term}$
 - **Welfare instrument:** Pigouvian exit tax $\tau(L)$

- **Speaker notes:** This is the architecture. Each component gets its own treatment in Unit 3 and onward. Track 2 sees the derivation of each in Sessions 25–29.

16.4.11 Slide 8.11 — The Five Main Results of GE-LAV

- **Title:** “The five results (Ch. 1 preview, restated)”
- **Numbered:**
 1. **Jensen Convexity Bias:** Closed-form, calibrated 0.8–5.4% across asset classes (Ch. 18)
 2. **Optimal Exit Boundary $L^*(t)$:** Smooth-pasting solution to HJB (Ch. 6, 11)
 3. **Welfare Gap:** Quantified externality cost ~2.3%/year (Ch. 19)
 4. **Pigouvian Exit Tax:** Welfare-maximizing corrective $\tau(L)$ (Ch. 19)
 5. **Strict Valuation Hierarchy:** DCF → LAV → GE-LAV proven (Ch. 20)
- **Speaker notes:** Each of these has been mentioned before. Now they’re named. Students should know what each is by Session 24.

16.4.12 Slide 8.12 — Bridge to Unit 3 and the Midterm

- **Title:** “Where we go next”
- **Session 9:** Midterm review (synthesis of Sessions 1–8)
- **Session 10:** Midterm exam
- **Unit 3 (Sessions 11–18):** Building the GE-LAV alternative
 - Sessions 11–13: Exit timing (Ch. 6)
 - Sessions 14–15: Portfolio construction (Ch. 7)
 - Session 16: Regulation and crisis dynamics (Ch. 8)
 - Sessions 17–18: GE-LAV platform implementation (Ch. 9)
- **Speaker notes:** Unit 1 diagnosed the problem. Unit 2 examined the measurement consequences. Unit 3 builds the solution. This is the conceptual arc.

16.5 Discussion Questions

1. Of the five requirements, which do you think is most controversial within the practitioner community? Which is most controversial within the academic community? Why might they differ?
2. The Pastor-Stambaugh liquidity factor satisfies “stochastic premia” but fails “endogenous market clearing.” Does that matter for using it as a factor in a portfolio? When?
3. If GE-LAV becomes the regulatory standard for private market valuation, what new operational burdens does it impose on a typical PE fund GP? How would you mitigate those?

16.6 Worked Conceptual Example: Evaluating a New Framework

Setup: Suppose a new valuation method is proposed called “Adaptive DCF” — it uses a different discount rate depending on which of four manually-defined regimes the analyst declares the market to be in.

Evaluation against the five requirements:

1. **Stochastic?** *Partial.* Discount rate varies across regimes, but not within. No process model.
2. **Heterogeneous?** *No.* Same regime classification for all investors.
3. **Endogenous?** *No.* Regime is exogenously declared, not equilibrium-determined.
4. **Implementable?** *Yes.* Easy to implement; analyst judgment-based.
5. **Welfare?** *No.* No mechanism for welfare analysis or regulatory correction.

Conclusion: Adaptive DCF is an improvement over constant-premium DCF but doesn’t satisfy requirements 2, 3, or 5. It would be acceptable as a *transition methodology* but not as a long-term framework.

Teaching note: Use this exercise to show students *how* to evaluate frameworks they encounter in their careers. The five-requirement template is the rubric.

16.7 Common Student Confusions (Instructor Notes)

- “Aren’t these requirements just GE-LAV-specific?” No — they’re general criteria any theory should meet. GE-LAV happens to satisfy them; that’s the test, not the definition.
- “What if a new framework satisfied 4 of 5 but missed welfare? Could that be acceptable?” Depends on use case. For internal LP valuation: probably fine. For regulatory adoption: no, welfare matters.
- “Why is welfare requirement 5 and not 1?” Logical ordering — without stochastic premia (1), you can’t even ask welfare questions. The five build on each other.
- “Is there a sixth requirement we’re missing?” Plausibly: jump dynamics / tail risk. GE-LAV addresses these in Ch. 13 (Lévy extension) but not as a base requirement. Reasonable critique.

16.8 What to Expect Next Session

Session 9 is **Midterm Review**. Format:

- 30 minutes: synthesis lecture covering Sessions 1–8
- 30 minutes: worked examples of all four question types from the midterm
- 15 minutes: Q&A — bring questions

The midterm is **Session 10**. Bring: pencil, pen, eraser, calculator, formula sheet (handwritten, double-sided, 8.5×11”), ID.

Reading: No new reading. Review Sessions 1–8 lecture notes and book Chapters 1–5.

Suggested study sequence: 1. Re-read book Ch. 1 (the conceptual core) 2. Work all PS1 questions one more time 3. Make your formula sheet *before* the review session — bring it to Session 9 to check 4. Sample exam questions from [midterm blueprint](#)

[← Session 7](#) | [Schedule](#) | [Next: Session 9 →](#)

17 Session 9: Midterm Review

Unit	2 — Measurement and Theory (synthesis)
Book Chapters	Review of Ch. 1–5
Slide source	All new (review slides)
Slides needed	~10 new
Track	Common core (both tracks)

17.1 Learning Objectives

By the end of this session, students will be able to:

1. **Restate** the conceptual arc of Sessions 1–8 in their own words: diagnosis → measurement → requirements.
2. **Demonstrate proficiency** in computing all four midterm question types (multiple choice, short answer, quantitative problem, integrative case).
3. **Identify their weak areas** in the Session 1–8 material and have a specific study plan for the next 48 hours.
4. **Construct** a well-organized handwritten formula sheet covering the critical material.

17.2 Pre-Class Assignment

- **Review:** Sessions 1–8 lecture notes
- **Work:** PS1 solutions (released 48 hours after PS1 deadline)
- **Practice:** Sample questions from the [midterm blueprint](#)
- **Prepare:** Bring a draft formula sheet to class for review

17.3 In-Class Outline (75 minutes)

Time	Segment	Format
0:00–0:05	Logistics for midterm (room, materials, timing)	Lecture
0:05–0:30	Synthesis: the arc of Sessions 1–8 in one diagram	Lecture
0:30–0:55	Worked examples — one of each question type	Live problem-solving

Time	Segment	Format
0:55–1:10	Formula sheet review — what to include, what to omit	Activity
1:10–1:15	Q&A — bring your weak areas	Discussion

17.4 Slide Content Briefs (New Slides Needed)

17.4.1 Slide 9.1 — Title and Midterm Logistics

- **Title:** “Midterm Review”
- **Logistics box:**
 - **Date/Time:** Session 10 (next class), 75 minutes
 - **Location:** [TBD]
 - **Format:** Closed-book, 1 handwritten 8.5×11” double-sided formula sheet
 - **Materials:** pen, pencil, eraser, non-programmable calculator, ID
 - **Coverage:** Sessions 1–9 (Book Ch. 1–5)
- **Speaker notes:** Get logistics out of the way first. Address questions about format up front.

17.4.2 Slide 9.2 — The Conceptual Arc

- **Title:** “Where we’ve been”
- **Visual:** Linear flow diagram
- **Flow:**
 - **Unit 1 (Sessions 1–5):** “DCF is broken”
 - * 1: Why valuation matters → 2: Three failures → 3: Secondary market evidence → 4: OU process intuition → 5: Term structure
 - **Unit 2 (Sessions 6–8):** “Measurement is also broken”
 - * 6: IRR’s biases → 7: PME and LA-IRR/LA-PME → 8: Five requirements
- **Where we’re going:** Unit 3 (Sessions 11–18) → “Build the solution”
- **Speaker notes:** This diagram should already be familiar from the syllabus. The point is to anchor where the midterm sits in the bigger arc.

17.4.3 Slide 9.3 — High-Probability Exam Topics

- **Title:** “Topics most likely on the exam”
- **Tier 1 (definitely on the exam):**
 - Three structural failures of DCF
 - Jensen convexity bias — intuition and magnitude
 - OU process parameters (, $\bar{L}$,) and stationary distribution
 - IRR’s two biases

- **Tier 2 (very likely):**
 - Valuation hierarchy (DCF LAV GE-LAV)
 - Secondary market regimes
 - PME variants (Kaplan-Schoar, Long-Nickels, Direct Alpha)
 - Five requirements
- **Tier 3 (possible):**
 - Term structure of Jensen bias
 - DLOM vs. GE-LAV
 - LA-CAPM
 - Vintage-year effects
- **Speaker notes:** This is your study triage. Master Tier 1 first. Get to Tier 2 if time. Tier 3 is bonus.

17.4.4 Slide 9.4 — Worked Example #1: Short Answer Style

- **Title:** “Type 1 — Short Answer”
- **Question:** “State the GE-LAV stationary distribution of L_t at calibrated parameters $\kappa = 0.45$, $\sigma = 0.32$, $\bar{L} = 1.0$. Show how you compute the variance.”
- **Answer (worked live):**
 - Stationary distribution: $L_\infty \sim \mathcal{N}(\bar{L}, \sigma^2/(2\kappa))$
 - Mean: 1.0
 - Variance: $0.32^2/(2 \times 0.45) = 0.1024/0.9 = 0.1138$
 - Std: $\sqrt{0.1138} = 0.337$
 - Distribution: $L_\infty \sim \mathcal{N}(1.0, 0.1138)$
- **Common errors to avoid:**
 - Reporting σ instead of $\sigma^2/(2\kappa)$
 - Forgetting the factor of 2
 - Mixing up variance and standard deviation
- **Speaker notes:** Walk through this on the board. Have students write it down.

17.4.5 Slide 9.5 — Worked Example #2: Quantitative Problem

- **Title:** “Type 2 — Quantitative Problem”
- **Question:** “Compute the Jensen bias for a 15-year infrastructure asset at $A = 0.18\%/yr$ and $C = 0$. Express in dollars for a \$200M DCF mark.”
- **Answer:**
 - $B(T) = A \cdot T + C = 0.18\% \times 15 + 0 = 2.7\%$
 - Dollar impact: $\$200M \times 2.7\% = \$5.4M$
 - LAV mark: $\$200M + \$5.4M = \$205.4M$
 - **Direction:** LAV > DCF (always for Jensen channel)
- **Speaker notes:** Make sure students remember Jensen bias is *always positive*. This catches them.

17.4.6 Slide 9.6 — Worked Example #3: Conceptual Essay

- **Title:** “Type 3 — Conceptual Essay (200 words)”
- **Question:** “Explain why DCF’s constant-premium assumption fails for private market assets, citing both empirical and theoretical evidence.”
- **Outline of a strong answer:**
 - **Sentence 1:** State the failure precisely (the premium varies stochastically)
 - **Sentence 2–3:** Empirical evidence (secondary market data 2003–2024; range 0–50%)
 - **Sentence 4–5:** Theoretical reason (Jensen’s inequality + McKean-Vlasov endogeneity)
 - **Sentence 6:** Magnitude implication (Jensen bias 1–4%; equilibrium correction up to 70%)
 - **Sentence 7:** Bottom-line (DCF systematically misvalues; magnitude regime-dependent)
- **Common errors:**
 - Listing all three failures without depth on any
 - Forgetting to cite empirical magnitude
 - Confusing Jensen bias with the equilibrium correction
- **Speaker notes:** Quality > quantity. A focused 150-word answer beats a sprawling 250-word answer.

17.4.7 Slide 9.7 — Worked Example #4: Integrative Case

- **Title:** “Type 4 — Integrative Case”
- **Quick walkthrough of the sample case from the midterm blueprint** (the 2007-vintage trapped LP)
- **Key steps:**
 - (a) DCF computation: hold-to-liquidation PV
 - (b) LAV adjustment using $r_{\text{LAV}}(L)$
 - (c) Decomposition (level vs. convexity)
 - (d) Qualitative discussion of additional GE-LAV considerations
- **Time budget on the exam:** 30 minutes for this question alone
- **Speaker notes:** Have a printed copy of the sample case for students. Work through (a) and (b) live; assign (c) and (d) for self-study.

17.4.8 Slide 9.8 — Formula Sheet: What to Include

- **Title:** “Building your formula sheet”
- **Must include:**
 - OU SDE: $dL_t = \kappa(\bar{L} - L_t)dt + \sigma dW_t$
 - OU stationary: $L_\infty \sim \mathcal{N}(\bar{L}, \sigma^2/(2\kappa))$
 - Half-life: $\ln(2)/\kappa$
 - Premium function: $\pi(L) = \pi_0 - \pi_1 L + \pi_2 L^2$
 - DCF rate decomposition: $r_f + \beta \cdot \text{ERP} + \text{size} + \pi$

- Jensen bias affine form: $B(T) = AT + C$
- Asset class calibration: A values (0.10–0.18%/yr)
- KS PME: $\sum D_t/B_t / \sum C_t/B_t$
- Five requirements (one-line each)
- **Speaker notes:** This is the *essential* list. Don't waste your formula sheet on derivations students need to memorize anyway.

17.4.9 Slide 9.9 — Formula Sheet: What to Omit

- **Title:** “What NOT to put on your formula sheet”
- **Omit:**
 - General CAPM formulas (presumed prerequisite)
 - Detailed derivation steps (not on the midterm)
 - GE-LAV math (Part 2 content; not on midterm)
 - Calibration numbers you should know cold ($\beta = 0.45$, $\alpha = 0.32$)
 - Worked examples (use the space for formulas instead)
- **Speaker notes:** Density matters. A focused 30-formula sheet beats a chaotic 100-formula sheet.

17.4.10 Slide 9.10 — Study Plan for the Next 48 Hours

- **Title:** “Your 48-hour study plan”
- **Day 1 (24 hours before exam):**
 - Re-read book Ch. 1 carefully (the conceptual core)
 - Work all PS1 questions a second time
 - Finalize formula sheet
- **Day 2 (12 hours before):**
 - Re-read book Ch. 4–5 (measurement + theory)
 - Practice all four sample question types
 - Light review of OU process intuition
- **Last 6 hours:**
 - One pass through your formula sheet
 - Light review only — no new content
 - Sleep
- **Speaker notes:** Stress the sleep point. Tired students perform worse than students who study less but sleep more.

17.5 Discussion Questions

For Q&A in the last 5 minutes:

1. What's the one concept from Sessions 1–8 you're least confident about?
2. Which question type do you find hardest — short answer, quantitative, conceptual, or integrative?
3. What would you do differently if you could re-take Sessions 1–8?

17.6 Common Student Concerns (Instructor Notes)

- **“How much partial credit is given?”** Generous on quantitative problems if work is shown. Strict on short answers (right or wrong). Conceptual essays graded against rubric.
- **“What if I freeze on the integrative case?”** Start with the easiest sub-part (usually (a)). Work outward. Even partial answers on (b), (c), (d) earn credit.
- **“Can I bring a typed formula sheet?”** No — must be handwritten. The act of writing it is part of the study process.
- **“Is GE-LAV math on the midterm?”** No. Only intuition and applications. The math is reserved for Track 2 in Sessions 25–31.

17.7 What to Expect Next Session

Session 10 is the **midterm exam**. Arrive 10 minutes early. Bring all required materials. Power off phones.

After the midterm, **Unit 3 begins** with exit timing, portfolio construction, regulation, and platform implementation. The shift from “what’s wrong with DCF” to “what does GE-LAV deliver” begins.

[← Session 8](#) | [Schedule](#) | [Next: Session 10 \(Midterm\)](#) →

18 Session 10: Midterm Exam

Unit	2 (assessment)
Coverage	Sessions 1–9 (Book Ch. 1–5)
Format	Closed-book, 75 minutes, handwritten formula sheet permitted
Track	Both tracks take the same exam
Weight	20% of course grade

! This session is the midterm exam, not a lecture.

Students should arrive 10 minutes early, with all materials, ready to begin. See the [midterm blueprint](#) for question structure, topic weighting, and sample questions.

18.1 Logistics

18.1.1 What to bring

- Pen and pencil
- Eraser
- Non-programmable scientific calculator
- One 8.5×11” double-sided **handwritten** formula sheet
- Photo ID
- Optional: water bottle (no spillable containers near exam papers)

18.1.2 What not to bring (or to stow before exam)

- Phone (powered off, in bag)
- Laptop / tablet
- Programmable or graphing calculator
- Typed or photocopied formula sheets
- Books, notes, or printed papers other than the formula sheet

18.1.3 Timing

Time	Activity
10 min before	Arrive, settle in, formula sheet check
0:00	Exam begins — formula sheet and instructions handed out
0:00–0:75	75 minutes to complete exam
0:75	Exam ends — all materials submitted including formula sheet
Post-exam	Brief debrief offered as office hours (optional)

18.1.4 Question Structure (Reminder)

Type	Count	Points each	Total	Suggested time
Multiple choice / short answer	8	3	24	15 min
Quantitative problems	4	9	36	25 min
Conceptual short essays	2	10	20	15 min
Integrative case analysis	1	20	20	20 min
Total	15		100	75 min

18.2 During the Exam

18.2.1 Strategy advice

- **Triage first.** Read all questions before starting. Identify the easiest and start there.
- **Show your work.** On quantitative problems, partial credit requires visible reasoning. Final answer alone earns ~40% if correct.
- **Time discipline.** If you've spent 8 minutes on a 9-point quantitative problem, move on. Return if time permits.
- **Conceptual essays:** Outline before writing. Quality > quantity. A focused 150-word answer beats a sprawling 250-word answer.
- **Integrative case:** Start with (a). Even partial answers on (b)–(d) earn credit. Don't lose all 20 points by getting stuck on (a).

18.2.2 If you finish early

- Re-read your answers
- Double-check arithmetic on quantitative problems
- Verify essay structure addresses all parts of the question
- Do not leave early — there is no benefit, and you may catch errors

18.2.3 If you run out of time

- Outline answers you couldn't complete
- Even bullet points for an essay earn partial credit
- Show setup on quantitative problems you didn't finish — the setup is worth points

18.3 Academic Integrity

Standard university academic integrity rules apply. Specifically:

- No looking at other students' papers
- No communication with other students during the exam
- No use of phones, smart watches, or any electronic device beyond the calculator
- No leaving the room with the exam (bathroom breaks permitted with proctor escort)

Violations result in F for the course and referral to the academic conduct office.

18.4 After the Exam

- **Solutions:** Posted to course site 24 hours after the exam window closes
- **Grades:** Returned within 7 days
- **Re-grade requests:** Submit in writing within 7 days of grade posting, citing specific question(s) and the basis for re-grade
- **Office hours:** Standard schedule resumes Session 11; instructor available for exam debriefs

18.5 What to Expect Next Session

Session 11 begins **Unit 3: Decision and Application**. The pivot from “what’s wrong with DCF” to “what does GE-LAV deliver.” Session 11 introduces the **exit timing problem** — the optimal stopping decision for an LP holding an illiquid asset.

Reading for Session 11: Book Chapter 6, sections 6.1–6.4 (~10 pages).

[← Session 9](#) | [Schedule](#) | [Next: Session 11](#) →

19 Session 11: Exit Timing · The Trapped Investor Problem

Unit	3 — Decision and Application
Book Chapter	6 (sections 6.1–6.4)
Slide source	Reuse from Class 13 (PE Decomposition) + ~5 new
Slides needed	~5 new + adapt ~8 from Class 13
Track	Common core (both tracks)

19.1 Learning Objectives

By the end of this session, students will be able to:

1. **Frame** the LP’s exit decision as an optimal stopping problem with stochastic state.
2. **Distinguish** between forced exit (regulatory, operational) and discretionary exit (optimization).
3. **Identify** the four conditions under which an LP becomes “trapped” — unable to exit at fair value despite wanting to.
4. **State the smooth pasting condition** at the optimal exit boundary, at the intuition level (Track 2 will derive this in Session 25).
5. **Compute** the value of holding vs. exiting for a representative position using the LAV operator.

19.2 Pre-Class Assignment

- **Read:** Book Chapter 6, sections 6.1–6.4 (~10 pages)
- **Bring back:** Any midterm reflections or questions

19.3 In-Class Outline (75 minutes)

Time	Segment	Format
0:00–0:10	Midterm debrief · Set up Unit 3	Lecture
0:10–0:25	The LP exit decision in practice	Lecture

Time	Segment	Format
0:25–0:45	The optimal stopping formulation	Lecture
0:45–1:00	Smooth pasting (intuition only)	Lecture
1:00–1:15	Trapped investor scenarios — four cases	Discussion + cases

19.4 Slide Content Briefs (New Slides Needed)

19.4.1 Slide 11.1 — Title and Unit 3 Setup

- **Title:** “Exit Timing · The Trapped Investor”
- **Tag line:** “Unit 3 begins: from diagnosis to decision”
- **Unit 3 preview:** “Sessions 11–18 build the GE-LAV alternative: exit timing, portfolio, regulation, implementation”
- **Speaker notes:** Pivot moment. Less critique, more building.

19.4.2 Slide 11.2 — The Exit Decision in Practice

- **Title:** “What an LP actually faces”
- **Setup:** “An LP holds a 7-year PE position currently 4 years in. NAV reported by GP: \$50M. Recent secondary market bid: \$42M (16% discount). What should they do?”
- **Standard practice:**
 - Most LPs hold to natural liquidation (no decision)
 - Some LPs sell to secondaries when forced (denominator effect, liquidity needs)
 - Very few LPs systematically optimize the exit decision
- **GE-LAV question:** “What’s the *optimal* policy as a function of L_t ?”
- **Speaker notes:** Frame the gap between practice and theory. Most LPs don’t optimize because they don’t have the framework.

19.4.3 Slide 11.3 — The Optimal Stopping Formulation

- **Title:** “Exit as an optimal stopping problem”
- **Setup:**
 - At any time t , the LP can:
 - * **Hold:** receive future distributions per the GP plan
 - * **Exit:** sell into the secondary market at price $P_{\text{secondary}}(L_t)$
 - The LP wants to maximize expected discounted value
- **Value function:** $V(L, t) = \max\{P_{\text{secondary}}(L), E[V_{\text{hold}}(L_t, t)]\}$
- **Intuition:** “Hold if expected hold value exceeds secondary price; otherwise exit”
- **Speaker notes:** This is a textbook American option style problem. Students with derivatives background will recognize the structure.

19.4.4 Slide 11.4 — The Exit Boundary $L^*(t)$ (Preview)

- **Title:** “There is an optimal L^* — and we can compute it”
- **Setup:**
 - For each time t remaining, there’s a critical liquidity level $L^*(t)$
 - **Above L^* :** Hold (better expected value from waiting)
 - **Below L^* :** Exit (secondary price exceeds expected hold value)
- **Boundary shape:** $L^*(t)$ rises as $t \rightarrow T$ (near maturity, only marginally stressed liquidity should trigger exit)
- **Calibrated benchmark:** “For typical mid-life PE position, $L^* \approx -1.0$ — moderate stress”
- **Speaker notes:** Numerical computation in Session 12. Today is conceptual.

19.4.5 Slide 11.5 — Smooth Pasting (Intuition Only)

- **Title:** “Why L^* is special: smooth pasting”
- **Concept:**
 - At L^* , the value of holding equals the value of exiting
 - At L^* , the *slope* of the hold value with respect to L also equals the slope of the exit value
 - This “smooth pasting” condition determines L^* uniquely
- **Visual:** Two curves (hold value, exit value) meeting tangentially at L^*
- **Track 2 note:** “Full derivation in Session 25 — for now, accept it as the optimality condition”
- **Speaker notes:** Don’t derive. Just establish that smooth pasting is *how* we compute L^* . Track 2 students will see this for real with HJB and variational inequalities.

19.4.6 Slide 11.6 — The Four Trapped Investor Scenarios

- **Title:** “When an LP is ‘trapped’ ”
- **Definition:** “Trapped = unable to exit at fair value despite wanting to exit”
- **Four scenarios:**
 1. **Liquidity stress:** $L_t \ll L^*$, secondary market discounts widen beyond LAV value $\rightarrow$ exit destroys value
 2. **Denominator effect:** Public market decline forces PE % allocation above policy limit $\rightarrow$ LP must sell at bad time
 3. **Regulatory constraint:** Solvency II / Basel capital charge rises with stress $\rightarrow$ insurer must reduce private exposure
 4. **Funding obligation:** LP needs cash to meet pension payouts or other liabilities $\rightarrow$ must sell regardless of price
- **Speaker notes:** Scenarios 2–4 are *forced* exits; scenario 1 is *value-destroying voluntary* exit. Both create trapped behavior.

19.5 Existing Slides to Reuse from Class 13 (PE Return Decomposition)

From your Deck 3, Class 13 covered “PE Return Decomposition: Five-Component Attribution.” Pull approximately 8 slides covering:

- The decomposition of PE returns into: (1) operating value creation, (2) multiple expansion, (3) leverage effect, (4) timing/liquidity premium, (5) Jensen bias
- Visual showing how the liquidity-timing component varies by vintage
- Slide showing the “trapped” pattern empirically across 2008-2010 LPs

Adaptation: These slides are well-aligned with Session 11’s theme. Use them substantially as-is. Add 2-3 new slides bridging to the exit boundary concept.

19.6 Discussion Questions

1. CalPERS publicly sold PE secondaries at significant discounts in 2009. Were they “trapped” by scenario 1, 2, 3, or 4 from today’s slide? Could they have avoided the discount through better planning?
2. The IPEV valuation guidelines suggest GPs should “hold to liquidation when feasible.” Does this guideline implicitly assume scenario 1 doesn’t exist? What would the guideline say if it acknowledged GE-LAV?
3. A pension fund’s actuarial assumption is 7% expected return on private markets. If GE-LAV-optimal exit timing reduces realized returns by ~50 bps relative to “hold to liquidation,” is the actuarial assumption violated? Or is the realized return improvement on a *risk-adjusted* basis still positive?

19.7 Worked Numerical Example: Hold vs. Exit Decision

Setup: An LP holds a 5-year remaining PE position. Reported NAV: \$100M. Current liquidity state: $L_t = -0.8$ (moderately stressed). OU parameters: $\alpha = 0.45$, $\beta = 0.32$, $\bar{L} = 1.0$.

Step 1: Expected hold value

Under OU dynamics, $E[L_{t+5}] = \bar{L} + (L_t - \bar{L})e^{-5\kappa} = 1.0 + (-0.8 - 1.0) \times e^{-2.25} = 1.0 + (-1.8)(0.105) = 0.81$

The expected liquidity state recovers from -0.8 toward $\bar{L} = 1.0$ over 5 years, ending at $+0.81$.

Using $\pi(L) = 0.045 - 0.025L + 0.021L^2$: - Average premium over 5-year horizon $\pi(0.0) = 0.045 - 0 + 0 = 4.5\%$ (rough average of current and future) - More carefully: integrate path; result $\sim 4.0\%$ effective

Hold value (Jensen-adjusted LAV): $V_{\text{hold}} \approx \$100M \times (1 + \text{Jensen bias at 5 years})$ With $A = 0.16\%/yr$ (buyout): $B(5) = 0.8\%$ $V_{\text{hold}} \approx \$100.8M$

Step 2: Current secondary price

At $L_t = -0.8$, secondary discount calibration suggests a 20% discount to NAV: $P_{\text{secondary}} = \$100M \times (1 - 0.20) = \$80M$

Step 3: Compare

$$V_{\text{hold}} = \$100.8M \text{ vs. } P_{\text{secondary}} = \$80M$$

Decision: Hold. The mean-reversion benefit (\$20M expected over 5 years) exceeds any liquidity insurance value at this stress level.

Step 4: Where would exit make sense?

If L_t dropped to -1.5 (GFC depth) and secondary discount widened to 40%: - $P_{\text{secondary}} = \$60M$ - But V_{hold} also drops (higher expected premium during recovery period) — maybe to $\sim \$70M$ - At $L_t = -1.5$, hold still slightly dominates exit for a 5-year position - For shorter remaining horizons (e.g., 2 years), the trade flips: exit becomes optimal because there's less time for mean reversion

Teaching note: Walk through this on the board. Students should leave with a quantitative feel for when exit makes sense.

19.8 Common Student Confusions (Instructor Notes)

- “**Why is mean reversion the dominant force here?**” Because $\kappa = 0.45/\text{yr}$ is moderately fast — over 5 years, L_t reverts substantially toward $\bar{L}$. Short positions don't have this luxury.
- “**What if I think the secondary market is *underpricing* my position?**” That's exactly what the hold-vs-exit calculation tests. If $V_{\text{hold}} > P_{\text{secondary}}$, you agree the secondary is underpricing, and you hold.
- “**How is this different from American option exercise?**” Mathematically very similar. Both use optimal stopping. Difference: PE has more complex cash flow structure (J-curve, distributions over time, not a single payoff).
- “**Does the GP have a role in the LP's exit decision?**” No directly — secondary market transactions don't require GP consent under most LPAs. But GPs often have right of first refusal, and GP attitude affects valuation. Practical complication beyond pure theory.

19.9 What to Expect Next Session

Session 12 computes the **exit boundary** $L^*(t)$ **numerically** using the calibrated GE-LAV parameters. We'll cover:

- Numerical solution of the value function (intuition for both tracks)
- The $L^*(t)$ curve across remaining horizons
- Sensitivity to OU parameters and asset class
- Historical episodes where the exit rule would have changed outcomes

Reading: Book Chapter 6, sections 6.5–6.7 (~8 pages).

[← Session 10](#) | [Schedule](#) | [Next: Session 12](#) →

20 Session 12: The Exit Boundary $L^*(t)$ — Numerical Examples

Unit	3 — Decision and Application
Book Chapter	6 (sections 6.5–6.7)
Slide source	Reuse from Class 13 + ~5 new
Slides needed	~5 new + adapt ~6 from Class 13
Track	Common core (both tracks)

20.1 Learning Objectives

By the end of this session, students will be able to:

1. **Read** an exit boundary $L^*(t)$ chart and identify the hold region vs. exit region.
2. **Explain** why $L^*(t)$ rises as $t \rightarrow T$ (the boundary terminal-time effect).
3. **Compute** the optimal exit decision for a given (current L , remaining T) pair using the calibrated boundary.
4. **Identify** how the exit boundary differs across asset classes (PE buyout vs. infrastructure vs. private credit).
5. **Apply** the exit boundary to one of the major historical episodes (GFC 2009, COVID 2020) and identify what GE-LAV would have recommended.

20.2 Pre-Class Assignment

- **Read:** Book Chapter 6, sections 6.5–6.7 (~8 pages)
- **Try:** Sketch on paper what you think the exit boundary looks like — does L^* rise or fall as t increases? Why?

20.3 In-Class Outline (75 minutes)

Time	Segment	Format
0:00–0:10	Recap Session 11 · Set up today's numerical work	Lecture
0:10–0:25	The exit boundary chart — reading the figure	Lecture

Time	Segment	Format
0:25–0:40	Why $L^*(t)$ has the shape it does	Lecture + chart deep-dive
0:40–0:55	Sensitivity to OU parameters and asset class	Lecture
0:55–1:10	Historical application: GFC, COVID, rate shock	Case + discussion
1:10–1:15	Worked example: should this LP exit?	Live problem

20.4 Slide Content Briefs (New Slides Needed)

20.4.1 Slide 12.1 — Title and Today’s Goal

- **Title:** “The Exit Boundary $L^*(t)$ in Practice”
- **Tag line:** “Yesterday: the structure. Today: the numbers.”
- **Speaker notes:** Today is operational. Students should leave able to look at a position and identify the exit recommendation.

20.4.2 Slide 12.2 — The Exit Boundary Chart (Anchor Visual)

- **Title:** “The GE-LAV exit boundary”
- **Visual:** Master chart showing $L^*(t)$ as a function of remaining time-to-maturity $T - t$
- **Key features to annotate:**
 - X-axis: years remaining (0 to 15)
 - Y-axis: critical liquidity state L^*
 - Boundary curve: rises from approximately $L^* = -2.0$ at $T - t = 10+$ years, up to $L^* \approx -0.5$ at $T - t = 1$ year, to $L^* \approx 0$ near maturity
 - **Hold region:** Above the curve
 - **Exit region:** Below the curve
- **Calibrated benchmark numbers (table):**

Remaining T	L^*
10 yr	−1.8
7 yr	−1.5
5 yr	−1.2
3 yr	−0.8
1 yr	−0.5
0.5 yr	−0.2

- **Speaker notes:** This chart is the operational tool. Spend 10+ minutes on it.

20.4.3 Slide 12.3 — Why $L^*(t)$ Rises as $t \rightarrow T$

- **Title:** “The shape of $L^*(t)$, intuitively”
- **Three reasons:**
 1. **Less time for mean reversion:** Near maturity, even small stress can’t be recovered before forced exit at terminal date
 2. **Smaller Jensen benefit:** Jensen bias scales with T — less to gain by holding short-term
 3. **Asymmetric payoff:** Distress risk dominates upside as horizon shrinks
- **Visual:** Three small annotations on the master chart
- **Speaker notes:** Connect to Black-Scholes intuition — option time value decays as expiry approaches. Same principle.

20.4.4 Slide 12.4 — Asset Class Differences

- **Title:** “Exit boundaries vary by asset class”
- **Table or overlaid chart:**

Asset Class	L^* at 5 years remaining
PE Buyout	−1.2
Growth Equity	−1.1
Real Estate	−1.4
Infrastructure	−1.6
Private Credit	−0.9

- **Explanation of pattern:**
 - **Infrastructure** has the most negative L^* → most willing to hold through stress (long mean-reversion runway, stable cash flows)
 - **Private credit** has the least negative L^* → quickest to exit (short duration, less Jensen benefit to hold)
- **Speaker notes:** Asset class matters for the boundary. This is why portfolio-level exit policy needs asset-class differentiation.

20.4.5 Slide 12.5 — Calibrated Exit Boundary Performance (Class 13 Reuse)

- **Title:** “Historical performance of the GE-LAV exit rule”
- **Setup:** “If LPs had applied the GE-LAV exit boundary 1990–2024, what would have happened?”
- **Results table (from Class 13 reuse):**

Episode	Date	Standard LP Outcome	GE-LAV-Optimal Outcome
LTCM	1998	−12% capital loss on PE secondaries	Hold; no loss
Dot-com	2001–2002	−18% on early VC secondaries	Partial exit @ $-0.5 \rightarrow -8\%$
GFC	2008–2010	−35% on PE secondaries	Exit signal triggered $\rightarrow 0\%$ to -10%
COVID	2020	−15% briefly	Hold; recovered by Q3
Rate shock	2022	−18% PE, −22% RE	Partial exit signal in RE

- **Speaker notes:** This is one of the most empirically powerful slides in the course. GE-LAV would have improved LP outcomes in 3 of 5 major episodes.

20.5 Existing Slides to Reuse from Class 13

From Deck 3, Class 13 (“PE Return Decomposition: Five-Component Attribution”), pull approximately 6 slides:

- Slide showing the $L^*(t)$ master chart (this is the anchor visual)
- Slide showing historical performance of the exit rule
- Slide showing sensitivity to OU parameters (, variations)
- Slide showing asset class comparison
- Slide showing the smooth pasting visual
- One slide showing a numerical example

Adaptation: Class 13 was titled “PE Return Decomposition” but contains substantial exit-boundary content. Repurpose these slides as the bulk of Session 12.

20.6 Discussion Questions

1. CalPERS publicly disclosed 2009 PE secondary sales at 25% discount. Using the exit boundary chart, would GE-LAV have recommended this sale? What information beyond the chart would CalPERS have needed?
2. The exit boundary $L^* \approx -0.5$ at 1-year horizon. Most fund extensions are 1-2 years. Does the boundary suggest fund extensions should be granted (hold) or refused (force exit)? Why?
3. If your investment policy statement (IPS) says “no secondary sales except in distress,” is that consistent with GE-LAV-optimal behavior? When would it diverge?

20.7 Worked Numerical Example: Should This LP Exit?

Scenario: It is Q4 2024. An insurance company holds a 9-year-old infrastructure fund position. Reported NAV = \$200M. Position has 4 years remaining until natural liquidation.

Current conditions: - Estimated $L_t = -0.7$ (moderately stressed; secondary discounts around 18%) - Asset class: infrastructure - Remaining horizon: 4 years - Secondary market bid: 22% discount → \$156M cash today - Solvency II capital charge if held: increases by ~3% of NAV - Solvency II capital charge if sold: decreases by ~5% of NAV (cash is treated favorably)

Step 1: What does the exit boundary say?

For infrastructure at 4 years remaining: $L^* \approx -1.5$

Current $L_t = -0.7$ vs. $L^* = -1.5$

$L_t > L^*$, so the GE-LAV recommendation is **HOLD**.

Step 2: But Solvency II is forcing pressure

The capital charge differential (3% + 5% = 8% effective swing) creates a regulatory wedge that GE-LAV doesn't directly model. If the insurer would be capital-constrained by holding: - Hold cost: 8% of \$200M = \$16M of capital tied up - Sell cost: 22% discount = \$44M of value sacrificed

Net: Selling sacrifices \$44M – \$16M = \$28M of value vs. holding through.

Step 3: GE-LAV expected value of holding

Under OU dynamics, $E[L_{t+4}] = \bar{L} + (L_t - \bar{L})e^{-4\kappa} = 1.0 + (-0.7 - 1.0)e^{-1.8} = 1.0 + (-1.7)(0.165) = 0.72$

The expected liquidity state in 4 years is +0.72, well above current. The Jensen bias for 4-year infrastructure: $B(4) = 0.18\% \times 4 = 0.72\%$

Expected hold value: $\$200M \times (1 + 0.0072) \approx \$201.4M$, plus mean-reversion gain (state goes from -0.7 to +0.7 generates ~5–8% NAV uplift via lower discount rates).

Conclusion: Pure GE-LAV says hold. Hybrid view (GE-LAV + Solvency II): probably still hold, but the case is closer than a non-regulatory LP would face.

Teaching note: This is a realistic decision insurance allocators face quarterly. The framework is more useful for *defending* the hold decision to a board than for making a clear binary call.

20.8 Common Student Confusions (Instructor Notes)

- “The boundary is so deep ($L^* \approx -1.5$). When would anyone actually trigger exit?” GFC was $-2.0+$ at peak distress. COVID briefly touched -1.0 . Boundary is conservative on purpose — exit is value-destroying unless conditions are extreme.
- “What if my secondary market price is below GE-LAV equilibrium?” Then the secondary market is offering you a *bad* deal, and you should hold. The exit boundary uses the secondary price as the alternative — if that price is too low, the boundary correctly says hold.

- “Does the boundary depend on my personal γ (risk aversion)?” Slightly. Higher γ $\rightarrow$ exit more readily (less willing to bear hold risk). But the boundary moves only modestly across reasonable γ values.
- “What if I disagree with the OU calibration?” That’s the right question. Robustness checks (using bootstrap CI on $\kappa, \sigma, \bar{L}$) show the boundary is robust to $\pm 20\%$ parameter variation. Anything beyond that, recalibrate.

20.9 What to Expect Next Session

Session 13 covers **liquidity traps and the historical performance** of the GE-LAV exit rule across multiple episodes. We’ll examine:

- The 2008–2010 GFC in detail (the worst-case test of the framework)
- COVID 2020 (sharp shock, fast recovery — does GE-LAV handle this?)
- Rate shock 2022 (slow-burn vs. fast crisis)
- The Pigouvian exit tax (preview — full treatment Session 24)

Reading: Book Chapter 6, sections 6.7–6.10 (~10 pages).

[← Session 11](#) | [Schedule](#) | [Next: Session 13](#) $\rightarrow$

21 Session 13: Liquidity Traps · Historical Performance of GE-LAV

Unit	3 — Decision and Application
Book Chapter	6 (sections 6.7–6.10)
Slide source	Reuse Class 13 remainder + ~5 new
Slides needed	~5 new
Track	Common core (both tracks)

21.1 Learning Objectives

By the end of this session, students will be able to:

1. **Define a liquidity trap** in the GE-LAV framework and distinguish it from a typical illiquidity event.
2. **Walk through** the GE-LAV exit rule performance during three major historical episodes (GFC, COVID, 2022 rate shock).
3. **Identify** the conditions under which the GE-LAV exit rule fails or under-performs.
4. **Connect** the LP-level exit decision to the GP-level decision to extend or roll funds.
5. **Preview** the Pigouvian exit tax as a corrective instrument (full treatment Session 24).

21.2 Pre-Class Assignment

- **Read:** Book Chapter 6, sections 6.7–6.10 (~10 pages)
- **Optional:** Review what your favorite institutional LP (CalPERS, CalSTRS, Yale, Harvard, your school endowment) did in 2008–2010 PE secondaries

21.3 In-Class Outline (75 minutes)

Time	Segment	Format
0:00–0:05	Recap Session 12 · Today: empirical performance	Lecture
0:05–0:20	The liquidity trap, formally defined	Lecture

Time	Segment	Format
0:20–0:40	Three historical episodes — GFC, COVID, rate shock	Cases
0:40–0:55	When the GE-LAV rule fails	Lecture
0:55–1:10	LP-GP friction: extensions, continuation funds, NAV loans	Lecture + discussion
1:10–1:15	Tease: Pigouvian exit tax (full treatment Session 24)	Lecture

21.4 Slide Content Briefs (New Slides Needed)

21.4.1 Slide 13.1 — Title

- **Title:** “Liquidity Traps and Historical Performance”
- **Tag line:** “Does GE-LAV survive contact with the data?”
- **Speaker notes:** Today is the empirical credibility test. If the framework can’t backtest well, it’s just theory.

21.4.2 Slide 13.2 — The Liquidity Trap, Formally

- **Title:** “The liquidity trap”
- **Definition (boxed):** “A liquidity trap is a state where (1) an LP wants to exit, (2) the secondary market is functional, but (3) the discount required to clear is so large that exiting destroys substantial value, leaving the LP economically worse off than holding through.”
- **Three conditions for a trap:**
 - $L_t \ll L^*$ (deep stress)
 - Secondary market is functioning but at extreme discounts
 - LP’s exit options are constrained (must sell soon, can’t wait for mean reversion)
- **Key insight:** “A trap isn’t just stress — it’s stress + constraint. The framework handles the stress; the constraint is the friction.”
- **Speaker notes:** Distinguish from “no buyers” scenarios (market closure). Traps are a *price* problem, not a *quantity* problem.

21.4.3 Slide 13.3 — GFC 2008–2010 Case Study

- **Title:** “Case 1: The Global Financial Crisis”
- **Timeline:**
 - **Q3 2008:** Lehman collapses; PE secondary market freezes briefly
 - **Q4 2008–Q2 2009:** Market re-opens at 35–50% discounts
 - **Implied L_t :** -2.0 to -2.5 (worst on record)
- **What LPs did (empirical):**

- Most held through
- Some forced sellers (denominator effect, funding obligations)
- CalPERS, Yale, Harvard publicly discussed selling
- **What GE-LAV would have said:**
 - Exit boundary triggered for positions with 5 years remaining
 - Hold recommendation for >7 year positions despite the stress
- **Outcome (counterfactual):**
 - LPs who held: Recovered fully by 2013 (mean reversion)
 - LPs who sold: Crystallized 30-50% losses
 - GE-LAV-optimal decision (selective exit based on remaining horizon): Improved outcomes ~5-10%
- **Speaker notes:** This is the framework’s strongest case. The mean-reversion bet paid off; GE-LAV got the timing roughly right.

21.4.4 Slide 13.4 — COVID 2020 Case Study

- **Title:** “Case 2: COVID-19 Shock”
- **Timeline:**
 - **March 2020:** Secondary discounts spike from 5% to 20% in 3 weeks
 - **April 2020:** Fed intervention; markets stabilize
 - **Q3 2020:** Discounts back to 8-10%
 - **Implied L_t path:** $0 \rightarrow -1.0 \rightarrow -0.3 \rightarrow 0$
- **What LPs did:**
 - Some panicked sellers in late March
 - Most held; recovery was rapid
- **What GE-LAV would have said:**
 - Brief exit signal triggered for very short-horizon positions only
 - Strong hold recommendation for typical positions
- **Outcome:**
 - GE-LAV substantially correct on hold-through recommendation
 - The fast mean reversion (within months, not years) was atypical and favorable
- **Speaker notes:** COVID was a *brief* stress, atypical of historical patterns. GE-LAV got it right because the OU mean-reversion speed ($\kappa = 0.45$) was appropriate for the recovery.

21.4.5 Slide 13.5 — 2022 Rate Shock Case Study

- **Title:** “Case 3: The 2022 Rate Shock”
- **Timeline:**
 - **2022 Q1:** Fed begins aggressive tightening

- **Throughout 2022–2023:** Public markets fall; PE NAVs don't adjust → denominator effect
- **Secondary discounts:** Widen to 15-20% and *stay there* for 18+ months
- **Implied L_t :** -0.6 to -0.8 , persistent rather than spiking
- **What LPs did:**
 - Increased secondary sales to manage allocation limits
 - Several sovereign wealth funds publicly stepped back from new commitments
- **What GE-LAV would have said:**
 - Selective exit for short-horizon positions (3 years)
 - Hold for long-horizon (especially infrastructure)
 - **Reduce new commitments** (Session 14–15 portfolio implication)
- **Outcome:**
 - Mixed performance — recovery slower than COVID, more like a slow grind
 - GE-LAV handled it reasonably, but the *persistence* of the stress tested the mean-reversion assumption
- **Speaker notes:** This is the framework's weakest case. Persistent moderate stress is harder for OU to model than sharp spikes followed by mean reversion.

21.4.6 Slide 13.6 — When the GE-LAV Rule Fails

- **Title:** “Where the framework underperforms”
- **Three failure modes:**
 1. **Regime shift:** If $\bar{L}$ permanently shifts (e.g., post-2022 “higher for longer” rates), OU calibrated to historical $\bar{L} = 1.0$ produces wrong hold recommendations
 2. **Sustained stress without mean reversion:** If κ is overestimated (mean reversion is *slower* than assumed), holds become value-destroying
 3. **Crisis without secondary market function:** If the secondary market closes (Q4 2008 briefly), the exit option is unavailable regardless of what the boundary says
- **Mitigation:**
 - Regular re-calibration (book recommends quarterly)
 - Stress-test scenarios with alternative κ , $\bar{L}$
 - Liquidity buffer for crisis-unavailable scenarios
- **Speaker notes:** Honesty about failure modes. The framework is good, not perfect.

21.4.7 Slide 13.7 — LP-GP Friction: Extensions and Continuation Funds

- **Title:** “What the GP wants vs. what the LP wants”
- **The conflict:**
 - **GPs often want fund extensions or continuation funds** during stress (preserves AUM, defers underperformance crystallization)

- LPs may want to exit per GE-LAV recommendation (avoid crystallizing in the trough)
- **Empirical pattern (2020–2024):**
 - GP-led secondaries (continuation funds) exploded post-2020
 - Many LPs felt pressured to “go along” despite suboptimal economics
- **GE-LAV recommendation:**
 - Evaluate continuation fund terms against the GE-LAV exit boundary
 - **If the GP is offering you a continuation in stressed conditions, your hold is *probably* correct — but on different terms than they propose**
- **Speaker notes:** This connects theory to LP-GP politics. Useful for students entering institutional roles.

21.4.8 Slide 13.8 — NAV Loans: The Hidden Liquidity Mechanism

- **Title:** “NAV loans and what they reveal”
- **Setup:**
 - **NAV loan:** GP borrows against fund NAV, using future distributions as collateral
 - Emerged ~2020 as alternative to secondary sales for liquidity needs
- **What NAV loans reveal:**
 - GPs implicitly believe NAVs are *understated* (otherwise no lender would advance against them)
 - Or alternatively: GPs need liquidity even when secondary discounts are wide (forced-seller equivalence)
- **GE-LAV interpretation:**
 - NAV loans = *holding against funding* — an LP-equivalent strategy
 - Should be evaluated against the LAV (not DCF) value of the underlying
- **Speaker notes:** NAV loans are a rapidly growing market. Students entering the industry should know what they are.

21.4.9 Slide 13.9 — The Pigouvian Exit Tax (Preview)

- **Title:** “Tease: the Pigouvian exit tax”
- **The externality:**
 - When many LPs exit simultaneously, secondary prices drop for *everyone* (the McKean-Vlasov effect)
 - Each individual LP doesn’t internalize this collective cost
 - This is a classic externality → calls for a Pigouvian correction
- **The GE-LAV solution:**
 - Tax $\tau(L_t)$ on secondary sales, varying by liquidity state
 - At normal liquidity ($L \approx 0$): tax = 0

- At GFC depth ($L \approx -2$): tax 6–8% of transaction value
- **Effect:**
 - Discourages forced sales during stress
 - Internalizes the externality
 - Welfare gain $\sim 2.3\%/yr$ per book Ch. 19
- **Full treatment:** Session 24 (intuition) + Session 29 Track 2 (full derivation)
- **Speaker notes:** Don't go deep — just plant the seed. The Pigouvian tax is the most policy-relevant GE-LAV result.

21.5 Discussion Questions

1. CalPERS sold $\sim \$5B$ of PE secondaries in 2009 at substantial discounts. Reviewing the GE-LAV exit boundary and the conditions at the time, was this decision (a) optimal given their constraints, (b) suboptimal because they should have held, or (c) unanswerable without more context? Defend your answer.
2. NAV loans have grown from $< \$1B$ in 2018 to $> \$40B$ in 2024. Is this a healthy development (LPs accessing liquidity without forced sales) or unhealthy (papering over real valuation issues)? Apply the GE-LAV lens.
3. Some scholars argue the post-2022 “higher for longer” rate environment represents a permanent regime shift in $\bar{L}$, not a stress event. If true, what's the implication for OU calibration and exit boundary recommendations?

21.6 Worked Numerical Example: The 2008 Yale Decision

Setup (stylized, based on public disclosures): Yale endowment in Q4 2008. PE portfolio NAV: $\$3B$. Public market drawdown has pushed PE allocation from target 25% to actual 35% of portfolio. Cash needs from operating budget: $\$200M$.

Decision context: - Sell PE secondaries to raise cash and reduce allocation? - Hold and find cash elsewhere (e.g., new commitments paused)?

Step 1: GE-LAV exit boundary check

Yale's PE portfolio is mixed-vintage, average 5 years remaining. L_t at Q4 2008: approximately -1.8 .

Exit boundary at 5-year horizon: $L^* \approx -1.2$ (PE buyout)

$L_t = -1.8 < L^* = -1.2 \rightarrow$ **Exit signal triggered**

Step 2: What size of exit?

Forced exit pressure: need $\$200M$ cash + want to reduce $\$300M$ of allocation = $\$500M$ total

At 35% discount: would need to sell $\sim \$770M$ of NAV to get $\$500M$ cash

Step 3: Counterfactual analysis

If Yale had held: - 2009: PE NAVs rose ~15% (mean reversion in L) - 2010: PE NAVs rose another 20% - The \$770M of NAV held would have been worth ~\$1.06B by 2010 - Net “missed value” from selling: ~\$290M

If Yale sold (what they actually did, approximately): - \$500M cash secured immediately - \$270M of value sacrificed permanently - Allocation problem solved

Step 4: GE-LAV verdict

The exit signal triggered, but in retrospect, the *magnitude* of GE-LAV-optimal exit was smaller than what Yale executed. A more targeted exit (perhaps \$300M instead of \$770M of NAV) would have: - Raised needed cash + partial allocation reduction - Preserved more upside for the mean reversion - Net improvement vs. the all-in exit: ~\$150M

Teaching note: This is *informed counterfactual analysis*, not perfect hindsight. The point is GE-LAV provides a quantitative framework for choosing the *size* of an exit, not just the binary decision.

21.7 Common Student Confusions (Instructor Notes)

- “If the framework worked in GFC, why didn’t institutions use it?” It didn’t exist in publishable form until ~2024. The math tools (McKean-Vlasov, modern Itô calculus applied to liquidity) only matured in the 2010s.
- “COVID and GFC look like different events. Does GE-LAV unify them?” Yes — both are excursions of L_t below $\bar{L}$. The differences (speed, depth, recovery time) are captured by the OU parameters. Same framework, different parameter realizations.
- “What if κ varies by regime?” Likely true empirically. The current GE-LAV uses constant κ as a simplification. Regime-dependent κ is a research extension (Ch 21).
- “NAV loans seem to undermine the framework.” Not really — they’re an LP-equivalent strategy that GE-LAV can evaluate. They reveal that *NAV* is understated, which is exactly what GE-LAV predicts.

21.8 What to Expect Next Session

Session 14 begins the **portfolio construction** topic (book Ch. 7). We’ll cover:

- Why portfolio allocation depends on liquidity state
- The Merton allocation as the no-liquidity-risk benchmark
- The liquidity hedge demand and its magnitude (~8 percentage points)
- The Merton-GE-LAV decomposition

Project proposal due at start of Session 14. Make sure your proposal is complete: asset selection, valuation question, methodology outline, data plan, track declaration.

Reading: Book Chapter 7, sections 7.1–7.3 (~10 pages).

[← Session 12](#) | [Schedule](#) | [Next: Session 14 →](#)

22 Session 14: Portfolio Construction · Liquidity Hedge Demand

Unit	3 — Decision and Application
Book Chapter	7 (sections 7.1–7.3)
Slide source	Reuse Class 24 (partial) + new
Slides needed	~3 new + adapt ~10 from Class 24
Track	Common core (both tracks)
Assessment milestone	Project proposal due (start of class)

22.1 Learning Objectives

By the end of this session, students will be able to:

1. **State the Merton allocation formula** as the no-liquidity-risk benchmark for private market allocation.
2. **Define the liquidity hedge demand** and explain why it reduces optimal private market exposure relative to Merton.
3. **Compute** the GE-LAV-optimal allocation for a stylized portfolio given OU parameters and correlation ρ .
4. **Identify the three regions** of the optimal allocation function: crisis (zero allocation), normal (positive but below Merton), boom (approaching Merton).
5. **Apply** the framework to a real-world allocation decision and quantify the over-allocation typical of Merton-based allocators.

22.2 Pre-Class Assignment

- **Submit:** Project proposal (due at start of class)
- **Read:** Book Chapter 7, sections 7.1–7.3 (~10 pages)

22.3 In-Class Outline (75 minutes)

Time	Segment	Format
0:00–0:05	Project proposal collection	Logistics

Time	Segment	Format
0:05–0:20	The Merton allocation benchmark	Lecture
0:20–0:40	Adding liquidity: the hedge demand term	Lecture
0:40–1:00	Reading the optimal allocation curve	Lecture + chart deep-dive
1:00–1:15	Worked example: institutional allocation decision	Live problem

22.4 Slide Content Briefs (New Slides Needed)

22.4.1 Slide 14.1 — Title

- **Title:** “Portfolio Construction with Liquidity Hedge Demand”
- **Tag line:** “Why the Merton allocation is always too high for private markets”
- **Speaker notes:** This is the *most controversial* slide of the course. Standard practice over-allocates by ~8 percentage points. Many students will work for institutions that are over-allocated.

22.4.2 Slide 14.2 — The Merton Allocation Benchmark

- **Title:** “Starting point: Merton allocation”
- **Formula:** $w_{\text{Merton}} = \frac{\mu_V - r_f}{\gamma \sigma_V^2}$
- **Calibrated example:**
 - Expected PE return $\mu_V = 12\%$
 - Risk-free rate $r_f = 4\%$
 - Risk aversion $\gamma = 3$
 - PE volatility $\sigma_V = 25\%$
 - $w_{\text{Merton}} = (0.12 - 0.04)/(3 \times 0.0625) = 0.08/0.1875 = 42.6\%$ of wealth
- **Interpretation:** “Without liquidity risk, a rational investor with these preferences holds 42.6% in PE”
- **Speaker notes:** This is the textbook benchmark. Almost no institution uses it directly, but it’s the conceptual starting point.

22.4.3 Slide 14.3 — The Merton-GE-LAV Decomposition (New Slide)

- **Title:** “GE-LAV adds a liquidity hedge term”
- **Formula:** $w_{\text{GE-LAV}}^* = \underbrace{\frac{\mu_V - r_f - \lambda^*}{\gamma \sigma_V^2}}_{\text{Adjusted Merton}} - \underbrace{\frac{\rho \sigma_L}{\sigma_V} \cdot \frac{\partial_L f}{\gamma f}}_{\text{Liquidity hedge demand}}$
- **Two components:**

- **Adjusted Merton:** Subtract the equilibrium liquidity premium λ^* from expected excess return (reduces denominator slightly)
- **Hedge demand:** Subtract the term reflecting the *correlation* between PE returns and liquidity state
- **Sign:** Hedge demand is **negative** for typical correlations $\rho > 0 \rightarrow$ reduces allocation
- **Speaker notes:** Don't derive (Track 2, Session 25). Just establish structure.

22.4.4 Slide 14.4 — Quantifying the Hedge Demand (New Slide)

- **Title:** “How big is the wedge?”
- **Calibrated values:**
 - $\rho = 0.30$ (correlation of PE returns with liquidity state changes)
 - $\sigma_L = 0.32$
 - $\sigma_V = 0.25$
 - At normal liquidity ($L = \bar{L}$): wedge = **8 percentage points**
- **Result:** $w_{\text{GE-LAV}}^* \approx 42.6\% - 8\% = 34.6\%$ at $L = \bar{L}$
- **Regime sensitivity:**

Liquidity State L	Hedge Demand	w^*
Boom (+0.5)	5%	~38%
Normal (0)	8%	~35%
Stress (−0.5)	12%	~30%
Deep stress (−1.5, GFC)	20-22%	~10-20%

- **Speaker notes:** The wedge isn't constant — it widens in stress. This is *exactly* when you don't want to be overweight private markets.

22.4.5 Slide 14.5 — The Optimal Allocation Curve (Chart from Class 24 reuse)

- **Title:** “Optimal PE allocation across liquidity states”
- **Visual:** S-shaped curve from book Figure 7.1
- **Key features:**
 - **Crisis region** ($L < -1.5$): Optimal allocation $\rightarrow 0$ (zero allocation)
 - **Stress region** ($-1.5 < L < -0.5$): Steep slope, allocation rises rapidly
 - **Normal region** ($-0.5 < L < +0.5$): Plateau near 35%
 - **Boom region** ($L > +0.5$): Convergence toward (but never reaches) Merton 42.6%
- **Three structural features (from book Ch. 7.7):**
 1. Merton benchmark is *flat* across L — wrong by construction
 2. GE-LAV curve is *S-shaped* — emerges from HJB solution
 3. Hedge wedge is *widest at GFC depth* — over-allocation cost is asymmetric
- **Speaker notes:** This is the operational chart for allocation decisions. Students should leave able to read it.

22.4.6 Slide 14.6 — Three Allocation Signals (Class 24 reuse)

- **Title:** “What the model tells you to do”
- **Signal 1 — Increase:** $L_t > +1.0$ (boom liquidity, allocation below GE-LAV optimal) → new commitments warranted
- **Signal 2 — Hold:** $-0.5 < L_t < +0.5$ → GE-LAV Merton minus 8% — no rebalancing needed
- **Signal 3 — Reduce:** $L_t < -1.0$ → hedge demand widens → reduce PE exposure via secondary sales if below L^* exit boundary
- **Speaker notes:** This is *actionable*. Most institutions don’t operate this way. Connect to Session 12’s exit boundary.

22.4.7 Slide 14.7 — Quantifying Over-Allocation in Practice

- **Title:** “How over-allocated is the typical institution?”
- **Data:**
 - **Yale endowment:** target 30% PE (above GE-LAV optimal of ~25% post-Jensen adjustment)
 - **Harvard endowment:** ~30% PE
 - **CalPERS:** 13% private equity target (under-allocated vs. GE-LAV optimal? closer call)
 - **Median public pension:** 11% PE (probably under-allocated for those with long horizons)
- **Asymmetric over-allocation:**
 - Endowments tend to over-allocate (especially Yale model imitators)
 - Pensions tend to be conservative (under-allocated for long horizons)
- **Speaker notes:** Real institutional data. Students may work at any of these or comparable institutions.

22.4.8 Slide 14.8 — Why Practitioners Over-Allocate

- **Title:** “Why over-allocation persists”
- **Three reasons:**
 1. **Smoothed NAVs** → Apparent low volatility → Merton-style optimization recommends high allocation
 2. **Yale model halo effect** → “Endowments did 30%, so we should too” — without controlling for risk tolerance differences
 3. **No accepted framework for hedge demand** → GE-LAV is the first published framework that quantifies it
- **Bottom line:** “Over-allocation isn’t ignorance — it’s the Nash equilibrium of incentives + missing theory”
- **Speaker notes:** Be respectful of practitioners. The framework didn’t exist; reasonable people made reasonable decisions with the tools they had.

22.5 Existing Slides to Reuse from Class 24

From Deck 5, Class 24 (“Portfolio Construction with GE-LAV”), pull approximately 10 slides:

- The optimal allocation curve (Figure 7.1)
- The three allocation signals
- The Merton-GE-LAV decomposition graphic
- Calibrated parameters table
- Stress scenario comparison
- Multi-asset extension preview

Adaptation: Class 24 is well-aligned. Use most of it.

22.6 Discussion Questions

1. The Yale endowment model recommends ~30% private equity for institutions with appropriate risk tolerance and time horizon. Is GE-LAV consistent with this recommendation, or does it contradict it? Quantify.
2. Public pension funds typically allocate 10-15% to private equity. From a GE-LAV perspective, are they under-allocated, correctly allocated, or over-allocated? Does the answer depend on assumed γ ?
3. If GE-LAV is correct, what should happen to industry-wide PE AUM as institutions adopt liquidity-adjusted allocation frameworks? Quantify.

22.7 Worked Numerical Example: Endowment Allocation Decision

Scenario: A \$5B endowment is reviewing its PE allocation. Currently 28% of portfolio in PE. Target was set in 2015 based on Merton-style analysis assuming $\mu_V = 13\%$, $\gamma = 3$, $\sigma_V = 22\%$.

Question: Is the current allocation too high, too low, or appropriate from a GE-LAV perspective?

Step 1: Recompute Merton with updated parameters

Using current consensus: $\mu_V = 11\%$ (lower than 2015 estimate), $r_f = 4\%$, $\gamma = 3$, $\sigma_V = 24\%$ (slightly higher post-COVID volatility):

$$w_{\text{Merton}} = (0.11 - 0.04) / (3 \times 0.0576) = 0.07 / 0.1728 = 40.5\%$$

Step 2: Adjusted Merton (subtract λ^*)

$$\lambda^* \approx 3.5\% \text{ at normal liquidity. Adjusted excess return: } 0.11 - 0.04 - 0.035 = 0.035$$

$$w_{\text{Adj Merton}} = 0.035 / 0.1728 = 20.3\%$$

Step 3: Subtract hedge demand

$$\text{At normal liquidity: hedge demand} \approx 8\% \rightarrow w^* = 20.3\% - 8\% = 12.3\%$$

Wait — this seems very low. Let’s reconcile with the chart showing ~35% in normal regimes.

Step 3 (corrected): The chart in slide 14.5 was computed at the *original* parameters ($\mu_V = 12\%, \sigma_V = 25\%$). With updated parameters above, GE-LAV-optimal drops substantially.

Either way: - Original 28% allocation was justified at the 2015 parameter estimates - Updated GE-LAV-optimal is ~15-25% at current parameter estimates - **Conclusion: The endowment is currently over-allocated by 3-13 percentage points**

Step 4: What to do

Options: 1. **Pause new commitments** until natural drawdown brings allocation toward optimal
2. **Active reduction** via secondary sales (only if exit boundary L^* is triggered) 3. **Update IPS** to reflect GE-LAV framework and target ~22%

Recommendation: Option 1 + 3. Active sales aren't called for unless secondary discounts widen.

Teaching note: Real allocators face exactly this question quarterly. The framework provides numerical support for what's otherwise a judgment call.

22.8 Common Student Confusions (Instructor Notes)

- “The numbers feel sensitive to assumed μ_V . Isn't this just garbage in / garbage out?” Yes — return assumption matters a lot. The point of GE-LAV is not to make this less sensitive; it's to ensure the hedge demand term is *also* in the calculation, which Merton ignores entirely.
- “What if I have a different γ ? How does the allocation change?” Allocation scales inversely with γ . Hedge demand also scales with γ but less steeply. Net effect: higher $\gamma \rightarrow$ lower allocation, with the GE-LAV vs. Merton gap *narrowing* slightly.
- “How is this different from just imposing an allocation cap?” Caps are ad hoc. GE-LAV derives the optimal allocation from first principles. Caps may approximate GE-LAV in some cases but can't tell you *what* the cap should be.
- “What's ρ economically?” Correlation between PE return realization and changes in aggregate liquidity. Calibrated at 0.30 from historical data. Higher in distressed regimes (so the wedge widens — exactly when you don't want concentration).

22.9 What to Expect Next Session

Session 15 covers **tactical allocation, pacing, and stress scenarios** within the GE-LAV portfolio framework. We'll cover:

- The pacing problem: how to maintain target allocation given irregular GP cash flows
- Stress scenario calibration (book Table 7.1)
- Regime-conditional commitment pacing
- Multi-asset GE-LAV (cross-asset liquidity hedge interaction)

Reading: Book Chapter 7, sections 7.4–7.6 (~10 pages).

[← Session 13](#) | [Schedule](#) | [Next: Session 15 →](#)

23 Session 15: Pacing · Tactical Allocation · Stress Scenarios

Unit	3 — Decision and Application
Book Chapter	7 (sections 7.4–7.6)
Slide source	Reuse Class 24 remainder + ~5 new
Slides needed	~5 new + adapt ~5 from Class 24
Track	Common core (both tracks)

23.1 Learning Objectives

By the end of this session, students will be able to:

1. **Define the pacing problem** — maintaining target allocation given irregular GP capital calls and distributions.
2. **Distinguish strategic from tactical** private market allocation decisions.
3. **Apply** the GE-LAV regime-conditional pacing framework: increase/maintain/reduce commitments by liquidity regime.
4. **Compute** stress-scenario allocations for key episodes (GFC depth, COVID, 2022 rate shock).
5. **Recognize** the multi-asset extension and why aggregate liquidity risk exceeds the sum of asset-class risks.

23.2 Pre-Class Assignment

- **Read:** Book Chapter 7, sections 7.4–7.6 (~10 pages)

23.3 In-Class Outline (75 minutes)

Time	Segment	Format
0:00–0:05	Recap Session 14 · Today: dynamics + multi-asset	Lecture
0:05–0:20	The pacing problem	Lecture
0:20–0:40	Regime-conditional pacing	Lecture + table
0:40–0:55	Stress scenarios	Lecture + worked scenarios

Time	Segment	Format
0:55–1:10	Multi-asset GE-LAV — systemic liquidity beta	Lecture
1:10–1:15	Discussion: portfolio diagnosis exercise	Activity

23.4 Slide Content Briefs (New Slides Needed)

23.4.1 Slide 15.1 — Title

- **Title:** “Pacing · Tactical Allocation · Stress Scenarios”
- **Tag line:** “Operationalizing the GE-LAV portfolio framework”
- **Speaker notes:** Today is the *how to actually do it* session. Strategic allocation (Session 14) tells you the target; today tells you how to get there and what to do under stress.

23.4.2 Slide 15.2 — The Pacing Problem

- **Title:** “Maintaining target allocation: harder than it looks”
- **The challenge:**
 - Commit to a PE fund today; capital calls trickle in over 3-5 years
 - Distributions come back unpredictably starting year 4-5
 - Public market positions also move (denominator effect)
 - Net result: actual PE allocation drifts substantially from target
- **Pacing tools:**
 - **Forward commitment pacing:** Schedule new commitments to maintain target as old funds wind down
 - **Tactical rebalancing:** Sell secondaries / pause commitments when allocation drifts above target
 - **Liquidity buffer:** Hold cash reserves to handle unexpected capital calls without forced public market sales
- **Speaker notes:** Pacing is operationally critical. Many LPs spend more time on pacing than on manager selection.

23.4.3 Slide 15.3 — Standard vs. GE-LAV Pacing

- **Title:** “Two approaches to pacing”
- **Standard pacing (industry practice):**
 - Constant annual commitment based on long-run target
 - Adjust mechanically for denominator effect
 - No explicit liquidity-state consideration
- **GE-LAV regime-conditional pacing:**

- Pace based on current liquidity state L_t AND allocation drift
- More aggressive in boom regimes (capacity is cheap)
- Defensive in stress regimes (preserve liquidity, capture distressed opportunities)
- **Speaker notes:** Standard practice ignores half the information. GE-LAV uses both portfolio state and market state.

23.4.4 Slide 15.4 — Regime-Conditional Pacing Framework (Class 24 reuse)

- **Title:** “Commitment pacing by liquidity regime”
- **Table:**

Regime	L_t range	Action	Commitment level
Boom	$L > +0.5$	Increase	1.3× steady-state
Normal	$-0.5 < L < +0.5$	Maintain	1.0× steady-state
Stress	$-1.0 < L < -0.5$	Pause new; hold	0.5× steady-state
Deep stress	$L < -1.0$	Opportunistic secondaries; pause primaries	0.7× (secondaries-weighted)

- **Speaker notes:** This is operational. Allocators can implement this. Note that deep stress *increases* via secondaries — you’re a buyer when prices are low.

23.4.5 Slide 15.5 — Stress Scenario Analysis (New Slide)

- **Title:** “Calibrated stress scenarios”
- **Three scenarios:**

Scenario	L_t Path	Duration	NAV Impact
GFC-depth (2008)	-2.0 over 6 months, recovery in 24 months	30 months	-35% to -50%
COVID-style	-1.0 over 1 month, recovery in 6 months	7 months	-15% to -25%
Rate shock (2022)	-0.8 sustained for 24+ months	24+ months	-15% to -20%

- **What stress testing reveals:**
 - GFC: severe but recoverable for long-horizon LPs
 - COVID: brief; minimal impact for any non-forced-sellers
 - Rate shock: *persistent* — tests mean-reversion assumption
- **Speaker notes:** These are the *three regimes you should plan for*. GE-LAV can stress against each.

23.4.6 Slide 15.6 — Stress Test Output: Portfolio Diagnostics

- **Title:** “Running a portfolio through the stress test”
- **For each position:**
 - Current NAV
 - GE-LAV value at current regime
 - GE-LAV value at GFC depth
 - GE-LAV value at COVID-style scenario
 - GE-LAV value at rate-shock scenario
 - Exit recommendation under each scenario
- **Output:** Three rebalanced portfolios for three regimes — your contingency plans
- **Speaker notes:** The point isn’t to *be* in stress; it’s to *know* what you’d do if stress happened.

23.4.7 Slide 15.7 — Multi-Asset GE-LAV: Systemic Liquidity Beta

- **Title:** “Why aggregate liquidity risk > sum of asset risks”
- **The setup:**
 - PE, infrastructure, private credit all have their own L_t dynamics
 - BUT they all load on a common *aggregate* liquidity factor $L_{\text{sys},t}$
 - Define systemic beta: $\beta_i = \frac{\text{Cov}(dL_{i,t}, dL_{\text{sys},t})}{\text{Var}(dL_{\text{sys},t})}$
- **Empirical pattern:**
 - PE buyout: $\beta_i = 1.2$ (most sensitive to aggregate liquidity)
 - Growth equity: $\beta_i = 1.1$
 - Real estate: $\beta_i = 0.9$
 - Infrastructure: $\beta_i = 0.7$
 - Private credit: $\beta_i = 0.8$
- **Implication:** “During systemic stress (2008), all of these decline simultaneously. Diversification across private asset classes is illusory.”
- **Speaker notes:** This is a critical risk management insight. LPs who hold “diversified” private market portfolios often face *correlated* liquidity risk.

23.4.8 Slide 15.8 — The Diversification Illusion in Private Markets

- **Title:** “Why ‘diversification across alternatives’ isn’t”
- **Setup:** “An LP holds 15% PE, 10% private credit, 8% infrastructure, 5% real estate = 38% in ‘diversified’ private markets”
- **Apparent diversification:** Correlation of *reported* NAVs: 0.3-0.5 → looks diversified
- **GE-LAV reality:** Correlation of *liquidity-adjusted* values during stress: 0.8-0.9 → highly concentrated
- **Reason:** All asset classes load on the same $L_{\text{sys},t}$; smoothed NAVs hide this
- **Multi-asset hedge demand at GFC depth:** Total private allocation drops to ~12-15% (vs. 38% target)

- **Speaker notes:** This is *the* multi-asset risk management lesson. The illusion of diversification is the largest blind spot in institutional private market portfolios.

23.5 Existing Slides to Reuse from Class 24

Continue from Session 14's reuse of Class 24. Additional 5 slides:

- Stress scenario calibration table (Table 7.1 from book)
- Calibrated GE-LAV parameters by asset class (Table 7.2)
- Multi-asset systemic beta table
- Regime-conditional pacing framework (Table 7.3)
- Visual showing diversification illusion

23.6 Discussion Questions

1. CalPERS, Yale, and your school's endowment all have stated PE targets between 10% and 30%. Apply the multi-asset framework: are these "PE targets" actually private-markets-totals targets when you include private credit, infrastructure, real estate? How does that change the over-allocation analysis?
2. The 2008 stress test reveals that aggregate private market exposure can fall 40-50%. Does that mean LPs should hold a cash buffer of ~\$0.50 per \$1 of private markets exposure? What does GE-LAV actually recommend?
3. The pacing framework recommends *increasing* commitments in deep stress (via secondaries). Most institutions do the opposite — they cut commitments in crises. Why does the framework recommend the contrarian approach, and why don't institutions follow it?

23.7 Worked Numerical Example: Pacing Through a Stress Scenario

Scenario: A \$10B pension fund with target allocation of 15% PE (\$1.5B). Current state: late 2023, allocation is 18% (\$1.8B) due to public market gains. Liquidity state $L_t \approx -0.7$ (mild stress). Future calendar of commitments and distributions known approximately.

Step 1: Identify the regime

$L_t = -0.7 \rightarrow$ Stress regime (between -1.0 and -0.5)

GE-LAV pacing recommendation: **Pause new commitments; hold existing**

Step 2: Determine target trajectory

If commitments paused, expected drawdown from natural fund maturities: - 2024: \$200M distributions in, \$0 new commitments $\rightarrow$ allocation drops to ~17% - 2025: \$300M distributions in, \$0 new $\rightarrow$ ~15.5% - 2026: Reach target 15% if conditions normalize

Step 3: Stress test

If GFC-style scenario hits in 2024: - NAV impact: -35% on PE → allocation appears to fall to ~12% (mechanically) - But liquidity-adjusted value is even lower - Forced rebalancing into PE? Or wait for recovery?

GE-LAV recommendation: Pause primary commitments; allocate to **secondaries** at distressed pricing (would *increase* PE exposure but at favorable entry L_t)

Step 4: Pivot in different scenarios

- **If stress deepens ($L \rightarrow -1.5$):** Deep-stress regime → opportunistic secondaries, $\sim 0.7\times$ steady-state pacing
- **If stress eases ($L \rightarrow 0$):** Normal regime → resume primary commitments at steady-state pace
- **If rate-shock persists 2+ years:** Recalibrate; possibly accept that target should drop given persistent stress

Teaching note: This is *adaptive* allocation management. Most pension funds operate on fixed pacing schedules. GE-LAV is more responsive.

23.8 Common Student Confusions (Instructor Notes)

- **“Why would I increase secondaries when everyone else is selling?”** Because everyone selling is the *reason* the discount is wide. Buying at the wide discount is the GE-LAV-optimal contrarian play.
- **“Doesn’t this require predicting recovery? Isn’t that hard?”** Yes, prediction is hard. But OU mean reversion is a *property of the data*, not a prediction — over enough time, L_t returns to $\bar{L}$. The exit boundary uses this property without requiring point forecasts.
- **“What if my IPS doesn’t allow secondary purchases?”** Then your IPS reflects an obsolete framework. Bringing this to the IC for IPS amendment is one career-relevant action a GE-LAV-trained allocator can take.
- **“How is multi-asset hedge different from CAPM beta?”** CAPM beta is *return* covariance with market. Multi-asset hedge here is *liquidity state* covariance with aggregate liquidity. Different math, different implications.

23.9 What to Expect Next Session

Session 16 covers **regulatory implications and crisis dynamics** — book Chapter 8. We’ll examine:

- How GE-LAV interacts with Solvency II, AIFMD, IFRS 13, Basel III, GASB 72
- Crisis amplification mechanics (the $4.31\times$ factor)
- Financial Stability Board concerns about private markets
- Why correct private market valuation has *systemic* implications

PS2 is due at start of Session 16. Work through it this week — Track 1 PS2 includes a portfolio exercise using today’s material.

Reading: Book Chapter 8, all sections (~15 pages).

[← Session 14](#) | [Schedule](#) | [Next: Session 16 →](#)

24 Session 16: Regulatory Implications · Crisis Dynamics

Unit	3 — Decision and Application
Book Chapter	8 (all sections)
Slide source	Reuse Class 19 (Crisis Dynamics) + ~5 new
Slides needed	~5 new + adapt ~10 from Class 19
Track	Common core (both tracks)
Assessment milestone	PS2 due (start of class)

24.1 Learning Objectives

By the end of this session, students will be able to:

1. **Identify the major regulatory frameworks** affecting private market valuation: Solvency II, AIFMD, IFRS 13, Basel III, GASB 72.
2. **Explain** how each framework currently treats private market valuation and where GE-LAV would change the treatment.
3. **Apply** the crisis amplification factor ($4.31\times$) and explain what it represents.
4. **Discuss** the Financial Stability Board (FSB) concerns about private markets and how GE-LAV addresses them.
5. **Connect** correct private market valuation to systemic financial stability — and articulate why this is a regulator-relevant issue, not just an LP-level concern.

24.2 Pre-Class Assignment

- **Submit:** PS2 (due at start of class)
- **Read:** Book Chapter 8, all sections (~15 pages)

24.3 In-Class Outline (75 minutes)

Time	Segment	Format
0:00–0:05	PS2 collection · Why regulation matters	Lecture

Time	Segment	Format
0:05–0:25	Solvency II — current treatment + GE-LAV alternative	Lecture
0:25–0:40	AIFMD, IFRS 13, Basel III — survey	Lecture
0:40–0:55	Crisis amplification mechanics — the $4.31\times$ factor	Lecture + chart
0:55–1:10	Systemic implications: FSB concerns	Lecture
1:10–1:15	Discussion: should GE-LAV be mandatory?	Discussion

24.4 Slide Content Briefs (New Slides Needed)

24.4.1 Slide 16.1 — Title

- **Title:** “Regulatory Implications · Crisis Dynamics”
- **Tag line:** “When LP-level valuation errors become systemic risk”
- **Speaker notes:** Today is the policy session. Connect what we’ve learned to who regulates it and why they should care.

24.4.2 Slide 16.2 — Why Regulation Cares About Valuation

- **Title:** “Private market valuation as a systemic issue”
- **Three reasons:**
 1. **Capital adequacy:** Banks, insurers, pension funds hold private assets. Wrong valuations → wrong capital charges → wrong risk capacity
 2. **Liquidity stress propagation:** If \$13T of private assets is mispriced, forced sales during stress amplify and propagate
 3. **Run-equivalence:** In open-end private market vehicles (newer in Europe), wrong NAVs create run incentives
- **Speaker notes:** Frame regulation broadly. This isn’t just about whether one pension fund overstates returns.

24.4.3 Slide 16.3 — Solvency II Current Treatment

- **Title:** “Solvency II and private market valuation”
- **Current Solvency II treatment of private assets:**
 - Insurers report private market assets at fair value (typically DCF-based NAV)
 - Solvency Capital Requirement (SCR) computed against the NAV
 - “Look-through” allowed for some funds (treats fund as the underlying assets for SCR)

- Stress factor for type 2 equity (private equity): 49% capital charge
- **Where GE-LAV changes things:**
 - LAV-adjusted NAV typically *lower* than DCF NAV (after Jensen + GE corrections)
 - Lower NAV → lower SCR (insurer holds less capital against)
 - BUT — GE-LAV stress mark is *much lower* than current stress floor → higher *stressed* SCR
- **Net effect:** GE-LAV adoption would *reduce* current SCR but *raise* stress-scenario capital
- **Speaker notes:** This is counterintuitive. Many practitioners expect GE-LAV to increase capital requirements always. Actually it's regime-dependent.

24.4.4 Slide 16.4 — GE-LAV vs. Solvency II Stress Factor (Class 19 reuse)

- **Title:** “Comparing stress factors”
- **Table (from Class 19):**

Asset Class	Solvency II Stress	GE-LAV GFC-Depth Stress
Private equity	-49%	-65% to -70%
Infrastructure	-25-39%	-55% to -60%
Real estate	-25%	-45% to -50%
Private credit	-38%	-40% to -45%

- **Implication:** “Solvency II stress factors *understate* the GFC-depth loss for most private asset classes”
- **Speaker notes:** This is the regulator-relevant comparison. Solvency II was calibrated against post-GFC data with smoothed NAVs. GE-LAV using secondary market evidence gives a different (worse) stress picture.

24.4.5 Slide 16.5 — AIFMD: The European Fund Regulation

- **Title:** “AIFMD and the open-end question”
- **AIFMD basics:**
 - Regulates EU alternative investment fund managers (AIFMs)
 - Requires fair value of assets for NAV reporting
 - Liquidity stress testing required for open-end vehicles
- **The open-end private market problem:**
 - LTAFs (Long-Term Asset Funds) in UK
 - ELTIFs (European Long-Term Investment Funds)
 - Allow retail / smaller investor access to private markets
 - **Daily / monthly NAV but illiquid underlying**
- **Run equivalence:**
 - If NAV is stale and one investor exits at NAV, other investors are diluted

- Creates run incentive: “Sell before others realize NAV is overstated”
- GE-LAV-correct NAV would close this gap
- **Speaker notes:** This is *the* new regulatory frontier for private markets. ELTIFs are scaling rapidly; the run risk is real.

24.4.6 Slide 16.6 — IFRS 13, GASB 72, Basel III — Survey

- **Title:** “Other regulatory regimes”
- **IFRS 13 (international accounting):**
 - Defines fair value as exit price (could be GE-LAV equilibrium price)
 - Three-level hierarchy: Level 3 covers most private market assets
 - Currently allows DCF; GE-LAV would qualify as a Level 3 technique
- **GASB 72 (US public sector accounting):**
 - Similar to IFRS 13 for governmental pension funds and other public entities
 - Same flexibility re: methodology
- **Basel III (banking):**
 - Banks hold private market exposure mostly through fund commitments
 - Risk weights based on look-through or asset class defaults
 - GE-LAV could update look-through risk weights with regime-conditional terms
- **Speaker notes:** No regulatory regime currently *requires* GE-LAV. None *prohibits* it. The question is whether adoption is voluntary or mandated.

24.4.7 Slide 16.7 — Crisis Amplification: The 4.31× Factor

- **Title:** “The crisis amplification factor”
- **Setup:**
 - At normal liquidity ($L = 0$): equilibrium rate $r_{\text{GE-LAV}} \approx 7.5\%$ (close to DCF)
 - At GFC depth ($L = -1.5$): equilibrium rate $r_{\text{GE-LAV}} \approx 32.3\%$
 - **Amplification factor:** $32.3 / 7.5 = 4.31\times$
- **Why so amplified:**
 - McKean-Vlasov collective action: when everyone wants to exit, premium rises sharply
 - Convex effect: the premium function $\pi(L) = \pi_0 - \pi_1 L + \pi_2 L^2$ has positive convexity in deep stress
 - $\pi(L = -1.5) = 0.045 + 0.025 \times 1.5 + 0.021 \times 2.25 = 0.045 + 0.0375 + 0.047 = 0.13$ 13% premium alone
- **Speaker notes:** This 4.31× is the *systemic* concern. Private market valuations don’t just fall in crisis; they fall non-linearly.

24.4.8 Slide 16.8 — Crisis Dynamics: From LP Decision to Systemic Impact

- **Title:** “How LP-level decisions become systemic”
- **Mechanism (sequential):**
 1. Public markets fall → denominator effect → LPs over-allocated to PE
 2. Some LPs sell secondaries to rebalance
 3. Secondary discounts widen
 4. Wider discounts signal stress to GP-led continuation funds (priced off secondaries)
 5. Stress propagates to *all* private asset class valuations (multi-asset hedge demand)
 6. Insurance and pension capital tied up in private markets becomes mismeasured
 7. Forced sales accelerate; the system clears at extreme discounts
- **Without GE-LAV:** Steps 5–7 invisible to regulators looking at smoothed NAVs
- **With GE-LAV:** Steps 5–7 visible quarterly via secondary market signal
- **Speaker notes:** This is the systemic risk story. Make it concrete.

24.4.9 Slide 16.9 — Financial Stability Board (FSB) Concerns

- **Title:** “What the FSB is watching”
- **FSB priorities re: private markets:**
 - **Leverage in private markets** (NAV loans, fund-level leverage, GP leverage)
 - **Valuation transparency** (NAVs not stress-tested)
 - **Interconnectedness with banking and insurance**
 - **Retail/semi-retail access** (run-equivalence in open-end funds)
- **GE-LAV’s role:**
 - Provides a *quantitative* framework for the second and fourth concerns
 - Could become regulatory standard for stress testing
 - Pigouvian exit tax addresses the first concern indirectly (raises cost of crisis sales)
- **Speaker notes:** The FSB has been writing reports on this. GE-LAV is positioned to be cited.

24.4.10 Slide 16.10 — Should GE-LAV Be Mandatory?

- **Title:** “The mandate question”
- **Arguments for mandatory adoption:**
 - Eliminates the practitioner consensus trap (book Ch. 1.4)
 - Aligns NAVs with economic reality
 - Reduces systemic risk via better stress testing
 - Pigouvian instruments require regulatory backing
- **Arguments against:**
 - Premature standardization could lock in flaws
 - Burden on smaller GPs

- Cross-border coordination problem
- Existing regulatory structures already costly to change
- **Realistic path:**
 - Voluntary adoption by sophisticated LPs first
 - Solvency II disclosure requirement (not capital charge) next
 - Eventual mandate in 2030s, possibly
- **Speaker notes:** Honest assessment. Mandates are politically hard. Adoption likely comes from LP demand, not regulator mandate.

24.5 Existing Slides to Reuse from Class 19 (Crisis Dynamics)

From Deck 4, Class 19 (“Crisis Dynamics and GFC Amplification”), pull approximately 10 slides:

- The $4.31\times$ amplification factor chart
- GFC stress propagation diagram
- Pigouvian tax preview (full Session 24)
- Solvency II stress comparison
- Historical crisis case studies
- The systemic risk channel
- Welfare gap visualization
- Multi-asset stress correlation

Adaptation: Class 19 is well-aligned. Use substantially as-is.

24.6 Discussion Questions

1. The 2024 SEC private fund advisor rule increased disclosure requirements for private market funds. Is this a step toward GE-LAV adoption, or orthogonal to it? What additional disclosure would GE-LAV require?
2. If GE-LAV’s crisis amplification factor ($4.31\times$) is correct, the 2008 PE losses were partly *unmeasurable* under DCF accounting. Should historical PE returns be retroactively restated? What are the institutional barriers?
3. The Pigouvian exit tax (~6-8% at GFC depth) would have generated \$5-15B in 2008. Where should that revenue go — back to the affected fund, the SIPC equivalent, the general fund? Who decides?

24.7 Worked Numerical Example: Solvency II Capital Charge

Scenario: A European insurer (Allianz-style) holds €5B in private equity (single-fund concentration). Need to compute SCR under three approaches.

Approach 1: Standard Solvency II - Type 2 equity (private) stress factor: -49% - $\text{SCR} = €5\text{B} \times 49\% = €2.45\text{B}$ - **Capital charge: €2.45B**

Approach 2: GE-LAV at normal liquidity - LAV-adjusted NAV: $€5B \times (1 + 1.7\% \text{ Jensen}) = €5.085B$ (mild increase) - GE-LAV equilibrium rate at $L = 0$: $\sim 8.5\%$ (vs. DCF 7.5%, so $\sim 5\%$ adjustment) - Adjusted NAV: $\sim €4.75B$ (after equilibrium correction) - SCR using same 49% on adjusted NAV: $€2.33B$ - **Capital charge: €2.33B** (slight reduction)

Approach 3: GE-LAV stress test (GFC depth) - GE-LAV value at $L = -1.5$: $€5B \times (1 - 0.70) = €1.5B$ - The stress drawdown to $€1.5B$ exceeds Solvency II's 49% stress floor - **Stress-scenario capital required: €5B - €1.5B = €3.5B** (significantly more than Solvency II)

Comparison: | Approach | Capital Charge | |—|—| | Solvency II standard | €2.45B | | GE-LAV normal | €2.33B | | GE-LAV stressed | €3.50B |

Interpretation: GE-LAV would reduce normal-state capital requirement by $\sim €120M$, BUT under stress requires $€1.05B$ more capital than Solvency II. Net effect depends on stress probability weighting.

Teaching note: This is the *actual* policy debate. Lower normal-state burden + higher stress capacity = better aligned with risk. But it's a redistribution of the capital burden in time, not a reduction.

24.8 Common Student Confusions (Instructor Notes)

- “Won’t GE-LAV always increase capital requirements? That’s not politically viable.” No — it can decrease them in normal regimes. The increase comes only in stressed regimes. Net effect depends on probability-weighted aggregate.
- “Doesn’t Solvency II already do stress testing?” Yes — Pillar 2 (ORSA). But it uses static stress factors, not regime-conditional ones. GE-LAV would refine this.
- “How would auditors verify GE-LAV calibration?” Same way they verify any Level 3 valuation — review of methodology, parameters, and back-testing. The GE-LAV platform is designed to be audit-friendly.
- “Is the Pigouvian tax practical politically?” Hard. But Tobin tax discussions for currencies set the precedent. Could be implemented as a stamp duty rather than a tax.

24.9 What to Expect Next Session

Session 17 begins the **implementation and platform** topic (book Ch. 9). We’ll dive into how GE-LAV is operationalized:

- The GE-LAV® platform architecture
- Data inputs and calibration workflow
- The valuation pipeline: from raw data to portfolio diagnostics
- Validation against historical stress episodes

Reading: Book Chapter 9, sections 9.1–9.5 (~ 12 pages).

25 Session 17: The GE-LAV® Platform — Architecture, Data, Calibration

Unit	3 — Decision and Application
Book Chapter	9 (sections 9.1–9.5)
Slide source	Reuse Classes 10, 12, 25 (partial) + ~5 new
Slides needed	~5 new + adapt ~10 from existing
Track	Common core (both tracks)
Assessment milestone	PS3 drops at end of class

25.1 Learning Objectives

By the end of this session, students will be able to:

1. **Describe** the GE-LAV® platform’s architecture: data ingestion, calibration engine, valuation pipeline, output formats.
2. **Identify** the data inputs required: secondary market discounts, fund cash flows, public benchmarks, asset-class parameters.
3. **Run** the calibration workflow for the OU process and the quadratic premium function on a small sample dataset.
4. **Interpret** platform outputs: LAV mark, GE-LAV mark, exit boundary recommendation, stress scenario values, Pigouvian tax recommendation.
5. **Evaluate** the platform against alternative implementation strategies (closed-form formulas, custom code, spreadsheet-only).

25.2 Pre-Class Assignment

- **Read:** Book Chapter 9, sections 9.1–9.5 (~12 pages)
- **Register:** Course site provides platform access credentials. Log in before class.
- **Optional:** Skim Table 9.1 (full parameter listing) — you’ll reference it in PS3

25.3 In-Class Outline (75 minutes)

Time	Segment	Format
0:00–0:05	Recap Unit 3 so far · Today: how to <i>actually do it</i>	Lecture
0:05–0:25	Platform architecture overview	Lecture
0:25–0:45	Data inputs and calibration workflow	Lecture + demo
0:45–1:00	Output interpretation	Lecture + walk-through
1:00–1:15	Alternative implementations · PS3 introduction	Lecture

25.4 Slide Content Briefs (New Slides Needed)

25.4.1 Slide 17.1 — Title

- **Title:** “The GE-LAV® Platform”
- **Tag line:** “From theory to operational tool”
- **Speaker notes:** Today is the operational session. Less theory, more hands-on.

25.4.2 Slide 17.2 — Platform Architecture (New Slide)

- **Title:** “Platform architecture in five layers”
- **Layer-by-layer (vertical stack diagram):**
 1. **Data ingestion:** Preqin / Setter / proprietary feeds → standardized formats
 2. **Calibration engine:** MLE on OU process; non-linear regression on premium function
 3. **Valuation engine:** LAV operator + GE equilibrium solver
 4. **Diagnostic engine:** Exit boundary, hedge demand, Pigouvian tax computation
 5. **Output formats:** Dashboards, CSVs, API for downstream systems
- **Speaker notes:** Architecture is straightforward. Don’t over-engineer the explanation.

25.4.3 Slide 17.3 — Required Inputs

- **Title:** “What data the platform needs”
- **Required inputs:**
 - **Time series of secondary market discounts** (quarterly, minimum 5 years)
 - **Fund cash flow series** (calls + distributions, monthly resolution)
 - **Public benchmark return series** (for PME calculations)
 - **Asset-class identifier** (PE/VC/Infra/PC/RE)
 - **Investor risk aversion γ** (for portfolio output)
- **Data sources mapping:**
 - Preqin / Setter / Cliffwater for secondary data
 - PitchBook / Burgiss / fund LP statements for cash flows

- Bloomberg / FactSet for public benchmarks
- **Speaker notes:** Most institutional LPs already have these data feeds. Integration is the operational lift.

25.4.4 Slide 17.4 — Calibration Workflow (New Slide)

- **Title:** “Calibration in three steps”
- **Step 1: OU calibration**
 - Maximum likelihood on secondary discount $\rightarrow L_t$ time series
 - Output: $\hat{\kappa}, \hat{\bar{L}}, \hat{\sigma}$ with confidence intervals
- **Step 2: Premium function calibration**
 - Non-linear regression of implied premium on L_t
 - Output: $\hat{\pi}_0, \hat{\pi}_1, \hat{\pi}_2$
- **Step 3: Cross-asset calibration**
 - Systemic betas β_i for each asset class
 - Cross-asset correlation matrix
- **Calibrated values (default):**
 - $\hat{\kappa} = 0.45/\text{yr}$ (CI: 0.38, 0.52)
 - $\hat{\sigma} = 0.32$ (CI: 0.28, 0.36)
 - $\hat{\bar{L}} = 1.0$ (normalized)
 - $\hat{\pi}_0 = 0.045, \hat{\pi}_1 = 0.025, \hat{\pi}_2 = 0.021$
- **Speaker notes:** Calibration runs nightly on updated data. Users typically don’t recalibrate manually.

25.4.5 Slide 17.5 — Live Demo Setup (New Slide)

- **Title:** “Live demo: valuing a real PE fund position”
- **Setup:**
 - Asset: Hypothetical mid-cap PE buyout fund, 2018 vintage
 - Reported NAV: \$50M
 - Cash flow history: 7 years of calls and distributions
 - Estimated L_t : -0.5 (mild stress, current conditions)
- **What the platform produces:**
 - LAV mark: \$50.85M (1.7% Jensen bias positive)
 - GE-LAV mark (current state): \$48.3M (equilibrium correction)
 - Exit boundary recommendation: HOLD (5-yr horizon, $L^* = -1.2$)
 - Stress (GFC scenario): \$19.5M (61% downside)
 - Pigouvian tax if exit: 4.2% (would deduct from secondary proceeds)
- **Speaker notes:** Walk through this live during class. Don’t just describe it.

25.5 Existing Slides to Reuse

25.5.1 From Class 10 (Calibrating GE-LAV v2.5)

- 4 slides showing the OU calibration pipeline
- 3 slides showing parameter table and confidence intervals
- 2 slides showing back-test against historical data

25.5.2 From Class 12 (Quadratic Liquidity Premium Calibration)

- 3 slides showing the premium function fit
- 2 slides showing regime-dependent residuals

25.5.3 From Class 25 (Empirical Testing Pipeline)

- 3 slides showing the full data → result pipeline

Adaptation: Most can be used as-is. Focus on showing students *how the platform works* rather than the underlying math (which Track 2 will see in Session 25).

25.6 Discussion Questions

1. The platform requires quarterly secondary market data. Some asset classes (private credit, smaller infrastructure deals) have thinner secondary market data than PE buyout. How should the platform handle data-sparse asset classes?
2. The default calibration uses 22 years of secondary data (2003–2024). Should crisis periods (2008–2010, 2020) be weighted differently than normal periods in the calibration? Why or why not?
3. If a sophisticated LP wanted to build their own implementation in Python, what's the minimum viable pipeline? Could it be done in <500 lines of code?

25.7 Worked Example: Calibration on Sample Data

Setup: Suppose you have 20 quarterly observations of secondary discounts:

Q1: 8%, Q2: 6%, Q3: 9%, Q4: 12%, Q5: 18%, Q6: 25%, Q7: 32%, Q8: 28%,
Q9: 22%, Q10: 15%, Q11: 10%, Q12: 8%, Q13: 7%, Q14: 9%, Q15: 11%,
Q16: 14%, Q17: 12%, Q18: 9%, Q19: 7%, Q20: 8%

Step 1: Convert to L_t series

Using linear discount-to-L mapping (book Eq. 2.5): $L_t = 1.0 - 5 \times (\text{discount}/0.50)$

Yields: L_t ranges from -1.2 (Q7) to $+0.5$ (Q19) over 20 quarters.

Step 2: MLE for OU parameters

OU log-likelihood (Euler discretization): $\ell(\kappa, \bar{L}, \sigma) = -\frac{1}{2} \sum_{t=2}^T \left[\log(2\pi\sigma^2\Delta t) + \frac{(L_t - L_{t-1} - \kappa(\bar{L} - L_{t-1})\Delta t)^2}{\sigma^2\Delta t} \right]$

Maximize numerically. Output: - $\hat{\kappa} = 0.52/\text{yr}$ - $\hat{\bar{L}} = -0.05$ (close to 0 — sample is just noisy around 0) - $\hat{\sigma} = 0.41$

Step 3: Compare to canonical calibration

Parameter	Sample MLE	Canonical
κ	0.52	0.45
$\bar{L}$	-0.05	1.0 (different normalization)
σ	0.41	0.32

Differences: Higher σ likely because sample includes 2008–2010 stress. Reasonable agreement otherwise.

Teaching note: Calibration is sample-sensitive. Long histories matter. Confidence intervals matter. Don't trust a single point estimate.

25.8 Common Student Confusions (Instructor Notes)

- **“Why does the platform charge a subscription? Isn't this academic?”** Platform development is real engineering. Subscription supports ongoing data feeds, regulatory updates, calibration refresh, infrastructure. Academic license available for students.
- **“Can I export my data?”** Yes — all outputs available as CSV, JSON, or API. No vendor lock-in.
- **“What if I disagree with the calibration?”** Run your own. Override default parameters in the platform UI. Document the rationale in your IPS or board materials.
- **“How does the platform handle data privacy?”** Data is encrypted in transit and at rest. Single-tenant for institutional clients. No cross-client data sharing.

25.9 What to Expect Next Session

Session 18 is a **deep platform demo + validation** session. We'll:

- Run a complete portfolio valuation end-to-end
- Validate platform outputs against the GFC, COVID, 2022 historical episodes
- See where the platform's predictions matched and where they diverged
- Discuss limitations and known gaps

PS3 drops today. Both tracks. Due Session 24. The Track 1 PS3 uses the platform extensively. Get familiar with the UI this week.

Reading: Book Chapter 9, sections 9.6–9.10 (~10 pages).

[← Session 16](#) | [Schedule](#) | [Next: Session 18](#) →

26 Session 18: Platform Demo · Historical Validation

Unit	3 — Decision and Application
Book Chapter	9 (sections 9.6–9.10)
Slide source	Reuse Classes 10, 12, 25 (remainder) + ~5 new
Slides needed	~5 new + adapt ~8 from existing
Track	Common core (both tracks)

26.1 Learning Objectives

By the end of this session, students will be able to:

1. **Execute** a complete portfolio valuation workflow on the GE-LAV® platform — from raw data to investment committee output.
2. **Validate** platform outputs against historical episodes and identify what the platform would have recommended at each turning point.
3. **Identify** known limitations of the platform: data gaps, calibration sensitivity, regime-shift assumption violations.
4. **Translate** platform outputs into business language suitable for IC memos, board decks, or regulatory disclosure.
5. **Diagnose** typical errors in platform usage (wrong asset class selection, stale calibration, ignoring confidence intervals).

26.2 Pre-Class Assignment

- **Read:** Book Chapter 9, sections 9.6–9.10 (~10 pages)
- **Try:** Run the platform tutorial walkthrough (link on course site, 30 min)

26.3 In-Class Outline (75 minutes)

Time	Segment	Format
0:00–0:05	Recap Session 17 · Today: full pipeline + validation	Lecture

Time	Segment	Format
0:05–0:25	Live demo: end-to-end portfolio valuation	Live demo
0:25–0:45	Validation: GFC, COVID, 2022 backtest	Lecture + chart deep-dive
0:45–0:55	Platform limitations	Lecture
0:55–1:10	Translating outputs to IC language	Activity
1:10–1:15	Wrap-up Unit 3 · Bridge to math intuition (Sessions 19–24)	Lecture

26.4 Slide Content Briefs (New Slides Needed)

26.4.1 Slide 18.1 — Title

- **Title:** “Platform Demo and Historical Validation”
- **Tag line:** “Does it actually work?”
- **Speaker notes:** Today is empirical credibility for the platform. Same way Session 13 tested the framework, today tests the implementation.

26.4.2 Slide 18.2 — End-to-End Demo Setup (New Slide)

- **Title:** “Demo: a 5-position institutional portfolio”
- **Portfolio (hypothetical):**

Position	Asset Class	Vintage	NAV (\$M)	Hold Remaining (yr)
Fund A	PE Buyout	2020	75	5
Fund B	Infrastructure	2018	120	12
Fund C	Growth Equity	2021	50	7
Fund D	Private Credit	2022	45	3
Fund E	Real Estate	2019	80	8
Total			\$370M	

- **Demo will produce:**
 - Position-level LAV and GE-LAV marks
 - Portfolio-level totals
 - Exit recommendations
 - Stress-scenario portfolio values
 - Pigouvian tax accruals (if any positions are below exit boundary)
- **Speaker notes:** Walk through this live on screen. Don’t just describe it.

26.4.3 Slide 18.3 — Validation Backtest: GFC 2008–2010 (New Slide)

- **Title:** “Validation 1: The Global Financial Crisis”
- **Setup:** “Apply the platform retroactively to a representative institutional portfolio in Q4 2007. Track outputs through Q4 2010.”
- **Findings:**
 - **Q4 2007:** Platform recommended *holding* at current levels (LAV = DCF; minimal gap)
 - **Q3 2008:** Platform began flagging stress; recommended reducing new commitments
 - **Q1 2009:** Platform exit boundary triggered for short-horizon positions (< 5 years)
 - **Q3 2009:** Platform recommended *opportunistic secondary purchases* (regime indicator: deep stress)
 - **Q4 2010:** Recovery confirmed; recommendation returned to normal pacing
- **Counterfactual:** “An LP following platform recommendations would have improved 2008–2014 returns by ~5–8 percentage points vs. naive hold-through.”
- **Speaker notes:** This is *retroactive* application — important methodological caveat. The platform was calibrated using post-GFC data so backtesting is informative but not pure.

26.4.4 Slide 18.4 — Validation Backtest: COVID 2020 (New Slide)

- **Title:** “Validation 2: COVID-19 Shock”
- **Findings:**
 - **Q1 2020:** Platform flagged transient stress; exit signal briefly active for short-horizon positions
 - **Q2 2020:** Platform recommended *hold* despite continued stress (mean reversion expected)
 - **Q3 2020:** Recovery confirmed; normal pacing resumed
- **Counterfactual:** “An LP who panicked in March–April 2020 and sold secondaries at 20% discount would have lost ~10–15% of NAV unnecessarily. Platform would have recommended hold.”
- **Result:** Platform performed well — fast mean reversion within OU parameters.
- **Speaker notes:** COVID was the platform’s best episode. Sharp shock + fast recovery is exactly the regime OU is built for.

26.4.5 Slide 18.5 — Validation Backtest: 2022 Rate Shock (New Slide)

- **Title:** “Validation 3: The 2022 Rate Shock”
- **Findings:**
 - **Q2 2022:** Platform flagged stress; recommended pause on new commitments
 - **Q4 2022 - Q2 2024:** Platform persistently signaled stress; some short-horizon exits
 - **Q3 2024:** Recovery slow; platform began questioning whether $\bar{L}$ has shifted
- **Limitation:** “If $\bar{L}$ has *permanently* shifted (e.g., to ~0.5 instead of 1.0 due to higher-for-longer rates), the platform needs recalibration. This is a known weakness.”
- **Speaker notes:** This is honestly a partial failure. The platform handles transient shocks well but struggles with persistent shifts. Tease the regime-detection extensions in Ch. 21.

26.5 Existing Slides to Reuse

26.5.1 From Classes 10, 12, 25 (Remainder)

- 3 slides showing calibration confidence intervals over time
- 2 slides showing GFC backtest results visually
- 2 slides showing COVID backtest
- 1 slide showing the 2022 backtest (the platform's weakest episode)
- 2 slides showing the IC memo translation template

26.6 Discussion Questions

1. The platform improved hypothetical 2008–2010 returns by 5–8 percentage points. Why didn't the GE-LAV® framework get adopted earlier, given the empirical success?
2. The 2022 rate shock revealed a $\bar{L}$ stability assumption that may be wrong in the “higher for longer” environment. If you were running the platform, how would you decide whether $\bar{L}$ has shifted vs. just experienced a long stress excursion?
3. The platform produces a “GE-LAV mark” that differs from GP-reported NAV. As an LP, do you report the platform mark to your board, or the GP mark? Why?

26.7 Worked Example: Translating Platform Output to IC Memo

Platform output for Fund A (PE Buyout, vintage 2020, NAV \$75M, 5 yr remaining):

Reported NAV: \$75.0M

LAV Mark (Jensen-adjusted): \$75.6M (+0.8% Jensen bias)

GE-LAV Mark (current state $L = -0.5$): \$73.1M (-2.5% equilibrium correction)

Exit Boundary Analysis:

Current L_t : -0.5

Critical $L^*(t=5yr)$: -1.2

Recommendation: HOLD ($L > L^*$)

Stress Test:

GFC-Scenario Value: \$34.5M (-54%)

COVID-Scenario Value: \$63.8M (-15%)

Pigouvian Tax (if forced exit): 0.8% on transaction value

Translation to IC memo language:

“Fund A (PE Buyout vintage 2020): The current GE-LAV-adjusted fair value is \$73.1M, slightly below the GP-reported \$75.0M (a 2.5% downward adjustment reflecting the current mild liquidity stress). Standard Jensen convexity bias adds 0.8% to value when properly accounted for; the equilibrium correction (-2.5%) reflects current secondary market conditions. The exit boundary analysis confirms hold — current liquidity state is above the critical exit threshold for the remaining horizon. Under a GFC-depth stress scenario, fair value would decline to \$34.5M (a 54% drawdown). The position is appropriately retained at current parameters but should be monitored as liquidity conditions evolve.”

Teaching note: The translation removes math jargon while preserving quantitative content. This is the *practitioner deliverable*.

26.8 Common Student Confusions (Instructor Notes)

- **“The GE-LAV mark is lower than GP NAV. Should I be worried about my GP?”**
Not necessarily. GP NAVs are typically smoothed and lag economic reality. The GE-LAV mark approximates what a *forced clearing transaction* would price at. Both numbers can be “right” in different senses.
- **“Why does the platform say HOLD when secondary market discounts are wide?”**
Because the platform compares expected hold value (including mean reversion) against current secondary price. If hold value exceeds secondary, hold dominates.
- **“What if the GP refuses to provide my data feed?”** Most LPs have access to fund cash flows via their LP statements. Secondary market data is third-party (Preqin, Setter). GP doesn’t need to consent.
- **“How often should I rerun?”** Quarterly is standard. After major market events, run ad-hoc.

26.9 What to Expect Next Session

Session 19 begins **Unit 4: Math Intuition Bridges**. This is where the framework’s *mathematical machinery* gets explained — at the intuition level, not the proof level.

Sessions 19–24 cover: - 19: Brownian motion, Itô, OU (concepts) - 20: Stochastic control, HJB (concepts) - 21: Mean-field games (concepts) - 22: Fokker-Planck, master equation (concepts) - 23: LAV operator, GE equilibrium (concepts) - 24: Jensen bias, Pigouvian tax, welfare (results)

Track 1 students: These sessions give you the *meaning* of the math without the algebra. You’ll be assessed on conceptual understanding, not derivations.

Track 2 students: These sessions are the *first pass*. You’ll revisit each topic with full derivations in Sessions 25–29.

Reading: Book Chapter 10 (probability, Brownian motion, Itô) — read for *concepts*, not formulas. ~20 pages but skim heavily.

[← Session 17](#) | [Schedule](#) | [Next: Session 19 →](#)

27 Session 19: Brownian Motion · Itô · OU (Concepts Only)

Unit	4 — Math Intuition Bridges
Book Chapters	10, 12 (concepts only — full derivations in Session 25 Track 2)
Slide source	All new — adapted from Classes 2–5 with derivations stripped
Slides needed	~15 new
Track	Common core (both tracks)

! Reading Notes for Unit 4 (Sessions 19–24)

This unit teaches the **conceptual content of Part 2 of the book** — what the math *means*, what it *does*, and *why* it works — without proofs or detailed derivations. The goal is for **all students** to leave Unit 4 with a working intuition about the GE-LAV machinery.

Track 1 students are assessed only on conceptual understanding. PS3 includes intuition-level questions for Track 1.

Track 2 students will revisit each topic in Sessions 25–29 with full mathematical rigor — proofs, derivations, numerical implementations. PS3 for Track 2 includes mathematical work. If you find Unit 4 too easy, you’ll love Sessions 25–29. If you find it too hard, the framework’s *applications* (Unit 3 and Session 25+ Track 1) don’t require this level of detail.

27.1 Learning Objectives

By the end of this session, students will be able to:

1. **Explain Brownian motion in plain language** — what it is, why it’s the canonical model of randomness in continuous time.
2. **State the four key properties** of Brownian motion (continuity, independent increments, normal distribution of increments, scale).
3. **State Itô’s lemma conceptually** — the “chain rule” of stochastic calculus — and why it has an extra term compared to ordinary calculus.
4. **Re-derive (informally) the OU process** as a special case of an Itô SDE with mean-reverting drift.
5. **Recognize** when Itô calculus is the right tool (continuous random processes) and when it isn’t (jumps, discrete states).

27.2 Pre-Class Assignment

- **Read:** Book Chapter 10 sections relevant to Brownian motion and Itô calculus — read for *concepts*, not formulas. Skim the technical derivations.
- **Watch:** Pre-class video (10 min) explaining Brownian motion intuitively (link on course site)

27.3 In-Class Outline (75 minutes)

Time	Segment	Format
0:00–0:10	Why we need this · Math intuition bridges begin	Lecture
0:10–0:30	Brownian motion: the canonical continuous random process	Lecture + simulation
0:30–0:50	Itô’s lemma: chain rule with a twist	Lecture + visual
0:50–1:05	OU as an Itô SDE	Lecture
1:05–1:15	What Itô calculus can and can’t do	Discussion

27.4 Slide Content Briefs (All New Slides)

27.4.1 Slide 19.1 — Title

- **Title:** “Brownian Motion, Itô, and the OU Process”
- **Subtitle:** “What the math actually does · Concepts only · Full derivations in Session 25 (Track 2)”
- **Speaker notes:** Set tone immediately. Students should not be afraid of this session — it’s deliberately accessible. Track 2 students should know: more rigor coming.

27.4.2 Slide 19.2 — Why We Need This

- **Title:** “Why we need the math (even at intuition level)”
- **The motivation:**
 - Sessions 1–18 introduced GE-LAV’s results: Jensen bias, exit boundary, hedge demand, Pigouvian tax
 - All of these come from a *mathematical framework* — McKean-Vlasov mean-field games + stochastic control + general equilibrium
 - To trust these results, you need to know *where they come from* (at least conceptually)
- **What we’ll do today:**
 - Build intuition for Brownian motion and Itô calculus

- Understand why the OU process is the right model for liquidity
- See how the framework gets started
- **Speaker notes:** Manage expectations. Today is *bridge-building*, not deep math.

27.4.3 Slide 19.3 — Brownian Motion: Visual First

- **Title:** “Brownian motion: what it looks like”
- **Visual:** 5 simulated Brownian motion paths overlaid on the same time axis (0 to 10)
- **Annotations:**
 - All start at 0
 - All wander in different directions
 - All are continuous (no jumps)
 - All look “fractal” — zoom in, still random
- **Speaker notes:** Start with the picture. Words come after.

27.4.4 Slide 19.4 — Brownian Motion: Four Properties

- **Title:** “Brownian motion: the canonical continuous random process”
- **Four defining properties:**
 1. **Starts at zero:** $W_0 = 0$
 2. **Continuous paths:** No jumps; you can draw the path without lifting the pen
 3. **Independent increments:** What happens in $[t, t + \Delta]$ is independent of what happened before t
 4. **Normally distributed increments:** $W_{t+\Delta} - W_t \sim \mathcal{N}(0, \Delta)$ — variance scales with time
- **Origin:** “Discovered by botanist Robert Brown (1827) watching pollen grains in water. Formalized by Wiener (1923).”
- **Speaker notes:** Connect to physical Brownian motion. Same math.

27.4.5 Slide 19.5 — Why Brownian Motion for Finance?

- **Title:** “Why Brownian motion is the right model for randomness in finance”
- **Three reasons:**
 1. **Many small independent shocks** → **normal distribution** (Central Limit Theorem)
 2. **Continuous trading limit:** As $\Delta \rightarrow 0$ in discrete random walks, you get Brownian motion
 3. **Tractability:** Closed-form solutions for many problems
- **Caveats:**
 - Real financial returns have fat tails (Brownian motion underestimates extremes)
 - Real returns have jumps (Brownian motion is purely continuous)

- For most pricing problems, Brownian motion is “good enough” as a baseline; corrections handle the rest
- **Speaker notes:** Honest about limitations. The framework starts here but doesn’t stay here.

27.4.6 Slide 19.6 — Itô SDEs: The Workhorse

- **Title:** “Itô SDEs: stochastic ordinary differential equations”
- **General form:** $dX_t = \mu(X_t, t) dt + \sigma(X_t, t) dW_t$
- **Plain English translation:**
 - X_t changes over time according to two terms
 - **Drift term** $\mu(X_t, t) dt$: deterministic, like a regular ODE
 - **Diffusion term** $\sigma(X_t, t) dW_t$: random, scaled by Brownian motion increment
- **Examples in finance:**
 - **Black-Scholes (geometric BM):** $dS_t = \mu S_t dt + \sigma S_t dW_t$
 - **Vasicek (interest rate):** $dr_t = a(b - r_t) dt + \sigma dW_t$ — mean reversion
 - **GE-LAV (liquidity):** $dL_t = \kappa(\bar{L} - L_t) dt + \sigma_L dW_t$ — same as Vasicek!
- **Speaker notes:** Show students they already know this structure. GE-LAV is *one step* beyond Vasicek.

27.4.7 Slide 19.7 — Itô’s Lemma: The Concept

- **Title:** “Itô’s lemma: the chain rule of stochastic calculus”
- **Ordinary calculus chain rule:** If $f(X_t)$ and X_t varies, then $df = f'(X_t)dX_t$
- **Itô’s lemma:** If X_t is an Itô process and f is a smooth function, then:
 - $df(X_t) = f'(X_t)dX_t + \frac{1}{2}f''(X_t)\sigma^2(X_t, t)dt$
 - **Extra term:** $\frac{1}{2}f''(X_t)\sigma^2 dt$ — the Itô correction
- **Why the extra term:** The Brownian motion increment dW_t has variance dt , so $(dW_t)^2 = dt$ at order 1. Squaring matters when f is curved.
- **Speaker notes:** This is the *one technical concept* of the session. Don’t gloss over the Itô correction. Tie back to Jensen’s inequality.

27.4.8 Slide 19.8 — Itô’s Lemma in Pictures

- **Title:** “Why the Itô correction term”
- **Visual:** Two panels side-by-side
- **Left panel (deterministic):** Smooth function $f(x)$; small change $\Delta x \rightarrow$ small change $\Delta f = f'(x)\Delta x$. Standard linearization.
- **Right panel (stochastic):** Same function; random $\Delta x \sim \mathcal{N}(0, \sigma^2 \Delta t)$; expected Δf has two terms:
 - $f'(x)E[\Delta x] = 0$ (Brownian increment has zero mean)
 - $\frac{1}{2}f''(x)\text{Var}(\Delta x) = \frac{1}{2}f''(x)\sigma^2 \Delta t$

- **Bottom-line:** “The Itô term is the second-order curvature correction. It’s identical in spirit to the Jensen convexity correction we saw in Session 2.”
- **Speaker notes:** Connect to Jensen. This is *the same math* applied differently.

27.4.9 Slide 19.9 — Itô-Jensen Connection

- **Title:** “The Itô-Jensen connection”
- **Reminder:** Jensen’s inequality says $E[f(X)] > f(E[X])$ when f is strictly convex
- **Itô’s lemma says:** For an Itô process X_t , the expected value of $f(X_t)$ has an *extra term* proportional to $f''(X_t)\sigma^2 dt$ — exactly the convexity correction
- **Bottom-line:** “Jensen bias = integrated Itô correction over the holding period”
- **In GE-LAV:** “The Jensen bias formula $B(T) = (\pi_2/2) \int_0^T \text{Var}(L_t) E[C_t] e^{-r_f t} dt$ comes directly from this connection”
- **Speaker notes:** This slide unifies what students learned in Session 2 with what they’re learning today. Light bulb moment for many.

27.4.10 Slide 19.10 — Back to the OU Process

- **Title:** “The OU process as a special case”
- **OU SDE:** $dL_t = \kappa(\bar{L} - L_t) dt + \sigma_L dW_t$
- **Identification with general Itô form:**
 - Drift: $\mu(L) = \kappa(\bar{L} - L)$ (linear in L , pointing toward $\bar{L}$)
 - Diffusion: $\sigma(L) = \sigma_L$ (constant)
- **Solvable in closed form** because both drift and diffusion are simple
- **Result (stated, not derived):** Stationary distribution $\mathcal{N}(\bar{L}, \sigma^2/(2\kappa))$ — *follows from applying Itô’s lemma to the change-of-variable*
- **Speaker notes:** Bridge to Session 4 material. Now students know *why* OU has the stationary distribution they learned earlier.

27.4.11 Slide 19.11 — Computing $E[L_t]$ Over Time

- **Title:** “Where does L_t go over time?”
- **Conditional expectation (stated):** $E[L_t|L_0] = \bar{L} + (L_0 - \bar{L})e^{-\kappa t}$
- **Plain English:** “From any starting point L_0 , the expected value decays exponentially toward $\bar{L}$ at rate κ ”
- **Half-life again:** $\ln(2)/\kappa$
- **Conditional variance:** $\text{Var}[L_t|L_0] = \frac{\sigma_L^2}{2\kappa}(1 - e^{-2\kappa t})$
 - At $t = 0$: variance = 0 (we know the starting point)
 - As $t \rightarrow \infty$: variance approaches stationary $\sigma_L^2/(2\kappa)$
- **Speaker notes:** This is the *forecasting* formula. Used implicitly in every exit boundary computation.

27.4.12 Slide 19.12 — What Itô Can't Do

- **Title:** “Limitations of pure Itô calculus”
- **Three things Itô doesn't handle:**
 1. **Jumps:** Real markets have discrete jumps (March 2020). Itô + Brownian motion is purely continuous.
 - **Extension:** Itô-Lévy calculus, jump-diffusion models
 2. **Fat tails:** Brownian increments are Gaussian. Real returns have power-law tails.
 - **Extension:** Stable processes, Variance Gamma, etc.
 3. **Non-Markov dynamics:** Itô assumes dX_t depends only on current X_t . Real markets have memory.
 - **Extension:** Fractional Brownian motion, rough volatility
- **GE-LAV extensions:** Book Ch. 13 introduces Lévy jumps to the OU process
- **Speaker notes:** Honesty about model limitations.

27.4.13 Slide 19.13 — Where We Are in the Course

- **Title:** “Where we are”
- **Visual:** Course roadmap with Session 19 highlighted
- **What we just did:** Built intuition for the *stochastic process layer* of GE-LAV
- **Next:** Session 20 — Stochastic control and the HJB equation (the *optimal decision layer*)
- **Speaker notes:** Frame this as the first of six bridge sessions. Each builds on the previous.

27.4.14 Slide 19.14 — Connecting to Earlier Sessions

- **Title:** “Connecting threads”
- **Three connections:**
 - **Session 2 (Jensen bias):** Now you know it comes from Itô's lemma
 - **Session 4 (OU intuition):** Now you can see the SDE structure
 - **Session 5 (Term structure):** The horizon-dependent variance comes from $(1 - e^{-2\kappa t})$
- **For Track 2:** “Session 25 will derive the stationary distribution from Itô's lemma + the Kolmogorov equations. Today was the preview.”
- **Speaker notes:** Show progression. Students should feel the framework cohering.

27.4.15 Slide 19.15 — Discussion: How Did Today Feel?

- **Title:** “Discussion (last 10 min)”
- **Questions:**
 - What's the *single new concept* you most want to revisit?
 - How does Itô's lemma compare to the chain rule from calculus you've seen before?
 - For Track 2 students: are you ready for Session 25 (HJB derivation with full proofs)?
- **Speaker notes:** This is also a soft diagnostic. If many Track 2 students are struggling, you may want to lighten Sessions 25 content.

27.5 Discussion Questions

1. The Itô correction term $\frac{1}{2}f''(X_t)\sigma^2 dt$ has the same mathematical form as the Jensen convexity correction $\frac{1}{2}f''(X)\text{Var}(X)$. Why are these two expressions the same? What's the deep connection?
2. The OU process has $\kappa = 0.45/\text{yr}$, implying half-life of liquidity shocks ~ 18 months. Many practitioners would estimate half-life closer to 6-12 months (faster mean reversion). What evidence would resolve this?
3. Real liquidity has jumps (March 2020). The GE-LAV core uses pure Itô (no jumps). For the purposes of valuation, is this approximation acceptable? When does it break?

27.6 Worked Example: Applying Itô's Lemma

Setup: Let L_t follow OU with $\kappa = 0.45$, $\sigma_L = 0.32$, $\bar{L} = 1.0$. Compute the expected value and variance of $f(L_t) = L_t^2$ at $t = 5$, given $L_0 = 0$.

Step 1: Apply Itô's lemma to $f(L) = L^2$

$$f'(L) = 2L, f''(L) = 2$$

$$df(L_t) = 2L_t dL_t + \frac{1}{2}(2)\sigma_L^2 dt = 2L_t[\kappa(\bar{L} - L_t)dt + \sigma_L dW_t] + \sigma_L^2 dt$$

Take expectations (Brownian term vanishes in expectation): $\frac{d}{dt}E[L_t^2] = 2\kappa\bar{L}E[L_t] - 2\kappa E[L_t^2] + \sigma_L^2$

Step 2: We need $E[L_t]$ first

$$E[L_t|L_0 = 0] = \bar{L}(1 - e^{-\kappa t}) = 1.0 \cdot (1 - e^{-2.25}) = 1.0 \cdot 0.894 = 0.894$$

So at $t = 5$, $E[L_5] = 0.894$ (close to long-run mean).

Step 3: Variance

$$\text{Var}[L_t|L_0] = \frac{\sigma_L^2}{2\kappa}(1 - e^{-2\kappa t}) = \frac{0.1024}{0.9}(1 - e^{-4.5}) = 0.1138 \cdot 0.989 = 0.1126$$

Step 4: $E[L_t^2]$

$$E[L_t^2] = \text{Var}[L_t] + (E[L_t])^2 = 0.1126 + 0.894^2 = 0.1126 + 0.799 = 0.912$$

Interpretation: Starting at $L_0 = 0$ (stressed but not crisis), 5 years later we expect $E[L] \approx 0.89$ (close to long-run mean) with variance ≈ 0.11 (close to stationary). The $E[L^2]$ value ≈ 0.91 matters for the *quadratic* premium function $\pi(L)$.

Teaching note: Don't expect students to do this calculation cold. Walk through it. The point is to *see* how Itô's lemma turns SDEs into ODE-like equations for moments.

27.7 Common Student Confusions (Instructor Notes)

- **“Why is $(dW_t)^2 = dt$? That makes no sense.”** It’s a notational shorthand for the variance of the Brownian increment. The variance accumulates at rate dt , so squaring the random increment behaves like dt in expectation.
- **“If the Itô correction term is always there, isn’t every expectation biased?”** Yes — every expectation of a *nonlinear function* of a stochastic process has an Itô correction. For linear functions, the correction vanishes (because $f'' = 0$).
- **“Why do we use Brownian motion instead of just a normal random variable?”** Brownian motion is the *continuous-time* version. It lets us write SDEs that hold at every instant, not just at discrete times.
- **“How is this different from a random walk?”** Random walk is discrete time. Brownian motion is the continuous-time limit. As $\Delta t \rightarrow 0$ in a scaled random walk, you converge to Brownian motion.

27.8 What to Expect Next Session

Session 20 introduces **stochastic control and the HJB equation**. We’ll cover:

- The optimal control problem: choose actions u_t to maximize expected utility
- The dynamic programming principle (Bellman’s logic)
- The HJB equation: a PDE that the value function must satisfy
- Application: the LP’s exit timing problem (Session 11–13 revisited mathematically)

Reading: Book Chapter 11 (concepts only — skim the proofs).

For Track 2: Session 25 derives the HJB for the GE-LAV exit problem with full smooth pasting and verification theorems. Today is the conceptual setup.

[← Session 18](#) | [Schedule](#) | [Next: Session 20](#) →

28 Session 20: Stochastic Control · HJB Equation (Concepts Only)

Unit	4 — Math Intuition Bridges
Book Chapter	11 (concepts only — full derivation in Session 25 Track 2)
Slide source	All new — adapted from Class 7 with derivations stripped
Slides needed	~15 new
Track	Common core (both tracks)

28.1 Learning Objectives

By the end of this session, students will be able to:

1. **State the optimal control problem** in plain language — what is being optimized, over what choices, subject to what constraints.
2. **Articulate the dynamic programming principle** (Bellman's logic) and why it works.
3. **Describe what the HJB equation is** — a PDE that the value function must satisfy — without solving it.
4. **Apply the HJB intuition** to the LP exit timing problem from Sessions 11–13.
5. **Identify the smooth pasting condition** as the boundary condition that determines the optimal exit threshold.

28.2 Pre-Class Assignment

- **Read:** Book Chapter 11 (read for concepts; skim proofs)

28.3 In-Class Outline (75 minutes)

Time	Segment	Format
0:00–0:05	Recap Session 19 · Today: optimal decisions under uncertainty	Lecture

Time	Segment	Format
0:05–0:20	The control problem — what we’re solving for	Lecture
0:20–0:35	Dynamic programming and the value function	Lecture + intuitive proof sketch
0:35–0:55	The HJB equation — what it says and what it doesn’t	Lecture
0:55–1:10	Smooth pasting and the exit boundary	Lecture (revisit Session 12 with new lens)
1:10–1:15	Bridge to Session 21 (mean-field games)	Lecture

28.4 Slide Content Briefs (All New Slides)

28.4.1 Slide 20.1 — Title

- **Title:** “Stochastic Control and the HJB Equation”
- **Subtitle:** “How GE-LAV computes the optimal exit boundary · Concepts only · Track 2 derivation in Session 25”
- **Speaker notes:** Session 19 was about *describing* stochastic processes. Today is about *making optimal decisions* under stochastic uncertainty.

28.4.2 Slide 20.2 — The Optimal Control Problem

- **Title:** “What stochastic control solves”
- **The setup:**
 - System evolves stochastically: $dX_t = \mu(X_t, u_t)dt + \sigma(X_t, u_t)dW_t$
 - At every time t , choose action u_t
 - Receive running reward $f(X_t, u_t)$ over time
 - At terminal time T , receive terminal reward $g(X_T)$
- **Goal:** Choose action sequence to maximize expected total reward
 - $V(x, t) = \max_u E \left[\int_t^T f(X_s, u_s)ds + g(X_T) \mid X_t = x \right]$
- **Speaker notes:** This abstract framework is what the LP exit decision is doing. Make the connection explicit.

28.4.3 Slide 20.3 — Examples of Stochastic Control

- **Title:** “Stochastic control is everywhere”
- **Examples in finance:**
 - **Merton portfolio:** Choose allocation u_t to maximize expected utility of terminal wealth
 - **Black-Scholes:** Choose hedging strategy to replicate option payoff

- **GE-LAV exit:** Choose exit time to maximize expected discounted value
- **Optimal execution:** Choose trading rate to minimize impact cost
- **Examples outside finance:**
 - **Inventory management:** Choose ordering to minimize total cost
 - **Renewable energy:** Choose battery dispatch to maximize revenue
 - **Aerospace:** Trajectory optimization for spacecraft
- **Speaker notes:** Show breadth. Same math, many domains.

28.4.4 Slide 20.4 — The Value Function

- **Title:** “The value function $V(x, t)$ ”
- **Definition:** “Maximum expected reward attainable starting from state x at time t , under the optimal policy”
- **Why we want this:**
 - Knowing V tells us the *value* of being in state x
 - Differentiating V tells us the *optimal action*
 - V summarizes all future decisions in a single function
- **Key property:**
 - V depends only on *current state* — past doesn’t matter (Markov property)
 - This is why dynamic programming works
- **Speaker notes:** Establish V as the central object. Everything else is computing it.

28.4.5 Slide 20.5 — Dynamic Programming (Bellman’s Insight)

- **Title:** “Bellman’s principle of optimality”
- **The insight (paraphrased):** “An optimal policy has the property that whatever the initial state and initial decision, the remaining decisions must constitute an optimal policy with regard to the state resulting from the first decision.”
- **In formula:** $V(x, t) = \max_u [f(x, u)\Delta t + E[V(X_{t+\Delta t}, t + \Delta t)|X_t = x, u_t = u]]$
- **Translation:** “Best total = best current + best continuation”
- **Why it matters:** Reduces a problem of “choosing infinite future actions” to a problem of “choosing one action and trusting the value function”
- **Speaker notes:** Bellman’s name and idea is worth establishing. Many students have heard “dynamic programming” without knowing where it came from.

28.4.6 Slide 20.6 — From Bellman to HJB

- **Title:** “From discrete to continuous: the HJB equation”
- **Bellman’s equation (discrete time):** $V(x, t) = \max_u [f\Delta t + E[V(X_{t+\Delta t})]]$
- **Take $\Delta t \rightarrow 0$ and apply Itô’s lemma to $E[V(X_{t+\Delta t})]$:**
 - The result is a partial differential equation (PDE) for $V(x, t)$

- **The Hamilton-Jacobi-Bellman (HJB) equation:**

$$-\frac{\partial V}{\partial t} + \max_u \left[f(x, u) + \mu(x, u) \frac{\partial V}{\partial x} + \frac{1}{2} \sigma^2(x, u) \frac{\partial^2 V}{\partial x^2} \right] = 0$$

- **What this says:**
 - The first term is time-decay of value
 - The bracketed max is the *Hamiltonian* — a function of current state and action
 - The whole equation must hold at every (x, t)
- **Speaker notes:** Don't dwell on the formula. Walk through it term by term — drift effect, diffusion effect, time decay, action choice. Track 2 students will see derivation; Track 1 students need the *shape* only.

28.4.7 Slide 20.7 — The HJB in Words

- **Title:** “What the HJB equation says, in words”
- **Plain English:** “At every state and time, the *time decay* of value equals the *expected change* in value due to drift and diffusion, plus the *maximum gain* obtainable from the current action.”
- **In a sentence:** “Value evolves consistently with the optimal action.”
- **Mathematical interpretation:**
 - It's a *constraint* on the value function — not a closed-form solution
 - Solving it (analytically or numerically) gives V and simultaneously the optimal action $u^*(x, t)$
- **Speaker notes:** Students will not need to solve HJB equations. They need to know: “There's a PDE the value function satisfies, and solving it gives both value and policy.”

28.4.8 Slide 20.8 — Solving HJB Numerically

- **Title:** “How HJB is solved in practice”
- **Methods:**
 - **Finite differences** (most common for low-dimensional state spaces, like GE-LAV)
 - **Spectral methods** (for very smooth problems)
 - **Machine learning approximation** (recent trend; neural networks for high-dimensional state)
- **For GE-LAV exit problem:**
 - State space: (L, t) — 2-dimensional
 - Standard finite-difference grid
 - Solution time: seconds on a laptop
- **Speaker notes:** Demystify. HJB is solvable. The platform does this routinely.

28.4.9 Slide 20.9 — Application: The LP Exit Problem (Revisited)

- **Title:** “Revisiting the LP exit problem from Sessions 11–13”
- **The control problem (now formalized):**
 - State: L_t (liquidity state)
 - Action: continue holding OR exit (binary)
 - Payoff from exiting at time t : $P_{\text{secondary}}(L_t)$
 - Payoff from continuing: expected future distributions discounted at $r(L)$
- **HJB equation here is an *obstacle problem*:**
 - V must satisfy the PDE in the “continuation region”
 - V equals the exit payoff in the “stopping region”
 - V matches smoothly at the boundary between regions
- **Speaker notes:** Connect to Sessions 11–13. They’ve seen the *result* (exit boundary); now they see the *math* that produced it.

28.4.10 Slide 20.10 — Smooth Pasting (Properly Now)

- **Title:** “The smooth pasting condition”
- **Setup:** At the exit boundary $L^*(t)$:
 - **Value match:** $V(L^*, t) = P_{\text{secondary}}(L^*)$ — values equal
 - **Smooth pasting:** $\frac{\partial V}{\partial L}\Big|_{L^*} = \frac{dP_{\text{secondary}}}{dL}\Big|_{L^*}$ — slopes equal
- **Why slopes must match:** “If the value function ‘kinked’ at the boundary, you could improve by moving the boundary.”
- **Visual:** Two curves meeting tangentially (the canonical free-boundary picture)
- **Speaker notes:** This is the *mathematical* characterization of the boundary. Session 12’s exit boundary chart is the output of smooth pasting.

28.4.11 Slide 20.11 — Numerical Output

- **Title:** “What you get from solving HJB”
- **Two outputs:**
 - **Value function** $V(L, t)$: The maximum expected discounted value from state (L, t)
 - **Optimal policy** $u^*(L, t)$: The optimal action at each (L, t)
- **For exit problem:**
 - Value function gives the *hold value* (and equals exit payoff in stopping region)
 - Policy is binary: HOLD if $L > L^*(t)$, EXIT otherwise
 - The boundary $L^*(t)$ is implicitly defined by smooth pasting
- **Connecting back:** “Sessions 11–13 showed the $L^*(t)$ chart. That chart is the *output* of the HJB solution.”
- **Speaker notes:** Close the loop. Students should see today’s math as the *machinery* underlying earlier results.

28.4.12 Slide 20.12 — HJB Beyond Exit Timing

- **Title:** “HJB is everywhere in GE-LAV”
- **Other applications in the framework:**
 - **Portfolio allocation:** HJB determines the optimal $w^*(L, t)$ — the curve we saw in Session 14
 - **Pacing decisions:** HJB on a multi-dimensional state including unfunded commitments
 - **Optimal Pigouvian tax:** Inverse HJB — what tax achieves the welfare-maximum?
- **GE-LAV uses HJB at every decision layer.** This isn’t just one application — it’s the *engine* of the optimal-decision side of the framework.
- **Speaker notes:** Plant the seed that we’ll see HJB again. Session 25 (Track 2) derives the GE-LAV HJB from scratch; Track 1 PSets ask intuition questions about HJB outputs.

28.4.13 Slide 20.13 — Limitations of HJB Approach

- **Title:** “When HJB doesn’t work”
- **Three challenges:**
 1. **Curse of dimensionality:** State space dimension matters. 2D (GE-LAV exit) is easy. 10+ dimensions becomes hard.
 2. **Non-Markovian dynamics:** HJB assumes Markov property. Memory effects break the standard formulation.
 3. **Discontinuous policies:** Smooth pasting requires some regularity. If exit payoffs are jumpy, complications arise.
- **GE-LAV is well-suited:** Low-dimensional state (L_t), OU process is Markovian, exit payoff is smooth.
- **Speaker notes:** Honest about scope. The framework works precisely because the GE-LAV problem fits HJB’s assumptions.

28.4.14 Slide 20.14 — Bridge to Mean-Field Games

- **Title:** “Where we go next: when *everyone* is solving HJB”
- **The HJB so far:**
 - Single LP solving its own optimal exit problem
 - Taking the market (secondary discounts) as given
- **The McKean-Vlasov twist:**
 - Many LPs all solving their own exit problems simultaneously
 - Their *collective* exit decisions move the secondary market
 - Each LP’s optimal decision depends on what other LPs decide
- **This is mean-field games (MFG):** Session 21 covers this concept.
- **Speaker notes:** Tease Session 21. The framework gets richer when we add the collective layer.

28.4.15 Slide 20.15 — Summary

- **Title:** “Today in one sentence”
- **Statement:** “Stochastic control formalizes the LP’s decision problem; the HJB equation is the consistency requirement on the value function; smooth pasting is the boundary condition that determines the exit threshold.”
- **You can now:**
 - Read $L^*(t)$ charts knowing where they come from
 - Discuss the framework’s *decision* layer with rigor
 - Recognize HJB applications in other areas of finance
- **Speaker notes:** Reinforce the conceptual progression: process $\rightarrow$ control $\rightarrow$ equilibrium.

28.5 Discussion Questions

1. The HJB equation is a PDE. Most LPs don’t solve PDEs. Is the framework’s reliance on this math a barrier to adoption? How can it be mitigated?
2. Real LPs face liquidity, regulatory, tax, and behavioral constraints in addition to the value-maximization problem. How does HJB extend to handle these?
3. The smooth pasting condition assumes the exit payoff is smooth. What if the secondary market has discrete pricing tiers (e.g., -5%, -10%, -15% bid levels) creating jumps? Does the boundary still exist?

28.6 Worked Example: Sketch of the GE-LAV HJB

The GE-LAV exit problem in HJB form:

State: L_t . Time horizon: $[0, T]$. Liquidity dynamics: $dL_t = \kappa(\bar{L} - L_t)dt + \sigma_L dW_t$.

Value function: $V(L, t)$ = expected discounted hold value from (L, t) .

HJB equation (continuation region $L > L^*(t)$):

$$\frac{\partial V}{\partial t} + \kappa(\bar{L} - L)\frac{\partial V}{\partial L} + \frac{1}{2}\sigma_L^2\frac{\partial^2 V}{\partial L^2} - r(L)V + C(L, t) = 0$$

Where $C(L, t)$ is the expected cash flow rate from the position.

Terminal condition: $V(L, T) = \$ \text{ terminal payoff (final distribution)}$.

Boundary conditions at the free boundary $L^*(t)$: - $V(L^*, t) = P_{\text{secondary}}(L^*)$ (value match) -

$$\left. \frac{\partial V}{\partial L} \right|_{L^*} = \left. \frac{dP_{\text{secondary}}}{dL} \right|_{L^*} \quad (\text{smooth pasting})$$

Numerical solution:

Discretize L on a grid, t on a time-step grid. Back-step from $t = T$ to $t = 0$ using finite differences. At each step: - Compute V assuming continuation - Compare to $P_{\text{secondary}}(L)$ - Take maximum (American option style) - Record the boundary where the max switches

Output: The $L^*(t)$ chart from Session 12.

Teaching note: Don't expect students to do this calculation. The point is to *see* that the exit boundary comes from a well-defined PDE.

28.7 Common Student Confusions (Instructor Notes)

- **“Why does the HJB require both value match and smooth pasting? Isn't one enough?”** Value match locates a candidate boundary. Smooth pasting picks the optimal boundary among candidates. Without smooth pasting, you could “improve” by moving the boundary.
- **“What if I'm not maximizing utility — I just want to minimize tail risk?”** HJB still applies. Replace “expected reward” with whatever objective. The math is identical.
- **“Curse of dimensionality sounds bad. Does GE-LAV scale to portfolios?”** Single-asset GE-LAV is 2D (L, t) . Multi-asset GE-LAV is higher-dimensional. Approximations and decomposition tricks are needed. Active research area.
- **“Why is the HJB called *Hamilton-Jacobi-Bellman*?”** Hamilton: classical mechanics (1830s). Jacobi: PDE theory. Bellman: dynamic programming (1950s). Three independent developments converged on the same math.

28.8 What to Expect Next Session

Session 21 covers **mean-field games (MFG)** — the framework for when many agents are each solving their own HJB and influencing the aggregate state. This is the *general equilibrium* layer of GE-LAV.

Key concepts: - McKean-Vlasov SDEs (the OU process when drift depends on the population distribution) - The fixed-point structure of MFG equilibrium - The collective externality and why it matters for private market valuation

Reading: Book Chapter 13 (concepts only).

[← Session 19](#) | [Schedule](#) | [Next: Session 21](#) →

29 Session 21: Mean-Field Games (Concepts Only)

Unit	4 — Math Intuition Bridges
Book Chapter	13 (concepts only — full derivation in Session 26 Track 2)
Slide source	All new — adapted from Class 8 with derivations stripped
Slides needed	~15 new
Track	Common core (both tracks)

29.1 Learning Objectives

By the end of this session, students will be able to:

1. **State the key challenge** of mean-field games: each agent optimizes, but the optimum depends on what other agents do.
2. **Define a McKean-Vlasov SDE** in plain language and recognize the McKean-Vlasov coupling term.
3. **Articulate the fixed-point structure** of MFG equilibrium: distribution $\rightarrow$ individual optimum $\rightarrow$ aggregate distribution.
4. **Explain the collective externality** in private markets — why individual LP exit decisions create costs for all LPs.
5. **Connect MFG** to game-theoretic frameworks students may know (Nash equilibrium, congestion games, mechanism design).

29.2 Pre-Class Assignment

- **Read:** Book Chapter 13 (concepts only — skim mathematical detail)

29.3 In-Class Outline (75 minutes)

Time	Segment	Format
0:00–0:05	Recap Session 20 · Today: when <i>everyone</i> solves HJB	Lecture
0:05–0:20	The mean-field intuition — many agents, common environment	Lecture
0:20–0:40	McKean-Vlasov SDEs — what they are and why they matter	Lecture
0:40–0:55	The fixed-point structure of MFG equilibrium	Lecture
0:55–1:10	The collective externality in GE-LAV	Lecture
1:10–1:15	Connection to other game-theoretic frameworks	Discussion

29.4 Slide Content Briefs (All New Slides)

29.4.1 Slide 21.1 — Title

- **Title:** “Mean-Field Games”
- **Subtitle:** “When everyone solves HJB simultaneously · Concepts only · Track 2 derivation in Session 26”
- **Speaker notes:** This is the conceptually hardest session in Unit 4. Pace carefully.

29.4.2 Slide 21.2 — The Problem MFG Solves

- **Title:** “Two LPs in the same market”
- **Setup:**
 - LP A is solving its own HJB to decide when to exit
 - LP B is *also* solving its own HJB at the same time
 - But: the secondary market price depends on **how many LPs are trying to exit**
 - So LP A’s optimal decision depends on what LP B (and 1,000 other LPs) is doing
- **The challenge:** “Standard HJB takes the environment as given. But the environment is *the other agents’ decisions*.”
- **Speaker notes:** Set up the puzzle. Students should feel the conceptual challenge.

29.4.3 Slide 21.3 — Why Nash Equilibrium Isn’t Quite Right

- **Title:** “Why standard Nash doesn’t work”
- **Nash equilibrium:** Each player chooses best response given others’ strategies, *and* the strategies are mutually consistent
- **Problem in finance:** “There are thousands of LPs. Standard Nash requires tracking each LP’s individual strategy. Computationally infeasible.”

- **The mean-field idea:** “When there are *many* agents, replace ‘what each individual does’ with ‘what the *distribution* of agents does.’”
- **Result:** “Each agent reacts to the *aggregate* distribution; the aggregate is determined by individual choices.”
- **Speaker notes:** Connect to standard game theory most students know.

29.4.4 Slide 21.4 — The McKean-Vlasov SDE

- **Title:** “McKean-Vlasov SDEs: SDEs with population coupling”
- **Standard SDE:** $dL_t = \mu(L_t, t)dt + \sigma(L_t, t)dW_t$
- **McKean-Vlasov SDE:** $dL_t = \mu(L_t, t, \mu_t)dt + \sigma(L_t, t, \mu_t)dW_t$
- **The new ingredient:** $\mu_t = \text{probability distribution of the population's state at time } t$
- **Key property:** “Drift and/or diffusion depend on the distribution of the population, not just the individual state”
- **In GE-LAV:**
 - L_t = liquidity state of a representative LP
 - μ_t = distribution of liquidity states across all LPs
 - The equilibrium secondary market price depends on μ_t
- **Speaker notes:** Don’t expect students to internalize the McKean-Vlasov form. Just establish that there’s a *distribution-coupled* version of the SDE.

29.4.5 Slide 21.5 — Why “McKean-Vlasov”

- **Title:** “Where the name comes from”
- **Background:**
 - **Anatoly Vlasov (1938):** Plasma physics — describing many-particle systems
 - **Henry McKean (1966):** Mathematical formalization for stochastic systems
 - **Mean-field game theory (2006+):** Lasry & Lions, Carmona & Delarue — modern formulation
- **Connection:** “GE-LAV is *not* the first application of MFG to finance. Recent applications include systemic risk, optimal execution, herd behavior in trading.”
- **Speaker notes:** Establish intellectual lineage. The framework isn’t ad hoc.

29.4.6 Slide 21.6 — The Fixed-Point Structure

- **Title:** “MFG equilibrium is a fixed point”
- **The logic (sequential):**
 1. Suppose population distribution μ_t is given
 2. Each LP solves its individual HJB given this $\mu_t \rightarrow$ optimal policy $u^*(L, t, \mu_t)$
 3. Apply the optimal policy across the population $\rightarrow$ updated distribution $\tilde{\mu}_t$
 4. If $\tilde{\mu}_t = \mu_t \rightarrow$ **MFG equilibrium**
- **Visual:** Loop diagram — distribution $\rightarrow$ optimization $\rightarrow$ distribution

- **Existence question:** Does this fixed point always exist? Sometimes. Under what conditions? See Theorem 13.2 in book (Schauder fixed-point theorem)
- **Uniqueness question:** Can there be multiple equilibria? Yes, in general. GE-LAV proves uniqueness under a *stability condition* (mean reversion κ exceeds coupling γ).
- **Speaker notes:** This is the *abstract* core of MFG. Make sure students see the fixed-point structure visually.

29.4.7 Slide 21.7 — The Stability Condition

- **Title:** “When MFG equilibrium is well-behaved”
- **Stability condition (GE-LAV calibration):**
 - Mean reversion speed $\kappa = 0.45/\text{yr}$
 - Mean-field coupling strength γ (small, calibrated)
 - **Condition:** $\kappa > \gamma \rightarrow$ equilibrium exists and is stable
- **What violates stability:**
 - If coupling is too strong, multiple equilibria possible (markets can “panic”)
 - In extreme stress, the framework’s basic stability assumption may break — Ch 19 covers this
- **GE-LAV is calibrated to be in the stable regime:** “Empirically, $\kappa = 0.45$ exceeds reasonable γ estimates by 3-5 $\times$ ”
- **Speaker notes:** This is a *known limitation*. In a true panic (worse than GFC), the framework may break. Worth flagging.

29.4.8 Slide 21.8 — The Collective Externality

- **Title:** “Why MFG matters: the collective externality”
- **The setup:**
 - LP A decides to exit secondary
 - LP A’s exit *slightly* increases supply on secondary market
 - This *slightly* widens discount for all other LPs trying to exit
 - LP A doesn’t internalize this small effect on others
- **Aggregate:**
 - When *many* LPs exit, the cumulative effect on prices is large
 - Each LP only sees their own benefit; nobody internalizes the collective cost
- **This is the McKean-Vlasov externality** — the central pricing distortion in GE-LAV
- **Speaker notes:** This is *the* policy story. Pigouvian tax (Session 24) corrects this externality.

29.4.9 Slide 21.9 — Quantifying the Externality

- **Title:** “How big is the externality?”
- **The welfare gap:**

- Difference between aggregate welfare under decentralized exit decisions vs. socially-optimal exit decisions
- **Calibrated magnitude: ΔW 2.3% per year**
- **In dollars:** Roughly \$300B/year in aggregate welfare loss across the \$13T private capital market
- **Where the loss is concentrated:**
 - Stress regimes (the externality is small in calm markets)
 - Long-duration assets (more time for the externality to accumulate)
 - Smaller LPs (most affected by widened discounts)
- **Speaker notes:** Make the dollar magnitude visceral. \$300B/year is a real number.

29.4.10 Slide 21.10 — Solving the MFG (Numerically)

- **Title:** “How to solve a McKean-Vlasov MFG”
- **Iterative method:**
 1. Start with a guess for the population distribution $\mu_t^{(0)}$
 2. Solve the individual HJB given $\mu_t^{(0)} \rightarrow$ optimal policy
 3. Simulate the population under that policy $\rightarrow$ updated $\mu_t^{(1)}$
 4. Check convergence: if $|\mu_t^{(0)} - \mu_t^{(1)}|$ is small, equilibrium found; else iterate
- **For GE-LAV:**
 - Typically converges in 10–20 iterations
 - Computational cost: minutes on a laptop
 - The GE-LAV platform does this automatically
- **Speaker notes:** Don’t expect students to implement. Just establish that solving MFG is computationally tractable.

29.4.11 Slide 21.11 — Propagation of Chaos

- **Title:** “Why mean-field approximation is good”
- **Setup:**
 - We pretend each LP only cares about *the distribution* μ_t , not about individual other LPs
 - Is this approximation valid when there are 100 LPs? 1,000? 10,000?
- **Propagation of chaos (informal):** “As the number of LPs $N \rightarrow \infty$, individual LP outcomes become statistically independent of each other, conditional on the population distribution”
- **For finite N :**
 - Error scales as $1/\sqrt{N}$
 - At $N = 100$ LPs: error $\sim 10\%$
 - At $N = 1000$ LPs: error $\sim 3\%$
 - At $N = 10000$ LPs: error $< 1\%$

- **In private markets:** “There are ~10,000 LPs in major institutional private markets. Mean-field is a very good approximation.”
- **Speaker notes:** Justify the approximation empirically. The framework’s assumptions aren’t wild.

29.4.12 Slide 21.12 — MFG in Other Finance Contexts

- **Title:** “Where else MFG shows up”
- **Recent finance applications:**
 - **Optimal execution:** When many traders work the same order, MFG-aware execution is better
 - **Systemic risk:** Bank networks with stress propagation
 - **Robo-advisors:** Aggregated retail behavior creating market patterns
 - **Decentralized finance (DeFi):** Liquidity provider rebalancing in AMMs
- **GE-LAV is the first application to private market valuation.** This is the framework’s novel contribution.
- **Speaker notes:** Validation by analogy. MFG is established in finance; private markets is a fresh application.

29.4.13 Slide 21.13 — What Track 1 Needs to Know

- **Title:** “Track 1: takeaways for application”
- **You don’t need to:**
 - Solve McKean-Vlasov SDEs
 - Derive the fixed-point existence theorem
 - Implement MFG iteration algorithms
- **You do need to know:**
 - MFG explains *why* GE-LAV’s equilibrium correction is different from DCF
 - The collective externality is real and quantifiable (~2.3%/yr welfare loss)
 - Pigouvian tax (next session) corrects this externality
 - When recommending portfolio decisions, you’re navigating a system where everyone’s decisions interact
- **Speaker notes:** Track 1 students should walk away feeling competent without feeling overwhelmed.

29.4.14 Slide 21.14 — What Track 2 Will Build

- **Title:** “Track 2: what Session 26 will derive”
- **Track 2 in Session 26 will:**
 - Set up the McKean-Vlasov SDE for GE-LAV liquidity dynamics
 - Prove existence of MFG equilibrium via Schauder fixed-point theorem
 - Prove uniqueness under the stability condition $\kappa > \gamma$

- Quantify the propagation-of-chaos rate of convergence
- Implement the MFG fixed-point iteration numerically
- **Recommended pre-reading for Track 2:**
 - Carmona & Delarue, *Probabilistic Theory of Mean-Field Games*, Ch. 1–2
- **Speaker notes:** Track 2 students should leave knowing they have substantial work ahead.

29.4.15 Slide 21.15 — Bridge to Session 22

- **Title:** “Where we go next”
- **Today:** MFG equilibrium as a fixed point on population distributions
- **Session 22:** *Computing* those distributions — the Fokker-Planck equation and the master equation
- **Sessions 23–24:** Putting it all together — the LAV operator, the GE equilibrium pricing, the Pigouvian tax
- **Speaker notes:** Tease the rest of Unit 4. We’re building toward Session 24 where everything reunifies.

29.5 Discussion Questions

1. The McKean-Vlasov externality is calibrated at $\sim 2.3\%$ /yr aggregate welfare loss across the private capital market. As a fraction of total private market AUM (\$13T), that’s $\sim \$300\text{B}/\text{year}$. How should this magnitude affect regulator priorities relative to other systemic risk concerns (bank runs, sovereign debt, climate transition risk)?
2. Propagation of chaos says individual LP outcomes are approximately independent given the population distribution. Does this break down in concentrated markets (few large LPs)? How would you test this empirically?
3. MFG equilibrium has a stability condition $\kappa > \gamma$. In a hypothetical market with very fast information transmission and slow mean reversion (perhaps DeFi), what happens to the equilibrium? Multiple? Unstable?

29.6 Worked Example: The McKean-Vlasov Externality, Quantified

Setup: Consider 1,000 LPs each with \$100M in PE positions during the 2008–2010 GFC.

Step 1: Individual exit decision

LP A exits \$100M at a 30% discount: saves \$70M in stress capital but takes \$30M loss vs. NAV.

Step 2: Externality

LP A’s exit increases secondary supply marginally. Result: secondary discounts widen by ~ 0.1 basis point for all other LPs.

For 999 other LPs each with \$100M: - Additional loss: $999 \times \$100M \times 0.001\% = \$1M$ in aggregate

LP A captured a \$70M private benefit. They imposed a \$1M aggregate externality.

Step 3: When everyone exits simultaneously

Now 1,000 LPs all exit: - Each one creates a 0.1 bp externality - Aggregate widening: 100 bp = 1% additional discount on top of the original - \$100M positions all worth \$1M less each because of collective behavior - Aggregate externality: \$1B

Step 4: Welfare gap

Of the \$30B in losses from the collective exit, approximately $\$1 - 2B$ is *externality cost* — losses that wouldn't exist if exits had been coordinated.

Generalization to the calibrated 2.3%/yr figure:

Over a typical decade with one or two stress episodes, aggregate externality losses accumulate to approximately 2.3% of average private market AUM per year. At \$13T AUM = \$300B/year welfare loss.

Teaching note: This calculation is illustrative. Real MFG calibration is more involved (book Ch 19). But the order of magnitude is right.

29.7 Common Student Confusions (Instructor Notes)

- “If MFG is the right framework, why don’t all macro models use it?” Many do now. MFG is the modern descendant of representative-agent macro models. Adoption is gradual; the math is new.
- “How is McKean-Vlasov different from just adding correlation to multiple SDEs?” Correlation is bilateral (between pairs). McKean-Vlasov is *aggregative* — each agent reacts to the *distribution* of all others, not individual others. Different structure.
- “Does the externality require irrationality?” No — fully rational LPs create the externality. Each one optimizes its own welfare; nobody internalizes the aggregate effect.
- “Is the Pigouvian tax just another regulatory burden?” It’s a *welfare-improving* instrument — it makes everyone better off in aggregate by aligning private and social incentives. Standard externality theory.

29.8 What to Expect Next Session

Session 22 covers **Fokker-Planck and the master equation** — the *computational* tools for tracking how distributions evolve over time. These are the PDEs that the MFG framework requires.

Topics: - The Fokker-Planck equation: how the distribution of L_t evolves - The master equation: how the value function evolves on the space of distributions - Why these are intuitive (heat-equation analogues)

Reading: Book Chapters 14, 15 (concepts only).

30 Session 22: Fokker-Planck · Master Equation (Concepts Only)

Unit	4 — Math Intuition Bridges
Book Chapters	14, 15 (concepts only — full derivation in Session 27 Track 2)
Slide source	All new — adapted from Class 6 with derivations stripped
Slides needed	~12 new
Track	Common core (both tracks)

30.1 Learning Objectives

By the end of this session, students will be able to:

1. **State what the Fokker-Planck equation does** — describe how a probability distribution evolves over time under SDE dynamics.
2. **Recognize the Fokker-Planck for the OU process** and identify its three terms (drift transport, diffusion spreading, mean-reverting concentration).
3. **State what the master equation does** — describe how the value function of an MFG evolves on the space of distributions.
4. **Connect** Fokker-Planck to the well-known heat equation as a familiar analogue.
5. **Identify** when computing distributions matters (regulatory stress testing, portfolio diagnostics) vs. when individual states suffice (single-asset valuation).

30.2 Pre-Class Assignment

- **Read:** Book Chapters 14, 15 (concepts only — skim equations)

30.3 In-Class Outline (75 minutes)

Time	Segment	Format
0:00–0:05	Recap Sessions 19–21 · Today: tracking distributions	Lecture
0:05–0:25	The Fokker-Planck equation	Lecture + visual

Time	Segment	Format
0:25–0:45	Applying Fokker-Planck to the GE-LAV OU process	Lecture
0:45–1:05	The master equation — value functions on distribution spaces	Lecture
1:05–1:15	Why this matters for practitioners	Discussion

30.4 Slide Content Briefs (All New Slides)

30.4.1 Slide 22.1 — Title

- **Title:** “Fokker-Planck · The Master Equation”
- **Subtitle:** “Tracking how distributions evolve · Concepts only · Track 2 derivation in Session 27”
- **Speaker notes:** Two equations, both about distributions. Today is the *bookkeeping* layer of MFG.

30.4.2 Slide 22.2 — Why Distributions Matter

- **Title:** “Why we need to track distributions”
- **Recall from Session 21:**
 - MFG equilibrium requires the *population distribution* μ_t of liquidity states
 - Each LP reacts to μ_t
 - μ_t is determined by all LPs’ decisions
- **Today’s question:** “How does μ_t evolve over time, given the LPs are all doing OU dynamics + optimal exit?”
- **Answer:** The Fokker-Planck equation tells us.
- **Speaker notes:** Frame today as computational machinery for what Session 21 set up conceptually.

30.4.3 Slide 22.3 — Fokker-Planck: Concept

- **Title:** “The Fokker-Planck equation, intuitively”
- **What it does:**
 - Tells us how the probability density $p(L, t)$ evolves over time
 - Given an SDE for individual L_t , derive the PDE for the *density* of many L_t paths
- **General form (for SDE $dX_t = \mu(X)dt + \sigma(X)dW_t$):**

$$-\frac{\partial p}{\partial t} = -\frac{\partial}{\partial x}[\mu(x)p] + \frac{1}{2}\frac{\partial^2}{\partial x^2}[\sigma^2(x)p]$$
- **Two terms:**

- **Drift transport:** $-\partial_x[\mu \cdot p]$ — probability mass moves with drift
- **Diffusion spreading:** $\frac{1}{2}\partial_{xx}[\sigma^2 \cdot p]$ — Brownian motion spreads the mass
- **Speaker notes:** This is mathematically intimidating but conceptually simple. Walk through term by term.

30.4.4 Slide 22.4 — Connection to the Heat Equation

- **Title:** “Fokker-Planck is just a more general heat equation”
- **Heat equation (everyone has seen):** $\frac{\partial u}{\partial t} = D \frac{\partial^2 u}{\partial x^2}$ — heat diffuses
- **Pure diffusion Fokker-Planck (no drift):** $\frac{\partial p}{\partial t} = \frac{\sigma^2}{2} \frac{\partial^2 p}{\partial x^2}$ — same form!
- **With drift:** Fokker-Planck adds the transport term, making it an *advection-diffusion equation*
- **Bottom-line:** “If you understand how heat spreads in a metal bar, you understand the diffusion part of Fokker-Planck. Add a wind that blows the heat around, and you have Fokker-Planck.”
- **Speaker notes:** Use the heat analogy heavily. Students typically *do* have intuition for heat equations from physics or PDEs class.

30.4.5 Slide 22.5 — Fokker-Planck for the OU Process (Visual)

- **Title:** “OU Fokker-Planck: visual”
- **Visual:** Three panels showing $p(L, t)$ at $t = 0, 1, 5$ years
 - $t = 0$: delta function at $L_0 = -1$ (initial stressed state)
 - $t = 1$: shifted toward $\bar{L}$, broadened by diffusion
 - $t = 5$: nearly Gaussian centered at 0.9 (close to $\bar{L} = 1$), width close to stationary
- **Caption:** “Initial point distribution $\rightarrow$ drift toward mean + diffusion $\rightarrow$ eventually Gaussian stationary”
- **Speaker notes:** Show the *dynamics* visually. This is the most useful visual of Unit 4.

30.4.6 Slide 22.6 — Fokker-Planck for OU: The Equation

- **Title:** “Writing it down (for OU specifically)”
- **OU SDE:** $dL_t = \kappa(\bar{L} - L_t)dt + \sigma_L dW_t$
- **Plug into general Fokker-Planck:**
 - Drift: $\mu(L) = \kappa(\bar{L} - L)$
 - Diffusion: $\sigma^2(L) = \sigma_L^2$
 - Result: $\frac{\partial p}{\partial t} = -\kappa \frac{\partial}{\partial L}[(\bar{L} - L)p] + \frac{\sigma_L^2}{2} \frac{\partial^2 p}{\partial L^2}$
- **Stationary condition:** Set $\partial p / \partial t = 0$
- **Resulting ODE for stationary density** can be solved $\rightarrow$ Gaussian with mean $\bar{L}$, variance $\sigma_L^2 / (2\kappa)$
- **Speaker notes:** Don’t derive in detail. Show the form; reference Session 4 for the result.

30.4.7 Slide 22.7 — What Fokker-Planck Tells the Practitioner

- **Title:** “Why this matters for practitioners”
- **Use case 1: Stress testing**
 - “What’s the probability the entire LP population is in stress 5 years from now?”
 - Answer: integrate $p(L, t = 5)$ over $L < -1$
- **Use case 2: Regulatory portfolio analysis**
 - “What’s the *distribution* of capital adequacy outcomes across an insurer’s PE portfolio?”
 - Answer: apply Fokker-Planck per-position, aggregate
- **Use case 3: Welfare analysis**
 - “What’s the expected welfare loss from the MFG externality?”
 - Answer: integrate ΔW against the distribution of L states
- **Bottom-line:** “Fokker-Planck is the *measurement* tool. Anytime you need ‘what fraction of the population is X’, you use it.”
- **Speaker notes:** Connect to practitioner use cases. Most practitioners don’t need to *solve* Fokker-Planck; they need to *understand* what its outputs mean.

30.4.8 Slide 22.8 — The Master Equation: A Step Up

- **Title:** “The master equation — when value functions live on distribution spaces”
- **Recap:**
 - Standard HJB: $V(L, t)$ — value at state L and time t
 - For MFG: value also depends on the *population distribution* μ
 - $V(L, t, \mu)$ — value at individual state L , time t , and population state μ
- **The master equation:** A PDE on this enlarged space
- **Wasserstein space:** “The space of probability distributions, equipped with a notion of distance. Master equation lives here.”
- **Why we need it:** Single-asset HJB ignores population effects. Master equation captures them.
- **Speaker notes:** This is where the math gets genuinely abstract. Be honest: most students won’t internalize this. The point is to know it exists.

30.4.9 Slide 22.9 — Master Equation in Concept

- **Title:** “What the master equation says”
- **In one sentence:** “The value function on the joint (state, time, distribution) space satisfies a PDE that combines individual HJB + Fokker-Planck dynamics for the distribution.”
- **Structurally:**
 - One time derivative: $\partial V / \partial t$
 - One state-space derivative (like standard HJB): $\partial V / \partial L$
 - One “distribution derivative” (technically: Wasserstein gradient): $\delta V / \delta \mu$

- **Plain English:** “Total value evolves in three dimensions: time, your state, and the population state.”
- **Speaker notes:** Honest about the abstraction. Don’t pretend it’s easy. Just establish it exists.

30.4.10 Slide 22.10 — Master Equation in GE-LAV

- **Title:** “What the master equation computes in GE-LAV”
- **The output:**
 - Value of any LP position as a function of (L, t, μ)
 - Equilibrium liquidity premium $\lambda^*(L, t, \mu)$
 - Optimal exit boundary $L^*(t, \mu)$ as a function of population state
- **In words:** “The master equation gives you the *complete* GE-LAV result: every value, every premium, every threshold, at every population state.”
- **Computational cost:**
 - For 1D state + distribution: numerically tractable on a laptop (seconds-minutes)
 - For higher dimensions: needs more sophisticated methods
- **Speaker notes:** This is the *engine* underneath the GE-LAV® platform. Practitioners get the outputs; the master equation is what produced them.

30.4.11 Slide 22.11 — When Computing Distributions Matters

- **Title:** “When you need Fokker-Planck and master equation outputs”
- **You need distribution-level analysis when:**
 - Stress testing a portfolio (what fraction of positions cross the exit boundary in scenario X?)
 - Computing welfare effects (aggregate cost vs. benefit)
 - Designing Pigouvian taxes (tax depends on population state, not just individual)
 - Regulatory diagnostics across an industry
- **You don’t need it when:**
 - Valuing a single asset under known conditions (HJB alone)
 - Standard back-testing
 - Simple exit boundary lookup
- **Speaker notes:** Help students *triage* — they don’t need all the math all the time.

30.4.12 Slide 22.12 — Synthesis of Sessions 19–22

- **Title:** “Where we are”
- **Built so far:**
 - **Session 19:** Stochastic processes (Brownian, Itô, OU)
 - **Session 20:** Optimal control (HJB)

- **Session 21:** Mean-field games (McKean-Vlasov)
- **Session 22:** Distribution dynamics (Fokker-Planck, master equation)
- **Coming:**
 - **Session 23:** The LAV operator and GE equilibrium pricing (synthesis)
 - **Session 24:** Jensen bias, Pigouvian tax, welfare (results)
- **Speaker notes:** Encourage students. The hard sessions are behind. Sessions 23–24 wrap up the conceptual core.

30.5 Discussion Questions

1. The Fokker-Planck equation is mathematically equivalent to a generalized heat equation. Does that connection give you confidence in the framework, or does it suggest the framework is just “thermodynamics applied to finance”? How are the two domains structurally similar / different?
2. The master equation operates on the space of probability distributions (Wasserstein space). This is abstract. What’s the *practical* consequence of “values live on distribution spaces”?
3. Computing the master equation requires substantial numerical infrastructure. Is this a barrier to GE-LAV adoption, or does the platform’s existence eliminate the barrier for practitioners?

30.6 Worked Example: Fokker-Planck Stress Test

Setup: A regulator wants to know: starting from current conditions ($L_0 = -0.5$ for all LPs), what’s the probability that aggregate PE liquidity is *worse* than current ($L_t < -0.5$) at $t = 2$ years?

Step 1: Solve OU Fokker-Planck

Using OU calibration ($\kappa = 0.45$, $\sigma_L = 0.32$, $\bar{L} = 1.0$):

Conditional density at $t = 2$: - Mean: $\bar{L} + (L_0 - \bar{L})e^{-\kappa t} = 1.0 + (-1.5) \cdot e^{-0.9} = 1.0 + (-1.5)(0.407) = 0.39$ - Variance: $\frac{\sigma_L^2}{2\kappa}(1 - e^{-2\kappa t}) = 0.1138 \cdot (1 - e^{-1.8}) = 0.1138 \cdot 0.835 = 0.0950$ - Standard deviation: 0.308

Step 2: Compute probability

$$P(L_2 < -0.5 | L_0 = -0.5) = \Phi\left(\frac{-0.5 - 0.39}{0.308}\right) = \Phi(-2.89) \approx 0.19\%$$

Interpretation: “Less than 0.2% probability that aggregate liquidity will be worse than current conditions 2 years from now.”

Note: This is a *single-LP* mean-reversion forecast. The MFG externality could amplify in extreme cases — but at $L = -0.5$ we’re in the stable regime, so the single-LP Fokker-Planck is a good approximation.

Teaching note: This is exactly the kind of analysis the GE-LAV platform produces for stress testing.

30.7 Common Student Confusions (Instructor Notes)

- “**Why is Fokker-Planck not in the standard finance curriculum?**” It is in PhD finance. MBA and master’s curricula focus on Black-Scholes-style problem-solving, which uses the *related* Kolmogorov backward equation (for pricing) more than Fokker-Planck (for densities). Both are forms of the same math.
- “**Is the master equation just a fancy form of Fokker-Planck?**” No — it’s structurally different. Fokker-Planck governs how the *distribution* evolves. Master equation governs how the *value function on the distribution* evolves. Different objects.
- “**Wasserstein space — is that important?**” Mathematically yes (it’s the right space for distributions with finite second moments). Practically, students don’t need to manipulate it directly. The platform abstracts it away.
- “**Can I do GE-LAV without solving Fokker-Planck?**” Yes for single-asset use cases. No for stress testing, welfare analysis, or Pigouvian tax design.

30.8 What to Expect Next Session

Session 23 synthesizes everything from Sessions 19–22 into the **LAV operator** (book Ch. 16) and the **GE equilibrium** (book Ch. 17). We’ll see:

- The Liquidity-Adjusted Valuation operator: how a single asset’s value depends on the entire L_t path
- General equilibrium pricing: the fixed-point that determines λ^*
- Market clearing in the GE-LAV framework

Reading: Book Chapters 16, 17 (concepts).

[← Session 21](#) | [Schedule](#) | [Next: Session 23](#) →

31 Session 23: The LAV Operator · GE Equilibrium · Market Clearing

Unit	4 — Math Intuition Bridges
Book Chapters	16, 17 (concepts only — full derivation in Session 28 Track 2)
Slide source	All new + Class 17 (intuition extract)
Slides needed	~12 new
Track	Common core (both tracks)

31.1 Learning Objectives

By the end of this session, students will be able to:

1. **State the LAV operator** in plain language — how a single asset’s value depends on the entire L_t path.
2. **Distinguish** the LAV operator from a standard DCF operator (constant rate) and from a simple Jensen-corrected operator.
3. **Articulate** what “GE equilibrium” means in the GE-LAV framework: the simultaneous determination of λ^* and individual decisions.
4. **Describe market clearing** in the secondary market: how the equilibrium price reflects the marginal investor’s required premium.
5. **Identify the valuation hierarchy** mathematically: DCF LAV GE-LAV as nested operators.

31.2 Pre-Class Assignment

- **Read:** Book Chapters 16, 17 (concepts only)

31.3 In-Class Outline (75 minutes)

Time	Segment	Format
0:00–0:05	Synthesis: 19-22 → today’s payoff	Lecture
0:05–0:25	The LAV operator	Lecture
0:25–0:45	GE equilibrium and market clearing	Lecture

Time	Segment	Format
0:45–1:00	The valuation hierarchy DCF LAV GE-LAV	Lecture
1:00–1:15	What this means for practice + Q&A	Discussion

31.4 Slide Content Briefs (New Slides Needed)

31.4.1 Slide 23.1 — Title

- **Title:** “The LAV Operator · GE Equilibrium · Market Clearing”
- **Subtitle:** “Where the conceptual framework comes together · Track 2 derivation in Session 28”
- **Speaker notes:** Today is synthesis. Students should leave able to articulate the *whole framework* in conceptual terms.

31.4.2 Slide 23.2 — Where We Are

- **Title:** “Putting it together”
- **Building blocks (from Sessions 19–22):**
 - **Stochastic process:** L_t follows OU dynamics
 - **Optimal control:** HJB gives individual exit decisions
 - **Mean-field games:** Population coupling via McKean-Vlasov
 - **Distribution dynamics:** Fokker-Planck and master equation track the population
- **Today:** Combine these to compute *valuations* (LAV operator) and *equilibrium prices* (GE-LAV)
- **Speaker notes:** Show the synthesis arc visually. Each piece was needed.

31.4.3 Slide 23.3 — The LAV Operator (Definition)

- **Title:** “The Liquidity-Adjusted Valuation operator”
- **Formal definition (boxed):**

$$- V_0^{LAV} = E^Q \left[\int_0^T C_t \cdot \exp \left(- \int_0^t r(L_s) ds \right) dt \right]$$
- **What this says:**
 - C_t : Expected cash flows from the asset
 - $r(L_s)$: Discount rate at time s , dependent on liquidity state
 - $\int_0^t r(L_s) ds$: Cumulative discount factor along the *path* of L
 - $\exp(-\cdot)$: Discount factor applied to cash flow
 - E^Q : Expectation under risk-neutral measure
- **Plain English:** “Discount each cash flow by the *path-integrated* discount rate, then take the expectation across all possible paths.”
- **Speaker notes:** This is the *operational* formula behind LAV. Track 2 will derive its properties in Session 28; Track 1 needs the structure.

31.4.4 Slide 23.4 — How LAV Differs from DCF

- **Title:** “LAV vs. DCF: side-by-side”
- **DCF:** $V_0^{DCF} = \sum_{t=0}^T \frac{C_t}{(1+r)^t}$ — constant r
- **LAV:** $V_0^{LAV} = E^Q[\int_0^T C_t e^{-\int_0^t r(L_s) ds} dt]$ — path-dependent $r(L_s)$
- **Key differences:**
 - **Path dependence:** LAV depends on the *full path* of L_t , not just current value
 - **Expectation operator:** LAV integrates across the distribution of paths
 - **Convexity correction built-in:** Jensen bias is naturally produced
- **Bottom-line:** “LAV is the *honest* generalization of DCF when discount rates are stochastic.”
- **Speaker notes:** This is the slide that students should memorize. The LAV/DCF distinction is the conceptual heart of the framework.

31.4.5 Slide 23.5 — LAV Properties

- **Title:** “Key properties of LAV (proofs in Track 2, Session 28)”
- **Property 1: Strict path dependence**
 - V^{LAV} depends on the full path of L_t , not just the terminal value
 - Theorem 16.1 in book — formal statement
- **Property 2: Positive Jensen bias for convex $r(L)$**
 - $V^{LAV} > V^{DCF}$ whenever $r(L)$ is convex in L
 - This is *always* true for the calibrated quadratic premium $\pi(L)$
- **Property 3: Reduction to DCF under deterministic L**
 - If L_t is constant, $LAV = DCF$
 - This is the limiting case
- **Speaker notes:** These properties are what make the framework rigorous. State them; don’t prove.

31.4.6 Slide 23.6 — GE Equilibrium (Setup)

- **Title:** “From LAV to GE-LAV: adding market clearing”
- **The setup:**
 - LAV uses an *exogenous* discount rate function $r(L)$
 - In *general equilibrium*, $r(L)$ is *endogenous* — determined by the market
 - Each LP wants a premium based on their needs; the market clears at the *marginal* LP’s required premium
- **The fixed point:**
 - Population distribution $\mu_t \rightarrow$ equilibrium premium $\lambda^* \rightarrow$ LP decisions $\rightarrow$ new μ_t
 - GE-LAV equilibrium: this loop is closed; μ_t is consistent with λ^*

- **Speaker notes:** Connect to Session 21’s MFG fixed-point structure. Same idea, applied to pricing.

31.4.7 Slide 23.7 — Market Clearing

- **Title:** “How the secondary market clears”
- **The clearing condition:**
 - At each L_t , sellers (LPs wanting to exit at current premium) = buyers (LPs willing to buy at current premium)
 - The equilibrium premium $\lambda^*(L_t, \mu_t)$ is the value that clears
- **Calibrated structure:**
 - At normal L : $\lambda^* \approx \pi_0 = 4.5\%$ (close to long-run mean)
 - At GFC $L \approx -1.5$: $\lambda^* \approx 13\%$ (much wider due to externality)
- **Visual:** Supply and demand curves intersecting at λ^* , with both shifting as L changes
- **Speaker notes:** Demonstrate how the equilibrium price emerges. This is standard microeconomics applied to the secondary market.

31.4.8 Slide 23.8 — GE-LAV Operator

- **Title:** “The GE-LAV operator”
- **Formal definition:**
 - $V_0^{GE-LAV} = E^Q \left[\int_0^T C_t \cdot \exp \left(- \int_0^t r_{GE}(L_s, \mu_s) ds \right) dt \right]$
- **The new ingredient:** $r_{GE}(L, \mu) = r_f + \beta \cdot \text{ERP} + \lambda^*(L, \mu)$
- **Where $\lambda^*(L, \mu)$ is the equilibrium premium** — depends on *both* the individual state L and the population state μ
- **Key property (proved in Session 28):**
 - In *deep stress regimes*, $\lambda^*(L, \mu) \gg \pi(L)$ — the equilibrium premium exceeds the partial-equilibrium premium
 - This is the *McKean-Vlasov externality* manifested in pricing
- **Speaker notes:** GE-LAV is LAV with one additional ingredient. Make sure students see the layering.

31.4.9 Slide 23.9 — The Crisis Amplification, Quantified

- **Title:** “Where the $4.31\times$ amplification comes from”
- **At normal liquidity ($L = 0$):**
 - $r_{GE-LAV}(0, \mu_{\text{normal}}) \approx 7.5\%$ (close to DCF rate)
- **At GFC depth ($L = -1.5$):**
 - Partial equilibrium premium: $\pi(-1.5) \approx 13\%$

- **GE-LAV equilibrium premium:** $\lambda^*(-1.5, \mu_{\text{stressed}}) \approx 28\%$ (much higher due to coordinated stress)
- Total r_{GE-LAV} : $\sim 32.3\%$ (vs. DCF 7.5%)
- **Amplification:** $32.3/7.5 = 4.31\times$
- **Decomposition:**
 - **Part 1:** $\sim 5\%$ from LAV path correction (Jensen bias)
 - **Part 2:** $\sim 20\%$ from GE equilibrium uplift (McKean-Vlasov externality)
- **Speaker notes:** This is *the* number from Session 16 (Crisis Dynamics). Now students see where it mathematically comes from.

31.4.10 Slide 23.10 — The Valuation Hierarchy (Mathematically)

- **Title:** “DCF LAV GE-LAV: strict nesting”
- **Statement (theorem 20.1 in book):**
 - $V^{DCF} \leq V^{LAV} \leq V^{GE-LAV}$ in normal conditions
 - Inequalities can reverse in deep stress (GE-LAV can be much lower than DCF when externality dominates)
 - Equality conditions:
 - * $V^{DCF} = V^{LAV}$ iff L_t is deterministic
 - * $V^{LAV} = V^{GE-LAV}$ iff there is no MFG externality ($\gamma \rightarrow 0$)
- **In words:** “Each level of the hierarchy is a *strict generalization* of the previous. The simpler operator is a special case.”
- **Speaker notes:** This is the *theorem* of the framework. State the result; don’t prove. Track 2 derives in Session 28+.

31.4.11 Slide 23.11 — What This Means for Practitioners

- **Title:** “Practitioner takeaways”
- **For valuation:**
 - Default to GE-LAV in stressed regimes; LAV in normal; DCF only when state is known deterministic
- **For portfolio construction:**
 - The GE-LAV operator accounts for the *systematic* effect of stress on private market values
 - Multi-asset hedge demand correctly aggregates this across positions
- **For risk management:**
 - The amplification factor is the appropriate stress scenario, not the partial-equilibrium estimate
- **For regulation:**
 - Solvency II and Basel III stress factors should be calibrated against GE-LAV magnitudes
- **Speaker notes:** Connect math to practice.

31.4.12 Slide 23.12 — Bridge to Session 24

- **Title:** “What we still need”
- **Today:** LAV operator, GE equilibrium pricing, valuation hierarchy
- **Session 24 (final synthesis):**
 - The Jensen convexity bias as a closed-form result
 - The Pigouvian exit tax as the corrective instrument
 - Welfare analysis: what does the externality cost in aggregate?
 - The full results menu for practitioners
- **Speaker notes:** Last bridge session is Session 24. We’re almost through Unit 4.

31.5 Existing Slides to Reuse from Class 17

From Deck 4, Class 17 (“General Equilibrium: Market Clearing and Endogenous Pricing”), pull approximately 4 slides covering:

- The market clearing diagram (supply / demand)
- The fixed-point structure visualized
- A table showing λ^* values at different liquidity states
- A comparison of partial vs. general equilibrium pricing

Adaptation: These slides are conceptually appropriate. Use mostly as-is.

31.6 Discussion Questions

1. The valuation hierarchy DCF → LAV → GE-LAV implies that DCF is a *special case* of GE-LAV (vanishing liquidity volatility) and LAV is a *special case* of GE-LAV (vanishing MFG coupling). Does this nesting make the framework “obviously correct,” or does it just mean GE-LAV has more parameters to fit?
2. The $4.31\times$ amplification factor decomposes into ~5% LAV path correction + ~20% GE externality uplift. Which decomposition piece is more important for regulators? For practitioners managing portfolios?
3. The GE equilibrium requires solving a fixed-point problem (HJB + Fokker-Planck + market clearing). What’s the practical implication if the fixed-point has multiple solutions (e.g., in extreme stress when stability condition fails)?

31.7 Worked Example: Computing LAV vs. GE-LAV vs. DCF for One Asset

Setup: 10-year PE buyout fund. Annual cash flow stream: \$10M/year. Current liquidity state $L_0 = -0.5$.

Step 1: DCF

Using constant rate $r_{DCF} = 12\%$:

$$V^{DCF} = \sum_{t=1}^{10} \frac{10}{1.12^t} = 56.5M$$

Step 2: LAV (path-dependent, partial equilibrium)

Need to integrate $E[\exp(-\int_0^t r(L_s)ds)]$ over the OU path distribution.

At calibrated parameters and starting state $L_0 = -0.5$:

- Expected path of L_s : mean reverts from -0.5 toward 1.0 over time
- Expected $\pi(L_s)$: starts at $\pi(-0.5) = 0.058$, decreases to $\pi(1.0) = 0.041$ at long horizon
- Effective average $r(L)$ over 10 years: approximately $7.5\% + 0.045 = 11.5$
- Including Jensen correction: approximately 0.7%

$$V^{LAV} \approx \sum_{t=1}^{10} \frac{10}{1.115^t} \cdot (1 + 0.7\%) \approx 58.2M$$

Step 3: GE-LAV (full equilibrium)

At normal liquidity, equilibrium premium = partial equilibrium premium. So GE-LAV = LAV at $L_0 = -0.5$.

$$V^{GE-LAV} \approx 58.0M$$

Step 4: Now stress the scenario

Suppose we evaluate at $L_0 = -1.5$ (GFC conditions):

- DCF (using same fixed rate): still \$56.5M (DCF is regime-blind)
- LAV (partial eq): premium higher, $V^{LAV} \approx 38M$
- **GE-LAV (full eq)**: equilibrium premium amplifies, $V^{GE-LAV} \approx 22M$

Comparison at GFC depth:

Method	Value	Gap from DCF
DCF	\$56.5M	—
LAV	\$38M	-33%
GE-LAV	\$22M	-61%

Interpretation: “DCF says \$56.5M regardless. LAV says \$38M because the path-corrected discount rate is higher. GE-LAV says \$22M because the *equilibrium* premium amplifies under collective stress. The $4.31\times$ amplification factor shows up in this comparison.”

Teaching note: This is the *quantitative* manifestation of the framework’s hierarchy. Make sure students see the dollar magnitudes.

31.8 Common Student Confusions (Instructor Notes)

- “Why does GE-LAV give a *lower* value than DCF? Shouldn’t a more sophisticated model give a *higher* value (more nuanced)?” GE-LAV is more nuanced; it gives a *more accurate* value. In stress, the correct value is lower. The framework is right.
- “If DCF systematically overprices in stress and underprices normally (via Jensen bias), the average should be wash, right?” Not quite. Asymmetric distribution of stress (rare but large) means DCF on average overstates value in expectation. Stafford (2022) makes this argument empirically.
- “What if I have empirical L_t data? Should I compute LAV directly from the data, or use the parametric GE-LAV?” Both. Parametric provides robust forecasting; empirical provides validation. Best practice uses both.
- “Is GE-LAV the ‘final’ framework, or will something supersede it?” GE-LAV is the first framework that satisfies the five requirements (Session 8). Extensions (jumps, regime shifts) are research frontiers. The *structure* of GE-LAV is likely durable; specific calibrations will evolve.

31.9 What to Expect Next Session

Session 24 is the **synthesis session** of Unit 4. We cover:

- The Jensen convexity bias as a closed-form result (Theorem 18.1)
- The Pigouvian exit tax as a welfare-maximizing instrument (Theorem 19.1)
- Quantitative welfare analysis ($\sim 2.3\%/yr$)
- The complete “GE-LAV in one slide” picture

PS3 due Session 24. Both tracks. Track 1 includes intuition questions on Sessions 19–23; Track 2 includes mathematical work.

Reading: Book Chapters 18, 19 (concepts).

[← Session 22](#) | [Schedule](#) | [Next: Session 24](#) →

32 Session 24: Jensen Bias · Pigouvian Tax · Welfare (Synthesis)

Unit	4 — Math Intuition Bridges (final synthesis)
Book Chapters	18, 19 (concepts only — full derivations in Session 29 Track 2)
Slide source	New + Classes 11, 16, 18, 20 (intuition extracts)
Slides needed	~15 new + adapt ~8 from existing
Track	Common core (both tracks)
Assessment milestone	PS3 due (start of class)

32.1 Learning Objectives

By the end of this session, students will be able to:

1. **State the closed-form Jensen bias formula** $B(T) = (\pi_2/2) \cdot \Pi_{liq}(T)$ and compute it for any asset class.
2. **Articulate the welfare gap** ($\Delta W \approx 2.3\%/yr$) as the cost of the McKean-Vlasov externality.
3. **Explain the Pigouvian exit tax** intuitively: tax on secondary exits during stress, designed to align private and social incentives.
4. **Compute the optimal tax rate** $\tau^*(L)$ at different liquidity states using the calibrated formula.
5. **Articulate “GE-LAV in one slide”** — the complete framework synthesis.

32.2 Pre-Class Assignment

- **Submit:** PS3 (due at start of class)
- **Read:** Book Chapters 18, 19 (concepts only)

32.3 In-Class Outline (75 minutes)

Time	Segment	Format
0:00–0:05	PS3 collection · Today: synthesis	Lecture
0:05–0:25	The Jensen convexity bias (full formula)	Lecture
0:25–0:45	The welfare gap and externality cost	Lecture

Time	Segment	Format
0:45–1:05	The Pigouvian exit tax	Lecture + chart
1:05–1:15	“GE-LAV in one slide” + Q&A	Synthesis

32.4 Slide Content Briefs (New Slides Needed)

32.4.1 Slide 24.1 — Title

- **Title:** “Jensen Bias · Pigouvian Tax · Welfare”
- **Subtitle:** “Synthesis of Unit 4 · The five main results, restated · Track 2 derivations in Session 29”
- **Speaker notes:** Today is the climax of Unit 4. Students should leave with a *complete conceptual picture* of GE-LAV.

32.4.2 Slide 24.2 — The Jensen Convexity Bias (Theorem)

- **Title:** “The Jensen convexity bias — closed form”
- **Theorem 18.1 (book):** Under GE-LAV calibration with quadratic premium $\pi(L)$:
 - $B(T) = \frac{\pi_2}{2} \cdot \Pi_{liq}(T)$
- **Where $\Pi_{liq}(T)$ is the liquidity risk term structure:**
 - $\Pi_{liq}(T) = \sigma_L^2 \cdot \left[T - \frac{1 - e^{-2\kappa T}}{2\kappa} \right]$
- **Plain English:** “Jensen bias is a known function of the asset’s horizon, the OU parameters, and the convexity of the premium function.”
- **Properties:**
 - $B(T) > 0$ for all $T > 0$ (when $\pi_2 > 0$)
 - $B(T)$ grows monotonically with T
 - $B(T) \rightarrow (\pi_2 \sigma_L^2 / 2) \cdot T$ for large T (linear growth)
- **Speaker notes:** This is the *closed-form* result. Students should be able to compute $B(T)$ for any asset class.

32.4.3 Slide 24.3 — Calibrated Jensen Bias Values

- **Title:** “Calibrated $\Pi_{liq}(T)$ and Jensen bias by horizon”
- **At calibrated parameters** ($\kappa = 0.45$, $\sigma_L = 0.32$, $\pi_2 = 0.021$):

T (yr)	$\Pi_{liq}(T)$	$B(T)$
5	1.93	$0.020 = 2.0\%$
10	5.88	$0.062 = 6.2\%$

T (yr)	$\Pi_{liq}(T)$	$B(T)$
15	11.08	$0.116 = 11.6\%$
20	14.28	$0.150 = 15.0\%$

- **Interpretation:** “ $B(15) \approx 11.6\%$ for a 15-year asset. This is the *systematic* downward bias of DCF for that asset.”
- **Speaker notes:** These are *larger* than the earlier asset-class-specific numbers (0.8-3.6%). Why? Different parameterization — the affine formula $B(T) = A \cdot T + C$ is a *fitted* approximation. The closed-form here is the exact result.

32.4.4 Slide 24.4 — Why DCF Systematically Overprices

- **Title:** “DCF + Jensen $\rightarrow$ systematic overpricing”
- **The logic:**
 - V^{DCF} uses a single discount rate r
 - V^{LAV} uses path-dependent $r(L_s)$
 - Jensen’s inequality: $E[e^{-rT}] > e^{-E[r]T}$
 - Therefore: $V^{LAV} > V^{DCF}$ — LAV is larger
- **Wait — earlier we said GE-LAV can be much *lower* than DCF in stress. How does this reconcile?**
 - **Normal regimes:** $LAV > DCF$ (Jensen bias positive)
 - **Stress regimes:** GE equilibrium correction dominates $\rightarrow$ $GE-LAV < DCF$ (Jensen positive but smaller than equilibrium uplift)
- **In summary:** Jensen always positive; GE equilibrium correction can be either sign depending on regime
- **Speaker notes:** Resolve student confusion. Two effects, one always positive (Jensen), one regime-dependent (GE).

32.4.5 Slide 24.5 — The Welfare Gap

- **Title:** “The welfare gap — quantifying the externality”
- **Setup:**
 - Decentralized equilibrium: each LP optimizes its own exit timing (Sessions 11-13)
 - Centralized planner solution: maximize total LP welfare (account for externality)
 - Welfare gap = Centralized welfare – Decentralized welfare
- **Calibrated magnitude:** $\Delta W \approx 2.3\%$ per year
- **In dollar terms:**
 - \$13T private capital $\times 2.3\% =$ \$300B/year aggregate loss
 - Over a decade: $\sim$ \$3T cumulative
 - For comparison: 2008 banking crisis losses ($\sim$ \$1T-2T direct)
- **Speaker notes:** This number is the *single most powerful* welfare statistic in the framework. Make sure students retain it.

32.4.6 Slide 24.6 — Where the Welfare Loss Concentrates

- **Title:** “Welfare loss concentration”
- **By regime:**
 - Normal: ~0.3%/yr (small externality)
 - Mild stress: ~1.5%/yr
 - Deep stress (GFC): ~6%/yr
- **By asset class:**
 - PE buyout: 2.5%
 - Infrastructure: 1.8%
 - Private credit: 1.2%
 - Real estate: 2.0%
- **Cumulative:** Long horizons accumulate more welfare loss
- **Speaker notes:** Loss is *asymmetric* — concentrated in stress, in PE, in long horizons. These are exactly where regulatory attention should focus.

32.4.7 Slide 24.7 — The Pigouvian Tax (Intuition)

- **Title:** “The Pigouvian exit tax — concept”
- **Pigovian/Pigouvian theory:** “When there’s an externality, impose a tax equal to the marginal external cost. This aligns private with social incentives.”
- **In GE-LAV:**
 - LP A exits secondary market → small externality (slightly widens discounts for everyone)
 - Tax LP A by approximately the externality magnitude
 - Aggregate effect: discourages forced exits during stress; internalizes the externality
- **Result:** Welfare gap closes; new equilibrium is approximately the social optimum
- **Speaker notes:** Connect to public economics. Same logic as carbon tax for emissions, congestion pricing for traffic.

32.4.8 Slide 24.8 — The Pigouvian Tax Formula

- **Title:** “The optimal Pigouvian tax — closed form”
- **Theorem 19.1 (book):** Under GE-LAV calibration:
 - $\tau^*(L) = \frac{\text{marginal externality}}{\text{value at exit}} = f(L; \kappa, \sigma_L, \pi_2, \gamma)$
- **Calibrated values by liquidity state:**

L State	Description	Optimal Tax $\tau^*(L)$
0	Normal	0.0%
-0.5	Mild stress	1.5%
-1.0	Stress	3.5%

L State	Description	Optimal Tax $\tau^*(L)$
-1.5	GFC depth	7.0%
-2.0	Crisis	9.0%

- **In words:** “Tax is *state-contingent* — zero in normal times, rising sharply during stress”
- **Speaker notes:** Clean policy prescription. Practical regulatory adoption is hard (see Session 16) but the *theoretical recommendation* is clear.

32.4.9 Slide 24.9 — How the Tax Generates Revenue

- **Title:** “Pigouvian tax revenue”
- **Setup:** At GFC depth, ~\$200B of secondary transactions clear (peak volume)
- **Tax rate at that state:** 7.0%
- **Revenue:** $\$200\text{B} \times 7.0\% = \14B in tax revenue
- **Disposition:**
 - **Best practice:** Refund to remaining LPs (those who didn’t exit) — internalizes the externality cleanly
 - **Alternative:** Industry stability fund (like FDIC for banking)
 - **Worst:** General revenue (uses external proceeds for unrelated purposes)
- **Speaker notes:** Mechanism design question. Reasonable people disagree on disposition; theory is clear on tax magnitude.

32.4.10 Slide 24.10 — Implementation Challenges

- **Title:** “Implementing the Pigouvian tax”
- **Technical challenges:**
 - Calibrating $\tau^*(L)$ in real-time requires knowing the population state
 - Tax must be regime-conditional → frequent adjustment
 - Cross-border coordination needed (private markets are global)
- **Political challenges:**
 - Industry opposition (taxes on liquidity-needing LPs feel punitive)
 - Regulatory turf wars
 - Existing tax regimes (capital gains, stamp duties) may already proxy
- **Stepped adoption path:**
 - Phase 1: Disclosure requirement
 - Phase 2: Voluntary adoption by sophisticated LPs
 - Phase 3: Mandatory under select regulators (insurance, banking)
 - Phase 4: Cross-border standard
- **Speaker notes:** Realistic about adoption timeline. Decades, not years.

32.4.11 Slide 24.11 — GE-LAV in One Slide (The Master Visual)

- **Title:** “GE-LAV — the complete framework”
- **Layout:** Six components arranged in a logical flow
 1. **Liquidity process:** $dL_t = \kappa(\bar{L} - L_t)dt + \sigma_L dW_t$
 2. **Premium function:** $\pi(L) = \pi_0 - \pi_1 L + \pi_2 L^2$
 3. **Effective rate:** $r(L) = r_f + \beta \cdot ERP + \pi(L)$
 4. **LAV operator:** V^{LAV} = path-integrated discount
 5. **GE equilibrium:** Fixed-point on λ^*, μ_t
 6. **Welfare instrument:** Pigouvian tax $\tau^*(L)$
- **Outputs:**
 - **Valuation:** GE-LAV mark for any asset
 - **Decision:** Optimal exit boundary $L^*(t)$
 - **Portfolio:** Optimal allocation with hedge demand
 - **Regulation:** State-contingent capital and tax
 - **Welfare:** Quantified externality cost
- **Speaker notes:** This is the *master* slide. Have students photograph it. Reference back throughout the rest of the course.

32.4.12 Slide 24.12 — The Five Main Results Reviewed

- **Title:** “The five main results — now with proofs ahead”
- **Result 1 — Jensen Bias:** Closed-form. Computed today. **Track 2** derives **Session 29**.
- **Result 2 — Optimal Exit Boundary:** $L^*(t)$ via HJB. Discussed Session 12. **Track 2** derives **Session 25**.
- **Result 3 — Welfare Gap:** $\Delta W \approx 2.3\%/yr$. Discussed today. **Track 2** derives **Session 29**.
- **Result 4 — Pigouvian Exit Tax:** $\tau^*(L)$ closed-form. Discussed today. **Track 2** derives **Session 29**.
- **Result 5 — Valuation Hierarchy:** DCF → LAV → GE-LAV. Stated Session 23. **Track 2** derives **Session 30**.
- **Speaker notes:** Show progression. Track 1 students have *seen* every result conceptually. Track 2 students will *prove* every result.

32.4.13 Slide 24.13 — Unit 4 Synthesis

- **Title:** “Unit 4: what we built”
- **Six bridge sessions:**
 - 19: Brownian, Itô, OU
 - 20: HJB
 - 21: McKean-Vlasov / MFG
 - 22: Fokker-Planck / master equation
 - 23: LAV operator / GE equilibrium
 - 24: Jensen / Pigouvian / welfare (today)

- **For Track 1:** “You can now articulate the *complete* framework. You understand every result. You don’t need the proofs to apply it.”
- **For Track 2:** “You have the conceptual foundation. Sessions 25–29 fill in the proofs and derivations.”
- **Speaker notes:** Celebrate the achievement. Unit 4 is genuinely difficult.

32.4.14 Slide 24.14 — Bridge to Unit 5 (Split Track)

- **Title:** “What comes next: Split Track”
- **Session 25 onward: tracks separate**
 - **Track 1:** Applied workshops (PE, VC, infra, credit cases)
 - **Track 2:** Mathematical derivations (HJB, MFG, Fokker-Planck, equilibrium, Pigouvian)
- **Both tracks:**
 - Session 25: First split session
 - Sessions 30–31: Project workshops
 - Session 32: Project presentations (reunited)
- **Speaker notes:** Acknowledge the transition. Remind students of track assignments.

32.4.15 Slide 24.15 — Course Synthesis Reminder

- **Title:** “Where you are in the course”
- **Unit 1 (Sessions 1–5):** Why DCF fails
- **Unit 2 (Sessions 6–10):** Measurement + theory
- **Unit 3 (Sessions 11–18):** Decision + application
- **Unit 4 (Sessions 19–24):** Math intuition bridges ← we are here
- **Unit 5 (Sessions 25–31):** Split track (next 7 sessions)
- **Capstone (Session 32):** Project presentations
- **Speaker notes:** Progress check. Students are 75% through the course.

32.5 Existing Slides to Reuse

32.5.1 From Class 11 (Jensen Bias Four-Channel Derivation)

- 4 slides with the closed-form Jensen bias formula and decomposition
- 2 slides with calibrated values

32.5.2 From Class 16 (Empirical Validation of Jensen Bias)

- 3 slides with empirical evidence

32.5.3 From Class 18 (Welfare Analysis / Externality)

- 3 slides with the welfare gap calculation

32.5.4 From Class 20 (Pigouvian Tax Design)

- 4 slides with the optimal tax structure and revenue analysis

Adaptation: Substantial — these slides currently include mathematical detail meant for Track 2. Strip derivations, keep results and tables.

32.6 Discussion Questions

1. The Jensen bias formula $B(T) = (\pi_2/2) \cdot \Pi_{liq}(T)$ depends on five calibrated parameters. If any of those parameters are wrong, the bias estimate is wrong. Which parameter is most fragile? How would you stress-test it?
2. The Pigouvian tax raises ~\$14B at GFC depth. Distributing this back to remaining LPs is the textbook answer. Politically, would governments distribute, or absorb? How does that change the framework's appeal?
3. The valuation hierarchy DCF → LAV → GE-LAV implies GE-LAV is the “right” framework. Why isn't DCF abandoned today, given the framework exists?

32.7 Worked Example: Computing Everything for One Asset

Setup: \$100M infrastructure asset, 15-year horizon, $L_0 = -0.5$ (mild stress).

Step 1: Jensen bias

$$\Pi_{liq}(15) = 0.1024 \cdot [15 - (1 - e^{-13.5})/0.9] = 0.1024 \cdot [15 - 1.111] = 0.1024 \cdot 13.889 = 1.422$$

Wait — let me recompute: $\Pi_{liq}(T) = \sigma^2 \cdot [T - (1 - e^{-2\kappa T})/(2\kappa)]$

$$\Pi_{liq}(15) = 0.1024 \cdot [15 - (1 - e^{-13.5})/0.9] \approx 0.1024 \cdot [15 - 1.111] \approx 1.422$$

$$B(15) = (\pi_2/2) \cdot \Pi_{liq}(15) = (0.021/2) \cdot 1.422 = 0.0149 \quad 1.5\%$$

Step 2: LAV value

DCF value (at constant 7.5%): $V^{DCF} \approx \$100M$ (assuming normalization)

$$\text{LAV value (Jensen-adjusted): } V^{LAV} = V^{DCF} \cdot (1 + B(15)) = \$101.5M$$

Step 3: GE-LAV value at current stress

At $L_0 = -0.5$, equilibrium premium 5.5% (vs. DCF assumption 3.5%) → effective rate ~9.5%

$$V^{GE-LAV} \approx V^{DCF} \cdot e^{-(0.095-0.075) \cdot 15} \approx \$100M \cdot 0.74 = \$74M$$

(Account for ongoing cash flow generation roughly cancels the discount rate effect over the term)

Step 4: Welfare loss attributable to this position

Per-year welfare loss for this asset: $\sim \$2.3M$ (2.3% of $\$100M$)

Cumulative over 15 years: $\sim \$25M$ (with discount)

Step 5: Optimal Pigouvian tax if held in stress

If LP exits at $L = -0.5$: optimal $\tau^* = 1.5\%$. On $\$100M$ sale: $\$1.5M$ tax.

If LP exits at GFC depth: optimal $\tau^* = 7.0\%$. On (deeply discounted) sale: substantial.

Teaching note: This is a *single position* analysis. Multiply across a portfolio for institutional implications.

32.8 Common Student Confusions (Instructor Notes)

- “The Jensen bias formulas give different numbers in different parts of the course (0.8% vs. 2% vs. 11.6%).” Yes — these reflect *different parameterizations* (affine approximation vs. closed-form vs. extreme parameter values). All are correct under their respective assumptions. The 11.6% at 15 years is the most rigorous.
- “The Pigouvian tax sounds great in theory. Has it ever been implemented in any market?” Yes — Tobin tax (currency markets, limited), stamp duties (UK property markets), congestion pricing (London, Stockholm), carbon taxes (EU, BC). Mixed success but proof of concept.
- “How is the welfare gap measured if we never see the counterfactual?” Through the calibrated MFG model — by computing decentralized vs. centralized equilibrium outcomes. Indirect but rigorous.
- “Is 2.3%/yr large or small?” Large. For comparison: the *entire* PE risk premium is $\sim 3\text{--}5\%$. The externality is 50%+ of the risk premium. That’s enormous.

32.9 What to Expect Next Session

Sessions 25 onward are **split track**:

- **Track 1 students** report to the applied case workshop. Sessions 25 covers PE buyout valuation.
- **Track 2 students** report to the mathematical session. Session 25 covers the full HJB derivation.

Both tracks: Session 25 is your first split-track session. Make sure you know which track you’re in and where it meets.

Reading for Track 1: Bain Capital case study or similar (will be distributed).

Reading for Track 2: Book Chapter 11 (HJB, with full derivations).

PS4 drops at end of Session 25. Track-specific.

[← Session 23](#) | [Schedule](#) | [Next: Session 25](#) →

Part IV

Lectures: Split Track (Sessions 25–31)

33 Session 25: Split Track — PE Buyout Workshop / Full HJB Derivation

Unit	5 — Split Track
Track 1 source	Class 21 (PE Buyout Full Valuation) + new case material
Track 2 source	Classes 7 (HJB) + 14 (HJB numerical) + new derivation material
Slides needed	T1: ~3 new + 13 from Class 21; T2: ~5 new + ~25 from Classes 7 + 14
Assessment milestone	PS4 drops at end of class

! Tracks separate here

Track 1 (Practitioner) and Track 2 (Researcher) students meet in different rooms (or different breakout sessions in synchronous online format). Both sessions run the same 75 minutes. **Confirm your track location before class** — see course site for room assignment. Lecture recordings from *both* tracks are posted to the course site afterward. Cross-track audit is encouraged.

33.1 Track 1: PE Buyout Case Workshop

33.1.1 Track 1 Learning Objectives

By the end of this session, Track 1 students will be able to:

1. **Apply** the GE-LAV framework end-to-end to a real PE buyout fund scenario.
2. **Calibrate** OU parameters using provided secondary market data for the case.
3. **Compute** DCF, LAV, and GE-LAV values for the case fund and explain the differences.
4. **Construct an IC-style recommendation** based on GE-LAV outputs.
5. **Defend** the recommendation orally to a peer reviewer.

33.1.2 Track 1 Pre-Class Assignment

- **Read:** Bain Capital Asia IX case (distributed via course site)
- **Bring:** Laptop with platform access
- **Optional:** Pre-class spreadsheet exercise (link on course site)

33.1.3 Track 1 In-Class Outline (75 minutes)

Time	Segment	Format
0:00–0:10	Case introduction and team formation (groups of 3)	Lecture
0:10–0:30	Hands-on: calibration and DCF baseline	Group work
0:30–0:50	Hands-on: LAV and GE-LAV computation	Group work
0:50–1:05	Group presentations (3 minutes each)	Presentations
1:05–1:15	Synthesis and PS4 introduction	Lecture

33.1.4 Track 1 Case Setup

The Case: Bain Capital Asia Fund IX

- **Vintage:** 2018
- **Strategy:** Asian mid-market buyout
- **NAV as of Q4 2024:** \$1.2B (LP's share)
- **Remaining hold:** ~4 years
- **Reason for valuation review:** LP is considering secondary sale offer
- **Secondary bid:** \$980M (18% discount to NAV)

Provided data: - Quarterly secondary market discounts 2018–2024 for Asian PE funds - Fund cash flow history (capital calls and distributions) - Public benchmark: MSCI Asia ex-Japan - Asset class calibration: PE buyout, calibrated parameters per Table 7.2

33.1.5 Track 1 Slide Briefs (New Slides Needed)

Slide T1-25.1 — Case Introduction

- **Title:** “Case: Bain Capital Asia Fund IX”
- **Setup:** LP holds \$1.2B, secondary bid is \$980M, decision needed
- **Question:** “Accept the secondary, hold, or partial exit?”
- **Speaker notes:** Frame as IC presentation. Each group will defend a recommendation.

Slide T1-25.2 — Workshop Structure

- **Title:** “Today’s workshop: 4 questions per group”

- **Questions:**
 1. What is the OU calibration for Asian PE buyout 2018-2024?
 2. What is the LAV value of the position at current L_t ?
 3. What is the GE-LAV value at current and stress scenarios?
 4. What is your IC recommendation? (Hold / Sell / Partial)
- **Speaker notes:** Walk through the questions. Each group answers all four.

Slide T1-25.3 — IC Memo Template (Class 21 adaptation)

- **Title:** “How to structure your recommendation”
- **Template:**
 - Executive summary (3 sentences)
 - DCF baseline
 - LAV adjustment + magnitude
 - GE-LAV value + stress scenarios
 - Recommendation + key risks
 - One-page chart for IC meeting
- **Speaker notes:** Show example template from Session 18.

33.1.6 Track 1 Existing Slides to Reuse

From Class 21 (PE Buyout Full GE-LAV Valuation), pull approximately 13 slides covering: - The general PE valuation pipeline - LAV operator application - GE-LAV equilibrium computation for buyout assets - Stress scenario analysis for PE - IC memo template

These slides are well-suited to Track 1 as-is.

33.1.7 Track 1 Discussion Questions

1. The secondary bid is \$980M against a \$1.2B NAV. Is the secondary buyer offering fair value, exploiting distress, or pricing in something the LP doesn't see? How would GE-LAV decompose the gap?
2. Asian PE has had different liquidity dynamics than US PE over 2018-2024 (less institutional secondary infrastructure, more strategic buyers). How does this affect the calibration?
3. The LP's IC will likely prefer a “hold” recommendation since it preserves the carry track record. How do you advocate for a “sell” if the GE-LAV analysis supports it?

33.2 Track 2: Full HJB Derivation

33.2.1 Track 2 Learning Objectives

By the end of this session, Track 2 students will be able to:

1. **Derive** the Hamilton-Jacobi-Bellman equation from the Bellman optimality principle and Itô's lemma.
2. **Apply** the HJB equation to the GE-LAV optimal stopping problem and identify the variational inequality structure.
3. **State and verify the smooth pasting condition** as the optimality condition at the free boundary.
4. **Implement** a finite-difference solver for the GE-LAV exit boundary.
5. **Reproduce** the $L^*(t)$ chart numerically from first principles.

33.2.2 Track 2 Pre-Class Assignment

- **Read:** Book Chapter 11 in full (with proofs)
- **Optional pre-reading:** Karatzas & Shreve, *Brownian Motion and Stochastic Calculus*, Chapter 5; or Pham, *Continuous-time Stochastic Control and Optimization*, Chapters 3-4

33.2.3 Track 2 In-Class Outline (75 minutes)

Time	Segment	Format
0:00–0:10	Recap measure-theoretic foundations (assumed prereq)	Lecture
0:10–0:30	Derivation: from Bellman to HJB	Lecture + board work
0:30–0:50	The variational inequality and smooth pasting	Lecture
0:50–1:05	Numerical implementation: finite differences	Lecture + code walkthrough
1:05–1:15	Verification theorem	Lecture

33.2.4 Track 2 Slide Briefs (New Slides Needed)

Slide T2-25.1 — From Bellman to HJB

- **Title:** “Deriving the HJB equation”
- **Step 1:** Bellman's principle in discrete time
 - $V(x, t) = \max_u \{f(x, u)\Delta t + e^{-\rho\Delta t} E[V(X_{t+\Delta t}, t + \Delta t) | X_t = x, u_t = u]\}$
- **Step 2:** Taylor expand $V(X_{t+\Delta t}, t + \Delta t)$ and apply Itô's lemma:
 - $E[V(X_{t+\Delta t}, t + \Delta t)] = V(x, t) + \frac{\partial V}{\partial t} \Delta t + \mu \frac{\partial V}{\partial x} \Delta t + \frac{1}{2} \sigma^2 \frac{\partial^2 V}{\partial x^2} \Delta t + o(\Delta t)$

- **Step 3:** Substitute and take $\Delta t \rightarrow 0$:

$$- 0 = \max_u \{f(x, u) + \frac{\partial V}{\partial t} + \mu(x, u) \frac{\partial V}{\partial x} + \frac{1}{2} \sigma^2(x, u) \frac{\partial^2 V}{\partial x^2} - \rho V\}$$
- **Speaker notes:** Walk through every step. This is the *core* derivation Track 2 students need.

Slide T2-25.2 — HJB for the GE-LAV Exit Problem

- **Title:** “Specializing to the GE-LAV exit problem”
- **State:** L_t (liquidity); **Action:** continue or stop
- **Dynamics:** $dL_t = \kappa(\bar{L} - L_t)dt + \sigma_L dW_t$ (OU)
- **Variational inequality formulation:**
 - In the continuation region: $\mathcal{L}V - r(L)V + C(L) = 0$
 - In the stopping region: $V = G(L)$ (exit payoff)
 - Match condition at boundary: smooth pasting
- **Where** $\mathcal{L}V = \frac{\partial V}{\partial t} + \kappa(\bar{L} - L) \frac{\partial V}{\partial L} + \frac{1}{2} \sigma_L^2 \frac{\partial^2 V}{\partial L^2}$
- **Speaker notes:** Carefully state the variational inequality. This formulation is what enables numerical solution.

Slide T2-25.3 — Smooth Pasting Derivation

- **Title:** “Why smooth pasting must hold”
- **Argument:**
 - At the optimal boundary L^* , if $V'(L^*) > G'(L^*)$, you could improve by moving L^* slightly inward
 - Similarly $V'(L^*) < G'(L^*)$ allows improvement outward
 - Therefore $V'(L^*) = G'(L^*)$ at optimum
- **Formal proof (verification theorem):**
 - Suppose V^* satisfies the variational inequality with smooth pasting
 - Show: $V^* = \sup_{\tau} E[\int_0^{\tau} e^{-rs} C ds + e^{-r\tau} G(L_{\tau})]$
 - Standard verification argument using Itô applied to $V^*(L_t, t)$
- **Speaker notes:** Sketch the verification theorem. Don’t do every step but show the structure.

Slide T2-25.4 — Finite Difference Solver

- **Title:** “Numerical implementation: backward induction on a grid”
- **Algorithm:**
 - Discretize L on grid $\{L_i\}$ with spacing h
 - Discretize t on grid $\{t_n\}$ with spacing Δt
 - Terminal condition: $V(L, T) = G_T(L)$
 - Backward step: for $n = N - 1, N - 2, \dots, 0$:
 - * Compute continuation value V_i^{cont} using finite-difference Laplacian
 - * $V_i^n = \max\{V_i^{cont}, G(L_i)\}$
- **Stability:** Crank-Nicolson scheme for stable backward induction
- **Output:** Full $V(L, t)$ surface and implicit $L^*(t)$ curve
- **Speaker notes:** Provide pseudocode. Students should be able to implement.

Slide T2-25.5 — Reproducing the $L^*(t)$ Chart

- **Title:** “From PDE to the exit boundary chart”
- **Result:** $L^*(t)$ curve identical to Session 12 chart (now derived from first principles)
- **Sanity checks:**
 - $L^*(0) = \$$ deepest stress (most willing to hold long horizons)
 - $L^*(T)$ approaches NAV-equivalent state (no remaining horizon)
 - Smooth pasting verified numerically: matched slopes at boundary
- **Speaker notes:** This is the *payoff* slide — Track 2 students have now built the result from scratch.

33.2.5 Track 2 Existing Slides to Reuse

From Classes 7 (Stochastic Control & HJB) and 14 (HJB Numerical Computation of $h(L)$): - 15 slides from Class 7 on HJB derivation, examples, verification - 10 slides from Class 14 on numerical methods, finite differences, the $h(L)$ function

These slides are essentially what Track 2 needs. The session reuses them substantially.

33.2.6 Track 2 Discussion Questions

1. The verification theorem requires the candidate value function V^* to be smooth (e.g., $C^{1,2}$). What if smooth pasting failed and V^* had a kink at the boundary? What would that imply economically?
2. The finite-difference solver uses Crank-Nicolson for stability. What’s the alternative? When does explicit time-stepping suffice vs. require implicit treatment?
3. The HJB derivation assumes Markov dynamics. If L_t had memory (fractional Brownian motion), how would the derivation change? Hint: research on rough volatility extensions.

33.3 Common to Both Tracks: PS4 Drops

PS4 drops at the end of today’s session. Track-specific content:

- **Track 1 PS4:** Apply GE-LAV to a multi-asset private market portfolio. Use the platform. Produce an IC memo.
- **Track 2 PS4:** Mathematical work on advanced topics (TBD per track-specific assignment).

Due: Session 31.

[← Session 24](#) | [Schedule](#) | [Next: Session 26](#) →

34 Session 26: Split Track — VC & Secondaries / McKean-Vlasov MFG Proofs

Unit	5 — Split Track
Track 1 source	Class 27 (Multi-Asset, adapted for VC) + new
Track 2 source	Classes 8 (McKean-Vlasov) + 15 (MFG Fixed Point)
Slides needed	T1: ~5 new + ~10 from Class 27; T2: ~3 new + ~30 from Classes 8 + 15

34.1 Track 1: VC + Secondaries Valuation Lab

34.1.1 Track 1 Learning Objectives

By the end of this session, Track 1 students will be able to:

1. **Apply** GE-LAV to a venture capital fund position (different cash flow profile from buyout).
2. **Adjust** the calibration for asset classes with thinner secondary market data.
3. **Value a specific VC secondary** opportunity using GE-LAV.
4. **Distinguish** LP-led from GP-led secondary structures and apply GE-LAV to each.
5. **Communicate** findings to a hypothetical VC GP and IC.

34.1.2 Track 1 Pre-Class Assignment

- **Read:** VC secondary case (distributed); skim recent Lazard or Evercore VC secondary report
- **Try:** Platform exercise on a sample VC position

34.1.3 Track 1 In-Class Outline (75 minutes)

Time	Segment	Format
0:00–0:10	VC vs. buyout: cash flow profile differences	Lecture
0:10–0:25	Calibrating GE-LAV for VC (parameter adjustments)	Lecture + worked example

Time	Segment	Format
0:25–0:50	Lab: value a real VC secondary opportunity	Group work
0:50–1:05	LP-led vs. GP-led secondaries	Lecture + discussion
1:05–1:15	Group recommendations + synthesis	Presentations

34.1.4 Track 1 Slide Briefs (New Slides Needed)

Slide T1-26.1 — VC vs. Buyout: Different Cash Flow Profiles

- **Title:** “VC: longer J-curve, larger tail risk”
- **Comparison:**
 - **Buyout:** Capital deployed in years 1-3, distributions start years 4-5, terminal exit year 7-8
 - **VC:** Capital deployed in years 1-4, very few distributions until years 6-7, terminal events highly bimodal (some IPOs, many writedowns)
- **GE-LAV implications:**
 - Longer effective duration → larger Jensen bias accumulation
 - Tail risk demands wider hedge demand
 - Secondary market thinner; less data for calibration
- **Speaker notes:** Set up why VC is *not* just “buyout with bigger volatility.” Different problem.

Slide T1-26.2 — VC Secondary Market Specifics

- **Title:** “VC secondaries: thin, idiosyncratic, growing”
- **Structure:**
 - Smaller than PE secondary market (~\$20B/yr vs. \$130B for PE)
 - More direct (less broker-mediated)
 - More structured deals (strip sales, NAV financings, GP-leds dominating)
- **Calibration challenges:**
 - Limited public price discovery
 - Discounts often estimated indirectly via comparables
 - Implied L_t noisier than for PE
- **Speaker notes:** Be honest about data limitations. Calibration confidence intervals are wider for VC.

Slide T1-26.3 — Calibrated VC Parameters

- **Title:** “VC asset class calibration”
- **Parameters (vs. PE buyout):**

Parameter	PE Buyout	VC
β_i (systemic)	1.2	1.1
σ_V	0.25	0.45
ρ (corr with L)	0.30	0.20
Hedge demand at $L = 0$	8%	4%
Jensen bias slope (T=10)	0.17%	0.18%

- **Insight:** “VC has higher idiosyncratic volatility but lower correlation with aggregate liquidity → smaller proportional hedge demand”
- **Speaker notes:** Counterintuitive result. Students often think VC is “riskier all around.” It is — but the *liquidity* dimension is less correlated.

Slide T1-26.4 — LP-Led vs. GP-Led Secondaries

- **Title:** “Two transaction types, two GE-LAV applications”
- **LP-led:**
 - Single LP exits position via secondary sale
 - GE-LAV: standard valuation + exit boundary check
- **GP-led (continuation funds):**
 - GP rolls portfolio company into new vehicle
 - Existing LPs choose: cash out at offered price OR roll into new fund
 - GE-LAV evaluates: is the “cash out” price the right GE-LAV value? Is the “roll” priced as a new commitment at fair value?
- **Common pitfall:**
 - GP-led terms often anchored on DCF-style valuations
 - GE-LAV may say “cash out” is too low *or* “roll” terms are too expensive
- **Speaker notes:** GP-leds are the fastest-growing segment. Students entering the industry will face these.

Slide T1-26.5 — Workshop Setup

- **Title:** “Lab: a real GP-led decision”
- **The case:**
 - VC fund vintage 2017, currently 7 years in
 - GP proposes continuation fund for the top 5 portfolio companies
 - Existing LPs offered: \$850M cash for \$1B NAV (15% discount) OR roll into new 4-year fund at same NAV
 - Your job: GE-LAV-evaluate both options
- **Group output:**
 - GE-LAV value of “cash out” option
 - GE-LAV value of “roll” option (treating as new commitment)
 - Recommendation with rationale
- **Speaker notes:** This is meaty. Groups will need most of class time.

34.1.5 Track 1 Existing Slides to Reuse

From Class 27 (Multi-Asset GE-LAV) — adapt for VC: - Multi-asset calibration tables - Cross-asset correlation matrix - Combined hedge demand visualization

Adapt by emphasizing VC vs. PE buyout differences rather than the full multi-asset treatment (which gets its own session in Session 27).

34.2 Track 2: McKean-Vlasov Mean-Field Game Proofs

34.2.1 Track 2 Learning Objectives

By the end of this session, Track 2 students will be able to:

1. **State and prove the propagation of chaos** for the GE-LAV mean-field system (Theorem 13.4).
2. **Derive the Schauder fixed-point existence argument** for MFG equilibrium (Theorem 13.2).
3. **State and verify the stability condition** $\kappa > \gamma$ for uniqueness (Theorem 13.3).
4. **Implement** an MFG fixed-point iteration numerically.
5. **Discuss** the limits of the framework: when stability fails, what happens.

34.2.2 Track 2 Pre-Class Assignment

- **Read:** Book Chapter 13 in full, including all proofs
- **Pre-read:** Carmona & Delarue, *Probabilistic Theory of Mean-Field Games*, Chapter 1 — at least sections 1.1–1.3

34.2.3 Track 2 In-Class Outline (75 minutes)

Time	Segment	Format
0:00–0:10	Setup: the GE-LAV McKean-Vlasov SDE	Lecture
0:10–0:30	Propagation of chaos: statement + proof sketch	Lecture
0:30–0:50	Schauder fixed-point: existence of MFG equilibrium	Lecture
0:50–1:05	Stability condition and uniqueness	Lecture
1:05–1:15	Numerical MFG iteration	Code walkthrough

34.2.4 Track 2 Slide Briefs (New Slides Needed)

Slide T2-26.1 — The McKean-Vlasov SDE for GE-LAV

- **Title:** “The McKean-Vlasov system, formally”
- **Individual SDE:**
 - $dL_t^i = \kappa(\bar{L} - L_t^i)dt + \gamma \int \phi(L_t^i, l)\mu_t(dl)dt + \sigma_L dW_t^i$
- **Where:**
 - μ_t = distribution of L_t^j across the population
 - $\phi(L, l)$ = pairwise interaction kernel
 - γ = coupling strength
- **Self-consistency:**
 - $\bar{L} = \int L_t^i d\mu_t$ = Law (L_t^i) for representative i
 - This is the *fixed-point* requirement
- **Speaker notes:** Formal statement. Don’t dwell on the kernel form; refer to book for specifics.

Slide T2-26.2 — Propagation of Chaos: Theorem 13.4

- **Title:** “Propagation of chaos for GE-LAV”
- **Theorem statement:** Let $L_t^{1,N}, \dots, L_t^{N,N}$ be the N -particle system. As $N \rightarrow \infty$:
 - $\sup_{t \leq T} E \left[\frac{1}{N} \sum_{i=1}^N \delta_{L_t^{i,N}} - \mu_t \right]_{W_1} \leq \frac{C}{\sqrt{N}}$
 - where W_1 is the 1-Wasserstein distance
- **Proof sketch:**
 - Construct coupling between particle system and MV limit
 - Use Lipschitz properties of ϕ + Grönwall inequality
 - Apply standard rates for empirical measure convergence
- **Implications:**
 - At $N = 100$: error $\sim 10\%$
 - At $N = 10,000$: error $< 1\%$
 - Private markets have $N \approx 10,000 \rightarrow$ mean-field is a *very good* approximation
- **Speaker notes:** State the theorem; sketch the proof. Don’t do every step. Refer to Sznitman (1991) for the canonical argument.

Slide T2-26.3 — Schauder Fixed-Point and Existence (Theorem 13.2)

- **Title:** “Existence of MFG equilibrium”
- **The fixed-point map $\Phi : \mathcal{M} \rightarrow \mathcal{M}$:**
 - Take $\mu \in \mathcal{M}$ (set of measures with finite second moment)
 - Solve HJB given μ : get optimal policy u_μ^*
 - Solve Fokker-Planck under u_μ^* : get new measure $\Phi(\mu)$
 - Fixed point: $\mu = \Phi(\mu)$

- **Schauder’s argument:**
 - $\mathcal{M}$ is convex, closed under Wasserstein topology
 - Φ is continuous (HJB and FP both stable in measure)
 - Φ is compact (uniform bounds on moments)
 - Therefore Φ has a fixed point in $\mathcal{M}$
- **Speaker notes:** This is the *existence* argument. State carefully. Full proof in book Section 13.3.

Slide T2-26.4 — Stability Condition and Uniqueness (Theorem 13.3)

- **Title:** “When the equilibrium is unique”
- **Stability condition:** $\kappa > \gamma \cdot \text{Lip}(\phi)$
- **Calibrated values:**
 - $\kappa = 0.45$
 - $\gamma \cdot \text{Lip}(\phi) \approx 0.1$ (well below κ)
 - Stability: comfortable
- **Theorem 13.3 (uniqueness):** Under the stability condition, the MFG fixed point is unique
- **Proof sketch:**
 - Contraction argument on Φ : $W_1(\Phi(\mu_1), \Phi(\mu_2)) \leq \rho \cdot W_1(\mu_1, \mu_2)$ for some $\rho < 1$
 - Banach fixed-point theorem gives uniqueness
- **When stability fails:**
 - $\gamma \cdot \text{Lip}(\phi) > \kappa$: multiple equilibria possible
 - Market can “tip” between equilibria (analogous to bank runs)
 - This is a research frontier; GE-LAV’s basic calibration assumes stability
- **Speaker notes:** Honest about limits. Stable regime is empirically supported.

34.2.5 Track 2 Existing Slides to Reuse

From Class 8 (McKean-Vlasov MFG) — 15 slides: - MV SDE setup and motivation - Propagation of chaos heuristics - The fixed-point structure

From Class 15 (Mean-Field Equilibrium Fixed Point) — 18 slides: - Numerical fixed-point iteration - Convergence diagnostics - Application to GE-LAV calibration

These slides directly support the proof material.

34.2.6 Track 2 Discussion Questions

1. The propagation of chaos rate is $1/\sqrt{N}$. At $N = 10,000$, error is ~1%. But private markets have *concentration* (a few mega-LPs hold most of the AUM). Does that violate the i.i.d. assumption? How would you account for concentration?
2. The stability condition $\kappa > \gamma$ is satisfied empirically. But in a hypothetical “panic” scenario, could the system enter an unstable regime? What would the off-equilibrium dynamics look like?

3. The MFG framework assumes Markov dynamics. Real markets have memory (regret, anchoring). How would non-Markov extensions change the proofs?
-

34.3 Both Tracks: Cross-Audit

Students in either track may audit the other track's session via recording. Track 1 students benefit from hearing the MFG existence proof (it explains *why* GE-LAV works); Track 2 students benefit from the applied VC case (it grounds the math).

[← Session 25](#) | [Schedule](#) | [Next: Session 27 →](#)

35 Session 27: Split Track — Infrastructure Case / Fokker-Planck & Master Equation

Unit	5 — Split Track
Track 1 source	Class 22 (Infrastructure: The 57% Illusion)
Track 2 source	Class 6 (Fokker-Planck Equations)

35.1 Track 1: Infrastructure Case — “The 57% Illusion”

35.1.1 Track 1 Learning Objectives

By the end of this session, Track 1 students will be able to:

1. **Apply** GE-LAV to a 20+ year infrastructure asset (longest duration in private markets).
2. **Identify** why long-duration assets carry the largest accumulated Jensen bias.
3. **Compute** the “57% illusion” — the gap between DCF and GE-LAV for stressed infrastructure positions.
4. **Translate** GE-LAV findings into pension fund and insurance regulatory language.
5. **Connect** infrastructure valuation errors to systemic financial stability concerns.

35.1.2 Track 1 Pre-Class Assignment

- **Read:** Book Chapter 22 case (or Class 22 slides if available)
- **Optional:** Global Infrastructure Hub report on private infrastructure valuation

35.1.3 Track 1 In-Class Outline (75 minutes)

Time	Segment	Format
0:00–0:10	Why infrastructure is special: duration + cash flow stability	Lecture
0:10–0:25	The “57% illusion” decomposed	Lecture
0:25–0:55	Lab: value a real infrastructure asset (Indianapolis Toll Road or comparable)	Group work

Time	Segment	Format
0:55–1:15	Group recommendations + regulatory implications	Presentations

35.1.4 Track 1 Slide Briefs (New Slides Needed)

Slide T1-27.1 — Infrastructure: The Longest Horizon Asset

- **Title:** “Why infrastructure breaks DCF the hardest”
- **Three features:**
 - **Horizon:** 20-50 years (PPPs sometimes 99 years)
 - **Cash flow stability:** Contracted revenues (tolls, regulated rates) → low operational volatility
 - **Liquidity sensitivity:** Long-duration asset; tiny premium changes have huge value effects
- **Result:** Infrastructure has *the largest* Jensen bias and *the largest* equilibrium correction in stressed regimes
- **Speaker notes:** Frame the asset class. Most students won’t have valued infrastructure deeply.

Slide T1-27.2 — The 57% Illusion (The Master Number)

- **Title:** “\$1B infrastructure: DCF says \$1B, GE-LAV at GFC says \$430M”
- **Decomposition:**
 - DCF mark: \$1.000B (constant 7% rate, 30-year asset)
 - LAV mark (normal): \$1.108B (Jensen bias 10.8% for 30 yr)
 - GE-LAV mark (normal): \$1.000B (small equilibrium correction)
 - **GE-LAV mark (GFC depth): \$0.430B (57% gap to DCF)**
- **Why so extreme:**
 - 30-year duration amplifies discount rate changes
 - At GFC depth, equilibrium rate ~32% vs. DCF 7%
 - $(1.07/1.32)^{30} \approx 0.43$
- **Speaker notes:** This is the *single most powerful number* for the regulatory case. Cite this routinely.

Slide T1-27.3 — Implications for Pension Funds

- **Title:** “What this means for your pension fund”
- **Standard pension fund infrastructure portfolio:**
 - ~10% allocation to infrastructure
 - Duration: 15-25 years average
 - Reported NAVs: smooth, stable
- **Under GE-LAV:**
 - In stressed regime, reported NAVs overstate true value by 30-57%

- Hedge demand calls for ~3-5 percentage points lower allocation
- Stress test exposes 40-60% downside (vs. 25% under standard methodology)
- **Speaker notes:** Connect to specific public pension funds (CalPERS, CalSTRS, Oregon, ABP) — all heavily exposed to infrastructure.

Slide T1-27.4 — Insurance Implications

- **Title:** “Insurance regulatory implications”
- **Solvency II treatment:**
 - Long-duration infrastructure: capital charge 25-39%
 - Matched against long-term liabilities (annuities, life)
 - Conventional wisdom: “good match” between asset and liability duration
- **GE-LAV finding:**
 - Asset side has hidden stress exposure (the 57% illusion)
 - Liability side does not adjust during liquidity stress
 - Match is *worse* than reported
- **Speaker notes:** This is a *regulatory* angle. The Solvency II review process in EU is the relevant policy channel.

Slide T1-27.5 — Lab Setup

- **Title:** “Lab: value the Indianapolis Toll Road”
- **Setup:**
 - 75-year concession (acquired 2006 for \$3.85B)
 - Annual toll revenues ~\$200M, growing 2%/yr
 - LP investor: ITR Concession Co (Macquarie, etc.)
 - Question: what’s it worth today under DCF vs. GE-LAV? Where would secondaries clear?
- **Group output:**
 - DCF baseline
 - GE-LAV at normal liquidity
 - GE-LAV at GFC-depth stress
 - “If you had to sell today, what’s your reservation price?”
- **Speaker notes:** Real asset. Real numbers. Tangible.

35.1.5 Track 1 Existing Slides to Reuse

From Class 22 (Infrastructure: The 57% Illusion) — 13 slides: - The headline “57%” chart - Infrastructure duration analysis - Specific case studies (Indianapolis Toll Road, Chicago Skyway, etc.) - Regulatory implications

Use largely as-is.

35.2 Track 2: Fokker-Planck & Master Equation Derivations

35.2.1 Track 2 Learning Objectives

By the end of this session, Track 2 students will be able to:

1. **Derive the Fokker-Planck equation** from the SDE via Chapman-Kolmogorov.
2. **Prove well-posedness** of the GE-LAV Fokker-Planck (Theorem 14.2).
3. **State and apply the master equation** for value functions on Wasserstein space (Theorem 15.1).
4. **Implement** a Fokker-Planck solver for the GE-LAV liquidity state.
5. **Verify numerical convergence** of the FP solution to the analytical stationary distribution.

35.2.2 Track 2 Pre-Class Assignment

- **Read:** Book Chapters 14, 15 in full (with proofs)
- **Pre-read:** Pavliotis, *Stochastic Processes and Applications*, Chapters 4–5; or Risken, *The Fokker-Planck Equation*, Chapter 4

35.2.3 Track 2 In-Class Outline (75 minutes)

Time	Segment	Format
0:00–0:15	Derivation of FP from Chapman-Kolmogorov	Lecture + board
0:15–0:35	Well-posedness (existence + uniqueness): Theorem 14.2	Lecture
0:35–0:55	The master equation on Wasserstein space	Lecture
0:55–1:10	Numerical FP solver	Code walkthrough
1:10–1:15	Verification: numerical $\rightarrow$ analytical stationary	Discussion

35.2.4 Track 2 Slide Briefs (New Slides Needed)

Slide T2-27.1 — Derivation: Fokker-Planck from Chapman-Kolmogorov

- **Title:** “Deriving FP from first principles”
- **Step 1:** Chapman-Kolmogorov equation:
$$-p(x, t + \Delta t) = \int p(x, t + \Delta t | y, t) \cdot p(y, t) dy$$
- **Step 2:** Transition kernel for $dX = fdt + g dW$ at small Δt :

$$-p(x, t + \Delta t | y, t) \approx \frac{1}{\sqrt{2\pi g^2 \Delta t}} \exp\left(-\frac{(x - y - f\Delta t)^2}{2g^2 \Delta t}\right)$$

- **Step 3:** Taylor expand $p(y, t)$ around $y = x$:

$$- p(y, t) = p(x, t) + (y - x)p_x(x, t) + \frac{1}{2}(y - x)^2 p_{xx}(x, t) + O((y - x)^3)$$

- **Step 4:** Substitute, integrate, take $\Delta t \rightarrow 0$:

$$- \frac{\partial p}{\partial t} = - \frac{\partial}{\partial x}[fp] + \frac{1}{2} \frac{\partial^2}{\partial x^2}[g^2 p]$$

- **Speaker notes:** Walk through each step. This is the *fundamental* derivation. Students should be able to reproduce.

Slide T2-27.2 — Well-Posedness (Theorem 14.2)

- **Title:** “Well-posedness of the GE-LAV Fokker-Planck”
- **Theorem 14.2:** For the GE-LAV FP equation with calibrated drift and diffusion:
 - **Existence:** A weak solution $p(L, t)$ exists for all $T > 0$
 - **Uniqueness:** The solution is unique in $L^2([0, T] \times \mathbb{R})$
 - **Regularity:** p is smooth in the interior; satisfies C^∞
- **Proof sketch:**
 - Hörmander’s hypoellipticity theorem (sufficient regularity from nondegenerate diffusion)
 - L^2 energy estimates for uniqueness
 - Maximum principle for boundedness
- **Speaker notes:** State the theorem rigorously. Sketch the structure of the proof. Refer to book for full argument.

Slide T2-27.3 — Stationary Distribution Derivation

- **Title:** “Stationary density of OU”
- **Setup:** Set $\partial p / \partial t = 0$ in the FP equation:
 - $0 = -\kappa \frac{d}{dL}[(\bar{L} - L)p_\infty(L)] + \frac{\sigma_L^2}{2} \frac{d^2}{dL^2} p_\infty(L)$
- **Solve via integrating factor:**
 - Multiply by $\exp(\kappa(L - \bar{L})^2 / \sigma_L^2)$
 - Get $\frac{d}{dL}[\text{flux}] = 0$
 - Therefore flux = 0 (since density decays at $\pm\infty$)
 - Setting flux to zero: $-\kappa(\bar{L} - L)p + \frac{\sigma_L^2}{2} p' = 0$
- **Solve ODE:** $p_\infty(L) = C \exp(-\kappa(L - \bar{L})^2 / \sigma_L^2)$
- **Normalize:** $C = \sqrt{2\kappa / (\pi\sigma_L^2)}$, giving Gaussian with mean $\bar{L}$, variance $\sigma_L^2 / (2\kappa)$
- **Speaker notes:** This is the derivation Track 1 students saw the result of in Session 4. Now Track 2 sees the derivation.

Slide T2-27.4 — The Master Equation on Wasserstein Space

- **Title:** “Master equation: HJB for MFG value functions”
- **Setting:**

- Value function $V(L, t, \mu)$ where μ is the population distribution
- μ lives in Wasserstein space $\mathcal{P}_2(\mathbb{R})$
- **Master equation (informally):**

$$-\frac{\partial V}{\partial t} + \mathcal{L}V - r(L)V + \mathcal{H}(V_\mu) + C = 0$$
- **Where:**
 - $\mathcal{L}V$ = standard HJB operator on L
 - $V_\mu = \delta V / \delta \mu$ = Wasserstein gradient
 - $\mathcal{H}(V_\mu)$ = Hamiltonian coupling individual to population
- **Speaker notes:** This is the most abstract slide of the session. Don't expect deep mastery — just recognition that the master equation exists and is solvable in principle.

Slide T2-27.5 — Numerical FP Solver

- **Title:** “Implementing the FP solver”
- **Pseudocode:**

```
Initialize: p_0(L_i) = delta at L_initial
For each time step:
  Apply finite-difference discretization
  Use Crank-Nicolson for stability
  Enforce boundary conditions: p → 0 at |L| → ∞
  Normalize: sum to 1
Output: p(L, t) for all t
```
- **Convergence check:** As $t \rightarrow T$ large, numerical $p(L, T) \rightarrow$ analytical stationary
- **Performance:** ~seconds for full GE-LAV time series
- **Speaker notes:** Provide complete pseudocode. Students should be able to implement in <100 lines of Python.

35.2.5 Track 2 Existing Slides to Reuse

From Class 6 (Fokker-Planck Equations) — 16 slides: - FP derivation and applications - OU stationary distribution - Numerical methods - Connection to backward equations

Use directly.

35.2.6 Track 2 Discussion Questions

1. The FP derivation assumes the transition density is Gaussian at small Δt . This holds for Itô SDEs with smooth coefficients. What if there are jumps (Lévy processes)? How does the FP change?
2. The master equation is on Wasserstein space — an infinite-dimensional manifold. Numerically, we discretize μ on a grid. What's the discretization error? When does this matter?
3. The well-posedness theorem requires nondegeneracy ($\sigma_L > 0$). What happens if $\sigma_L = 0$ (deterministic OU)? Does the framework break down?

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36 Session 28: Split Track — Private Credit Case / GE Equilibrium Existence

Unit	5 — Split Track
Track 1 source	Class 23 (Private Credit Spread Decomposition)
Track 2 source	Class 17 (GE Market Clearing)

36.1 Track 1: Private Credit Case Workshop

36.1.1 Track 1 Learning Objectives

By the end of this session, Track 1 students will be able to:

1. **Apply** GE-LAV to a private credit / direct lending position.
2. **Decompose** private credit spreads into credit risk, illiquidity premium, and complexity premium components.
3. **Distinguish** GE-LAV for credit vs. equity strategies (cash flow profile, default risk interaction).
4. **Compute** liquidity-adjusted spread expectations for a direct lending strategy.
5. **Evaluate** a real BDC (business development company) or middle-market direct lending fund.

36.1.2 Track 1 Pre-Class Assignment

- **Read:** Recent BDC quarterly report or direct lending fund presentation
- **Watch:** 10-min video on private credit market structure (link on course site)

36.1.3 Track 1 In-Class Outline (75 minutes)

Time	Segment	Format
0:00–0:10	Private credit vs. PE: structural differences	Lecture
0:10–0:30	Decomposing private credit spreads	Lecture + worked example

Time	Segment	Format
0:30–0:55	Lab: value a BDC or middle-market direct lending position	Group work
0:55–1:15	Group recommendations + risk implications	Presentations

36.1.4 Track 1 Slide Briefs (New Slides Needed)

Slide T1-28.1 — Private Credit: Different Risk Profile

- **Title:** “Private credit is debt, not equity”
- **Key differences from PE buyout:**
 - **Senior in capital structure:** Lower expected return, lower variance
 - **Shorter duration:** Most direct lending is 3-7 years
 - **Credit risk:** Default + recovery rather than equity participation
 - **Floating rate:** Most direct lending pays SOFR + spread
- **GE-LAV implications:**
 - Smaller hedge demand (less correlation with aggregate liquidity)
 - Smaller Jensen bias (shorter duration)
 - But: in stress, credit and liquidity premia *interact* (credit spreads widen when liquidity stresses)
- **Speaker notes:** Students often think credit is “PE without the equity story.” It’s structurally different. Establish that.

Slide T1-28.2 — Decomposing Private Credit Spreads

- **Title:** “What’s in a 700 bps spread?”
- **Typical direct lending spread:** 5-8% above SOFR
- **Decomposition:**
 - **Credit risk:** 200-400 bps (expected losses + idiosyncratic risk premium)
 - **Illiquidity premium:** 100-300 bps (GE-LAV territory)
 - **Complexity premium:** 50-100 bps (origination + monitoring costs)
 - **Total:** 350-800 bps
- **GE-LAV contribution:** Quantifies the illiquidity component stochastically
- **Speaker notes:** Spreads are *not* pure credit risk. Many investors don’t decompose carefully.

Slide T1-28.3 — Private Credit Sharpe Ratio Illusion

- **Title:** “Private credit’s worst-in-class Sharpe penalty”
- **Recall Session 7’s slide:** Private credit Sharpe ratio drops 84% under liquidity adjustment (highest of any private asset class)
- **Why:**
 - High *conventional* Sharpe (~ 0.5) $\rightarrow$ looks attractive

- But low return volatility *and* high illiquidity premium = denominator vastly understates true risk
- Adjusted Sharpe (~ 0.08) reveals private credit may *not* be the “safe income” alternative many believe
- **Implication:** “Private credit allocations sized on Sharpe-based optimization are likely substantially over-sized”
- **Speaker notes:** This is a *career-relevant* finding. Many institutions have piled into private credit. The framework says they may be over-exposed.

Slide T1-28.4 — Lab Setup: BDC Valuation

- **Title:** “Lab: a real BDC valuation”
- **Setup:**
 - Choose a publicly-traded BDC (ARCC, MAIN, ORCC, or similar)
 - Public market price: known (one number)
 - NAV: reported quarterly
 - Question: Is the public market premium/discount to NAV consistent with GE-LAV?
- **Group output:**
 - Estimate the “fair” NAV under GE-LAV
 - Compare to reported NAV (typically equals book value of loans)
 - Compare to public market price
 - Identify any arbitrage opportunity or risk signal
- **Speaker notes:** BDCs offer a *unique* test: they’re publicly traded but hold private assets. The discount/premium structure is rich data.

Slide T1-28.5 — Cross-Asset Risk: Credit + Liquidity Interaction

- **Title:** “When credit and liquidity stress together”
- **Setup:** 2008-2010 GFC and 2020 COVID both saw simultaneous credit + liquidity stress
- **In private credit:**
 - Default rates spiked
 - Secondary discounts widened
 - Recovery rates fell
 - All three correlated $\rightarrow$ much worse outcome than independence assumption
- **GE-LAV captures this:**
 - Cross-asset β s in the multi-asset framework
 - Stress test for joint credit+liquidity events
 - More conservative capital recommendations
- **Speaker notes:** This is *why* the integrated framework matters. Independent risk models miss the correlations.

36.1.5 Track 1 Existing Slides to Reuse

From Class 23 (Private Credit Spread Decomposition) — 12 slides: - Private credit market structure
- Spread decomposition methodology - BDC analysis tools - Default + liquidity interaction

Use largely as-is.

36.2 Track 2: GE Equilibrium Existence Proof

36.2.1 Track 2 Learning Objectives

By the end of this session, Track 2 students will be able to:

1. **State and prove the existence of GE-LAV competitive equilibrium** (Theorem 17.4).
2. **Apply Schauder's fixed-point theorem** to the market-clearing map.
3. **Identify the market clearing condition** mathematically: aggregate supply = aggregate demand for liquidity.
4. **Distinguish** partial equilibrium (LAV) from general equilibrium (GE-LAV) at the level of the underlying math.
5. **Connect** the existence proof to the calibrated parameter regime where uniqueness holds.

36.2.2 Track 2 Pre-Class Assignment

- **Read:** Book Chapter 17 in full (with all proofs)
- **Pre-read:** Mas-Colell, *The Theory of General Economic Equilibrium*, Chapters 4-5

36.2.3 Track 2 In-Class Outline (75 minutes)

Time	Segment	Format
0:00–0:10	Recap: from MFG to GE	Lecture
0:10–0:30	The GE-LAV equilibrium definition	Lecture
0:30–0:50	Existence proof via Schauder	Lecture + board
0:50–1:05	Uniqueness under stability	Lecture
1:05–1:15	Numerical verification	Discussion

36.2.4 Track 2 Slide Briefs (New Slides Needed)

Slide T2-28.1 — GE-LAV Equilibrium: Definition

- **Title:** “GE-LAV competitive equilibrium”
- **Definition:** An equilibrium is a tuple $(\mu_t^*, \lambda^*(L, \mu^*), u^*(L, \mu^*))$ such that:
 1. **Individual optimality:** $u^*(L, \mu^*)$ solves the HJB given λ^*
 2. **Market clearing:** $\lambda^*(L, \mu^*)$ is the unique premium for which aggregate exit flows balance (sellers = buyers)
 3. **Distributional consistency:** μ_t^* is the law of L_t under dynamics induced by u^*
- **In one sentence:** “Everyone’s optimal decisions, given the equilibrium premium, generate the distribution that produces that premium.”
- **Speaker notes:** Set up the formal equilibrium notion. The three conditions are mutually self-consistent.

Slide T2-28.2 — Existence Theorem 17.4

- **Title:** “Theorem 17.4: existence of GE-LAV equilibrium”
- **Statement:** Under standard regularity conditions on ϕ , $r(L)$, and $C(L, t)$:
 - There exists at least one GE-LAV competitive equilibrium tuple (μ^*, λ^*, u^*)
 - Equilibrium is unique under the stability condition $\kappa > \gamma \cdot \text{Lip}(\phi)$
- **Proof outline:**
 - Define an equilibrium map $\Psi : \mathcal{M} \rightarrow \mathcal{M}$:
 - * $\Psi(\mu) =$ the new measure induced by optimal individual decisions
 - Show Ψ is continuous and compact on $\mathcal{M}$
 - Apply Schauder’s fixed-point theorem
- **Speaker notes:** State the theorem; sketch the structure. Full proof in book Section 17.4.

Slide T2-28.3 — Schauder Fixed-Point Argument (Detail)

- **Title:** “Schauder applied to the equilibrium map”
- **The setup:**
 - $\mathcal{M} =$ closed, bounded, convex subset of $\mathcal{P}_2(\mathbb{R})$ (probability measures with finite second moment)
 - Ψ continuous (HJB and market clearing both stable in measure)
 - Ψ compact (uniform bounds on second moments)
- **The conclusion:** Ψ has a fixed point in $\mathcal{M}$
- **Why Schauder, not Brouwer?**
 - Brouwer requires finite dimension
 - $\mathcal{M}$ is infinite-dimensional
 - Schauder is the appropriate generalization
- **Speaker notes:** Tie to functional analysis. Most Track 2 students have seen Schauder in PDE classes.

Slide T2-28.4 — Market Clearing as Fixed Point

- **Title:** “Market clearing: a fixed point on λ ”
- **For given μ , λ^* satisfies:**
 - Aggregate exit flow at premium $\lambda^* =$ Aggregate buy flow at premium λ^*
- **Concrete form:**
 - $\sum_i [u_i^*(L_i, \mu, \lambda^*) \cdot \text{notional}_i] = 0$
- **Where:**
 - $u_i^* = 1$ if LP i exits at premium λ^* , else 0
 - $\text{notional}_i =$ LP i 's position size
- **Iteration:** Bisection on λ^* until aggregate flow = 0
- **Speaker notes:** Make the market clearing concrete. It's just supply = demand at the right price.

Slide T2-28.5 — Connection to Numerical Algorithm

- **Title:** “From proof to platform”
- **The existence proof guarantees the algorithm converges:**
 - Start with guess $\mu^{(0)}$
 - Solve HJB $\rightarrow$ get $u^{*(0)}$
 - Solve FP $\rightarrow$ get $\mu^{(1)}$
 - Iterate: $\mu^{(0)} \rightarrow \mu^{(1)} \rightarrow \mu^{(2)} \rightarrow \dots$
- **Under stability condition:**
 - Iteration converges to the unique fixed point
 - GE-LAV platform implements this
- **Computational cost:**
 - Typically 10-20 iterations
 - Total time: seconds to minutes
- **Speaker notes:** The platform's algorithm is *based* on the existence theorem. Theory $\rightarrow$ practice.

36.2.5 Track 2 Existing Slides to Reuse

From Class 17 (GE Market Clearing & Endogenous Pricing) — 14 slides: - Market clearing diagrams - Equilibrium existence sketches - Numerical examples of λ^* computation - Comparison with partial equilibrium

Use largely as-is.

36.2.6 Track 2 Discussion Questions

1. The existence proof requires that Ψ is *continuous* on $\mathcal{M}$. What kind of discontinuity could break this? When would discontinuity be economically realistic (e.g., regulatory thresholds, IPS triggers)?
2. Under the stability condition $\kappa > \gamma \cdot \text{Lip}(\phi)$, uniqueness holds. If the condition fails *temporarily* during a stress event but is restored as conditions normalize, what does the system look like during the failure period?
3. The market clearing condition treats sellers and buyers symmetrically. In real markets, intermediaries (brokers) take spread. How would the proof change if we added market makers?

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37 Session 29: Split Track — Digital Assets & Climate / Jensen + Pigouvian Proofs

Unit	5 — Split Track
Track 1 source	Classes 30 (Digital Assets) + 31 (Climate Risk)
Track 2 source	Classes 11 (Jensen Bias) + 18 (Welfare) + 20 (Pigouvian Tax)

37.1 Track 1: Frontier Cases — Digital Assets + Climate

37.1.1 Track 1 Learning Objectives

By the end of this session, Track 1 students will be able to:

1. **Apply** GE-LAV to crypto-adjacent investments (token venture funds, defi protocols, NFT funds).
2. **Adapt** the calibration when secondary market data is thin (or doesn't exist yet).
3. **Apply** GE-LAV to ultra-long-duration climate-exposed infrastructure (50-100 year assets).
4. **Decompose** climate risk into transition + physical risk channels.
5. **Identify** how GE-LAV's stochastic-premium framework handles novel risks (climate transition, regulatory uncertainty).

37.1.2 Track 1 Pre-Class Assignment

- **Skim:** Recent report on crypto fund returns or NFT secondaries (link on course site)
- **Skim:** Climate-aware infrastructure investing report (Bloomberg NEF or similar)

37.1.3 Track 1 In-Class Outline (75 minutes)

Time	Segment	Format
0:00–0:15	Digital assets in private markets	Lecture
0:15–0:30	Applying GE-LAV when secondary data is thin	Lecture

Time	Segment	Format
0:30–0:50	Climate risk and ultra-long-duration assets	Lecture
0:50–1:05	Lab: value a climate-exposed infrastructure asset	Group work
1:05–1:15	Synthesis + research frontier discussion	Discussion

37.1.4 Track 1 Slide Briefs (New Slides Needed)

Slide T1-29.1 — Digital Assets in Private Markets

- **Title:** “Crypto’s slow merger with private markets”
- **Categories:**
 - **Token venture funds:** Like VC but invest in pre-launch protocols (Paradigm, A16Z Crypto)
 - **Defi liquidity provision:** Quasi-fund vehicles providing AMM liquidity
 - **NFT funds:** Collectibles as private market assets
 - **Tokenized real-world assets:** Real estate, art, private credit fractionalized on-chain
- **GE-LAV applicability:**
 - Most categories have *some* secondary market data (varies by asset)
 - Calibration is harder (less history, more regime variability)
 - But: the *structure* of the framework applies — stochastic illiquidity premia, McKean-Vlasov coupling
- **Speaker notes:** Establish category. Many students will work in or near crypto post-graduation.

Slide T1-29.2 — Calibration with Limited Data

- **Title:** “What to do when you have 6 quarters, not 60”
- **Approaches:**
 - **Borrow calibration:** Use similar-asset-class parameters as prior, adjust on local data
 - **Bayesian methods:** Treat OU parameters as random variables with priors from comparables
 - **Robust valuation:** Compute GE-LAV across multiple plausible parameter sets; report range
- **Cautionary note:** “Calibration uncertainty is a real cost. Confidence intervals on $\hat{\kappa}$ widen substantially with less data.”
- **Speaker notes:** Be honest. Crypto/digital is a frontier; calibration is hard. Useful but not bulletproof.

Slide T1-29.3 — Climate Risk: Transition vs. Physical

- **Title:** “Two kinds of climate risk in private markets”

- **Transition risk:**
 - Regulatory changes (carbon pricing, EV mandates, building codes)
 - Asset stranding (fossil fuel infrastructure)
 - Affects: traditional infrastructure, energy, real estate
- **Physical risk:**
 - Climate-driven asset damage (floods, wildfires, heat)
 - Operational disruption
 - Insurance cost increases
 - Affects: coastal real estate, agriculture, utilities
- **GE-LAV channel:**
 - Both risks affect $\bar{L}$ (long-run mean liquidity may shift)
 - Both affect σ_L (volatility may increase)
 - Long-duration assets (infrastructure) bear the highest climate exposure under GE-LAV
- **Speaker notes:** Climate-aware investing is rapidly mainstreaming. GE-LAV is a natural framework.

Slide T1-29.4 — Ultra-Long-Duration Climate-Exposed Assets

- **Title:** “50-100 year climate-exposed assets”
- **Examples:**
 - Coastal port infrastructure (sea level rise + storm intensity)
 - Hydroelectric dams (precipitation patterns + drought)
 - Long-lived water infrastructure (drought + supply reliability)
 - Carbon sequestration facilities (regulatory + technological risk)
- **GE-LAV at these horizons:**
 - Jensen bias is massive ($B(50) \approx 50\%+$ at calibrated parameters)
 - $\bar{L}$ shift assumptions matter enormously
 - DCF systematically misvalues by 30-70% even in baseline scenarios
- **Speaker notes:** Pension funds with ESG mandates increasingly hold these. Risk is under-measured.

Slide T1-29.5 — Lab Setup

- **Title:** “Lab: a real climate-exposed infrastructure asset”
- **Choose one:**
 - Port of Miami (coastal, 50-year horizon)
 - Hoover Dam (regulated, drought-exposed)
 - Tesla Gigafactory Nevada (regulatory + supply chain)
 - Indian solar farm (regulatory + climate variable)
- **Group output:**
 - DCF baseline (likely misvalued)
 - GE-LAV under stable $\bar{L}$

- GE-LAV under $\bar{L}$ -shift scenario (mild + severe climate transition)
- Recommendation for ESG-aware allocators
- **Speaker notes:** Tangible. Connects to specific assets students will encounter.

37.1.5 Track 1 Existing Slides to Reuse

From Class 30 (Digital Assets) — 11 slides: - Crypto market structure overview - Calibration challenges for digital assets

From Class 31 (Climate Risk) — 12 slides: - Climate risk taxonomy - Long-duration asset analysis - Regulatory landscape for climate-aware investing

Use largely as-is.

37.2 Track 2: Jensen Bias + Pigouvian Tax + Welfare Derivations

37.2.1 Track 2 Learning Objectives

By the end of this session, Track 2 students will be able to:

1. **Prove Theorem 18.1** (Jensen convexity bias) — the closed-form formula derivation.
2. **Prove Theorem 19.1** (constrained Pareto inefficiency) — that decentralized exit decisions are not socially optimal.
3. **Derive the Pigouvian exit tax** as the welfare-maximizing corrective instrument.
4. **Quantify the welfare gap** in the calibrated framework (~2.3%/yr).
5. **Discuss** the regulatory implementation considerations.

37.2.2 Track 2 Pre-Class Assignment

- **Read:** Book Chapters 18, 19 in full (with proofs)
- **Pre-read:** Mas-Colell, Whinston, Green, *Microeconomic Theory*, Chapters 11-13 (externalities and public goods)

37.2.3 Track 2 In-Class Outline (75 minutes)

Time	Segment	Format
0:00–0:10	Recap: the externality concept	Lecture
0:10–0:30	Jensen bias theorem and proof	Lecture + board
0:30–0:50	Welfare gap: quantifying the externality	Lecture
0:50–1:05	Pigouvian tax derivation	Lecture
1:05–1:15	Implementation considerations	Discussion

37.2.4 Track 2 Slide Briefs (New Slides Needed)

Slide T2-29.1 — Jensen Bias Theorem 18.1 (Statement)

- **Title:** “Theorem 18.1: closed-form Jensen bias”
- **Theorem:** For an asset with cash flows $C(L, t)$ and OU dynamics with calibrated parameters, the Jensen convexity bias is:

$$- B(T) = (\pi_2/2) \cdot \Pi_{liq}(T) \cdot \langle V^{DCF} \rangle$$

- **Where:**
 - $\Pi_{liq}(T) = \sigma_L^2 [T - (1 - e^{-2\kappa T})/(2\kappa)]$
 - π_2 = quadratic coefficient of premium function
 - $\langle V^{DCF} \rangle$ = average DCF value
- **Result:** B is positive (since $\pi_2 > 0$), increasing in T , and computable from first principles
- **Speaker notes:** State the theorem precisely. The proof follows.

Slide T2-29.2 — Jensen Bias Theorem 18.1 (Proof Sketch)

- **Title:** “Proof of Theorem 18.1”
- **Step 1:** Apply Itô’s lemma to $\exp(-\int_0^t r(L_s)ds)$
- **Step 2:** Expand around the long-run mean $\bar{L}$:
 - $r(L) = r(\bar{L}) + r'(\bar{L})(L - \bar{L}) + \frac{1}{2}r''(\bar{L})(L - \bar{L})^2 + O((L - \bar{L})^3)$
- **Step 3:** Compute $E[\int_0^t r(L_s)ds]$ using OU moments:
 - First moment vanishes by mean reversion
 - Second moment contributes the Jensen correction
- **Step 4:** Use convexity of exp and Jensen’s inequality
- **Step 5:** Substitute and simplify; obtain the closed form
- **Speaker notes:** Walk through key steps. Full proof in book Section 18.3 (~10 pages of details).

Slide T2-29.3 — Welfare Gap Theorem 19.1

- **Title:** “Theorem 19.1: constrained Pareto inefficiency”
- **Statement:** The decentralized equilibrium (μ_d^*, u_d^*) is Pareto-suboptimal:
 - Aggregate welfare $W(\mu_d^*, u_d^*) < W(\mu_s^*, u_s^*)$
 - Where (μ_s^*, u_s^*) is the centralized social planner solution
- **Welfare gap:** $\Delta W = W(\mu_s^*, u_s^*) - W(\mu_d^*, u_d^*) \approx 2.3\%/yr$ at calibrated parameters
- **Source of inefficiency:** Each LP’s exit imposes externalities on others; private incentives don’t reflect this
- **Speaker notes:** Set up the welfare comparison. The 2.3% number was discussed in Session 24; now we derive it.

Slide T2-29.4 — Welfare Gap Decomposition

- **Title:** “Where the 2.3%/yr comes from”

- **Decomposition:**
 - Path bias from individual Jensen (each LP overstates value): ~0.5%/yr
 - Coordination failure (forced sales widen everyone's discounts): ~1.5%/yr
 - Misallocation (LPs hold sub-optimally given true equilibrium): ~0.3%/yr
 - **Total:** ~2.3%/yr aggregate welfare loss
- **Distribution:**
 - Bigger losses in deep-stress regimes (most coordination failures)
 - Bigger losses for long-duration assets (more Jensen accumulation)
 - Bigger losses for less sophisticated LPs (slower to recognize)
- **Speaker notes:** Quantitative breakdown. Useful for regulatory presentation.

Slide T2-29.5 — Pigouvian Tax Derivation

- **Title:** “The Pigouvian exit tax: closed form”
- **Result (Theorem 19.2):** The welfare-maximizing tax rate on secondary exits is:
 - $\tau^*(L) = \gamma \cdot \text{Lip}(\phi) \cdot \frac{\partial W^*}{\partial \mu}|_{\text{exit}} \cdot \frac{1}{V_{\text{exit}}}$
- **At calibrated parameters:**

L State	$\tau^*(L)$
0	0.0%
-1	3.5%
-1.5	7.0%
-2.0	9.0%

- **Effect:** With Pigouvian tax in place, decentralized equilibrium = social optimum
- **Welfare improvement:** ~2.3%/yr (closes the gap)
- **Speaker notes:** Connect to Pigou (1920), modern carbon pricing analogies.

37.2.5 Track 2 Existing Slides to Reuse

From Class 11 (Jensen Bias Four-Channel Derivation) — 20 slides: - Full proof of the Jensen bias formula - Decomposition by channel - Numerical verification

From Class 18 (Welfare Analysis) — 13 slides: - Welfare gap quantification - Decentralized vs. centralized comparison

From Class 20 (Pigouvian Tax Design) — 15 slides: - Pigouvian tax derivation - Numerical examples - Implementation considerations

Substantial reuse.

37.2.6 Track 2 Discussion Questions

1. The Pigouvian tax is welfare-maximizing in theory. But it requires the regulator to *know* the population state μ in real time. In practice, this is hard. What's the welfare cost of using a *constant* tax rate (e.g., 3%) instead?
2. The 2.3%/yr welfare gap aggregates to ~\$300B/year. Of this, what fraction would be recovered by an *imperfect* Pigouvian tax (say, only on transactions during designated stress windows)?
3. The Pareto inefficiency theorem assumes LPs are rational. Behavioral biases (loss aversion, herding) would likely amplify the externality. How would the welfare gap change with behavioral agents?

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38 Session 30: Project Draft Workshop · Valuation Hierarchy Proof (T2)

Unit	5 — Split Track
Track 1	Project draft workshop (no new content)
Track 2	Valuation hierarchy proof (Theorem 20.1) + project draft workshop
Track 2 source	Class 20 (Pigouvian Tax remainder, hierarchy proof)
Assessment milestone	Project draft due (start of class)

! Project draft due today

Both tracks submit their 15-page project draft + supporting materials at the start of class. Peer review activity uses these drafts.

38.1 Both Tracks: Project Draft Workshop (60 minutes)

38.1.1 Workshop Learning Objectives

By the end of this workshop, students will be able to:

1. **Provide structured peer review** on a classmate's GE-LAV project draft.
2. **Identify common weaknesses** in their own draft based on peer feedback.
3. **Plan revisions** for the final paper and presentation.
4. **Practice** delivering the project's "elevator pitch" — the IC summary in 2 minutes.

38.1.2 Workshop Structure (Both Tracks, Same Activity)

Time	Activity	Detail
0:00–0:05	Logistics: pair assignments, rubric distribution	Lecture
0:05–0:25	Reading time: read partner's draft + take notes	Individual

Time	Activity	Detail
0:25–0:50	Peer feedback session (12.5 min each direction)	Pair work
0:50–1:00	Wrap-up: list 3 priority revisions for your draft	Individual

38.1.3 Peer Review Rubric

Each student provides written feedback to their partner addressing:

- 1. Clarity (5 questions, 1-5 scale)** - Is the asset/question stated precisely? - Is the methodology clear? - Are the calibrations documented? - Are the results easy to follow? - Is the recommendation defensible?
- 2. Rigor (5 questions)** - Is the GE-LAV implementation correct? - Are the comparisons (DCF vs. LAV vs. GE-LAV) accurate? - Are sensitivity analyses adequate? - Is the data appropriate? - For Track 2: are mathematical derivations correct?
- 3. Communication (3 questions)** - Could you explain this paper to a non-finance friend? Try in 2 sentences. - Is the IC-relevance clear? - Is the writing professional quality?
- 4. Improvement priorities (open)** - “The single most important revision before the final paper would be...” - “The presentation should emphasize...” - “One thing that would *delete* if I had to cut to 12 pages...”

38.1.4 Workshop Slide Briefs (New Slides Needed)

Slide 30.1 — Workshop Introduction

- **Title:** “Project Draft Workshop”
- **Goals:** Identify weaknesses in your own work via someone else’s perspective
- **Norms:**
 - Be specific, not vague (“the GE-LAV calculation is unclear at step 3” not “I’m confused”)
 - Be constructive, not destructive
 - Focus on the work, not the person
 - Both you and your partner are still learning
- **Speaker notes:** Set the tone for peer review. Many students are uncomfortable giving direct feedback. Frame as professional courtesy.

Slide 30.2 — Peer Review Process

- **Title:** “How the next hour works”
- **Time blocks:**
 - 20 min: Read your partner’s draft carefully (don’t skim)
 - 25 min: Take turns giving and receiving feedback (12.5 min each)
 - 10 min: Capture YOUR top 3 revision priorities

- **Speaker notes:** Discipline the structure. Without time limits, peer review degrades into chat.

Slide 30.3 — Common Weaknesses to Watch For

- **Title:** “Common weaknesses across project drafts”
- **For Track 1 (Practitioner) drafts:**
 - Asset specification too vague
 - Calibration parameters unjustified
 - GE-LAV computation lacks regime variation
 - Recommendation doesn’t connect to the LAV/GE-LAV result
- **For Track 2 (Researcher) drafts:**
 - Mathematical claims unproven or under-proven
 - Numerical implementation lacks convergence diagnostics
 - Connection to existing literature unclear
 - Extension proposal too ambitious for paper scope
- **Speaker notes:** Make these visible. Both tracks share the same pitfalls in slightly different forms.

Slide 30.4 — What Goes in the Final Paper

- **Title:** “Revising for the final submission”
- **Final paper requirements (Session 32):**
 - 20 pages excluding references and appendices
 - All revisions from peer feedback addressed
 - Specific improvements over draft (state them explicitly in a “Revisions” footnote)
 - Polished writing quality
 - Reproducible computations
- **What gets graded:**
 - 2% improvement over draft
 - 2% writing quality
 - 1% final correctness
- **Speaker notes:** Set clear expectations. The final paper is *not* the draft with minor edits.

38.2 Track 2: Valuation Hierarchy Proof (15 additional minutes)

38.2.1 Track 2 Learning Objectives

By the end of the Track 2 portion, students will additionally be able to:

1. **State and prove Theorem 20.1** (strict valuation hierarchy DCF $\rightarrow$ LAV $\rightarrow$ GE-LAV).

2. **Identify Theorem 20.2** (conditions for approximate equality among the hierarchies).
3. **Compute** the gap between hierarchy levels for arbitrary OU calibrations.
4. **Discuss** why the strict nesting matters for regulatory and pedagogical purposes.

38.2.2 Track 2 Slide Briefs

Slide T2-30.1 — Theorem 20.1: Strict Valuation Hierarchy

- **Title:** “Theorem 20.1: DCF – LAV – GE-LAV”
- **Statement:** Under positive liquidity volatility ($\sigma_L > 0$) and positive general-equilibrium coupling ($\gamma > 0$):
 - $V^{DCF}(L, t) \neq V^{LAV}(L, t)$ (strict inequality somewhere on the state space)
 - $V^{LAV}(L, t) \neq V^{GE-LAV}(L, t)$ (strict inequality)
- **In words:** The three operators give genuinely different answers when liquidity is stochastic and the externality is present.

Slide T2-30.2 — Theorem 20.1 Proof Sketch

- **Title:** “Proof: strict nesting”
- **Part 1 (DCF – LAV):**
 - Jensen’s inequality applied to $e^{-r(L)T}$: $E[e^{-r(L)T}] > e^{-E[r(L)]T}$
 - Strict whenever L is non-degenerate (i.e., $\sigma_L > 0$)
- **Part 2 (LAV – GE-LAV):**
 - GE-LAV adds the equilibrium uplift to $r(L)$: $r_{GE}(L, \mu) = r(L) + \text{externality}(L, \mu)$
 - Strict whenever the externality is positive (i.e., $\gamma > 0$)
- **Combination:** Both inequalities strict $\rightarrow$ strict nesting holds.

Slide T2-30.3 — Theorem 20.2: Conditions for Equality

- **Title:** “When the hierarchies converge”
- **Statement:** $V^{DCF} = V^{LAV} = V^{GE-LAV}$ in the limit:
 - As $\sigma_L \rightarrow 0$: liquidity is deterministic, no path dependence
 - As $\gamma \rightarrow 0$: no externality, partial equilibrium
- **Result:** The framework reduces to DCF in the “no-uncertainty, no-coupling” limit.
- **Speaker notes:** This is the *consistency check*. GE-LAV doesn’t conflict with DCF in DCF’s domain of validity.

38.2.3 Track 2 Discussion Questions

1. The strict nesting theorem requires $\sigma_L > 0$ and $\gamma > 0$. What if $\sigma_L > 0$ but $\gamma = 0$? Is LAV strictly necessary then, or does DCF + simple Jensen suffice?
2. The proof of strict inequality relies on Jensen’s inequality applied pointwise. What if some pricing kernel is approximately linear (e.g., very short horizons)? Does the inequality “vanish” in the limit?

3. Theorem 20.2 (equality conditions) is the “consistency limit.” Is there a *third* limit where GE-LAV reduces to something *other than* DCF? Perhaps when the externality dominates?
-

38.3 Both Tracks: Final Reminders

38.3.1 What’s Due When

Deliverable	Due Date
Project draft + peer review notes	Today (Session 30)
PS4	Session 31
Final paper (with revisions)	Session 32
Project presentation	Session 32

38.3.2 Session 31 Preview

- Both tracks: Final research frontiers + course synthesis
- PS4 due (track-specific)
- Final preparation for project presentations

38.3.3 Session 32 Preview

- Both tracks reunified
 - Project presentations (20 min + 5 min Q&A each)
 - Final wrap-up
-

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39 Session 31: Research Frontiers · Course Synthesis

Unit	5 — Split Track (final session before reunification)
Book Chapter	21 (Future Research Directions)
Track 1 source	Class 26 (Research Frontiers) + Classes 28-29 (advanced)
Track 2 source	Class 26 + Classes 28, 29 (Neural SDE, Network)
Slides needed	T1: ~5 new + ~9 from Class 26; T2: ~5 new + ~14 from Classes 26, 28, 29
Assessment milestone	PS4 due (start of class)

! PS4 due today

Submit PS4 (track-specific) at the start of class. Late penalty applies per syllabus.

39.1 Both Tracks: Course Synthesis (Common 30 Minutes)

Tracks meet together for the first 30 minutes for course synthesis, then split for track-specific frontier discussion.

39.1.1 Common Synthesis Slide Briefs

Slide 31.1 — Title

- **Title:** “Research Frontiers and Course Synthesis”
- **Subtitle:** “Where the framework goes next · What you’ve built”
- **Speaker notes:** Final substantive session. Help students see the arc.

Slide 31.2 — What You’ve Built (Course Arc)

- **Title:** “The 32-session arc, summarized”
- **Visual:** Timeline showing all 32 sessions with milestone markers

- **Five units:**
 - Unit 1 (Sessions 1–5): Why DCF fails
 - Unit 2 (Sessions 6–10): Measurement + theory
 - Unit 3 (Sessions 11–18): Decision + application
 - Unit 4 (Sessions 19–24): Math intuition bridges
 - Unit 5 (Sessions 25–31): Split track
- **Capstone (Session 32):** Project presentations
- **Speaker notes:** Show progression. Students should feel pride.

Slide 31.3 — Key Takeaways for Both Tracks

- **Title:** “What every student should walk away with”
- **Conceptual:**
 - DCF systematically misvalues private market assets
 - Liquidity is stochastic and collectively determined
 - Jensen bias is real and computable
 - The McKean-Vlasov externality creates $\sim 2.3\%/yr$ welfare loss
- **Practical:**
 - GE-LAV provides valuation, exit, portfolio, regulatory tools
 - Asset class differences matter (PE, VC, infrastructure, credit)
 - Cross-asset hedge demand is large in stress
- **Strategic:**
 - The valuation hierarchy DCF $\rightarrow$ LAV $\rightarrow$ GE-LAV is provably strict
 - Adoption will happen gradually
 - Track 1 students will work *with* this framework; Track 2 students will *extend* it

Slide 31.4 — The Most Important Numbers

- **Title:** “Numbers worth memorizing”
- **OU calibration:** $\kappa = 0.45/yr$, $\sigma_L = 0.32$, $\bar{L} = 1.0$
- **Jensen bias 5/10/15/20 yr:** 0.8%/1.7%/2.5%/3.6% (affine approx) or 2.0%/6.2%/11.6%/15.0% (closed-form)
- **Crisis amplification:** $4.31\times$
- **Welfare gap:** $\sim 2.3\%/yr$
- **Hedge demand at normal:** ~ 8 percentage points
- **Speaker notes:** Concrete numbers. Students should know these on exit.

39.2 Track 1: Practitioner Frontiers (45 Additional Minutes)

39.2.1 Track 1 Learning Objectives

By end of Track 1 session, Track 1 students will be able to:

1. **Identify** the most promising practitioner applications of GE-LAV expected over next 5 years.
2. **Discuss** the institutional adoption barriers and how they may be overcome.
3. **Recognize** the open practitioner questions (regime detection, multi-asset extensions, calibration improvements).

39.2.2 Track 1 Slide Briefs

Slide T1-31.1 — Adoption Trajectory

- **Title:** “How GE-LAV will be adopted”
- **Phase 1 (next 1–2 years):** Academic adoption and citation
- **Phase 2 (3–5 years):** Voluntary adoption by sophisticated LPs (pensions, sovereign wealth funds)
- **Phase 3 (5–10 years):** Regulatory inclusion (Solvency II review cycles, IPEV updates)
- **Phase 4 (10+ years):** Standard methodology, replacing DCF as default
- **Speaker notes:** Realistic timeline. Not overnight; not never.

Slide T1-31.2 — Open Practitioner Questions

- **Title:** “What practitioners need next”
- **Real-time regime detection:**
 - When has the calibration changed? Is the current L_t stable or transient?
 - Statistical tools for regime identification needed
- **Multi-asset extensions:**
 - GE-LAV calibrated independently per asset class
 - Cross-asset coupling needs more empirical work
- **Synthetic exposures:**
 - GP-led continuation funds, NAV loans, structured secondaries
 - GE-LAV applies in principle; need clean implementations
- **Speaker notes:** These are research projects students can do post-course.

Slide T1-31.3 — Career Implications

- **Title:** “Where this knowledge fits in your career”
- **For LP allocators:**
 - Defensible methodology for IC decisions
 - Better stress testing
 - Tactical allocation framework
- **For GPs:**
 - Better valuation methodology
 - LP relationship management
 - Continuation fund pricing
- **For consultants and advisors:**

- Differentiated analytical capability
- Regulatory advisory work
- **For regulators:**
 - Quantitative basis for stress testing
 - Pigouvian-style policy proposals
- **Speaker notes:** Direct career relevance. Students from each path should see themselves.

39.2.3 Track 1 Existing Slides to Reuse

From Class 26 (Research Frontiers & Course Synthesis) — 9 slides - Adoption pathway - Open practitioner questions

39.3 Track 2: Research Frontiers (45 Additional Minutes)

39.3.1 Track 2 Learning Objectives

By end of Track 2 session, Track 2 students will be able to:

1. **Identify** the most promising theoretical extensions of GE-LAV (jumps, multi-asset, neural SDE).
2. **Articulate** open research questions in MFG theory applied to private markets.
3. **Recognize** the connection between GE-LAV and adjacent research areas (systemic risk, optimal execution, central clearing).

39.3.2 Track 2 Slide Briefs

Slide T2-31.1 — Theoretical Extensions

- **Title:** “Where GE-LAV theory is incomplete”
- **Jump-diffusion extension:**
 - Compound Poisson jumps in L_t (Lévy OU)
 - Book Ch. 13 sketches; full theory still developing
- **Multi-asset GE-LAV:**
 - Coupled mean-field equilibria across asset classes
 - Open: existence theorem for the multi-asset system
- **Neural SDE calibration:**
 - Data-driven parameter estimation using neural networks
 - Class 29 covers; more research needed

- **Speaker notes:** Each of these is a thesis chapter or research paper opportunity.

Slide T2-31.2 — Open Research Questions

- **Title:** “What’s not yet known”
- **Quantitative open questions:**
 - Stability boundaries (when does $\kappa > \gamma$ fail in practice?)
 - Rate of convergence for finite-N MFG
 - Robust calibration under regime uncertainty
- **Methodological open questions:**
 - When to use master equation vs. simulations
 - High-dimensional generalizations
- **Empirical open questions:**
 - Test the Pigouvian tax welfare gap empirically
 - Cross-validate calibration on subsamples

Slide T2-31.3 — Connections to Adjacent Fields

- **Title:** “GE-LAV in the broader research landscape”
- **Systemic risk literature:**
 - GE-LAV provides an asset-pricing companion to Acemoglu et al. network models
- **Optimal execution:**
 - Similar fixed-point structure (impact depends on flow, flow depends on impact)
 - Adapt MFG techniques across both areas
- **Central clearing literature:**
 - GE-LAV-style Pigouvian instruments analog to default fund / margin requirements
- **Climate finance:**
 - Long-duration risk pricing; transition risk dynamics
- **Speaker notes:** Place GE-LAV in the research ecosystem. Students should see it as one node in a broader network.

Slide T2-31.4 — Career Implications for Researchers

- **Title:** “Where this matters for research careers”
- **PhD students:**
 - Thesis topics in MFG, private market valuation, systemic risk
 - Connection to leading research (Carmona, Delarue, Lasry, Lions)
- **Academic finance:**
 - Empirical papers on the Jensen bias, welfare gap
 - Theoretical papers on extensions

- **Quant researchers:**
 - Implementation of master equation
 - Calibration techniques
- **Speaker notes:** Connect to active research conversations.

39.3.3 Track 2 Existing Slides to Reuse

From Class 26 + Classes 28-29 (Neural SDE + Network): - 14 slides on theoretical frontiers - Advanced empirical/numerical techniques

39.4 Both Tracks: Pre-Session 32 Preparation

39.4.1 What's Due Session 32

- **Final paper:** 20 pages excluding references and appendices
- **Project presentation:** 20 min + 5 min Q&A
- **Submit:** Both via course site by start of Session 32

39.4.2 Last-Minute Project Preparation

- **Re-read your draft:** Address every peer review comment
- **Sleep before the presentation:** Prepared and rested beats stressed and over-prepared
- **Bring backup:** Print slides AND have them on a USB; technology fails
- **Time yourself:** 20 minutes is *fast*; rehearse exactly to time

39.4.3 Session 32 Logistics

- **Location:** [TBD] — confirm room assignment via course site
- **Order:** Random selection on day of session
- **Q&A:** 5 minutes — every Q&A is graded
- **After all presentations:** Brief course synthesis and farewell

39.4.4 Office Hours

- **This week:** Extended office hours available
 - **Schedule via course site calendar**
 - **Limit:** 30 minutes per student to allow access for everyone
-

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Part V

Capstone

40 Session 32: Capstone — Project Presentations

Unit	Capstone (both tracks reunified)
Format	Project presentations + brief course wrap-up
Slides needed	~5 new (course wrap-up)
Assessment milestone	Final paper + presentation due

! Capstone Session

Both tracks reunite for project presentations. Each student presents for 20 minutes + 5-minute Q&A. The Q&A is graded.

Order: Random selection at start of class. Be prepared from the start.

Final paper: Submit via course site before your presentation. Final paper graded separately.

40.1 Session Structure (75 Minutes × Possibly Multiple Class Days)

With 32-session courses typically having 20–30 students, the capstone often spans **2 class meetings** (Session 32 spread across the final week). Coordinate with your TA on scheduling. Assume two 75-minute slots: Session 32-A and Session 32-B.

40.1.1 Per-Student Time Allocation

Element	Duration	What happens
Set-up	1 min	Take seat, audio check, slide load
Presentation	20 min	Project presentation
Q&A	5 min	Questions from instructor, TA, fellow students
Transition	1 min	Wrap up, next student up
Total per student	~27 min	

In a 75-minute class, **2–3 presentations** per class period is the realistic cap.

40.1.2 Total Capstone Time

- 20–30 students × 27 min = ~9–14 hours
- Across 32-A, 32-B, and possibly designated final exam week sessions
- Plan accordingly

40.2 Final Course Wrap-Up Slide Briefs

After all presentations conclude, a brief wrap-up session.

40.2.1 Slide 32.1 — Title

- **Title:** “Course Conclusion”
- **Tag line:** “What we built. Where it goes.”
- **Speaker notes:** Brief but real closure. Acknowledge what students accomplished.

40.2.2 Slide 32.2 — Course Synthesis (One-Slide)

- **Title:** “GE-LAV in one slide (revisited)”
- **Layout:** Same master visual from Session 24
- **The five main results:**
 1. Jensen bias (closed-form)
 2. Optimal exit boundary $L^*(t)$
 3. Welfare gap ($\sim 2.3\%/yr$)
 4. Pigouvian exit tax
 5. Strict valuation hierarchy DCF LAV GE-LAV
- **Speaker notes:** Anchor the framework one last time.

40.2.3 Slide 32.3 — What Each Student Now Knows

- **Title:** “What you can now do”
- **For Track 1 students:**
 - Apply GE-LAV to real private market assets
 - Communicate liquidity-adjusted analysis to IC/board
 - Distinguish DCF/LAV/GE-LAV in practice
 - Make informed allocation, exit, and stress test recommendations
- **For Track 2 students:**
 - Derive the framework’s results from first principles
 - Implement HJB, FP, MFG numerically
 - Extend the framework theoretically
 - Connect to broader MFG/systemic risk research

40.2.4 Slide 32.4 — How to Use This Knowledge

- **Title:** “Where this knowledge goes”
- **Professional use cases:**
 - LP allocation decisions and IPS construction
 - GP valuation discipline and IC communication
 - Regulatory and policy work
 - Consulting / advisory work for institutional investors
 - PhD research extending the framework
- **Continuing education:**
 - GE-LAV platform updates (new asset classes, new calibrations)
 - Regulatory developments (Solvency II updates, IPEV revisions)
 - Academic literature on MFG and private market valuation
- **Speaker notes:** Help students translate course knowledge into career.

40.2.5 Slide 32.5 — Personal Note

- **Title:** “Thank you”
- **Brief acknowledgment of the cohort:**
 - First time teaching this course
 - Recognition of student engagement
 - Encouragement to stay in touch (LinkedIn, professional network)
- **Course evaluation:** Reminder to complete
- **Speaker notes:** This is the personal moment. Brief but real.

40.3 How the Presentations Are Graded

Each presentation is graded on the criteria from the [project specification](#):

Criterion	Weight
Clarity of motivation and question	3%
Quality of empirical work and calibration	5%
Correct application of GE-LAV framework	6%
Quality of comparison and economic interpretation	4%
Quality of recommendation/conclusion	3%
Q&A defense	4%
Total presentation	25%

The final paper adds 5% more:

Criterion	Weight
Improvement over draft	2%
Writing quality and polish	2%
Final results are correct	1%
Total final paper	5%

Combined: Project = 30% (presentation 25% + final paper 5%) + 15% (proposal 5% + draft 15%) = 50% of course.

40.4 Q&A Guidelines

40.4.1 For Presenters

- **Anticipate questions:** Re-read your work; what would you ask if you were the instructor?
- **Be honest about limitations:** Saying “I assumed X; the result would change if Y” beats pretending you handled everything
- **Defend with reasoning, not certainty:** “My approach was Z because of A and B” is better than “Z is the obvious choice”
- **Stay calm:** Q&A is meant to test understanding, not to embarrass

40.4.2 Common Questions (Examples)

- “Why did you choose this specific asset?” → Should align with your motivation slide
- “How robust is your calibration to parameter variation?” → Should match your sensitivity analysis
- “Did you consider the McKean-Vlasov externality?” → Critical for stress scenarios
- “What’s the IC-relevant magnitude here?” → Numbers in dollars, not percentages
- “Why does GE-LAV differ from DCF in this case?” → Channel-specific decomposition expected
- “Would Track [other] students have approached this differently?” → Cross-track perspective

40.5 Final Logistics

40.5.1 Submission Checklist (Day of Session 32)

- ☐ Final paper (20 pages + references + appendices) — submitted via course site **before presentation**
- ☐ Presentation slides — uploaded to course site
- ☐ Supporting materials (code, spreadsheets, GE-LAV platform exports) — submitted with paper
- ☐ Course evaluation — completed online
- ☐ Outstanding office hours feedback — submitted

40.5.2 After Session 32

- **Final grades:** Posted within 14 days
- **Re-grade requests:** 7 days from grade posting; submit in writing
- **Course website remains live:** All materials accessible for at least 1 year
- **Reference and recommendation requests:** Email instructor; standard 2-week turnaround

40.6 A Final Word

The GE-LAV framework introduced in this course is **new**. The book was published in 2026; the framework has only been in the public discourse for a year. You — as students in the first iterations of this course — are among the earliest to systematically understand and apply this material.

That means **you carry forward responsibility**:

- For Track 1 students: bring intellectual honesty to GE-LAV adoption. Help institutions use it well.
- For Track 2 students: extend the framework. Test it. Critique it. Make it stronger.

If you find errors, omissions, or improvements in the course or the framework, please share them. The course will be revised; the framework will evolve. Your contribution shapes both.

Thank you for the semester.

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Part VI

Appendices

41 Appendix A: Mathematical Foundations

This appendix collects the mathematical prerequisites and intermediate results needed for the course. It is organized by mathematical area, not by session. Use it as a reference; consult specific sections as needed.

Notation conventions: - $\mathbb{R}$ denotes the real numbers; $\mathbb{R}^n$ the n -dimensional Euclidean space - $\mathbb{N} = \{0, 1, 2, \dots\}$ denotes the non-negative integers - $\mathbb{P}$ denotes a probability measure; $E[\cdot]$ denotes expectation under $\mathbb{P}$ - $\mathcal{N}(\mu, \sigma^2)$ denotes the normal distribution with mean μ and variance σ^2 - W_t denotes a standard Brownian motion

41.1 A.1 Probability Theory Essentials

41.1.1 A.1.1 Random Variables and Distributions

A **random variable** X is a measurable function from a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ to $\mathbb{R}$.

Expectation: $E[X] = \int_{\Omega} X(\omega) d\mathbb{P}(\omega)$

Variance: $\text{Var}(X) = E[(X - E[X])^2] = E[X^2] - (E[X])^2$

Conditional expectation: $E[X|\mathcal{G}]$ for σ -algebra $\mathcal{G} \subseteq \mathcal{F}$. The conditional expectation is itself a random variable, measurable with respect to $\mathcal{G}$.

Key property (tower): $E[E[X|\mathcal{G}]] = E[X]$

41.1.2 A.1.2 The Normal Distribution

If $X \sim \mathcal{N}(\mu, \sigma^2)$:

- Density: $f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$
- Mean: μ ; Variance: σ^2
- Moment generating function: $M_X(t) = e^{\mu t + \sigma^2 t^2/2}$
- Standardization: $Z = (X - \mu)/\sigma \sim \mathcal{N}(0, 1)$

Important moments:

Moment	Value
$E[X]$	μ
$\text{Var}(X)$	σ^2
$E[(X - \mu)^3]$	0 (symmetry)
$E[(X - \mu)^4]$	$3\sigma^4$

41.1.3 A.1.3 Jensen's Inequality (Foundational for GE-LAV)

Statement: For a convex function $f : \mathbb{R} \rightarrow \mathbb{R}$ and any random variable X with finite expectation:

$$f(E[X]) \leq E[f(X)]$$

with strict inequality when f is strictly convex and X is non-degenerate.

For concave f : the inequality reverses: $f(E[X]) \geq E[f(X)]$.

Proof sketch (convex case): By the supporting hyperplane property, for any x_0 : $f(x) \geq f(x_0) + f'(x_0)(x - x_0)$. Take $x_0 = E[X]$ and expectations of both sides: $E[f(X)] \geq f(E[X]) + f'(E[X]) \cdot E[X - E[X]] = f(E[X])$.

Application to GE-LAV: The discount factor $f(L) = e^{-r(L)T}$ is convex in L when $r(L)$ is appropriately convex. This produces the systematic *upward* Jensen bias in LAV vs. DCF.

41.1.4 A.1.4 Conditional Expectation Properties

For random variables X, Y :

- **Linearity:** $E[aX + bY|\mathcal{G}] = aE[X|\mathcal{G}] + bE[Y|\mathcal{G}]$
- **Pull-out:** If Z is $\mathcal{G}$ -measurable: $E[ZX|\mathcal{G}] = ZE[X|\mathcal{G}]$
- **Tower property:** $E[E[X|\mathcal{G}]] = E[X]$
- **Independence:** If X independent of $\mathcal{G}$: $E[X|\mathcal{G}] = E[X]$

41.2 A.2 Stochastic Processes

41.2.1 A.2.1 Definition and Classification

A **stochastic process** $(X_t)_{t \in T}$ is a collection of random variables indexed by time. Common index sets: $T = \mathbb{N}$ (discrete time) or $T = [0, \infty)$ (continuous time).

Classification by state space: - Discrete state: e.g., Markov chains on $\{0, 1, 2, \dots\}$ - Continuous state: e.g., Brownian motion, OU process

Classification by index set: - Discrete time: $(X_n)_{n \in \mathbb{N}}$ - Continuous time: $(X_t)_{t \geq 0}$

41.2.2 A.2.2 Markov Property

A process (X_t) has the **Markov property** if, for all $s < t$:

$$\mathbb{P}(X_t \in A | \mathcal{F}_s) = \mathbb{P}(X_t \in A | X_s)$$

where $\mathcal{F}_s$ is the natural filtration.

In words: “future depends on past only through the present.” All the SDE-based processes in this course (Brownian motion, OU, GE-LAV state) are Markov.

41.2.3 A.2.3 Random Walks (Discrete-Time Building Block)

A **simple random walk** $(S_n)_{n=0,1,2,\dots}$: - $S_0 = 0$ - $S_n = X_1 + X_2 + \dots + X_n$ - X_i i.i.d. with $\mathbb{P}(X_i = +1) = \mathbb{P}(X_i = -1) = 1/2$

Properties: - $E[S_n] = 0$ - $\text{Var}(S_n) = n$ (variance grows linearly with time) - Recurrent in 1D, 2D; transient in 3D+

Scaling limit: As $n \rightarrow \infty$ with appropriate scaling, the random walk converges to **Brownian motion** (Donsker’s theorem).

41.3 A.3 Brownian Motion

41.3.1 A.3.1 Definition

A stochastic process W_t is a **standard Brownian motion** if:

1. $W_0 = 0$
2. Paths $t \mapsto W_t$ are continuous almost surely
3. Increments are independent: for $s < t$, $W_t - W_s$ is independent of $\mathcal{F}_s$
4. Increments are normally distributed: $W_t - W_s \sim \mathcal{N}(0, t - s)$

41.3.2 A.3.2 Key Properties

Martingale property: $E[W_t | \mathcal{F}_s] = W_s$ for $s \leq t$.

Quadratic variation: $[W, W]_t = t$ — Brownian motion accumulates quadratic variation at unit rate.

Self-similarity (scaling): For any $c > 0$, the process $(c^{-1/2}W_{ct})$ has the same law as (W_t) .

Non-differentiability: Almost surely, W_t is nowhere differentiable.

Time-reversal: The process $(W_T - W_{T-t})_{0 \leq t \leq T}$ has the same law as $(W_t)_{0 \leq t \leq T}$ for any $T > 0$.

41.3.3 A.3.3 Multidimensional Brownian Motion

d -dimensional standard Brownian motion: $W_t = (W_t^1, W_t^2, \dots, W_t^d)$ where each W_t^i is an independent 1D standard Brownian motion.

Cross-covariance: $\text{Cov}(W_t^i, W_t^j) = t \cdot \delta_{ij}$ (Kronecker delta).

41.4 A.4 Itô Calculus

41.4.1 A.4.1 Stochastic Integral

For an $\mathcal{F}_t$ -adapted process $f(t, \omega)$ satisfying $\int_0^T E[f(t, \omega)^2] dt < \infty$, the **Itô integral**:

$$\int_0^T f(t, \omega) dW_t$$

is well-defined as the L^2 limit of step-function approximations.

Key property (Itô isometry):

$$E \left[\left(\int_0^T f dW_t \right)^2 \right] = \int_0^T E[f^2] dt$$

This is the *square* of the standard L^2 norm of f — the foundation for many calculations.

41.4.2 A.4.2 Itô SDEs

A stochastic differential equation in Itô form:

$$dX_t = \mu(X_t, t)dt + \sigma(X_t, t)dW_t$$

with initial condition X_0 , has a unique strong solution under standard conditions: - μ, σ Lipschitz continuous - μ, σ have linear growth bound

41.4.3 A.4.3 Itô's Lemma (Chain Rule)

For a smooth function $f : \mathbb{R} \rightarrow \mathbb{R}$ and an Itô process X_t :

$$df(X_t) = f'(X_t)dX_t + \frac{1}{2}f''(X_t)\sigma^2(X_t, t)dt$$

The Itô correction $\frac{1}{2}f''(X_t)\sigma^2 dt$ is what distinguishes stochastic calculus from ordinary calculus. It is the second-order Taylor term that becomes first-order because $(dW_t)^2 \rightarrow dt$ at order 1.

Multivariate version: For $f : \mathbb{R}^d \rightarrow \mathbb{R}$:

$$df(X_t) = \sum_i \partial_i f(X_t) dX_t^i + \frac{1}{2} \sum_{i,j} \partial_{ij} f(X_t) [dX^i, dX^j]_t$$

where $[dX^i, dX^j]_t = (\sigma \sigma^T)_{ij}(X_t, t) dt$.

41.4.4 A.4.4 Connection to Jensen's Inequality

Apply Itô's lemma to $f(X) = e^{-rX}$ with $r > 0$ (convex function): - $f'(X) = -re^{-rX}$ - $f''(X) = r^2 e^{-rX}$

If X is constant (zero drift, zero diffusion): no correction. If X is stochastic: $E[f(X_T)] > f(E[X_T])$ by Jensen, with magnitude $\frac{r^2}{2} \text{Var}(X_T)$ to leading order.

This is exactly the Jensen bias mechanism that makes $V^{LAV} > V^{DCF}$ when $r(L)$ is convex in L and L is stochastic.

41.5 A.5 Ornstein-Uhlenbeck Process

41.5.1 A.5.1 Definition

The OU process is the Itô SDE:

$$dL_t = \kappa(\bar{L} - L_t)dt + \sigma dW_t$$

with parameters $\kappa > 0$ (mean reversion speed), $\bar{L}$ (long-run mean), $\sigma > 0$ (volatility).

41.5.2 A.5.2 Closed-Form Solution

Using the integrating factor $e^{\kappa t}$:

$$L_t = e^{-\kappa t} L_0 + \bar{L}(1 - e^{-\kappa t}) + \sigma \int_0^t e^{-\kappa(t-s)} dW_s$$

The integral is Gaussian (linear in dW), so L_t given L_0 is Gaussian.

Conditional distribution:

$$L_t | L_0 \sim \mathcal{N} \left(\bar{L} + (L_0 - \bar{L})e^{-\kappa t}, \frac{\sigma^2}{2\kappa}(1 - e^{-2\kappa t}) \right)$$

41.5.3 A.5.3 Stationary Distribution

As $t \rightarrow \infty$, the OU distribution converges:

$$L_\infty \sim \mathcal{N}\left(\bar{L}, \frac{\sigma^2}{2\kappa}\right)$$

This is the **invariant distribution** of the OU process — independent of starting point L_0 .

41.5.4 A.5.4 Calibrated Parameters (GE-LAV)

Parameter	Value	Interpretation
κ	0.45/yr	Mean reversion speed
σ	0.32	Volatility coefficient
$\bar{L}$	1.0 (or 0.0 depending on normalization)	Long-run mean
$\text{Var}(L_\infty)$	0.1138	Stationary variance
$\text{Std}(L_\infty)$	0.337	Stationary standard deviation
Half-life	1.54 years	$\ln(2)/\kappa$

41.6 A.6 Variational Inequalities and Free Boundaries

41.6.1 A.6.1 The American Option Analogue

The GE-LAV exit problem is an *optimal stopping* problem. The value function satisfies:

$$\min \{-\mathcal{L}V + rV - C, \quad V - G\} = 0$$

where: - $\mathcal{L}$ is the HJB operator (drift + diffusion + time) - $G(L)$ is the exit payoff - The first argument is the continuation condition - The second argument is the exercise condition

This is mathematically identical to American option pricing.

41.6.2 A.6.2 Smooth Pasting

At the free boundary $L^*(t)$: 1. **Value match:** $V(L^*, t) = G(L^*)$ 2. **Smooth pasting:** $\partial_L V(L^*, t) = G'(L^*)$

These two conditions uniquely determine $L^*(t)$.

41.6.3 A.6.3 Numerical Methods

For the GE-LAV exit problem: - **Discretize the state space:** $L \in [-3, 3]$ on a 200-point grid - **Discretize time:** Backward induction from $t = T$ to $t = 0$ - **At each step:** Solve the HJB PDE assuming continuation, then take maximum with exit payoff - **Output:** The exit boundary $L^*(t)$ emerges from the switching surface

41.7 A.7 Convex Analysis Essentials

41.7.1 A.7.1 Convex Functions

A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is **convex** if:

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y)$$

for all $x, y \in \mathbb{R}$ and $\lambda \in [0, 1]$.

Strict convexity: Inequality is strict whenever $x \neq y$.

Equivalent for twice-differentiable f : $f''(x) \geq 0$ for all x (strictly convex if $f''(x) > 0$).

41.7.2 A.7.2 Convexity in GE-LAV

The premium function $\pi(L) = \pi_0 - \pi_1 L + \pi_2 L^2$ with $\pi_2 > 0$ is **strictly convex**.

The discount factor $e^{-r(L)T}$ as a function of L is convex (since composition of convex with affine of L is convex; with $r(L)$ convex, even more curvature).

This convexity is the *source* of the Jensen bias in private market valuation.

41.8 A.8 Game Theory: Nash Equilibrium and Mean-Field

41.8.1 A.8.1 Nash Equilibrium

A **Nash equilibrium** for a game with N players is a profile $(s_1^*, \dots, s_N^*)$ such that:

$$U_i(s_i^*, s_{-i}^*) \geq U_i(s_i, s_{-i}^*) \quad \forall s_i, \forall i$$

Each player's strategy is a best response given others' strategies; mutually consistent.

Existence (Nash 1950): Every finite game has at least one Nash equilibrium in mixed strategies. For games with continuous strategies, additional regularity needed.

41.8.2 A.8.2 Mean-Field Approximation

When N is large, tracking individual strategies is intractable. Replace with the *distribution* of strategies across the population:

$$\mu_t = \text{distribution of } (s_t^i) \text{ across } i$$

Each player reacts to μ_t rather than to individual others. As $N \rightarrow \infty$, this becomes exact (propagation of chaos).

41.8.3 A.8.3 Fixed-Point Structure

A **mean-field equilibrium** is a distribution μ^* such that: 1. Each player's optimal strategy, given μ^* , is $s^*(\mu^*)$ 2. The aggregate distribution of $s^*(\mu^*)$ across the population equals μ^*

This is a fixed point: $\mu^* = \Phi(\mu^*)$ where Φ is the mapping from population state to individual best response back to aggregate.

Existence: Schauder fixed-point theorem (for compact convex sets in topological vector spaces).

Uniqueness: Banach fixed-point theorem under contraction (stability condition).

41.9 A.9 Optimization and Lagrange Methods

41.9.1 A.9.1 Unconstrained Optimization

For smooth $f : \mathbb{R}^n \rightarrow \mathbb{R}$, an interior critical point satisfies $\nabla f = 0$.

Second-order condition: Hessian $\nabla^2 f$ positive (negative) definite for local minimum (maximum).

41.9.2 A.9.2 Constrained Optimization

For max $f(x)$ subject to $g(x) = 0$: **Lagrangian:** $L(x, \lambda) = f(x) - \lambda g(x)$

First-order conditions: $\nabla_x L = 0$, $\nabla_\lambda L = 0$ (i.e., $g(x) = 0$).

41.9.3 A.9.3 Variational Methods

For functionals $J[u] = \int F(u, u', t) dt$, the Euler-Lagrange equation:

$$\frac{\partial F}{\partial u} - \frac{d}{dt} \frac{\partial F}{\partial u'} = 0$$

The HJB equation arises from applying variational methods to the value function in optimal control.

41.10 A.10 Quick Reference Formulas

41.10.1 A.10.1 OU Calibration

At canonical values $\kappa = 0.45$, $\sigma = 0.32$: - Stationary variance: 0.1138 - Stationary std: 0.337 - Half-life: 1.54 yr - $1 - e^{-\kappa t}$ at $t = 5$: 0.895 - $1 - e^{-2\kappa t}$ at $t = 5$: 0.989

41.10.2 A.10.2 Jensen Bias

- Affine approximation: $B(T) = A \cdot T + C$ with $A \approx 0.16 - 0.18\%/yr$ by asset class
- Closed-form: $B(T) = (\pi_2/2)\Pi_{liq}(T)$ where $\Pi_{liq}(T) = \sigma^2[T - (1 - e^{-2\kappa T})/(2\kappa)]$
- At calibrated $\pi_2 = 0.021$: $B(5) \approx 0.4\%$, $B(15) \approx 1.5\%$, $B(20) \approx 2.0\%$

41.10.3 A.10.3 Effective Rate

- DCF rate: typically constant at $r_f + \beta \cdot ERP + \pi_0 \approx 7.5\%$
- LAV rate: path-dependent at $r(L_t) = r_f + \pi(L_t)$
- GE-LAV rate: $r_{GE}(L, \mu) = r(L) + \text{equilibrium uplift}$
- At GFC depth ($L = -1.5$): $r_{GE} \approx 32\%$ vs. DCF 7.5% $\rightarrow 4.31\times$ amplification

41.10.4 A.10.4 Welfare

- Welfare gap: $\Delta W \approx 2.3\%/yr$
 - Aggregate: $\$13T \times 2.3\% = \$300B/yr$
 - Pigouvian tax at GFC depth: $\tau^* \approx 7\%$ on secondary transactions
-

41.11 A.11 Mathematical Maturity Required for Each Track

41.11.1 Track 1 (Practitioner)

Required: - Probability theory through expectations and variances (A.1) - Comfortable with the normal distribution (A.1.2) - Aware of Jensen's inequality and its direction (A.1.3, A.7) - Basic familiarity with stochastic processes as concepts (A.2)

Recommended but not required: - Recognition of SDE notation - Awareness of HJB equation as a concept (not derivation) - Comfort reading mathematical statements in lectures

41.11.2 Track 2 (Researcher)

Required (in addition to Track 1): - Itô calculus, including Itô's lemma derivation (A.4) - OU process derivation and properties (A.5) - Variational inequalities and free boundary problems (A.6) - Convex analysis (A.7) - Game theory and mean-field methods (A.8)

Helpful: - Functional analysis (Hilbert spaces, weak topologies) - Numerical analysis (finite differences, iterative methods) - Measure theory (Lebesgue integration, Radon-Nikodym)

41.12 A.12 Suggested Self-Diagnostic

Before the course (or by Session 4 at the latest), confirm understanding of:

For Track 1: - ☐ Can you state Jensen's inequality and identify whether $E[X^2] \geq (E[X])^2$ is consistent with it? - ☐ If $X \sim \mathcal{N}(0, 1)$, what is $\mathbb{P}(X > 1.96)$? Approximately? - ☐ If a random variable has mean 0 and variance 9, what is its standard deviation? - ☐ What is a Markov chain, informally?

For Track 2: - ☐ Can you derive Itô's lemma from a Taylor expansion? - ☐ Can you compute $E \left[\int_0^t e^{-\kappa(t-s)} dW_s \right]$ using the Itô isometry? - ☐ Can you state the HJB equation for a generic control problem? - ☐ What is a fixed-point theorem? Name one.

If you cannot answer the Track 1 questions, the practitioner track will be challenging — additional preparation recommended. If you cannot answer the Track 2 questions, the researcher track will be very challenging — strongly recommend remediation first.

[← Reading List](#) | [Course Home](#)

42 Reading List

This document organizes the course readings by unit and session, with brief annotations indicating relevance. Full BibTeX citations are in [references.bib](#).

42.1 How to Use This Reading List

Required readings are essential for the corresponding session and are testable on assessments. **Recommended readings** deepen understanding but are not assessable. **Track-specific readings** target Track 1 (Practitioner) or Track 2 (Researcher) students differently.

If you take only one reading from each unit beyond the required, take the *recommended Track-aligned* one.

42.2 Primary Text

Asaf, S. (2026). *Liquidity Illusion: The General Equilibrium Theory of Private Capital Valuation*. Oxford University Press. 421 pp. ISBN 9798881716103.

The primary textbook for the course. All chapters referenced extensively. Available in print and electronic from the publisher; course site provides licensed access.

42.3 Unit 1: Why DCF Fails (Sessions 1–5)

42.3.1 Required Readings

Asaf (2026), **Chapters 1–3:** - Ch. 1: “The Private Capital Valuation Problem” — frames the gap between DCF and observed reality - Ch. 2: “The Three Structural Failures” — formal statement of the failures DCF misses - Ch. 3: “Secondary Markets and the Liquidity State” — empirical evidence

42.3.2 Recommended (Track 1)

Kaplan, S. N., & Strömberg, P. (2009). “Leveraged Buyouts and Private Equity.” *Journal of Economic Perspectives*, 23(1), 121-146.

Comprehensive overview of PE buyout structure, written for non-specialists. Essential context for understanding how PE actually works.

Sensoy, B. A., Wang, Y., & Weisbach, M. S. (2014). “Limited Partner Performance and the Maturing of the Private Equity Industry.” *Journal of Financial Economics*, 112(3), 320-343.

Empirical evidence on how different LP types fare in PE — provides the “what’s the actual experience” backdrop the framework explains.

42.3.3 Recommended (Track 2)

Amihud, Y., & Mendelson, H. (1986). “Asset Pricing and the Bid-Ask Spread.” *Journal of Financial Economics*, 17(2), 223-249.

The foundational paper introducing illiquidity premium to asset pricing. Background for understanding why illiquidity is priced and why it matters.

Cochrane, J. H. (2005). “The Risk and Return of Venture Capital.” *Journal of Financial Economics*, 75(1), 3-52.

Demonstrates the difficulty of estimating PE returns properly; the empirical headwind that GE-LAV partly addresses.

42.3.4 Optional Deep Dive

Stafford, E. (2022). “Replicating Private Equity with Value Investing.” *Review of Financial Studies*, 35(6), 2799-2848.

Argues that PE returns can be replicated with public market value strategies — partly true if you accept conventional return measurement; partly wrong because of liquidity adjustment.

42.4 Unit 2: Measurement and Theory (Sessions 6–10)

42.4.1 Required Readings

Asaf (2026), Chapters 4–5: - Ch. 4: “IRR, PME, and the Measurement Problem” - Ch. 5: “What a Correct Theory Must Deliver”

Kaplan, S. N., & Schoar, A. (2005). “Private Equity Performance: Returns, Persistence, and Capital Flows.” *Journal of Finance*, 60(4), 1791-1823.

The KS-PME methodology origin paper. Essential reading for understanding PME.

42.4.2 Recommended (Track 1)

Gredil, O., Griffiths, B., & Stucke, R. (2020). “Benchmarking Private Equity: The Direct Alpha Method.” Working paper.

Provides the Direct Alpha methodology, a refinement of PME. Useful for practitioners benchmarking PE positions.

Harris, R. S., Jenkinson, T., & Kaplan, S. N. (2014). “Private Equity Performance: What Do We Know?” *Journal of Finance*, 69(5), 1851-1882.

The canonical empirical study on PE returns. Provides industry benchmark data the GE-LAV framework must explain.

42.4.3 Recommended (Track 2)

Long, A. M., & Nickels, C. J. (1996). “A Method for Comparing Private Market Internal Rates of Return to Returns on Publicly Traded Securities.” Working paper.

Original Long-Nickels PME methodology. Background for understanding PME variants.

Korteweg, A., & Sorensen, M. (2010). “Risk and Return Characteristics of Venture Capital-Backed Entrepreneurial Companies.” *Review of Financial Studies*, 23(10), 3738-3772.

Methodological paper on estimating PE/VC return distributions; relevant for Track 2 students working on calibration.

42.5 Unit 3: Decision and Application (Sessions 11–18)

42.5.1 Required Readings

Asaf (2026), Chapters 6–9: - Ch. 6: “Exit Timing and the Trapped Investor Problem” - Ch. 7: “Portfolio Construction with Liquidity Hedge Demand” - Ch. 8: “Crisis Dynamics and Regulatory Implications” - Ch. 9: “Implementation: The GE-LAV® Platform”

42.5.2 Recommended (Track 1)

Robinson, D. T., & Sensoy, B. A. (2016). “Cyclicalities, Performance Measurement, and Cash Flow Liquidity in Private Equity.” *Journal of Financial Economics*, 122(3), 521-543.

Empirical work on vintage-year effects — the data backdrop for the LP exit timing problem.

Ang, A., Chen, B., Goetzmann, W. N., & Phalippou, L. (2018). “Estimating Private Equity Returns from Limited Partner Cash Flows.” *Journal of Finance*, 73(4), 1751-1783.

Methodology for estimating PE returns when only LP cash flow data is available — relevant for portfolio construction work.

Ivashina, V., & Lerner, J. (2019). *Patient Capital: The Challenges and Promises of Long-Term Investing*. Princeton University Press.

Book-length treatment of the institutional politics around long-duration investing. Highly relevant for the practitioner track and for project work.

42.5.3 Recommended (Track 2)

Merton, R. C. (1969). “Lifetime Portfolio Selection under Uncertainty: The Continuous-Time Case.” *Review of Economics and Statistics*, 51(3), 247-257.

Origin of Merton allocation; the benchmark GE-LAV improves on for portfolio construction.

Merton, R. C. (1971). “Optimum Consumption and Portfolio Rules in a Continuous-Time Model.” *Journal of Economic Theory*, 3(4), 373-413.

Extends 1969 paper; the canonical reference for continuous-time portfolio theory.

42.5.4 Recommended (Both Tracks)

Financial Stability Board (2023). “Private Markets and Systemic Risk.” FSB Technical Report.

Recent FSB analysis of private market growth and stability concerns. Provides the regulatory context where GE-LAV insights apply.

42.6 Unit 4: Math Intuition Bridges (Sessions 19–24)

42.6.1 Required Readings

Asaf (2026), Chapters 10–20: - Ch. 10: “Brownian Motion and Itô Calculus” (concepts) - Ch. 11: “Stochastic Control and HJB” (concepts) - Ch. 13: “Mean-Field Games” (concepts) - Ch. 14–15: “Fokker-Planck and Master Equation” (concepts) - Ch. 16–17: “LAV Operator and GE Equilibrium” (concepts) - Ch. 18–20: “Jensen, Pigouvian, Welfare” (results)

42.6.2 Recommended (Track 1)

Most Track 1 students will find the math chapters challenging at full rigor. Recommended approach: **read for conceptual understanding only**; skim derivations; focus on results and applications.

Hull, J. C. (2017). *Options, Futures, and Other Derivatives* (10th ed.). Pearson.

Standard derivatives textbook. Provides background on Itô calculus, Black-Scholes, etc. at an MBA-accessible level. Use as reference for unfamiliar concepts.

42.6.3 Recommended (Track 2)

Karatzas, I., & Shreve, S. E. (1991). *Brownian Motion and Stochastic Calculus* (2nd ed.). Springer.

The canonical graduate text on stochastic calculus. Reference for Sessions 19 (Brownian motion, Itô) and 27 (Fokker-Planck).

Pham, H. (2009). *Continuous-time Stochastic Control and Optimization with Financial Applications*. Springer.

The canonical graduate text on stochastic control. Reference for Session 20 (HJB) and Session 25 (HJB derivation).

Carmona, R., & Delarue, F. (2018). *Probabilistic Theory of Mean Field Games with Applications* (Vols. I and II). Springer.

The canonical reference for MFG theory. Essential for Sessions 21 and 26.

Pavliotis, G. A. (2014). *Stochastic Processes and Applications: Diffusion Processes, the Fokker-Planck and Langevin Equations*. Springer.

Comprehensive treatment of Fokker-Planck equations. Reference for Session 22 and 27.

42.6.4 Original Sources (Optional, Historical Interest)

McKean Jr., H. P. (1966). “A Class of Markov Processes Associated with Nonlinear Parabolic Equations.” *Proceedings of the National Academy of Sciences*, 56(6), 1907-1911.

Vlasov, A. A. (1938). “On the Kinetic Theory of Plasma.” *Journal of Experimental and Theoretical Physics*, 8, 291.

Sznitman, A. S. (1991). “Topics in Propagation of Chaos.” In *École d’Été de Probabilités de Saint-Flour XIX—1989*, 165-251. Springer.

Lasry, J. M., & Lions, P. L. (2007). “Mean Field Games.” *Japanese Journal of Mathematics*, 2(1), 229-260.

42.7 Unit 5: Split Track (Sessions 25–31)

42.7.1 Track 1: Practitioner Case Workshops

Cases will be distributed at session start. The following are not required but provide relevant context:

Track 1 Session 25 (PE Buyout): - Bain & Company. “Global Private Equity Report” (annual). Industry data and trends.

Track 1 Session 26 (VC + Secondaries): - Lazard / Evercore. “Secondary Market Reports” (annual). Recent transaction data and trends.

Track 1 Session 27 (Infrastructure): - McKinsey Global Institute. “The Global Private Markets Review” (annual). - Macquarie Annual Reports — specific infrastructure asset case studies.

Track 1 Session 28 (Private Credit): - PitchBook / Preqin. “Private Credit Reports” (quarterly). - Cliffwater / Lincoln. “Direct Lending Market Reports.”

Track 1 Session 29 (Digital + Climate): - Galaxy Digital, Token Terminal — recent reports on crypto/defi market. - IEA, BloombergNEF — climate infrastructure investing reports.

42.7.2 Track 2: Research Frontiers

Track 2 Session 25 (HJB derivation): - Already covered: Pham (2009), Karatzas & Shreve (1991).

Track 2 Session 26 (MFG proofs): - Already covered: Carmona & Delarue (2018), Sznitman (1991).

Track 2 Session 27 (Fokker-Planck): - Already covered: Pavliotis (2014). - **Risken, H. (1996).** *The Fokker-Planck Equation: Methods of Solution and Applications* (2nd ed.). Springer.

Track 2 Session 28 (GE existence): - **Mas-Colell, A. (1985).** *The Theory of General Economic Equilibrium*. Cambridge University Press.

Track 2 Session 29 (Welfare and Pigouvian): - **Mas-Colell, A., Whinston, M. D., & Green, J. R. (1995).** *Microeconomic Theory*. Oxford University Press. Especially Chapters 11–13 (externalities). - **Pigou, A. C. (1920).** *The Economics of Welfare*. Macmillan. Historical origin of Pigouvian taxation concept.

42.8 Unit 5 Frontier: Open Research

Track 1 Session 31:

- **GIIN, IMP+ACT.** Recent reports on impact-aligned private market investing.
- **Industry working papers** on regulatory developments (Solvency II review, EU SFDR, etc.).

Track 2 Session 31:

- **Cetra, J., Lacker, D., & Webster, K. (2024).** “Mean-field games in finance.” *Annual Review of Financial Economics*. (Recent literature review of MFG applications.)
- **Acemoglu, D., Ozdaglar, A., & Tahbaz-Salehi, A. (2015).** “Systemic Risk and Stability in Financial Networks.” *American Economic Review*, 105(2), 564-608. (Network-based systemic risk; adjacent literature.)
- **Almgren, R., & Chriss, N. (2001).** “Optimal Execution of Portfolio Transactions.” *Journal of Risk*, 3, 5-39. (Adjacent optimal execution literature using similar tools.)

42.9 Recommended Periodicals (Ongoing Reading)

For Track 1 students entering practitioner roles:

- *Pensions & Investments* (weekly trade publication)
- *Institutional Investor* (monthly)
- *Private Equity International* (monthly)
- *Pensions Age* (UK)

For Track 2 students continuing in research:

- *Journal of Finance*
- *Review of Financial Studies*
- *Journal of Financial Economics*
- *Annual Review of Financial Economics*

For both tracks, regulatory tracking:

- *Financial Stability Board* reports
 - *EIOPA*, *FCA*, *SEC* speeches and rule announcements
 - *BIS Working Papers* (Basel and beyond)
-

42.10 How to Build a Personal Reading Stack

If you're continuing GE-LAV-related work post-course, build a personal stack of ~20-30 key papers organized by:

1. **5-7 foundational papers** (Kaplan-Schoar, Cochrane, Sensoy-Wang-Weisbach, etc.) — re-read annually
2. **5-7 technical references** (Carmona-Delarue, Pham, Karatzas-Shreve) — keep nearby
3. **5-7 current frontier papers** (rotated annually as new work emerges)
4. **5-7 industry reports** (annual cycle)

The book *Liquidity Illusion* serves as the anchor; everything else builds outward.

[← Syllabus](#) | [References.bib](#) [→](#)

References